

Supplementary Information for *Dendritic morphology as a dictionary for local credit assignment*

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12 **How to use this supplement**

13 The notes follow the scientific questions in the main paper. They distinguish three roles for a tree: serial forward
14 computation, spatial delivery of learning signals, and a reconstructed cable model. A spatial dictionary is a set
15 of delivery profiles; the coefficients that combine them are a separate information resource. Exact transported
16 errors, best-fit projections and supplied reliability moments are identified as oracles when a restricted learner cannot
17 calculate them.

18 Each experiment specifies its independent unit. Synapses, sites, scans, data splits and channel draws remain
19 nested within cells, targets or training seeds. All scientific outcomes and exclusions are retained in Source Data,
20 including comparisons summarized numerically rather than drawn again. Tables S11 and S12 locate the cohorts and
21 their complete records. Negative, small and budget-dependent results are reported beside the claims they qualify.

Main figure	Scientific question	Supplementary support
1	Exact local rules and image credit	Sections S1–S2; Figs. S1–S8
2–3	Branch conflict and ancestry	Section S3; Figs. S9–S10
4	Interactions and credit resolution	Section S4; Figs. S11–S16
5	Local conductance gating	Section S5; Figs. S17–S20
22 6	Serial computation and budgets	Section S6; Figs. S21–S23
7	Anatomical dictionaries	Section S7; Figs. S24–S27
8	Conductance and route gain	Section S8; Figs. S28–S29
9	Measured-response evidence	Section S9; Figs. S30–S32
Design exten- sion	What can be selected from data	Section S10; Figs. S33–S35

S1 Exact gradients and local learning rules

1.1 Exact gradients in a directed conductance tree

To determine which factors a local learning rule can compute, we first derive the exact conductance gradient, following the factorization and LocalCA rule introduced in our predecessor preprint [1]. This requires the forward steady state, the local synaptic sensitivity and transport of the somatic loss derivative through the tree.

1.2 Steady-state compartment voltage

Consider a rooted dendritic tree with soma node 0. Let V_n and C_n be the voltage and capacitance of compartment n , and let $\mathcal{C}_n$ be its set of children. A non-somatic compartment n receives excitatory contacts $\mathcal{S}_n^E$ and inhibitory contacts $\mathcal{S}_n^I$, with $\mathcal{S}_n = \mathcal{S}_n^E \cup \mathcal{S}_n^I$. A contact of type $X \in \{E, I\}$ has nonnegative maximal conductance g_i^X , nonnegative presynaptic activity x_i^X , open conductance $g_i^X x_i^X$, and reversal E_X . We use $E_i^{\text{rev}} = E_E$ on $\mathcal{S}_n^E$ and $E_i^{\text{rev}} = E_I$ on $\mathcal{S}_n^I$; when contact type is not at issue, we write the generic g_i^{syn} , x_i , and E_i^{rev} . Define the aggregate open conductances

$$G_n^E = \sum_{i \in \mathcal{S}_n^E} g_i^E x_i^E, \quad G_n^I = \sum_{j \in \mathcal{S}_n^I} g_j^I x_j^I.$$

The compartment also receives child-to-parent coupling conductances $g_{c \rightarrow n}^{\text{den}}$, and has leak g_n^L with reversal E_L . A child transmits $a_c = f_c(V_c)$, where f_c is the branch output nonlinearity (named “reactivation” in the implementation) and is the identity when that mechanism is disabled. Before taking the steady state, the directed compartmental dynamics are

$$C_n \dot{V}_n = g_n^L (E_L - V_n) + G_n^E (E_E - V_n) + G_n^I (E_I - V_n) + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} (a_c - V_n). \quad (1)$$

The total child coupling is $G_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}$ and its voltage-weighted contribution is $J_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c$. Setting $\dot{V}_n = 0$ and collecting the coefficients of V_n then gives

$$g_n^{\text{tot}} = g_n^L + G_n^E + G_n^I + G_n^{\text{child}}, \quad (2)$$

$$V_n = R_n^{\text{tot}} (g_n^L E_L + G_n^E E_E + G_n^I E_I + J_n^{\text{child}}), \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}. \quad (3)$$

Here g_n^{tot} and R_n^{tot} are the total conductance and its reciprocal, respectively. The soma obeys the same current balance with $n = 0$, including its child couplings and any explicitly modeled somatic inputs, but has no parent term. The controlled regular-tree architectures used here have no external somatic input synapses unless a control is explicitly described otherwise. For nonnegative conductances and activities, Eq. 3 is a conductance-weighted average of leak, synaptic reversal potentials and child activities. This boundedness is useful for numerical stability, but is not required for the chain-rule factorization below. Both E and I increase g_n^{tot} ; they differ through the driving forces $E_E - V_n$ and $E_I - V_n$, not through a signed conductance. When $E_I < V_n < E_E$, the former is depolarizing and the latter hyperpolarizing. At $V_n = E_I$, an inhibitory conductance has zero net current but still reduces R_n^{tot} , producing a pure shunt at that operating point.

The directed artificial-tree simulations use the dimensionless convention $g_n^L = 1$, $E_L = E_I = 0$, and $E_E = 1$, so inhibition is denominator-only. The reconstructed focal-cable analysis uses $E_E = 1$ and $E_I = -0.2$, so its inhibitory perturbation is both hyperpolarizing and shunting. Neither convention is a fit in millivolts.

51 1.3 Local sensitivities and synaptic eligibility

Writing the numerator of Eq. 3 as $b_n = g_n^{\text{tot}} V_n$, quotient differentiation gives

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = \frac{x_i E_i^{\text{rev}} g_n^{\text{tot}} - x_i b_n}{(g_n^{\text{tot}})^2} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n), \quad (4)$$

$$\frac{\partial V_n}{\partial V_c} = f'_c(V_c) g_{c \rightarrow n}^{\text{den}} R_n^{\text{tot}}, \quad (5)$$

$$\frac{\partial V_n}{\partial g_{c \rightarrow n}^{\text{den}}} = R_n^{\text{tot}} (a_c - V_n). \quad (6)$$

52 For a differentiable task loss $\mathcal{L}$, the exact synaptic conductance gradient therefore factors as

$$\frac{\partial \mathcal{L}}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\text{compartment error}}. \quad (7)$$

53 The eligibility contains quantities available at the synapse or its host compartment. All remaining task dependence
 54 lies in the compartment error. In particular, excitatory and inhibitory contacts use $x_i^E R_n^{\text{tot}} (E_E - V_n)$ and $x_j^I R_n^{\text{tot}} (E_I -$
 55 $V_n)$, respectively. Near $V_n = E_I$, the inhibitory contact's own gradient can be small even though its conductance still
 56 reduces R_n^{tot} and changes the sensitivity of other inputs.

57 1.4 Transport of the somatic error

58 Suppose the loss depends on the tree through its somatic output. Let $p(n)$ be the unique parent of non-somatic node
 59 n , and define the somatic voltage error $\delta_0^V = \partial \mathcal{L} / \partial V_0$. The chain rule gives

$$\frac{\partial \mathcal{L}}{\partial V_n} = \frac{\partial \mathcal{L}}{\partial V_{p(n)}} f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}. \quad (8)$$

Because the graph is a tree, iterating Eq. 8 along the unique path from n to the soma yields

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial V_n} = \tilde{\alpha}_n \delta_0^V. \quad (10)$$

60 The effective path gain $\tilde{\alpha}_n$ includes every local transfer derivative. With identity reactivation, it reduces to the
 61 conductance-stage gain α_n^{cond} obtained by removing the f'_i factors. Equation 10 is an exact derivative, not a
 62 separately fitted model parameter.

63 1.5 Raw-parameter transforms

64 Trainable raw parameters θ are transformed to nonnegative conductances by a monotone map $g = \psi(\theta)$. The
 65 optimizer receives

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial g} \psi'(\theta). \quad (11)$$

66 Where $\psi'(\theta) > 0$, this transform preserves the sign of the conductance gradient. Active synapse masks are applied
 67 after forming the conductance-space gradient. The exact-gradient diagnostics compare gradients in the same
 68 parameter space and include the transform derivative.

1.6 General implicit-adjoint form

The preceding path product is a corollary of a more general steady-state identity. Let $\mathbf{V}$, $\mathbf{g}$ and $\mathbf{x}$ collect compartment voltages, conductances and inputs. Suppose their steady state is defined by the residual equations $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}$, with nonsingular voltage Jacobian $J_V = \partial \mathbf{F} / \partial \mathbf{V}$. Define the adjoint $\mathbf{q}$ by

$$J_V^\top \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}. \quad (12)$$

The implicit-function theorem gives

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^\top \frac{\partial \mathbf{F}}{\partial g_i}. \quad (13)$$

For a passive residual $\mathbf{F} = G\mathbf{V} - \mathbf{b}$, let G be the conductance matrix, $\mathbf{b}$ the vector of imposed current contributions and $\mathbf{e}_n$ the coordinate vector for compartment n . A conductance synapse at n has $\partial \mathbf{F} / \partial g_i = x_i(V_n - E_i^{\text{rev}})\mathbf{e}_n$, and therefore

$$\frac{\partial \mathcal{L}}{\partial g_i} = x_i(E_i^{\text{rev}} - V_n)q_n. \quad (14)$$

Both contact classes add positive diagonal loading to G , while their different reversals enter $\mathbf{b}$ and the local driving force in Eq. 14. Equations 12–14 recover the directed path product when J_V is triangular. For the reciprocal passive cable, $J_V = G$ is symmetric positive definite, so $G^{-\top} = G^{-1}$. A general nonsymmetric or recurrent operator instead requires the inverse transpose. Active steady states enter through their Jacobian whenever the required differentiability and nonsingularity conditions hold; no passive symmetry is assumed by the general identity.

1.7 Conductance-specific control of path gain

Holding coupling conductances and the forward state fixed, add effective inhibitory conductance $\Delta G_k^{\text{I}} \geq 0$ at compartment k . This is an activity-weighted aggregate conductance, not an individual maximal synaptic conductance. For a compartment n whose path crosses k ,

$$\frac{\alpha_n^{\text{cond}}(\Delta G^{\text{I}})}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^{\text{I}}}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^{\text{I}}} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n), \end{cases} \quad (15)$$

where $\mathcal{A}(n) = \{k : (i \rightarrow k) \in \text{path}(n \rightarrow 0)\}$ contains the parent endpoints of the path edges: the strict ancestors of n , including the soma if its conductance is varied. Equation 15 proves path-selective attenuation. It does not prove that attenuation improves optimization. No reversal potential appears in Eq. 15 because this calculation retains only the total-conductance denominator while holding the forward state fixed. Any positive conductance has the same direct denominator effect. It is specifically a shunt when E_1 is near the local voltage, so that its accompanying current is small; the full forward and gradient effects retain E_1 .

Let $z_n = \log \alpha_n^{\text{cond}}$ and let β_n be the accumulated log-gain attenuation along path n . The perturbed log gain is $z'_n = z_n - \beta_n$. Across any specified ensemble of paths,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta). \quad (16)$$

Shunting narrows this distribution only if $\text{Cov}(z, \beta) > \text{Var}(\beta)/2$. Path-gain compression is therefore an operating-point-dependent empirical outcome rather than a theorem.

1.8 Local rules and feedback objects

The exact compartment error derived above is generally unavailable to a local synapse. We therefore retain the exact local eligibility and ask what restricted teaching field can replace the transported error.

99 1.9 Three-factor update

100 The three-factor rule retains the exact local eligibility and substitutes an available voltage-space teaching field $\widehat{\delta}_n$
 101 for the exact compartment error:

$$\widehat{\nabla}_{g_i^{\text{syn}}} \mathcal{L} = \left\langle x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_n \right\rangle_B, \quad \widehat{\nabla}_{g_c^{\text{den}}} \mathcal{L} = \left\langle R_n^{\text{tot}} (a_c - V_n) \widehat{\delta}_n \right\rangle_B, \quad (17)$$

102 where B is the number of examples in a minibatch and $\langle \cdot \rangle_B$ denotes their average. Excitatory and inhibitory contacts
 103 use the same expression; their reversal potentials set the sign and magnitude of the driving force.

The four- and five-factor rules multiply Eq. 17 by slowly varying, bounded stage-level scalar reliability estimates. For stage ℓ , mini-batch t , and example b , the implementation averages the recorded layer voltage and corresponding somatic population activity over their feature axes, giving $\bar{V}_{\ell,b}^{(t)}$ and $\bar{V}_{0,b}^{(t)}$. The runs reported here used an uncentered dot product,

$$s_{\ell,t} = \frac{1}{B} \sum_{b=1}^B \bar{V}_{\ell,b}^{(t)} \bar{V}_{0,b}^{(t)}, \quad (18)$$

$$m_{\ell,1} = s_{\ell,1}, \quad m_{\ell,t} = 0.9m_{\ell,t-1} + 0.1s_{\ell,t} \quad (t > 1), \quad (19)$$

$$r_{\ell,t}^{\text{4F}} = \text{clip}_{[0.1,2.0]}(m_{\ell,t}). \quad (20)$$

104 Thus $r_{\ell,t}^{\text{4F}}$ is one scalar for a recorded stage, not a separate branch-wise signal, and is not a centered covariance or
 105 a Pearson correlation. Recorded voltages are detached from automatic differentiation, and the same minibatch
 106 supplies $s_{\ell,t}$ and the local update. The positive clamp prevents a negative or very small dot product from reversing
 107 or eliminating the update.

The five-factor rule retains this multiplier and adds a conditional variance-ratio factor. Let $Y_\ell^{(t)}$ be the recorded stage voltage and $X_\ell^{(t)}$ its source-resolved parent proxy, each centered across mini-batch t . Let $\mathcal{E}_{0.1}[\cdot]$ denote an exponential moving average (EMA) with update rate 0.1. Define

$$v_{Y,\ell,t} = \mathcal{E}_{0.1}[\langle (Y_\ell^{(t)})^2 \rangle], \quad v_{X,\ell,t} = \mathcal{E}_{0.1}[\langle (X_\ell^{(t)})^2 \rangle], \quad c_{YX,\ell,t} = \mathcal{E}_{0.1}[\langle Y_\ell^{(t)} X_\ell^{(t)} \rangle], \quad (21)$$

$$\beta_{\ell,t} = \frac{c_{YX,\ell,t}}{v_{X,\ell,t} + 10^{-3}}, \quad v_{\text{res},\ell,t} = \max\{v_{Y,\ell,t} - \beta_{\ell,t} c_{YX,\ell,t}, 10^{-12}\}, \quad (22)$$

$$\phi_{\ell,t} = \text{clip}_{[0.25,4.0]} \left(\frac{v_{Y,\ell,t}}{v_{\text{res},\ell,t}} \right), \quad \widehat{\nabla}^{\text{5F}} = r_{\ell,t}^{\text{4F}} \phi_{\ell,t} \widehat{\nabla}^{\text{3F}}. \quad (23)$$

108 If $v_{X,\ell,t} \leq 10^{-12}$, the implementation sets $\phi_{\ell,t} = 1$. These factors are empirical preconditioners. They are neither
 109 needed for the exact factorization nor extra task-error channels. Under exact transported error, the three-factor
 110 rule recovers the exact dendritic parameter gradient in the stated steady-state model. This separates feedback
 111 approximation from the validity of the local eligibility term; agreement of trained endpoints with backpropagation
 112 is tested empirically rather than assumed from the identity.

113 1.10 Feedback fields

114 For neuron u with N_u routed dendritic compartments, indexed $n = 1, \dots, N_u$ while the soma retains index $n = 0$, let
 115 $\delta_u^V = (\delta_{1,u}^V, \dots, \delta_{N_u,u}^V)^\top$ collect the exact compartment errors and let $\widehat{\delta}_u^V$ collect the values available to the local rule.
 116 Let $\mathbf{c}_u = (c_{u,1}, \dots, c_{u,K})^\top$ contain the signed values carried by K feedback channels. Let $A_u \in \mathbb{R}^{N_u \times K}$ be their fixed
 117 compartmental address matrix: column k is the spatial field produced by unit amplitude on channel k , and $(A_u)_{nk}$ is
 118 its delivery weight at compartment n . Linear superposition of the channel fields gives

$$\delta_u^V = A_u \mathbf{c}_u, \quad \widehat{\delta}_{n,u}^V = \sum_{k=1}^K (A_u)_{nk} c_{u,k}. \quad (24)$$

Equation 24 specifies the allowed feedback routes; it is a modeling assumption separate from the membrane current balance. The address matrix specifies where signals can be delivered; the coefficient vector specifies the values delivered on the current example. The model is exact for $A_u = I$ and $\mathbf{c}_u = \delta_u^V$. It becomes a neuron-wide broadcast for $K = 1$ and $A_u = \mathbf{1}_{N_u}$. A binary subtree column is one on descendants of its branch point and zero elsewhere. The equation leaves coefficient estimation open: projection analyses use best-fitting oracle coefficients to measure representational capacity, whereas learning experiments use the feedback rule specified for that task.

Here K counts fixed spatial delivery profiles, not an intrinsic number of independent external errors. Conditional on the forward state, a directed single-output tree has the scalar somatic error multiplied by its state-dependent path profile (Eq. 10). Exact path transport uses that scalar together with the sample-specific derivatives and is tested as a separate transport resource. A context gate can likewise reside in the local eligibility. The dimension of a field ensemble therefore measures the demand on a fixed dictionary across states; it does not exclude scalar error delivery combined with additional local transport or gating.

Table S2 defines the feedback objects without conflating bandwidth with routing. The strict scalar field used for the plotted MNIST ladder applies the scalar reduction defined below at every local-rule stage. The matched-width scalar-fallback field reuses a somatic vector only at a stage of the same width; at wider stages it uses the same robust scalar reduction as the strict scalar condition. For an N -dimensional somatic error vector δ_b in example b , the code first computes $\mu_b = N^{-1} \sum_u \delta_{b,u}$. With $\varepsilon = 10^{-6}$,

$$s_b = \begin{cases} \mu_b, & |\mu_b| > \varepsilon, \\ \delta_{b,u^*}, & u^* = \arg \max_u |\delta_{b,u}|, \quad |\mu_b| \leq \varepsilon. \end{cases} \quad (25)$$

For a one-dimensional error, the coordinate is retained. Selecting the largest-magnitude coordinate when the mean nearly cancels prevents the reduction from producing a near-zero teaching signal. The replacement count was not recorded in the initial runs and was therefore measured directly in the same-seed computational rerun. The matched-width field does not repeat each neuron’s coordinate over that neuron’s descendants. The neuron-specific feedback field does exactly that, using one communicated coordinate per neuron and the tree as a fixed coordinate-to-arbor map. Exact path transport supplies the entire path derivative and is an information upper bound.

In the regular $[3, 3]$ networks, the matched-width scalar-fallback field retained the 128-dimensional somatic vector only at the 128-wide stage. It reduced the field to one scalar at the 384- and 1,152-wide stages, so all 2,408,448 candidate feedforward synaptic parameters (92,160 active contacts) received strict scalar feedback. The exception comprised 384 final branch-to-soma couplings and 256 somatic gain and threshold parameters: 640 of 2,413,312 core parameters (0.027%). We therefore trained a separate 15-seed strict-scalar condition in each architecture, which supplies the scalar rung in the original ladder retained in Supplementary Fig. S7. The fresh six-rule ladder also uses strict scalar feedback. Despite its small parameter count, this near-somatic exception affected learning. Strict scalar reduced mean accuracy relative to matched-width scalar-fallback feedback by 4.859 percentage points in shunting networks (95% paired-seed interval, 3.994–5.669; fallback higher in 15/15 seeds) and by 2.274 points in raw-additive networks (1.796–2.699; fallback higher in 14/15 seeds, one tie). Both intervals lay well outside the prespecified ± 0.5 -point practical-equivalence margin. These results show that a small near-somatic parameter group can exert disproportionate control over learning; they do not imply that the remaining neurons collapse under strict scalar feedback, because their local eligibility traces remain distinct. The paired comparison is retained in the Source Data indexed in Table S12.

1.11 Conditional gradient alignment

For one tree and one example, suppose the correct somatic error is available but path gain is omitted. Write local eligibility as e_i and let $n(i)$ be the host compartment. Then

$$g_i^{\text{exact}} = e_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad g_i^{\text{shared}} = e_i \delta_0^V. \quad (26)$$

Let g^{exact} and g^{shared} collect these synaptic components, and let CV_w denote the weighted coefficient of variation defined below. For nonnegative path gains and nonzero gradient vectors,

$$\cos(g^{\text{shared}}, g^{\text{exact}}) = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha})^2}}, \quad w_i = (e_i \delta_0^V)^2. \quad (27)$$

Here

$$\bar{\alpha}_w = \frac{\sum_i w_i \tilde{\alpha}_{n(i)}}{\sum_i w_i}, \quad \text{CV}_w(\tilde{\alpha})^2 = \frac{\sum_i w_i [\tilde{\alpha}_{n(i)} - \bar{\alpha}_w]^2}{\bar{\alpha}_w^2 \sum_i w_i}, \quad (28)$$

so CV_w is the eligibility-energy-weighted coefficient of variation of the path gain. This identity explains why path-gain dispersion can matter after the correct somatic coordinate has been supplied. It does not solve the separate problem of supplying the correct coordinate across network layers.

1.12 Topology-matched credit capacity

A restricted field is useful only if its spatial supports approximate the required compartment-level credit. We therefore ask how much of that field can be represented by K ancestry-defined routes, and how this compares with a point-neuron partition at the same bandwidth.

1.13 Ancestry dictionary

Let A_T be the ancestry matrix of tree T , with one row per synaptic coordinate and one column per candidate route. Let β contain the state-dependent gains of these routes. A low-bandwidth teaching coefficient vector $\mathbf{c} \in \mathbb{R}^K$ generates a synaptic field

$$\hat{\mathbf{g}} = \Phi_{T,K} \mathbf{c}, \quad \Phi_{T,K} = D_e A_T \text{diag}(\beta) S_K, \quad (29)$$

where the diagonal matrix D_e holds local eligibility or anatomical weights and S_K assigns the K communicated channels to route sites. In this dictionary model, topology fixes binary support and β rescales route columns. Nonzero column rescaling preserves their span at a fixed selected route set, although it can affect selection, conditioning and coefficient dynamics. The focal cable interventions studied below are different: a physical shunt changes spatial transfer and local eligibility, and need not be a diagonal gain on these fixed columns. The compartment-level matrix A_u in Eq. 24 is the general address map. Here A_T is its anatomy-specific lift to synapse coordinates, and $\Phi_{T,K}$ is the resulting synaptic-update dictionary after eligibility, route gain and channel selection are included.

1.14 Address dimension and the point-neuron resource comparison

Separate the synapse-local eligibility from the task-dependent target credit field and let $\mathbf{t}_u \in \mathbb{R}^{N_u}$ denote the latter for neuron u . For this comparison, Φ acts on the chosen task-credit coordinates; eligibility-weighted update dictionaries use the same projection construction in their own coordinates. Let W be a positive diagonal coordinate-weight matrix, and let $\dagger$ denote the Moore–Penrose pseudoinverse. Define

$$\|\mathbf{v}\|_W^2 = \mathbf{v}^\top W \mathbf{v}, \quad P_{\Phi,W} = \Phi(\Phi^\top W \Phi)^\dagger \Phi^\top W. \quad (30)$$

This is the W -orthogonal projector onto $\text{col}(\Phi)$. Equivalently, with $\bar{\Phi} = W^{1/2} \Phi$, $\bar{P}_\Phi = \bar{\Phi} \bar{\Phi}^\dagger = W^{1/2} P_{\Phi,W} W^{-1/2}$ is the ordinary Euclidean projector in the transformed coordinates. For a prespecified nested family of dictionaries $\Phi_{u,K}$, a residual tolerance $0 \leq \varepsilon < 1$ and positive expected target energy, define

$$d_{\varepsilon,u} = \min \left\{ K : \frac{\mathbb{E} \|P_{\Phi_{u,K},W} \mathbf{t}_u\|_W^2}{\mathbb{E} \|\mathbf{t}_u\|_W^2} \geq 1 - \varepsilon \right\}. \quad (31)$$

We set $d_{\varepsilon,u} = +\infty$ if no tested dictionary attains the threshold. This definition concerns the task-dependent coefficient after eligibility has been factored out; heterogeneous presynaptic activity alone is therefore not counted as an extra teaching coordinate. It is conditional on the model, state ensemble, coordinate metric and dictionary family. It specifies a resolution at a stated residual tolerance, rather than an intrinsic task complexity or a unique optimal morphology.

For any fixed $\Phi \in \mathbb{R}^{N_u \times K}$, all fields generated as $\Phi \mathbf{c}$ lie in $\text{col}(\Phi)$ and hence occupy at most $\text{rank}(\Phi) \leq K$ dimensions. Conversely, the weighted least-squares field in that space is $P_{\Phi,W} \mathbf{t}_u$, and its expected weighted residual energy is

$$\mathbb{E} \|(I - P_{\Phi,W}) \mathbf{t}_u\|_W^2 = \text{tr}[(I - \bar{P}_\Phi) \bar{\Sigma}_t], \quad \bar{\Sigma}_t = \mathbb{E}[W^{1/2} \mathbf{t}_u \mathbf{t}_u^\top W^{1/2}]. \quad (32)$$

Equations 31–32 prove the fixed-dictionary dimension statement used in the main text; $\bar{\Sigma}_t$ is an uncentered second moment. The ordinary point-neuron local rule has $\Phi = \mathbf{1}$ and therefore one task-dependent coefficient per neuron. A K -route dendritic dictionary can use up to K structured fields inside that neuron. A point-neuron network is not mathematically forbidden from matching them, but doing so requires K separately routed teaching coefficients, K gated parameter groups, or an equivalent expansion into additional units. Accordingly, the comparison is about where address resources are represented, not about an absolute expressive separation between all possible dendritic and point-neuron networks.

1.15 Field capture

We now measure how much of a target field lies in a dictionary’s span. Let $\mathbf{g}$ denote that field in the coordinates under consideration, and use the positive weighting from Eq. 30. Set $\bar{\mathbf{g}} = W^{1/2} \mathbf{g}$, $\bar{\Phi}_{T,K} = W^{1/2} \Phi_{T,K}$, and let $\bar{P}_\Phi$ be the ordinary Euclidean orthogonal projector onto the columns of $\bar{\Phi}_{T,K}$. For a nonzero target field, capture is

$$\mathcal{C}_K(\mathbf{g}; \Phi, W) = \frac{\|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2} = 1 - \frac{\|\bar{\mathbf{g}} - \bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}. \quad (33)$$

The unweighted case has $W = I$. The MICrONS structural analysis uses summed mapped excitatory synaptic area on the diagonal of W . For the transformed uncentered second moment $\bar{\Sigma}_g = \mathbb{E}[\bar{\mathbf{g}} \bar{\mathbf{g}}^\top]$, aggregate energy capture for a fixed projector is

$$\mathcal{C}_K(\bar{\Sigma}_g; \Phi, W) = \frac{\text{tr}(\bar{P}_\Phi \bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)}. \quad (34)$$

This is the ratio of expected projected energy to expected total energy. It is generally different from the mean per-field fraction $\mathbb{E}[\mathcal{C}_K(\mathbf{g}; \Phi, W)]$, which weights nonzero fields equally. The two coincide for a fixed field norm, or when the second moment is formed after normalizing each field. The MNIST dictionary gallery averages per-field fractions; the cable-response operator analysis aggregates energy over its columns. The second moment equals a centered covariance only for mean-zero fields. Equation 34 states the topology–task matching condition: a sparse anatomical basis is useful only to the extent that task credit lies in its span.

Proposition 1 (Ky–Fan credit-capacity bound). *Let $\bar{\Sigma}_g \in \mathbb{R}^{p \times p}$ be positive semidefinite, with eigenvalues satisfying $\lambda_1 \geq \dots \geq \lambda_p \geq 0$ and $\text{tr}(\bar{\Sigma}_g) > 0$. For every rank- K orthogonal projector P ,*

$$\frac{\text{tr}(P \bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)} \leq \frac{\sum_{j=1}^K \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (35)$$

Equality is attained by any rank- K projector whose range is a Ky–Fan-maximizing invariant subspace: it contains every eigenspace with eigenvalue strictly greater than λ_K and may choose the remaining directions within the eigenspace at λ_K . The difference between the right side and anatomical capture is the rank-matched spectral regret.

Proof. Ky–Fan’s maximum principle [2] gives $\max_{P=P^\top=P^2, \text{rank}(P)=K} \text{tr}(P \bar{\Sigma}_g) = \sum_{j=1}^K \lambda_j$. Division by the fixed total spectral energy proves Eq. 35. $\square$

224 The spectral screen interpolates between two task-field second moments C_0 and C_1 with equal trace, using
 225 $C(\rho) = (1 - \rho)C_0 + \rho C_1$ for $0 \leq \rho \leq 1$. Thus, for two fixed projectors P and Q , the normalized capture difference
 226 has the sign of

$$\Delta(\rho) = \text{tr}[(P - Q)C_0] + \rho \text{tr}[(P - Q)(C_1 - C_0)], \quad (36)$$

227 so an interior crossover is determined exactly by its two endpoints. This affine result distinguishes alignment from
 228 rank: increasing K raises the oracle bound, whereas increasing ρ determines whether the anatomical subspace
 229 approaches it.

230 1.16 One-step descent bound

231 When the target field is a true loss gradient, capture also determines a one-step descent guarantee in the correspond-
 232 ing metric. Let $\mathbf{w}$ be the parameter vector and $\mathbf{g} = \nabla_{\mathbf{w}} \mathcal{L}$. To express the weighted calculation as Euclidean descent,
 233 define

$$\bar{\mathbf{w}} = W^{-1/2} \mathbf{w}, \quad \mathbf{w} = W^{1/2} \bar{\mathbf{w}}, \quad (37)$$

234 and define $\bar{\mathcal{L}}(\bar{\mathbf{w}}) = \mathcal{L}(W^{1/2} \bar{\mathbf{w}})$. Then $\nabla_{\bar{\mathbf{w}}} \bar{\mathcal{L}} = W^{1/2} \mathbf{g} = \bar{\mathbf{g}}$. If $\bar{\mathcal{L}}$ has L_W -Lipschitz gradients and η is the step size,
 235 the descent lemma gives

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} + \frac{L_W \eta^2}{2} \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (38)$$

236 Orthogonal projection implies $\bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} = \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2$, hence

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \left(1 - \frac{L_W \eta}{2}\right) \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (39)$$

237 At $\eta = 1/L_W$, the ratio between the projected- and full-gradient guarantees is Eq. 33. In the original coordinates,
 238 the projected step is

$$\mathbf{w}^+ = \mathbf{w} - \eta W^{1/2} \bar{P}_{\Phi} W^{1/2} \mathbf{g}, \quad (40)$$

239 which is a particular preconditioned update rather than an ordinary Euclidean gradient step in $\mathbf{w}$. The implementation
 240 was checked on 10,000 random positive-definite quadratic losses: there were no bound violations, minimum
 241 numerical slack was 2.67×10^{-7} , and the capture identity agreed to 1.89×10^{-15} . This numerical test checks the
 242 Euclidean projector algebra, which also applies after an explicit coordinate transformation; it does not validate
 243 synaptic-area weighting as a physical parameter metric. The result is a one-step guarantee only when the tested
 244 field is an exact gradient in the stated coordinates. It is not a convergence or generalization theorem and does not
 245 show that a biological circuit can infer the oracle coefficients.

246 1.17 Signal-noise theory of restricted credit routing

247 Capture alone favors increasing route resolution, but a broader operator also admits more stochastic error. We next
 248 formalize this signal-noise trade-off and ask when restricting the communicated credit improves expected progress.

249 1.18 Expected descent from routed-update moments

250 Let $\mathcal{L} : \mathbb{R}^p \rightarrow \mathbb{R}$ have L -Lipschitz gradient, with $L > 0$, on a neighborhood containing every update segment under
 251 consideration. At a fixed parameter state, write $\mathbf{g} = \nabla \mathcal{L}(\mathbf{w})$ and let $\mathbf{d}$ be a routed update estimator with finite
 252 second moment. Define $\mathbf{m} = \mathbb{E}[\mathbf{d}]$ and $V = \mathbb{E}[(\mathbf{d} - \mathbf{m})(\mathbf{d} - \mathbf{m})^T]$. The expectation is over the stated sampling or
 253 feedback noise, conditional on $\mathbf{w}$. The update is

$$\mathbf{w}^+ = \mathbf{w} - \eta \mathbf{d}. \quad (41)$$

254 **Proposition 2** (Routed-update moment bound). *For Eq. 41,*

$$\mathbb{E}\mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top \mathbf{m} + \frac{L\eta^2}{2} [\|\mathbf{m}\|_2^2 + \text{tr}(V)]. \quad (42)$$

255 Write $a = \mathbf{g}^\top \mathbf{m}$ and $b = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. For $b > 0$, minimizing the bound over nonnegative steps gives $\eta^* = [a]_+ / (Lb)$.

256 Provided the stated smoothness condition holds along the update segments at this step, the certified decrease is

$$U = \frac{[\mathbf{g}^\top \mathbf{m}]_+^2}{2L\{\|\mathbf{m}\|_2^2 + \text{tr}(V)\}}, \quad [a]_+ = \max(a, 0). \quad (43)$$

257 For $b = 0$, $\mathbf{d} = 0$ almost surely and we set $U = 0$. A zero score means that this bound certifies no positive one-step
258 decrease at the evaluated state; it does not establish failure of every later update or learning rule.

259 *Proof.* Apply the descent lemma [3] to $-\eta \mathbf{d}$, take expectations and use $\mathbb{E}\|\mathbf{d}\|_2^2 = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. Minimizing the
260 resulting scalar quadratic over $\eta \geq 0$ gives the claim. $\square$

261 A fixed linear routing operator gives a useful special case. If $\mathbf{d} = M(\mathbf{g} + \boldsymbol{\xi})$ with deterministic $M \in \mathbb{R}^{p \times p}$,
262 $\mathbb{E}\boldsymbol{\xi} = 0$ and $\text{Cov}(\boldsymbol{\xi}) = \Sigma$, then $\mathbf{m} = M\mathbf{g}$ and $V = M\Sigma M^\top$. Equations 42–43 then recover the credit-operator
263 expressions, denoted $U(M)$ below. The identity, orthogonal projectors and fixed block-gain screens use this
264 specialization.

265 The more general moment form is needed for sample-dependent eligibilities and gates. In the context-gated task,
266 changing an unselected distractor leaves the exact parameter gradient unchanged but changes a shared-feedback
267 update through that distractor’s eligibility. A fixed matrix of the exact parameter gradient therefore cannot represent
268 every routed update. The output-error-to-update map remains linear conditional on the complete forward state; that
269 conditional map is a different object. The factorial reanalysis measures routed means and second moments directly
270 and does not require a fixed map from exact parameter gradients to routed updates.

271 At fixed infinitesimal update norm, $-\mathbf{g}$ is the steepest Euclidean descent direction by Cauchy–Schwarz. Conse-
272 quently, no routed local operator provides a uniform first-order improvement over exact full-batch backpropagation
273 on the same parameters and objective. Equation 43 instead compares restricted or noisy estimators and exposes
274 when rejecting stochastic energy compensates for discarded signal.

Corollary 1 (Projected stochastic crossover). *Consider the identity-curvature quadratic $\mathcal{L}(\mathbf{w}) = \frac{1}{2}\|\mathbf{w} - \mathbf{w}^*\|_2^2$, initialize at $\mathbf{w} = 0$, set $\mathbf{g} = -\mathbf{w}^*$, use unit step size, and let P be an orthogonal projector. The expected post-update losses of the unprojected and projected unbiased gradients are*

$$\mathbb{E}\mathcal{L}[-(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} \text{tr}(\Sigma), \quad (44)$$

$$\mathbb{E}\mathcal{L}[-P(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} [\|(I - P)\mathbf{g}\|_2^2 + \text{tr}(P\Sigma P)]. \quad (45)$$

275 *The projected estimator is better precisely when*

$$\text{tr}[(I - P)\Sigma] > \|(I - P)\mathbf{g}\|_2^2, \quad (46)$$

276 *that is, rejected noise exceeds discarded signal. A local estimator using the same P with additional independent,*
277 *zero-mean coefficient noise cannot achieve a lower expected loss than backpropagation followed by P .*

278 For completeness, at a common step $0 < \eta < 2$ the unprojected-minus-projected expected loss is

$$\frac{\eta}{2} \{ \eta \text{tr}[(I - P)\Sigma] - (2 - \eta) \|(I - P)\mathbf{g}\|_2^2 \}. \quad (47)$$

279 Hence projection is better exactly when $\text{tr}[(I - P)\Sigma] > [(2 - \eta)/\eta] \|(I - P)\mathbf{g}\|_2^2$; the unit-step result above is the
280 special case $\eta = 1$. Smaller common steps therefore require proportionally more rejected noise to compensate for
281 the discarded signal.

1.19 Interior optimum along the nested route ladder

The scalar, neuron-specific and correctly routed ancestry fields in the controlled hierarchy analyses span nested subspaces. Their orthogonal projectors form the ladder $P_{\text{scalar}} \subset P_{\text{neuron}} \subset P_{\text{ancestry},K} \subset I$. Along such a ladder the capture of Eq. 33 is monotone by construction, but the guaranteed utility is not. For an orthogonal projector with retained signal energy $S = \mathbf{g}^\top P \mathbf{g} = \|P \mathbf{g}\|_2^2$ and admitted noise $N = \text{tr}(P \Sigma P)$, the optimal-step utility of the credit-operator bound is

$$U(S, N) = \frac{S^2}{2L(S + N)}. \quad (48)$$

Let one rung up the ladder add marginal signal $\Delta S \geq 0$ and marginal noise $\Delta N \geq 0$. Substituting into Eq. 48 and clearing denominators gives the exact marginal condition

$$U(S + \Delta S, N + \Delta N) > U(S, N) \iff \Delta S [(S + N)(2S + \Delta S) - S^2] > S^2 \Delta N. \quad (49)$$

The bracket is positive, so for small marginal additions the condition reduces to a threshold on the marginal signal-to-noise ratio, $\Delta S / \Delta N > S / (S + 2N)$: the utility keeps rising while each added route scale contributes proportionally more signal than the running threshold, and falls once the added scales admit mostly noise. When the signal spectrum decays faster than the admitted noise across ladder scales, U can attain an interior maximum. On a finite tested ladder, the marginal condition must become unfavorable before the last rung; spectral decay alone does not guarantee that crossing or a unique optimum. The displayed marginal condition assumes $S > 0$; if $S = 0$, a rung with positive retained signal has positive utility whenever its total second moment is finite and nonzero. Equations 48 and 49, together with the interior-maximum construction, are verified numerically in the released test suite (`tests/test_laminar_optimality.py`). Random, deranged and independently gained route families elsewhere in the paper need not lie on this same nested ladder.

Two experiments evaluate this optimum empirically (Table S12; analysis code is included in the release). In the hierarchy-depth routing experiment the initialization utility of the aligned tree, computed per seed and task depth, placed its argmax at the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched the trained optimum exactly at $H_c = 1, 2, 3$; at the full-rank boundary $H_c = 4$ the one-step bound was conservative and favoured $D_r = 2$, reflecting its single-step horizon.

The trained subtree factorial provides a second, retrospective test. Its full-batch moment score uses the global logistic curvature bound and the same four-control comparator as the accuracy contrast: deranged, depth-interleaved, random-sparse and random-rank routes. The mean ancestry-minus-best-control utility was positive at $K = 4$, negative at $K = 1, 2$ and zero at $K = 8$. Its contrast-maximizing budget was $K = 4$ in all 20 seeds, matching the observed accuracy-contrast maximum in 13 of 20. Mean accuracy-contrast selection regret was 1.94 percentage points. Adding the hypothetical minibatch variance made the mean $K = 4$ utility contrast negative, reduced optimum agreement to 8 of 20 and gave mean regret 2.75 points. Thus the score matched to the deterministic training recipe recovers the mean intermediate-bandwidth ancestry peak, while its seed-level selection remains imperfect. This retrospective result is distinct from the raw within-family performance optimum at $K = 8$ and from prospective morphology selection. The comparator definitions, scores and prediction errors are supplied in `source_data/review_evidence_reanalysis/`.

1.20 Laminar route families and hierarchical task second moments

The distinction between a point neuron and a dendritic tree can be stated as a constraint on the feasible credit operators at fixed rank. A point neuron addresses its synapses with the single shared column **1**: a rank-one operator with uniform support. An ancestry dictionary uses structured columns whose supports form a *laminar* family: any two supports are nested or disjoint. An unconstrained rank- K feedback matrix is the comparison class. For the balanced binary tree used in the factorial, the span of the K -route ancestry dictionary equals the span of the K coarsest vectors of the orthonormal tree-Haar basis $U = [\mathbf{h}_1, \dots, \mathbf{h}_n]$ (constant, then sibling-subtree differences, coarse to fine) for every admissible budget $K \in \{1, 2, 4, \dots, n\}$. Here n is the number of leaf coordinates. Let $P_{A,K}$ denote the orthogonal projector onto the selected ancestry span.

Proposition 3 (Laminar capture of a hierarchical second moment). *Let the task-credit second moment be $C = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ along the coarse-to-fine Haar order, with $\text{tr}(C) > 0$. Then for every admissible budget K the ancestry dictionary attains the Ky–Fan bound,*

$$\frac{\text{tr}(P_{A,K}C)}{\text{tr}(C)} = \frac{\sum_{i \leq K} \lambda_i}{\text{tr}(C)} = \max_{\substack{P=P^\top=P^2 \\ \text{rank}(P)=K}} \frac{\text{tr}(PC)}{\text{tr}(C)}. \quad (50)$$

For a general second moment C with eigenpairs $(\lambda_i, \mathbf{v}_i)$ ordered by decreasing eigenvalue, the shortfall from the Ky–Fan bound decomposes exactly as

$$\sum_{i \leq K} \lambda_i - \text{tr}(P_{A,K}C) = \sum_{i \leq K} \lambda_i \|(I - P_{A,K})\mathbf{v}_i\|_2^2 - \sum_{j > K} \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2, \quad (51)$$

the leading-eigenspace energy outside the laminar span minus the trailing energy that the span retains. It is therefore bounded above by the former term. The shortfall is zero exactly when $P_{A,K}$ is Ky–Fan maximizing; when $K < n$ and $\lambda_K > \lambda_{K+1}$, this is equivalent to the ancestry span being the unique leading- K eigenspace.

Proof. The ancestry span equals the span of the K coarsest Haar vectors, which under the stated ordering are the top- K eigenvectors of C ; a rank- K orthogonal projector maximizes $\text{tr}(PC)$ exactly on a top- K eigenspace (Ky–Fan). For general C , expanding $\text{tr}(P_{A,K}C) = \sum_j \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2$ and $\|(I - P_{A,K})\mathbf{v}_i\|_2^2 = 1 - \|P_{A,K}\mathbf{v}_i\|_2^2$ for $i \leq K$ gives the stated decomposition. Ky–Fan’s equality condition, including arbitrary choices inside a degenerate cutoff eigenspace, gives the last statement. $\square$

This ordered hierarchical second moment is one sufficient class for which ancestry attains the rank- K spectral optimum. At a fixed K , the equality condition is the leading-eigenspace condition in the proposition; it does not require the entire second moment to have a Haar eigenbasis and does not identify a unique physical tree. Hierarchical organization without the stated coarse-to-fine spectral ordering is not sufficient. The spectral-alignment experiment realizes both regimes: at full alignment and $K = 4$ ancestry attained the rank-matched Ky–Fan bound to numerical precision, whereas at random alignment its advantage over sampled random routes vanished (main text). The attainment, the shortfall identity and the failure case with a fine-scale-dominant second moment are verified numerically in `tests/test_laminar_optimality.py`.

1.21 Point-neuron partition residual and route-coefficient error

Let t_i be the exact task-dependent target remaining after local eligibility has been factored out, let $w_i > 0$ be coordinate weights, and let $\pi(i)$ assign synapse i to a group that must share one teaching coefficient. The best allowed field minimizes $\sum_i w_i [t_i - c_{\pi(i)}]^2$.

Proposition 4 (Shared-coefficient residual). *For every group G , the optimum is its weighted mean*

$$c_G^* = \frac{\sum_{i \in G} w_i t_i}{\sum_{i \in G} w_i}, \quad (52)$$

and the irreducible residual is

$$\mathcal{R}(\pi) = \sum_G \sum_{i \in G} w_i (t_i - c_G^*)^2. \quad (53)$$

Refining a partition cannot increase $\mathcal{R}$; it decreases the residual only when an old group contains coefficient variation resolved by the new groups.

Proof. Differentiate the separable weighted least-squares objective with respect to each c_G . The refinement statement follows because the old solution remains feasible when two or more refined coefficients are constrained to be equal. $\square$

The ordinary point-neuron rule has one group per neuron. Dendritic ancestry offers a particular nested sequence of refinements, but is helpful only if the task coefficient varies across those subtrees. An explicitly grouped point model can implement the same partition, which is why the resource-matched point controls in the main article coincide with the dendritic implementation.

For a general dictionary Φ and positive diagonal W , let $\mathbf{c}^* = \arg \min_{\mathbf{c}} \|\mathbf{t} - \Phi \mathbf{c}\|_W^2$ and let $\hat{\mathbf{c}}$ be the coefficients actually supplied by a local circuit. Weighted projection orthogonality yields the exact decomposition

$$\|\mathbf{t} - \Phi \hat{\mathbf{c}}\|_W^2 = \underbrace{\|\mathbf{t} - \Phi \mathbf{c}^*\|_W^2}_{\text{address / representation error}} + \underbrace{\|\Phi(\hat{\mathbf{c}} - \mathbf{c}^*)\|_W^2}_{\text{coefficient-estimation error}}. \quad (54)$$

If the coefficient error is unbiased with covariance Ω_c , the expected second term is $\text{tr}(\Phi^\top W \Phi \Omega_c)$. The Gram matrix $G_\Phi = \Phi^\top W \Phi$ therefore controls how coefficient error is expressed. Estimating $\mathbf{c}^*$ is sensitive to noise along directions with small positive Gram eigenvalues. Correlated route columns can likewise make individual coefficients poorly identifiable even when their combined field is stable. Address rank alone is not a complete learning metric.

1.22 Static route gains preserve span but can change conditioning

Proposition 5 (Route-wise static-gain invariance). *Let D_β be an invertible diagonal matrix of fixed nonzero route gains. Then*

$$\text{col}(\Phi D_\beta) = \text{col}(\Phi), \quad P_{\Phi D_\beta, W} = P_{\Phi, W}. \quad (55)$$

Thus route-wise static scalar gains cannot change address rank, projection residual or spectral capture. They can change the Gram matrix from $\Phi^\top W \Phi$ to $D_\beta^\top \Phi^\top W \Phi D_\beta$ and hence alter conditioning and coefficient dynamics.

This proposition concerns nonzero scalar rescaling of existing route columns. Spatially varying row gains, state-dependent gating and gains that reach zero can change the effective span or active support; those cases are not covered by Eq. 55. Physical shunts also change the cable Green’s function. For a somatic adjoint source on a passive tree, that change has an exact diagonal-gain representation in the baseline-weighted ancestry partition associated with the shunt site (Corollary 2). Its partition indicators are differences of nested subtree indicators. The diagonal form does not extend to an arbitrary selected overlapping dictionary, and the local synaptic driving force can still vary within each block.

1.23 Finite-time coefficient learning has a bias–variance crossover

Equal span need not give equal learning trajectories. Let $\mathbf{f}_t = \Phi \mathbf{c}_t$ be the field produced by coefficients $\mathbf{c}_t$ after update t , and let $\lambda_{\max} > 0$ be the largest eigenvalue of $\Phi \Phi^\top$. Normalizing ordinary coefficient descent by this eigenvalue gives $H_\Phi = \Phi \Phi^\top / \lambda_{\max}$. For normalized step size η and noisy rate-based target estimates $\mathbf{g} + \boldsymbol{\xi}_t$, with $\mathbb{E} \boldsymbol{\xi}_t = 0$ and $\text{Cov}(\boldsymbol{\xi}_t) = \sigma^2 I / n$, where n is the effective sample size and σ^2 the single-observation noise scale, the field iteration is

$$\mathbf{f}_{t+1} = \mathbf{f}_t - \eta H_\Phi (\mathbf{f}_t - \mathbf{g} - \boldsymbol{\xi}_t). \quad (56)$$

Proposition 6 (Exact finite-time coefficient risk). *Assume $\mathbf{f}_0 = 0$, $\mathbf{g} \in \text{col}(\Phi)$ and let $(\mu_j, \mathbf{u}_j)$ be the positive eigenpairs of H_Φ , with $\theta_j = \mathbf{u}_j^\top \mathbf{g}$. For independent isotropic noise and a stable step satisfying $|1 - \eta \mu_j| < 1$ for every positive mode, the expected population loss after T updates is*

$$\mathbb{E} \frac{\|\mathbf{f}_T - \mathbf{g}\|_2^2}{2} = \frac{1}{2} \sum_j \left[(1 - \eta \mu_j)^{2T} \theta_j^2 + \frac{\sigma^2}{n} \frac{(\eta \mu_j)^2 [1 - (1 - \eta \mu_j)^{2T}]}{1 - (1 - \eta \mu_j)^2} \right]. \quad (57)$$

For Gram-preconditioned descent, $H_\Phi = P_\Phi$ and all positive $\mu_j = 1$; therefore every dictionary with the same projector has the same field trajectory under common observations.

391 *Proof.* Project Eq. 56 onto each eigenvector. The deterministic error contracts by $1 - \eta\mu_j$ per step. Independent
 392 innovations have variance $(\eta\mu_j)^2\sigma^2/n$ and accumulate as a finite geometric series. Summing half the modal second
 393 moments gives Eq. 57. Gram preconditioning replaces $\Phi^\top\Phi$ by its pseudoinverse on the positive subspace, so the
 394 induced field operator is P_Φ . $\square$

395 Equation 57 makes the low-data boundary explicit. Small μ_j retain more finite-time bias but admit less estimator
 396 variance. For two operators with the same span, let ΔB be the difference in the bias term and ΔV the difference in the
 397 coefficient of the noise term. Their risk difference is $\Delta B + (\sigma^2/n)\Delta V$. If $\Delta B > 0$ and $\Delta V < 0$, the ordering reverses
 398 at $n^* = -\sigma^2\Delta V/\Delta B$: slow coordinates can help below n^* by implicit noise filtering, whereas well-conditioned
 399 coordinates help above it by reducing bias. This is a finite-horizon regularization effect, not new address capacity
 400 and not an unconditional advantage of ill-conditioning.

401 1.24 Branch reliability shrinkage

402 Partition the parameter coordinates into disjoint branch blocks b . Let $\mathbf{g}_b$ be the clean gradient in block b , Σ_b its
 403 noise covariance, and I_b the identity on that block. Define signal and noise energies $S_b = \|\mathbf{g}_b\|_2^2$ and $N_b = \text{tr}(\Sigma_b)$.
 404 With attenuation a_b in each block, $M = \text{blockdiag}(a_b I_b)$ and a fixed step $\eta = c/L$ for $c > 0$, Eq. 42 guarantees a
 405 loss decrease of

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{ca_b^2}{2} (S_b + N_b) \right]. \quad (58)$$

406 **Proposition 7** (Fixed-step reliability-optimal attenuation). *For each block with $S_b + N_b > 0$, Eq. 58 is maximized*
 407 *independently at*

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}, \quad (59)$$

408 *for a physical attenuator constrained to $0 \leq a_b \leq 1$. A block with $S_b = N_b = 0$ contributes zero for every gain*
 409 *and can be omitted from the ratios below. If at least one block has positive energy, the best common gain is*
 410 *$a_g^* = \min\{1, \sum_b S_b / [c \sum_b (S_b + N_b)]\}$, and the branch-specific guarantee is no smaller because its feasible set*
 411 *contains every common gain. At $c = 1$, Eq. 59 reduces to the reliability factor $S_b / (S_b + N_b)$; in that case*
 412 *$\mathcal{G}_{\text{branch}} = (2L)^{-1} \sum_b S_b^2 / (S_b + N_b)$ and $\mathcal{G}_{\text{global}} = (2L)^{-1} (\sum_b S_b)^2 / \sum_b (S_b + N_b)$, with equality when reliability is*
 413 *constant across positive-energy blocks. If all blocks have zero energy, every common gain is optimal and both*
 414 *guarantees are zero.*

415 *Proof.* Differentiate each concave quadratic in Eq. 58 and project its unconstrained maximizer onto $[0, 1]$. Restricting
 416 all coordinates to a common value gives a_g^* . The branchwise optimum cannot be worse than this restricted solution.
 417 For $c = 1$, the closed-form comparison also follows from Cauchy–Schwarz applied to $S_b / \sqrt{S_b + N_b}$ and $\sqrt{S_b + N_b}$.
 418 $\square$

419 This optimum assumes access to S_b and N_b . A point implementation provided with the same branch partition
 420 and multiplicative factors produces the same update. Its biological relevance depends on whether local conductance
 421 can estimate or realize those factors.

422 1.25 Why depth can help or hurt

423 Let $P_1 \preceq P_2 \preceq \dots \preceq P_{D_r}$ be nested tree-route projectors. Increasing depth reduces the address bias $\|(I - P_{D_r})\mathbf{g}\|^2$ but
 424 increases admitted noise $\text{tr}(P_{D_r}\Sigma P_{D_r})$ whenever the new modes carry stochastic energy. These competing terms
 425 can therefore produce an interior optimum when task signal is concentrated above one hierarchy level and noise
 426 continues into finer levels; its location depends on their magnitudes, as specified by Eq. 46. This is the controlled
 427 role isolated by the hierarchy-depth routing experiment. It is distinct from nonlinear forward composition, which
 428 can make deeper dendritic subunits representationally useful even when the backward address span is unchanged.

429 Depth can also impose a transport cost. In an idealized ensemble with equal local eligibility, let the route gain
 430 be $A = \prod_{\ell=1}^{D_p} G_\ell$ and assume independent Gaussian log gains, $\log G_\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2)$. The cosine between a shared

coefficient and the exact gain-weighted field in the large-route limit is

$$\frac{\mathbb{E}A}{\sqrt{\mathbb{E}A^2}} = \exp\left(-\frac{1}{2} \sum_{\ell=1}^{D_p} \sigma_\ell^2\right). \quad (60)$$

Additional levels can therefore add address resolution while simultaneously making uncompensated shared feedback less aligned. The balance depends on task hierarchy, gain variability, noise and the available coefficient estimator; “more depth” has no unconditional sign.

1.26 Controlled signal–noise experiment

The signal–noise experiment used 16 Euclidean coordinates represented in the orthonormal Haar basis of a balanced binary tree, ordered from the global mode (level 0) to individual sibling contrasts (level 4). The configuration, 50 analysis seeds 7000–7049, two preliminary validation seeds and numerical tolerances were specified before the analysis outcomes were inspected. The preliminary runs verified basis orthogonality, projector idempotence, static-gain span invariance and exact agreement with the explicit point-gain control. All planned observations were completed: 3,750 spectral, 1,800 depth, 3,750 projection and 1,500 reliability rows.

In the spectral screen, the task-gradient second moment assigned total energy (0.15, 0.35, 0.30, 0.20, 0) to tree levels 0–4. An alignment coefficient $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ mixed this second moment with an independently rotated endpoint having the same eigenvalues. Intermediate mixtures do not preserve the spectrum and jointly vary alignment and concentration. Ancestry, random rank-matched and dense principal-eigenspace projectors were compared at $K \in \{1, 2, 4, 8, 16\}$. The dense value was also calculated as the Ky–Fan sum of the leading K eigenvalues divided by total second-moment energy. Because the endpoint second moments had equal trace, ancestry-minus-random capture varied affinely with alignment. Endpoint differences therefore gave the exact per-seed crossover; seeds already favoring ancestry at zero alignment were assigned threshold zero. In these unweighted coordinates, let $\bar{\Sigma}_g$ denote the uncentered task-gradient second moment; energy-weighted capture was $\text{tr}(P\bar{\Sigma}_g)/\text{tr}(\bar{\Sigma}_g)$.

The same-span diagnostic compared raw nested indicators, their tree-Haar basis and indicators multiplied by fixed nonzero lognormal column gains. Rank, coherence, positive-spectrum Gram condition number and capture were calculated directly.

In the depth screen, a unit-norm target was sampled from modes at levels 0 through H_c , with $H_c \in \{1, 2, 3, 4\}$. The gradient-noise standard deviation at level ℓ was $0.65[1 + 1.5 \max(\ell - H_c, 0)]$. Starting from zero, 80 common-noise updates of size 0.12 used the projector onto levels 0 through $D_r \in \{1, 2, 3, 4\}$. A fixed leaf permutation provided the rewired control, and the identity provided stochastic backpropagation.

The projection screen independently set retained signal r and retained noise n to $\{0.25, 0.5, 0.75, 0.9, 1\}$. It compared an unprojected stochastic gradient, backpropagation followed by the same projection, and a routed estimator with an added total coefficient-noise fraction of 0.05. For a unit step on the identity-curvature quadratic, expected losses were calculated analytically and stochastic losses were recorded from paired draws.

In the reliability screen, eight disjoint blocks had signal and noise energies proportional to $\exp(hx_b)$ and $\exp(-hx_b)$, respectively, with equally spaced $x_b \in [-1, 1]$ and heterogeneity $h \in \{0, 0.5, 1, 1.5, 2\}$. We compared no attenuation, the best common gain, $S_b/(S_b + N_b)$, seed-shuffled gains, reversed gains and a point-gain control receiving the same branch factors. Intervals used 20,000 resamples of the 50 paired seeds, and paired tests were two-sided Wilcoxon signed-rank tests.

The resolved protocol is `configs/credit_phase_theory/confirmatory.json`. Source tables are in `source_data/credit_phase_theory/`. All full-rank projectors agreed to 4.44×10^{-16} , route-wise static gain rescaling preserved the projector to 1.33×10^{-15} , and the explicit point-gain control matched reliability-aligned attenuation exactly. This fixed-state quadratic experiment validates the signal–noise prediction but does not join oracle reliability to learned positive conductances.

1.27 Initialization score in the trained subtree factorial

The existing factorial used full-batch gradient descent. Its secondary stochastic diagnostic reconstructed the frozen training data, initialization and routes, then estimated routed-update variance from 64 hypothetical minibatches of 64 examples per seed. The full-data routed mean and that variance supplied Eq. 43. This variance was not noise experienced by the full-batch optimizer and cannot by itself explain the trained contrasts. The original released score used the largest logistic Hessian eigenvalue at initialization as its curvature scale. The corrected analysis uses the global logistic bound

$$L_{\text{glob}} = \max_c \lambda_{\text{max}} \left(\frac{8}{4N} X_c^T X_c \right),$$

where rows of X_c contain the selected-stream features for examples in context c , N is the total training-set size, and eight is the gradient context-rescaling factor. This bound is 1.00083–1.02125 times the initialization Hessian maximum across seeds. We scored the deterministic full-batch update with $V = 0$, the hypothetical minibatch estimator with its estimated V , and the deterministic bound at the actual training step $\eta = 0.3$.

Across 540 tree fits, deterministic utility correlated with norm-matched one-step progress at 0.9509 and final accuracy at 0.9242; hypothetical-minibatch values were 0.9373 and 0.9157. Intervals use 5,000 whole-seed resamples. These are retrospective associations within the factorial, not independent prospective validation.

To characterize the task-coordinate spectrum, the analysis used the uncentered second moment of $\delta_i \mathbf{e}_{c_i}$, where δ_i is the output error on example i and $\mathbf{e}_{c_i}$ is the coordinate vector for its context c_i . Eight balanced contexts and logits near zero make this spectrum nearly isotropic, giving the reported participation rank near eight. This value characterizes the chosen coefficient representation at initialization; it is not an independent estimate of task complexity or a morphology law. The mean within-family utility and accuracy maxima occur at $K = 8$ for all five nondegenerate route families. Across their 25 implementation–family cells, there are nine distinct nonconstant mean utility curves: four implementations coincide, whereas leaf rewiring changes four of the five family curves; the principal component analysis (PCA) oracle curve is unchanged. In the dendritic-tree implementation, deterministic utility selected an accuracy-maximizing bandwidth in 95 of 100 seed–family curves, the same success count as always choosing the largest K ; the hypothetical-minibatch score matched in 97 of 100. The 20 derangement curves had zero optimized-step utility throughout and were reported as degenerate. Repeated equivalent parameterizations do not provide independent tests of endpoint ordering. The distinct interior-optimum and ancestry-minus-control analyses, including their prediction failures, are reported in Section 1.19. The corrected scores, per-family selection errors and regret are provided in `source_data/review_evidence_reanalysis/`.

1.28 State-matched positive-conductance reliability experiment

We next tested whether positive conductances could realize the attenuation prescribed by Eq. 59 in a separate trained branch model (Table S12). Here b indexes branches, i examples and d inputs within a branch. Every input rate x_{bid} was nonnegative, and each excitatory conductance was parameterized by $g_{bd} = \text{softplus}(q_{bd})$, with trainable raw parameter q_{bd} . With leak conductance g^L , define total unshunted conductance and voltage on example i

$$G_{bi} = g^L + \sum_d x_{bid} g_{bd}, \quad V_{bi} = \frac{\sum_d x_{bid} g_{bd}}{G_{bi}}. \quad (61)$$

A fixed shunt $\kappa_b \geq 0$ had reversal zero. To isolate its effect on input resistance from its effect on the forward feature, an oracle current $\kappa_b V_{bi}$ was added on each example. The perturbed state is therefore

$$V'_{bi} = \frac{\sum_d x_{bid} g_{bd} + \kappa_b V_{bi}}{G_{bi} + \kappa_b} = V_{bi}, \quad \frac{\partial V'_{bi}}{\partial q_{bd}} = \frac{x_{bid}(1 - V_{bi})}{G_{bi} + \kappa_b} \text{sigmoid}(q_{bd}). \quad (62)$$

Relative to the unshunted eligibility, the example-specific physical input-resistance factor is $G_{bi}/(G_{bi} + \kappa_b)$. The compensating current was recomputed from the pre-shunt state and treated as a clamp, not differentiated as an independently learned pathway. Consequently this manipulation deliberately removes forward feature suppression while retaining conductance attenuation in the local update.

512 The binary task used eight branches, four inputs per branch, 512 training examples and 2,048 held-out examples.
 513 Input rates were generated from a positive baseline plus a class-dependent branch pattern and clipped at zero. Initial
 514 full-data gradients supplied $S_b = \|\mathbf{g}_b\|_2^2$. Coefficient-noise variance was then scaled so that

$$N_b = S_b \exp(-2hx_b), \quad x_b \in [-1, 1], \quad (63)$$

515 where x_b is the evenly spaced branch-order coordinate and $h \in \{0, 0.5, 1, 1.5, 2\}$ controls reliability heterogeneity.
 516 The statistical reliability was $r_b = S_b / (S_b + N_b)$, lower-bounded at 0.03 for numerical stability. Because the common
 517 step was $\eta = c/L$ with $c = 0.5$, the corrected fixed-step target was $a_b^* = \min\{1, r_b/c\}$ and the initial shunt was
 518 $\kappa_b = \bar{G}_b[(a_b^*)^{-1} - 1]$, where $\bar{G}_b$ is the training-example mean of G_{bi} at initialization. Shunts remained fixed during
 519 the 40 updates; the excitatory conductance parameters were trained, and the same coefficient-noise realizations
 520 were used across matched methods within seed.

521 Nine conditions separated the relevant alternatives: exact clean backpropagation, noisy credit without a shunt,
 522 a state-matched unattenuated-eligibility control, the step-consistent best single global shunt, reliability-aligned,
 523 seed-shuffled and anti-aligned shunts, an explicit point-gain control receiving the identical branch factors, and
 524 aligned shunting with clean credit. The point-gain control is a resource control, not a biological assertion: because
 525 it receives the same branch partition and factors, its update must equal Eq. 62. The clean-backpropagation condition
 526 retains the unattenuated exact gradient and is the optimization reference.

527 An initial implementation using r_b despite $c = 0.5$ was excluded; the corrected protocol preceded two excluded
 528 checks and fifty paired seeds 8100–8149 (2,250 outcomes). State mismatch was below 3×10^{-16} ; independently
 529 computed point-gain and unattenuated controls agreed to 1.4×10^{-14} , and fixed-compensating-current finite
 530 differences to relative error 3.60×10^{-8} .

531 At heterogeneity two, aligned-minus-best-global one-step test-loss reduction was 0.001817 (95% paired interval,
 532 0.001618–0.002020; 49/50 positive). Aligned attenuation also improved immediate progress over unshunted
 533 learning. After forty updates it remained better than global, shuffled and reversed attenuation, but its loss difference
 534 from no shunt was 0.000425 (95% interval, -0.002456 to 0.003071). Clean BP retained the lowest mean loss.
 535 Thus the state-clamped oracle factors improve relative placement without establishing sustained superiority to
 536 unshunted learning; compact endpoints and all seeds are indexed below.

537 1.29 Adaptive local reliability estimation

538 The preceding experiment supplied the reliability moments. We next asked whether they could be estimated from
 539 noisy local observations, specifying the adaptive-gain experiment before examining its outcomes (Table S12). At
 540 the current parameter state, branch b received two conditionally independent noisy local gradient observations, $\hat{\mathbf{g}}_{1b}$
 541 and $\hat{\mathbf{g}}_{2b}$, on the same examples. It formed

$$\hat{S}_b^{\text{obs}} = \langle \hat{\mathbf{g}}_{1b}, \hat{\mathbf{g}}_{2b} \rangle, \quad \hat{N}_b^{\text{obs}} = \frac{1}{2} \|\hat{\mathbf{g}}_{1b} - \hat{\mathbf{g}}_{2b}\|_2^2. \quad (64)$$

542 For independent zero-mean credit noise, these estimate the squared clean branch gradient and the noise energy of
 543 one observation. Eight paired observations initialized the moments; exponential averages with decay 0.9 were then
 544 updated before each of 40 training steps. Negative finite-sample signal estimates and moments below 10^{-12} were
 545 clipped only when the gain was constructed. The step-consistent gain followed Eq. 59, was bounded to $[0.03, 1]$,
 546 and was converted to a shunt using the branch’s current mean total conductance.

547 The task and state clamp are unchanged. At heterogeneity 0/1/2, seven rules compare clean BP, noisy no-shunt,
 548 fixed initial oracle, adaptive local/global/permutated shunts and the independently computed adaptive point gain.
 549 Update and probe noise are paired. Two implementation-check seeds are excluded; fifty paired seeds 8200–8249
 550 supply 1,050 outcomes. State, finite-difference and point-gain checks pass at the stated tolerances.

551 At heterogeneity two, adaptive local attenuation lowers final loss relative to global and shuffled estimates,
 552 yet exceeds no-shunt loss by 0.004250 (95% interval, 0.002499 – 0.005975) and fixed-oracle loss by 0.003449
 553 (0.002577 – 0.004305). Only 26/50 immediate local-versus-shuffled differences favored local, failing the predefined

direction criterion. Final gain ordering correlates with the initial oracle at 0.756, with mean absolute error 0.205. The supplied paired probes, optimizer step fraction and exact state clamp remain necessary assumptions; this is not a fully autonomous inhibitory-plasticity mechanism.

1.30 Same-span noisy coefficient-learning experiment

To isolate coefficient learning from address capacity, we compared three dictionaries with the same span in a rate-based factorial (Table S12). The 16-leaf balanced-tree Haar basis supplied the common rank-eight span through depth three. The tested dictionaries were the eight orthonormal Haar modes, all 15 nested indicators through depth three, and the same nested indicators scaled by positive gains evenly spaced on the log scale from $\exp(-3)$ to $\exp(3)$. The dictionaries' projectors agreed to 9.44×10^{-16} and their positive-spectrum Gram condition numbers were 1, 15 and 388.52.

Each independent seed drew a unit-norm target in the common span. At every update the estimator observed the target plus isotropic Gaussian noise with standard deviation $2/\sqrt{n_{\text{eff}}}$, where $n_{\text{eff}} \in \{4, 16, 64, 256\}$. The same 80 observation vectors were used for every matched parameterization. Vanilla coefficient descent used $0.2/\lambda_{\max}(\Phi^T \Phi)$; the oracle coordinate control used $0.2(\Phi^T \Phi)^+$. Population field loss was recorded after 1, 5, 20 and 80 updates. The target address residual was below 1.45×10^{-30} in every condition, so all reported loss is within-span coefficient-estimation error.

The predictions were specified before two implementation-check seeds, which were excluded from inference, and 50 paired seeds (9100–9149). The results contradicted the prediction that Haar coordinates would always reduce finite-time error. At $n_{\text{eff}} = 4$, raw-minus-Haar final loss was -0.2774 (95% paired interval -0.3260 to -0.2313) and scaled-minus-Haar was -0.1906 (-0.2500 to -0.1335): slower modes filtered high estimator noise. At $n_{\text{eff}} = 256$, the effects reversed to 0.02555 (0.02171 – 0.02952) and 0.17091 (0.14576 – 0.19686), positive in 50/50 seeds for both contrasts. Equation 57, derived after these outcomes to explain the reversal, predicted median crossovers at effective sample sizes 38.53 and 8.25. The Gram-preconditioned field trajectories agreed across dictionaries to 1.15×10^{-15} . Thus static parameterization can alter finite-data learning without altering the attainable credit field.

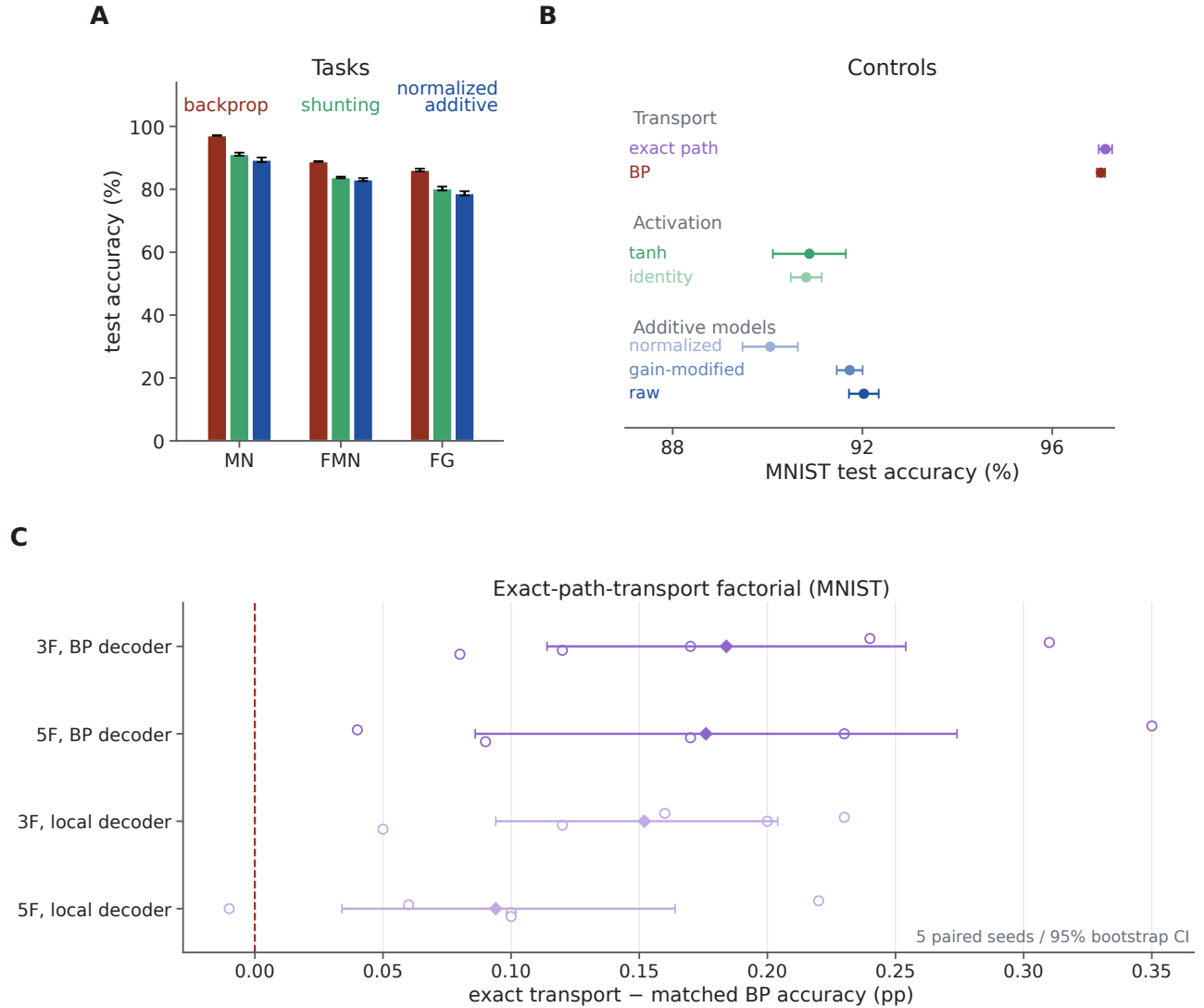


Figure S1: Local eligibility and exact transport reproduce the learning gradient. **A**, Matched shunting backpropagation and local learning on MNIST, Fashion-MNIST and figure-ground MNIST; blue denotes normalized-additive learning. **B**, Transport, activation and additive-model implementation controls on MNIST. **C**, Exact-transport-minus-backpropagation accuracy across three-/five-factor and local/backpropagated decoder choices; points are five paired seeds and bars are 95% paired-seed bootstrap intervals. Exact transport supplies an oracle compartment error. The local three-factor eligibility is sufficient when this error is supplied. Normalized- and raw-additive models are distinct controls; no noise-task shunting aggregate is used here.

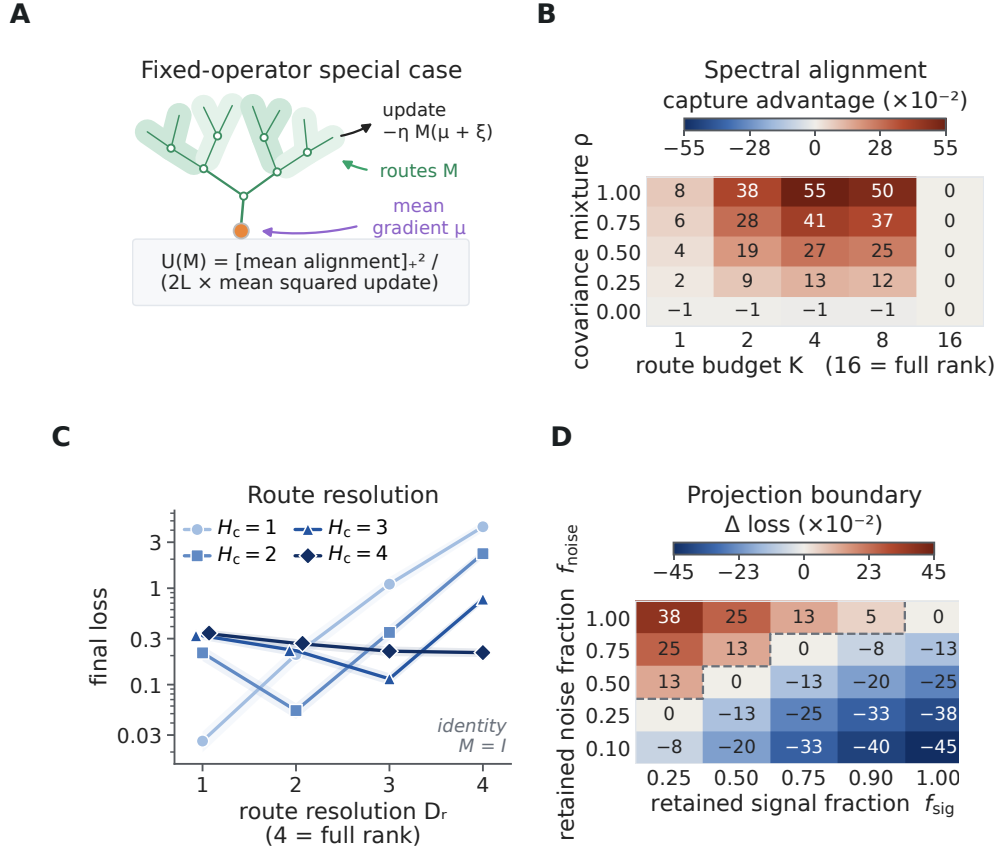


Figure S2: **Restricted routes trade signal retention against admitted noise.** **A**, Fixed-operator version of the one-step utility bound: the delivered update is $M(\mu + \xi)$, with mean gradient μ and zero-mean noise ξ . **B**, Subtree-minus-random capture across route budget and covariance-mixture coefficient; only the endpoints are isospectral. **C**, Final quadratic loss when the signal hierarchy and amplified fine-scale noise make an interior route resolution favorable by construction. **D**, Exact one-step projection boundary: retaining signal fraction f_{sig} and noise fraction f_{noise} changes normalized loss by $(f_{\text{noise}} - f_{\text{sig}})/2$. The related initialization-utility diagnostic matches 135 of 200 seed-task optima; the remaining cases are retained numerically. Panels B–D use abstract Haar coordinates, not a forward dendritic neuron. The 50 paired seeds per signal–noise screen and all four task depths are retained. The bound concerns local progress; the prospective endpoint-selection failure is reported in Fig. S33.

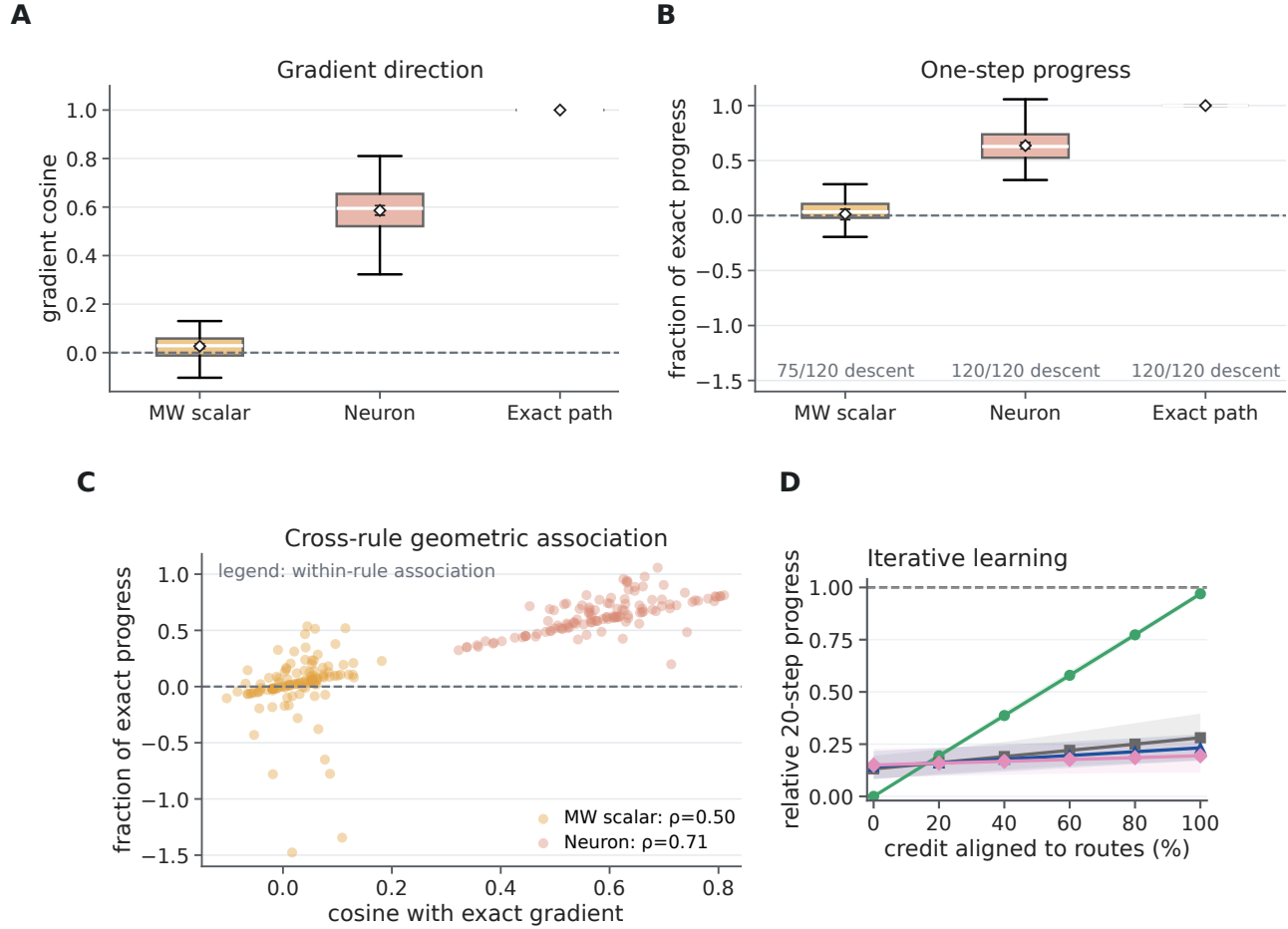


Figure S3: **Gradient geometry predicts a small matched step under stated conditions.** **A–C**, Candidate/exact gradient cosine, norm-matched one-step loss decrease and their within-rule association at 120 valid trained checkpoints. Boxes summarize checkpoints; diamonds and intervals show means and 95% checkpoint-bootstrap intervals. A single held-out batch and relative step 10^{-5} are used. **D**, A separate constructed positive control on eight reconstructed arbors: a field interpolates between the route span and its orthogonal complement, then undergoes twenty projected quadratic steps. Green denotes morphology-selected routes, gray random paths, blue depth bins and pink ancestry-shuffled routes. Symbols show cell means and hierarchical 95% bootstrap intervals. The imposed alignment and oracle projections explain this conditional relationship; it does not measure biological credit. Neither diagnostic validates a long-horizon initialization selector.

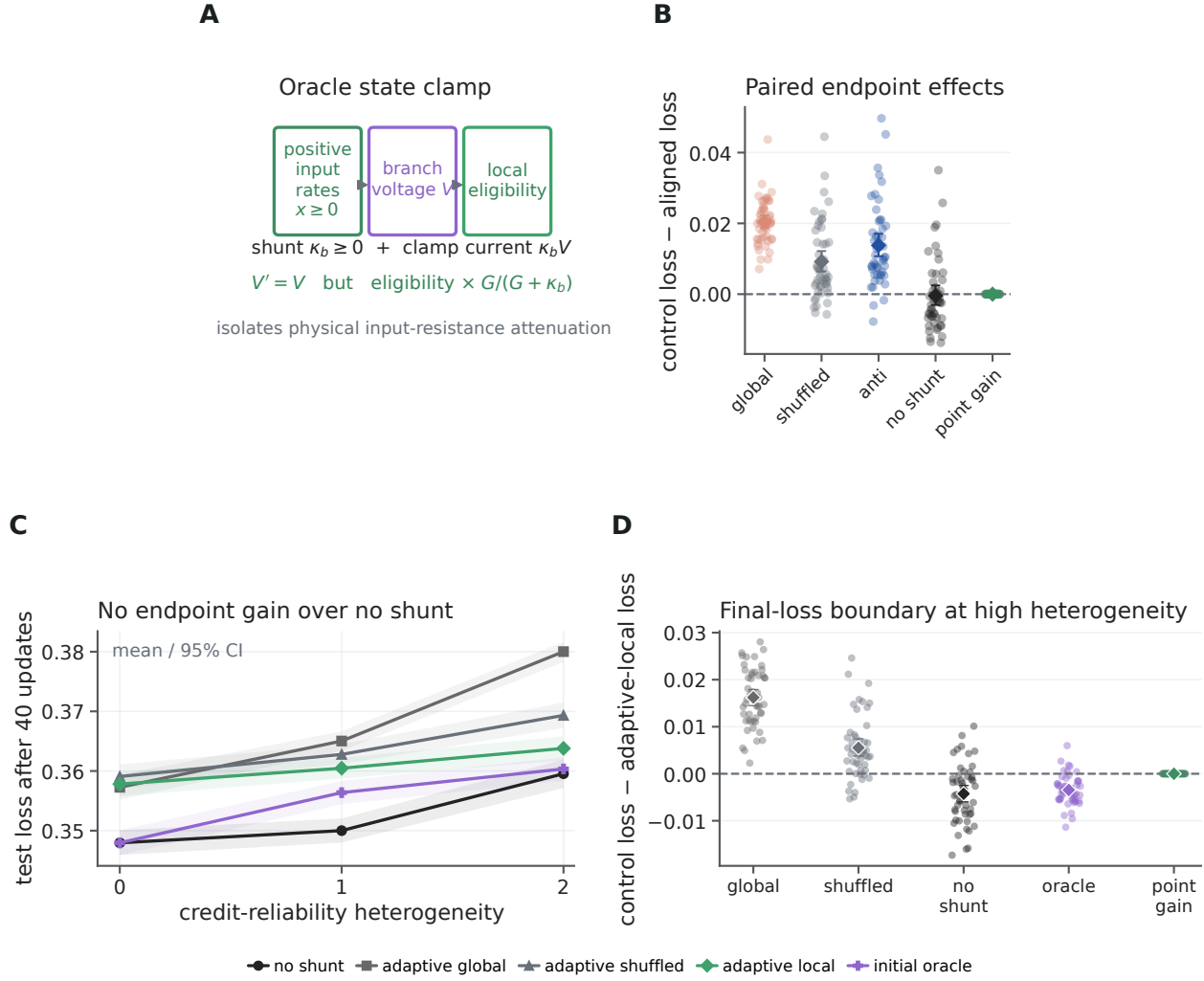


Figure S4: Fixed and estimated reliability gains have bounded learning benefits. All panels use positive-rate synthetic tasks with eight parallel, nonserial branch blocks. **A**, A supplied compensating current preserves branch voltage while a positive shunt reduces input resistance and eligibility. **B**, Paired final-loss contrasts at maximal heterogeneity; fixed aligned shunting has no reliable final advantage over unshunted noisy learning. **C,D**, Adaptive-rule final loss and paired control-minus-adaptive contrasts. Adaptive local gain beats global/shuffled attenuation but loses to no shunt and the fixed oracle. The point implementation given identical gains agrees numerically. Curves and contrast intervals are means and 95% paired-seed bootstrap intervals across 50 independent seeds in each study. Supplied state compensation and gradient-moment observations are explicit information resources.

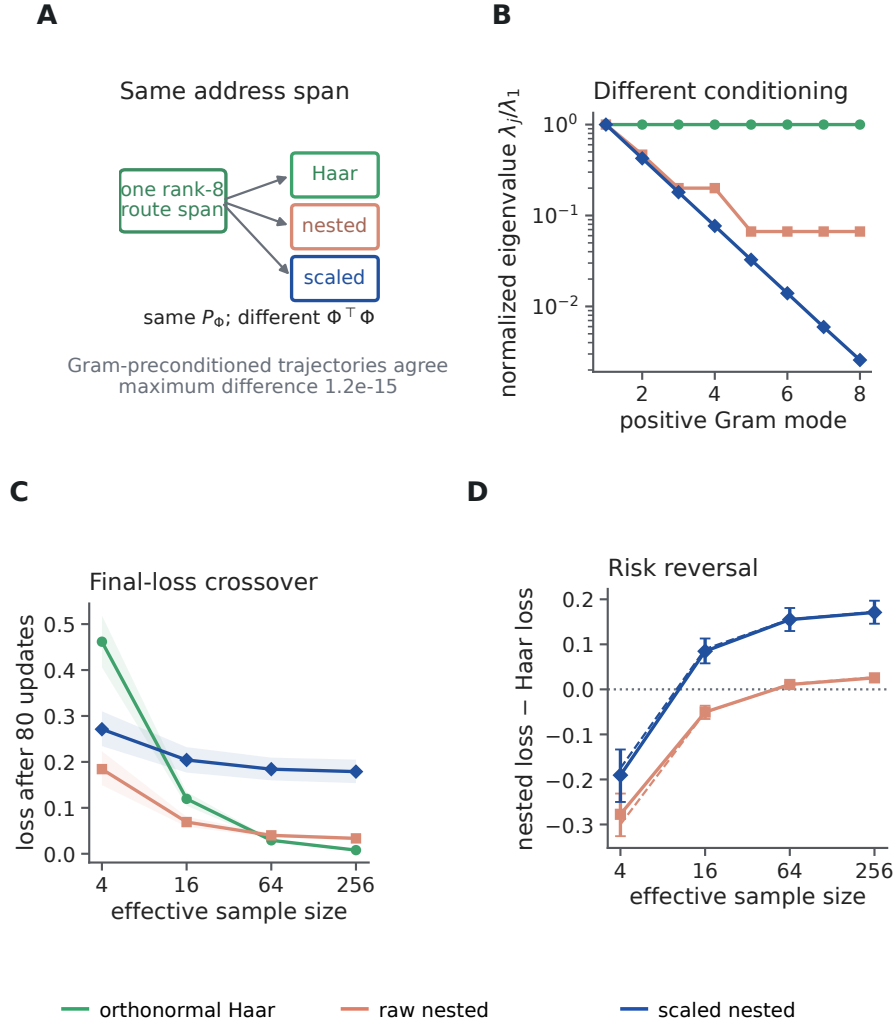


Figure S5: **Identical route spans can learn differently through conditioning.** **A**, Haar, redundant nested and statically scaled nested coordinates parameterize the same rank-eight span in sixteen abstract coefficients. **B**, Positive Gram eigenvalues; condition numbers are 1, 15 and 388.52. **C**, Population loss after eighty updates across effective sample size. **D**, Paired nested-minus-Haar contrasts; negative values favor nested coordinates. Points and bars are means and 95% paired-seed bootstrap intervals across fifty seeds; dashed curves are the exact finite-time risk. Slow modes reduce variance at low data and retain bias at high data. Gram-preconditioned controls agree to 1.15×10^{-15} . This is a coefficient-learning comparison within one span, not a simulation of forward dendritic morphology or a biological implementation of the preconditioner.

S2 Image benchmarks and feedback controls

2.1 Shared architecture, data and local optimization

The MNIST core contains one dendritic population of 128 neurons, with no separate inhibitory-cell population. Each neuron has a soma, three proximal compartments and nine distal compartments, denoted [3, 3]. Separate TopK arrays select 40 excitatory and 20 inhibitory contacts from the 784 pixels at each nonsomatic compartment. There are no somatic input synapses: 92,160 active contacts are selected from 2,408,448 candidate synaptic parameters. Inactive local weights are not updated. Softplus transforms enforce nonnegative synaptic and coupling parameters. A linear 128×10 decoder gives ten class scores. The complete network has 2,414,602 parameter slots, including 1,290 decoder parameters.

Pixels are divided by 255 and flattened without additional normalization or augmentation. Each seed divides the 60,000 official training images into 48,000 fitting and 12,000 validation examples; the official 10,000-image test set is held out. The shared MNIST recipe uses three-factor local eligibility, batches of 256, 180 epochs and Adam with $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$, without weight decay or mixed precision. After minibatch averaging, input, coupling, reactivation and decoder gradients are clipped elementwise to $[-5, 5]$. Their original learning rates are 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} , respectively. Early stopping is disabled and minimum validation loss selects the retained checkpoint. Deterministic PyTorch settings disable TF32 and cuDNN benchmarking and use the highest float32 matrix-multiplication precision.

Every compartment uses the learned output nonlinearity

$$f_n(V) = \frac{\tanh[m_n(V - b_n)] + 1}{2}, \quad f'_n(V) = \frac{m_n}{2} \{1 - \tanh^2[m_n(V - b_n)]\}, \quad (65)$$

where m_n and b_n are slope and midpoint. Shunting trees use analytical initialization. Raw-additive trees use the prespecified occupancy-quantile policy: empirical voltage quantiles 0.1 and 0.9 map to outputs 0.1 and 0.9, with minimum quantile width 10^{-3} , maximum slope 50 and reversion on an invalid fit. The comparisons therefore retain the architecture-specific initialization policies, not identical initial voltage scales.

2.2 Additive controls and gradient coordinates

The raw-additive alias `dendritic_additive` disables both shunting and additive normalization. Its current sums and local derivatives are

$$J_n^{\text{E,add}} = \sum_{i \in \mathcal{S}_n^{\text{E}}} g_i^{\text{E}} x_i^{\text{E}}, \quad J_n^{\text{I,add}} = \sum_{j \in \mathcal{S}_n^{\text{I}}} g_j^{\text{I}} x_j^{\text{I}}, \quad (66)$$

$$J_n^{\text{child,add}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c, \quad V_n^{\text{add}} = J_n^{\text{E,add}} - J_n^{\text{I,add}} + J_n^{\text{child,add}}, \quad (67)$$

with $a_n = f_n(V_n^{\text{add}})$, and

$$\frac{\partial V_n^{\text{add}}}{\partial g_i^{\text{E}}} = x_i^{\text{E}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_j^{\text{I}}} = -x_j^{\text{I}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_{c \rightarrow n}^{\text{den}}} = a_c, \quad \delta_c^V = \delta_n^V g_{c \rightarrow n}^{\text{den}} f'_c(V_c). \quad (68)$$

Inhibition contributes a subtractive current, not a negative conductance. Raw-parameter gradients include the softplus derivative. The inherited normalized-additive control instead standardizes each example across a stage's output coordinates. With $Z_{sn} = J_{sn}^{\text{E,add}} - J_{sn}^{\text{I,add}} + J_{sn}^{\text{child,add}}$, it uses

$$V_{sn}^{\text{norm}} = \frac{Z_{sn} - \langle Z_s \rangle_n}{\text{sd}_n(Z_s) + \epsilon}, \quad a_{sn} = \frac{\tanh(0.1 V_{sn}^{\text{norm}}) + 1}{2}. \quad (69)$$

Its fixed slope and midpoint are 0.1 and zero. Because this changes forward computation, its outcomes are not pooled with raw-additive results.

608 The fresh dictionary experiment routes errors of compartment output activation; f'_n remains in local eligibility,
609 including gate-parameter derivatives. Neuron-specific feedback repeats each exact somatic error within that tree.
610 The projected $K = 1$ rule averages exact activation errors over all twelve nonsomatic compartments; $K = 3$ averages
611 them separately within each proximal compartment and its three children. Both acquire coefficients from the same
612 exact field and preserve its common mean at identical weights, but need not preserve equal field norms. Their
613 somatic error remains exact. Strict scalar instead reduces the neuronal error vector at every core stage, including the
614 soma, using Eq. 25. Exact transport retains every compartment error. The linear decoder receives exact gradients in
615 all arms. Decoder-only learning suppresses core and gate gradient writeback; initial/final core hashes verify the
616 intended freeze. These projected comparisons test delivery capacity with oracle coefficients.

617 2.3 Fresh dictionary controls and retained image cohorts

618 The six-rule MNIST experiment compares common rate multipliers 0.3, 1, 3. Three development seeds per
619 architecture, rule and multiplier give 108 fits. A separately frozen addendum specified projected $K = 1$ before fresh
620 outcomes, with the same tuning procedure. Mean minimum validation loss selects each rule’s multiplier. Ten new
621 paired seeds per architecture and rule then test the selected rate and, when different, the original multiplier one.
622 Coincident conditions reuse a fit, giving 190 unique fresh fits; twelve three-epoch canaries are excluded. Pairing
623 covers initialization, contacts, splits and minibatches. Descriptive intervals use 50,000 whole-seed bootstrap draws.
624 Full configurations and seed endpoints are in `source_data/image_ladder_controls/`.

625 Capture was measured at initial and exact-trained checkpoints on the first 2,048 official test images. For each
626 example/neuron, the field comprises twelve nonsomatic errors. The primary endpoint averages nonzero field-wise
627 squared-energy ratios; aggregate energy capture is also retained. Activation-space capture matches the trained
628 projections, while voltage-space capture preserves the earlier diagnostic coordinate. Two historical checkpoints
629 reproduced their recorded voltage captures to numerical precision; instrumentation left outputs unchanged. In the
630 earlier fifteen-seed cohort, moving from one to three profiles increased mean voltage-credit capture from 0.480
631 to 0.625 in shunting trees and from 0.519 to 0.718 in additive trees. These values describe a separate cohort and
632 coordinate from the activation-error capture in main Fig. 1F.

633 The original MNIST ladder uses seeds 42–56: strict scalar, neuron-specific and exact credit in both architectures
634 give 90 fits; matched-width scalar fallback adds 30. The latter differs from strict scalar in a small near-somatic
635 parameter group, as quantified in Section 1.8. These consistently executed runs replace the superseded mixed-source
636 set; the replacement decision preceded outcome inspection. The independent direct-feedback-alignment factorial
637 uses fifteen paired seeds per dynamics and 120 new fits with frozen exact-readout references. A seed-fixed Gaussian
638 output-to-neuron matrix, scaled by inverse square root of output dimension, replaces the decoder derivative.
639 Ownership controls permute neuronal recipients without changing values or bandwidth.

640 Fashion-MNIST retains the MNIST architecture, contacts, optimizer, initialization and 180-epoch budget
641 without retuning. Seeds 10200–10209 pair 48,000/12,000 fitting/validation images and the 10,000-image official test
642 set. Sixty fits cross two architectures with matched-width scalar fallback, neuron-specific and exact fields. Neuron-
643 specific-minus-fallback accuracy is 4.46 percentage points (95% interval 4.09–4.88) for shunting and 3.78 (3.43–
644 4.14) for additive trees; all ten differences are positive in each. Exact-minus-neuron-specific differences are -0.11
645 (-0.21 to -0.01 ; 2/10 positive) and 0.13 (-0.08 to 0.35 ; 8/10). The prespecified neuronal-identity criterion required
646 at least 8/10 positive differences and an interval above zero in each architecture. All sixty fits were complete and
647 finite without feedback fallbacks. Full means and tests remain in `source_data/fashion_feedback_ladder/`.

648 2.4 CIFAR-10 and exact-gradient validation

649 The raw-additive CIFAR-10 cohort uses flattened images, no augmentation or normalization, a 90:10 train-
650 ing/validation split, one 20-soma [3, 3, 3, 3] population, 25 contacts of each sign per branch and a 20–32–16–10
651 decoder. The realized `input_mode=1` model has direct excitatory/inhibitory inputs and no inhibitory-cell popula-
652 tion. A validation-only BP screen jointly selected adaptive initialization and empirical stage calibration from three
653 batches; it cannot attribute their benefit separately. A provisional configuration with mean BP accuracy 38.98%

was excluded for an inadequate operating point. The selected three-seed calibration reached 49.960% validation and 49.723% test accuracy. A separate five-seed check reproduced normalized-additive and shunting reference endpoints to within 0.026 percentage points.

All four fresh rules use Adam at 10^{-3} , batch size 256, at most 200 epochs, patience 40 and the validation-best checkpoint. Weight decay is zero; the separate 0.01 configuration field controls sparse-weight maintenance. Seeds 10800–10819 give eighty finite, complete fits. Mean accuracies are 0.34472 (strict scalar), 0.50865 (neuron-specific), 0.50006 (exact path) and 0.50122 (matched BP). The exact-minus-neuron-specific difference is -0.00859 (95% interval -0.01171 to -0.00547 ; 2/20 positive), so the exact-path superiority criteria fail. Exact-minus-BP is -0.00115 (-0.00457 to 0.00226), with two one-sided tests supporting equivalence within ± 0.01 (90% interval -0.00398 to 0.00167 ; $P = 1.58 \times 10^{-5}$). This is a compact transfer assay, not a competitive vision benchmark.

A separate five-seed exact-transport factorial (42–46) crosses three-/five-factor rules with local/BP decoder updates over 200 epochs and a ReLU core output. Local rates for input, coupling, reactivation and decoder are 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} , 1.5×10^{-3} ; matched BP uses 10^{-3} , 5×10^{-4} , 5×10^{-4} , 10^{-3} . Analytic and automatic gradients were compared in 100 diagnostic runs: median relative error 7.43×10^{-8} , maximum 1.88×10^{-7} , with parameter tensors as diagnostic units. The architectures and method-specific rates above distinguish these protocols.

2.5 Depth, ownership and input-map controls

The original depth-by-feedback factorial crosses two tasks, two cores, stages D1–D4, three local fields plus BP and ten seeds: 640 planned fits. Completeness requires the specified seed/configuration, checkpoint and full curves, expected stages, nondegenerate couplings, finite metrics and no transport-shape fallback. All completed, but a later outcome-independent validity check excludes 160 synthetic-noise shunting fits with signed conductance drive. The retained 480 fits contain both cores on MNIST and additive noise-task models. Depth changes compartment and parameter count. The ownership comparison at D2/D4 retains 120/160 fits after the same exclusion. A 400-fit inhibitory-dose family is entirely excluded from inference because its primary shunting contrast uses invalid signed input, although configurations and exclusion records remain available.

The clean exact/BP rerun instead uses valid inputs: two tasks, two cores, four depths, two methods and seeds 42–51 give 320 fits. Shunting uses explicit ReLU transfer; additive noise inputs remain signed. Source commits are 74792ca9 for implementation and 4f3612a5 for the nested generator. All 320 fits are finite and complete; 42 use streaming evaluation of a terminal metric. Failed attempts with missing generator code, insufficient memory or rejected signed shunting have no included endpoint. Re-executed configurations retain their scientific settings. Comparisons first average four depths within each seed, then use 20,000 paired bootstrap draws and descriptive Wilcoxon tests. Across all 160 method pairs, mean exact-minus-BP accuracy is -0.0545 points; the largest absolute pair difference is 5.90 points. This does not establish pointwise equality.

The spatial-input control uses depth four, 21 contacts per distal branch and 320 planned fits. Spatial branches recursively partition the 28×28 feature plane into sixteen disjoint regions; random branches sample all 784 coordinates. The 240 conductance-valid fits cross maps, methods, cores and tasks; the noise task retains only additive models. Persistence under BP and random task projection identifies a forward input-coverage effect. A separate 320-fit noise comparison holds sixteen terminal branches and 960, 960, 968 or 960 active contacts per soma across D1–D4; it retains 160 additive fits. Compartments, couplings and reactivation parameters still increase with depth.

At 120 valid BP checkpoints, a fixed held-out batch compares scalar fallback, neuron-specific and exact fields using nonsomatic stages and trainable branch parameters only. Outcomes are optimally scaled field capture, eligibility-weighted gradient capture/cosine and norm-matched loss change. Relative step 10^{-5} is primary; 10^{-6} , 10^{-4} , 10^{-3} are sensitivity values. Checkpoints are neither updated nor selected by this diagnostic, and bootstrap resampling preserves paired fields. Complete controls and validity accounting accompany the figure-linked Source Data.

2.6 Identity of the historical noise tasks

The key `noise_resilience` named two generators. The clean rerun’s recovered source, commit 4f3612a5, draws class y uniformly from $\{0, 1, 2\}$ and constructs a square image L_y : blank, a uniformly selected horizontal row of ones, or a uniformly selected vertical column of ones. It returns

$$x = \text{vec}(L_y + 0.2\epsilon), \quad \epsilon_{ab} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (70)$$

The executed dispatcher’s default side length is 784, giving 614,656 inputs, independently confirmed by checkpoint sizes. Its 10,000 unnormalized, unclipped examples split into 8,000/1,000/1,000 fitting/validation/test samples using run-seeded generators. Projected-noise fields in launch configurations were ignored. Thus these rows test implementation agreement on three classes, not corrupted-MNIST robustness.

The documented prospective depth/routing/dose/fixed-budget protocol instead perturbs MNIST digits as

$$\tilde{x} = \text{clip}_{[0,1]}(x + 1.5Pz), \quad z \sim \mathcal{N}(0, I_{50}), \quad (71)$$

where $x \in [0, 1]^{784}$, $P \in \mathbb{R}^{784 \times 50}$ has independent normal entries and unit-norm columns, and digit labels are unchanged. Projection seed is zero; training/validation/test latent-noise seeds are 1/2/3. The intended splits are 48,000/12,000/10,000 without further gain or normalization. Frozen configurations and model sizes support this intended protocol, but no hash of the executed nested generator survives. The fixed-budget outcome/validity join also lacks raw resolved configurations and does not close that gap.

Inherited aggregate feedback/gradient panels lack a complete run-to-generator join. Twenty-six historical shunting points remain unresolved and are excluded from conductance-valid figures and inference; neither generator is assigned to them retrospectively. The fixed-budget audit retains 160 additive and excludes 160 signed-shunting outcomes. The release exposes `legacy_noisy_lines` and `projected_noise_mnist` separately and rejects the ambiguous key without explicit choice. The latter is a frozen implementation of the documented protocol, not recovered historical execution. See `source_data/release_task_identity/task_identity.json` and `source_data/review_curve_lineage/`.

Inherited regime sweeps use five seeds per condition; Gaussian broadcast noise is scaled relative to the original teaching signal, and depth varies at fixed somatic width. Their normalized-additive, raw-additive and shunting variants remain separate. They test operating regimes, not isolated backward effects of inhibition.

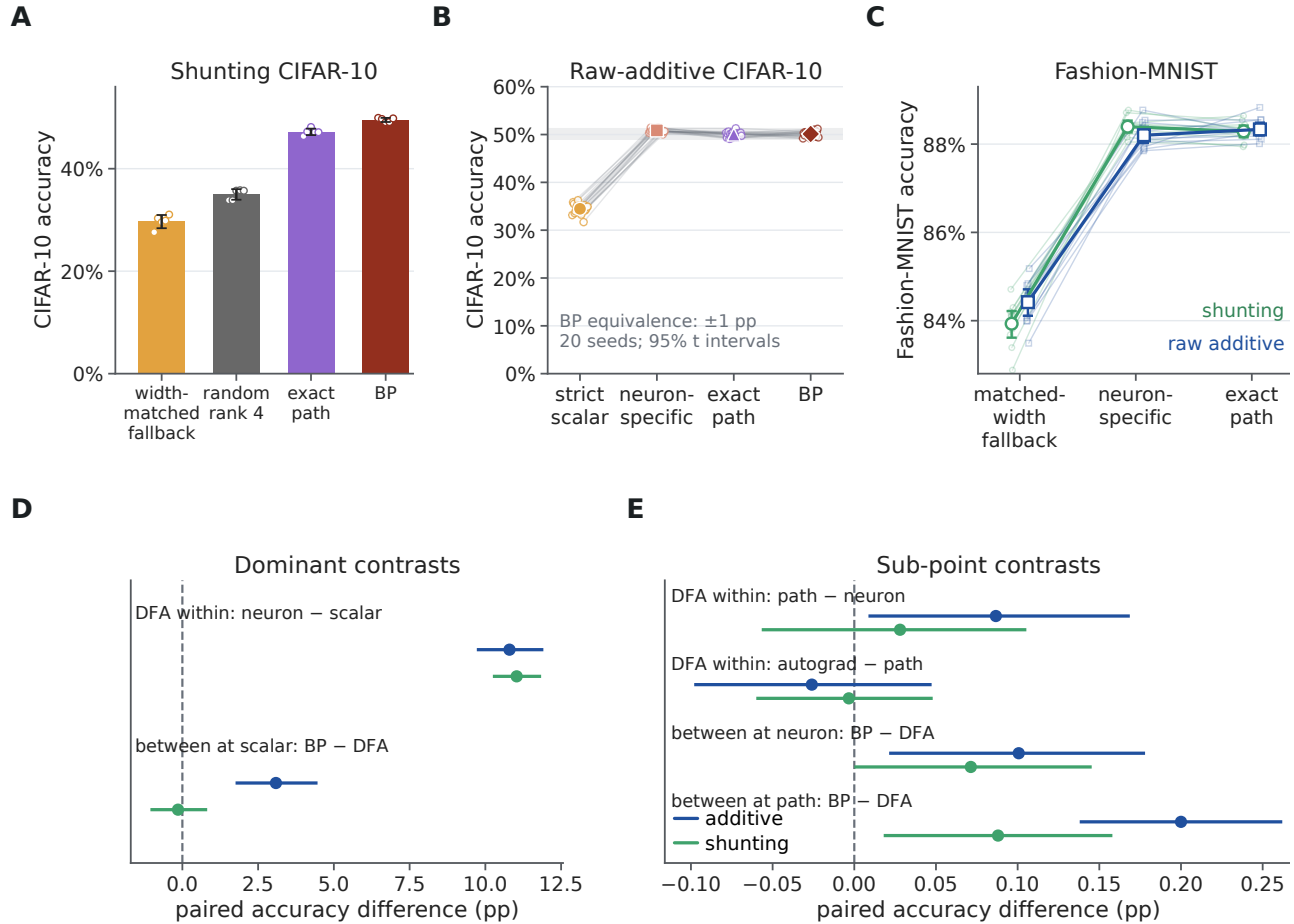


Figure S6: The neuron-identity bottleneck generalizes across image and feedback controls. **A**, Flattened CIFAR-10 in the shunting architecture, with matched-width scalar fallback, random rank-four, exact-path and backpropagation rules (five seeds). **B**, Raw-additive CIFAR-10 with strict scalar, neuron-specific, exact-path and backpropagation feedback (twenty paired seeds; 95% Student-*t* intervals). Exact path is 0.859 percentage points below neuron-specific feedback and equivalent to backpropagation within the prespecified one-point margin. **C**, Fashion-MNIST in shunting and raw-additive trees (ten paired seeds; 95% bootstrap intervals); path resolution adds no reliable benefit. **D,E**, Dominant and sub-point contrasts under fixed random soma feedback (DFA), with fifteen paired MNIST seeds and 95% bootstrap intervals. DFA preserves the neuron-versus-scalar benefit, but does not add a second dendritic layer or increase the dimensionality of the ten-class readout error. Percentage-point scales differ.

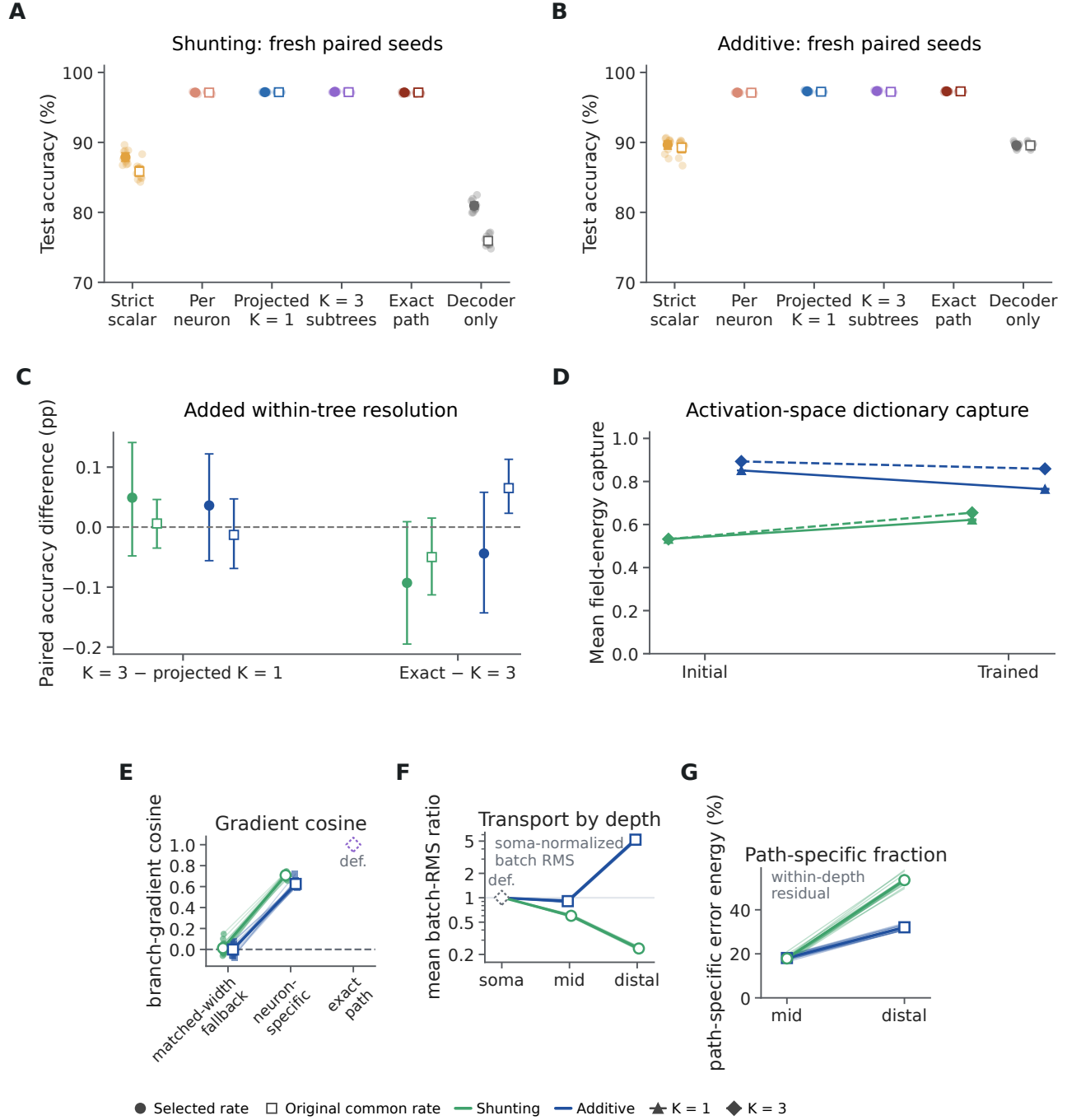


Figure S7: Intermediate dictionaries capture more image-task credit without improving accuracy. **A,B**, Six-rule MNIST ladder in shunting and raw-additive networks; ten fresh paired seeds per architecture. Filled/open symbols use development-selected/original common rates. Decoder-only learning freezes the initialized core. **C**, Paired accuracy differences for three profiles versus a projected common profile, and exact versus three-profile credit. **D**, Initial and trained activation-error capture by one or three profiles, averaged over 2,048 images within each seed. Projection coefficients use the exact field. Means and bars in A–D have descriptive 95% seed-bootstrap intervals. **E–G**, Independent fifteen-seed cohort: common-state branch-gradient cosine, soma-normalized exact voltage-error magnitude by depth, and within-depth path-specific error energy. These distinguish direction, amplitude and spatial variation; exact-path cosine one is definitional. Greater path-specific variation did not yield an image accuracy benefit. Voltage- and activation-space capture are different quantities. All image cohorts have one dendritic layer and a linear readout.

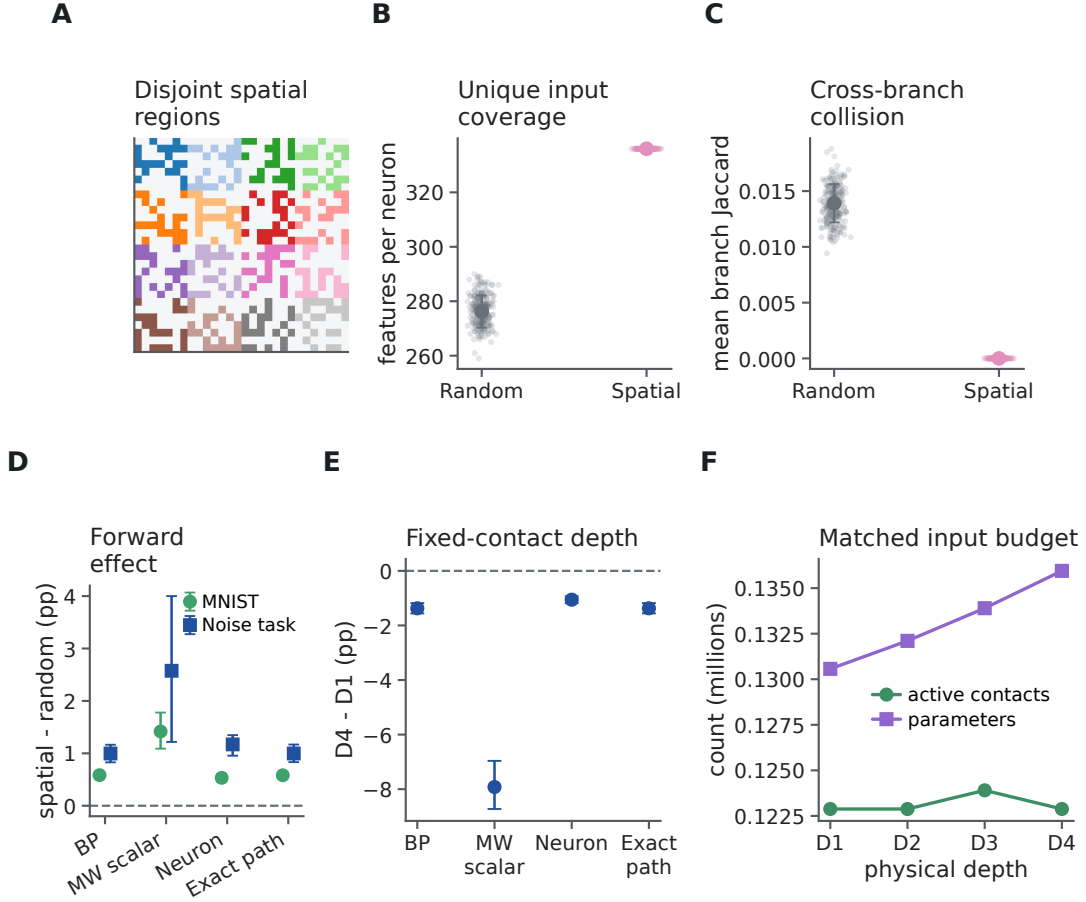


Figure S8: Input coverage and ordinary-task depth are distinct from credit resolution. **A–D**, Spatial versus randomly assigned contacts in depth-four $[2, 2, 2, 2]$ trees: example map, unique input coverage, cross-branch overlap and learning benefit under backpropagation and three local fields. Contact count is matched, but spatial maps reach more unique inputs and avoid collisions. The accuracy benefit persists under backpropagation and therefore reflects a forward sparsity prior. **E,F**, Separate raw-additive historical noise cohort with sixteen terminals: all four D4-minus-D1 accuracy contrasts are negative in 10/10 paired seeds; active contact counts are approximately fixed while parameter counts vary. Learning contrasts use means and 95% paired-seed bootstrap intervals; map summaries use mean $\pm$ s.d. The noise-task executed generator remains unresolved and its shunting runs fail the positive-conductance input criterion. These retained additive outcomes are a depth boundary, not evidence for shunting; exclusions are enumerated in Table S5.

S3 Branch conflict, ancestry routing and coefficient learning

3.1 Context-gated Fashion-MNIST protocol

The context-gated task used Fashion-MNIST classes 0 and 6. Inputs were average-pooled to 7×7 , standardized using training data and augmented with a constant coordinate. Each seed used 8,192 training and 2,000 test draws with replacement from the official binary splits, balanced across context and label. Branch counts were $B = 2, 4, 8$; conflict doses were $0, 0.25, 0.5, 4/7, 0.6, 2/3, 0.75, 1$. Thresholding one underlying uniform mask made conflict nested across doses. Correct, shared, cyclically deranged, BP and gated-point rules reused inputs and initialization. Full-batch descent used 250 updates at step 0.04 and an initial weight standard deviation (SD) of 0.01. Three exploratory seeds (100–102) selected a stable setting; an additional implementation-check seed was also excluded. Analysis seeds 11000–11019 were fixed before outcomes, with no outcome-dependent stopping.

The primary estimand was each seed’s least-squares slope of correct-minus-shared accuracy on conflict, separately for each B . Prespecified criteria required positive slopes in at least 16/20 seeds, at least ten percentage points of separation at full conflict, less than two points at zero conflict, correct routing above derangement and exact BP/gated-point agreement. All criteria passed. The maximum analytic-versus- autodifferentiation discrepancy was 3.43×10^{-7} ; equivalent-rule accuracy differences were zero. Intervals used 20,000 paired-seed bootstrap draws; tests were two-sided Wilcoxon tests. Mean-field assumptions and empirical chance crossings are distinguished below.

The same twenty seeds were replayed for post hoc gradient diagnostics at epochs 0, 1, 10, 50, 100 and 250, comparing both each method’s own state and a common exact-rule state. Official input-file checksums were verified; the input manifest and all 28,800 rows are retained in `source_data/review_branch_trajectories/`. Endpoint drift was below 5×10^{-9} in loss. At conflict 0.5, common-state shared-gradient cosine fell from 0.987 to -0.519 for four branches and from 0.952 to -0.701 for eight branches by epoch 250 (Supplementary Fig. S9). The initialization boundary below was retained without refitting; trained gradient direction and trained accuracy crossing are separate endpoints.

3.2 Mean-field boundary for context-gated branch selection

The Fashion-MNIST conflict experiment selects one branch c uniformly from B branches. Only $\mathbf{w}_c^\top \mathbf{x}_c$ enters the logit, so the exact derivative of every unselected branch is zero. Shared feedback omits that selector and updates every active eligibility. For the centered mean-field prediction, assume balanced class signs $s \in \{-1, 1\}$, class means $\mathbb{E}[\mathbf{x} | s] = s\mathbf{v}$, and zero logits, so the logistic error is $\delta = -s/2$. An unselected input copies the selected image with probability $1 - \chi$ and is independently drawn from the opposite class with probability χ . Its mean class evidence relative to the selected view is therefore $1 - 2\chi$. The resulting compatibility matrix is

$$C_\chi = (1 - 2\chi)\mathbf{1}\mathbf{1}^\top + 2\chi I.$$

Its shared-mode eigenvalue is $B - 2\chi(B - 1)$, which vanishes at $\chi_c = B/[2(B - 1)]$. Under these population and initialization assumptions, the expected shared update loses positive alignment with the exact update at that boundary. Logistic derivatives and class-conditional moments need not preserve this relation as weights change. The empirical trained chance crossing is consequently a separate descriptive endpoint, not a theorem about the entire optimization trajectory. The missing branch selector may be supplied in feedback or in local eligibility; it is not evidence that the unselected branches require opposite exact errors.

3.3 Two-branch subtree-address experiment

The minimal two-sibling experiment used ten paired seeds (4100–4109), following an excluded numerical-check seed 4099. The check verified nondegenerate shared capture, exact-route capture one and analytic exact/BP agreement. The subsequent eight-stream factorial varied route budget.

For seed-specific unit vectors $\mathbf{u}_A$ and $\mathbf{u}_B$, binary label sign $y \in \{-1, 1\}$ and context $c \in \{0, 1\}$, the two input means were

$$\mathbb{E}[\mathbf{x}_A | y, c] = y [\mathbb{I}(c = 0) - 0.75\mathbb{I}(c = 1)]\mathbf{u}_A, \quad (72)$$

$$\mathbb{E}[\mathbf{x}_B | y, c] = y [\mathbb{I}(c = 1) - 0.75\mathbb{I}(c = 0)]\mathbf{u}_B. \quad (73)$$

Each stream had 12 coordinates and independent Gaussian noise with standard deviation 1.15. The 2,048-example train and test splits contained exactly 512 examples in each context–label cell. The somatic logit selected branch *A* when $c = 0$ and branch *B* when $c = 1$. Both inputs nevertheless remained active, preventing the one-coordinate rule from acquiring branch specificity through a zero eligibility on the distractor.

For somatic logit z and logistic sigmoid σ , let $\delta = \sigma(z) - \mathbb{I}(y = 1)$ and $\mathbf{r}(c) = (\mathbb{I}[c = 0], \mathbb{I}[c = 1])$. Exact and correct-subtree coefficients were $\delta \mathbf{r}(c)$; within-neuron derangement reversed the two entries, giving $\delta \mathbf{r}(c)_{::-1}$; the shared neuronal and same-depth controls used $\delta(1, 1)$; and the random rank-two control used $\delta \mathbf{r}(c) R_s^T$ for a seed-fixed orthogonal rotation R_s . The gated point emulation used one somatic coefficient together with the context-gated features $(\mathbb{I}[c = 0]\mathbf{x}_A, \mathbb{I}[c = 1]\mathbf{x}_B)$, making its alternative address resource explicit. All conditions had 24 trainable weights and the same paired initialization and batch order. The random dense control had four routing nonzeros; the sparse two-route conditions had two.

Training used mini-batch stochastic gradient descent (SGD), learning rate 0.3, batch size 64 and 60 epochs. The switch endpoint applied another 12 epochs to 1,024 context-1 examples with balanced labels and measured the change in context-0 accuracy. For gradient geometry, exact and candidate per-example gradients were concatenated over both branches. Scaled capture was squared cosine after optimal scalar rescaling. The one-step candidate was normalized to the exact batch-gradient norm and evaluated at step 0.25. The Source Data indexed in Table S12 retain every tested condition; no condition was removed on the basis of its outcome.

All controls, including poor deranged and variable random-map outcomes, are retained. Correct routing, exact transport, BP and gated-point emulation coincide because the exact path coefficient is the binary selector. Thus this experiment isolates the supplied address resource; the gated-point control reproduces it.

3.4 Subtree-address bandwidth and architecture factorial

To vary feedback bandwidth, the complete factorial extended the two-sibling task to eight context-selected streams arranged in a balanced binary hierarchy. Context and binary label were exactly balanced in train and test sets, every stream remained active on every example, and the sign of the task signal reversed outside the selected context. Thus a shared coefficient could not recover the required address simply because off-route eligibilities were zero. The prespecified feedback budgets were $K \in \{1, 2, 4, 8\}$ nested routes. Twenty paired analysis seeds (4200–4219) crossed five representations and 27 method–budget combinations, giving 135 fits per seed and 2,700 fits in total.

The dendritic tree, explicitly gated point model, flat-compartment model and grouped-point model were implementation-equivalent controls: each contained 64 trainable weights, 64 active input contacts and one output nonlinearity, and each received the same routed coefficient field. The fifth representation was a degree- and depth-matched rewired tree. Its resource counts were unchanged, but its ancestry groups were permuted relative to the task hierarchy. Feedback families comprised correct ancestry, fixed within-neuron derangement, depth-interleaved groups, random sparse groups, a seed-fixed random rank- K map, and a task-fitted rank- K upper bound. Neuron-shared, exact-transport and analytic-backpropagation references completed the design. Route fields were unit-normalized per example. Data, initialization and every stochastic control map were keyed to seed, method family and K , so matched representations saw identical stochastic realizations.

Pre-analysis checks required balanced labels and contexts, nonzero activity in every stream, shared capture below 0.2, exact capture one, analytic exact/BP equality and no fallback. An initial 2,700-fit cohort used stochastic maps unpaired across representation labels and was excluded. Only the complete repeat with paired maps and cross-representation equivalence checks enters inference.

The primary bandwidth contrasts are retained in the Source Data indexed in Table S12. Correct nested-subtree routing strongly exceeded route derangement at $K = 2$ and $K = 4$, but its comparison with the best matched

non-anatomical control was deliberately more stringent: ancestry lost at $K = 1$ and $K = 2$, won by 0.0127 accuracy at $K = 4$ (95% paired seed-bootstrap interval 0.00593–0.01946; 15/20 seeds; paired Wilcoxon signed-rank Holm-adjusted $P = 0.0101$ across the four budgets; the Source Data retain the paired mean-sign-flip sensitivity test), and tied at full rank. Correct routing on the task-matched tree exceeded the rewired tree at $K = 2$ and $K = 4$ in all 20 seeds. The four matched representations were exactly equal in every condition (maximum held-out-accuracy difference zero). The experiment therefore supports an intermediate-bandwidth advantage for a task-aligned nested address, while showing that an explicitly routed point implementation can reproduce it.

3.5 Trained partition-residual reconstruction

Using the saved configurations and outputs, we quantified irreducible address error in a post hoc diagnostic of the complete factorial. For example b , let c_b denote context, δ_b the somatic error, $e_{c_b,i}$ the indicator selecting leaf i , and $\mathbf{x}_{bi}$ its input vector. We defined the exact coefficient field $c_{bi} = \delta_b e_{c_b,i}$, coordinate weight $w_{bi} = \|\mathbf{x}_{bi}\|_2^2$, and active route field ϕ_{bi} . Allowing the best single coefficient for that route on each example gives

$$z_b^* = \frac{\sum_i w_{bi} \phi_{bi} c_{bi}}{\sum_i w_{bi} \phi_{bi}^2}, \quad \mathcal{C}_{\text{part}} = 1 - \frac{\sum_{b,i} w_{bi} (c_{bi} - z_b^* \phi_{bi})^2}{\sum_{b,i} w_{bi} c_{bi}^2}. \quad (74)$$

This is the normalized complement of the shared-coefficient residual in Eq. 53. It grants an optimal route coefficient and therefore isolates address/support error from the coefficient scale actually used during training.

All 20 seeds and 135 conditions per seed were reconstructed at initialization and after 80 training epochs. Reconstructed losses agreed with the recorded values within 4.97×10^{-9} and accuracies within 5.01×10^{-11} , the precision of the saved decimal tables; the address-plus-coefficient Pythagorean decomposition closed within 7.28×10^{-12} . On the dendritic-tree representation, trained correct-ancestry capture was 0.162, 0.315, 0.547 and 1.000 at $K = 1, 2, 4, 8$. Fixed coordinate-to-arbor derangement had the common scalar value at $K = 1$ and zero capture at $K = 2, 4, 8$, because its active support excluded the target leaf (Supplementary Fig. S10A).

For each seed we averaged implementation-equivalent representations within feedback family and budget, then correlated trained capture with held-out accuracy across conditions. The mean within-seed Spearman correlation was 0.824 (95% seed-bootstrap interval 0.810–0.838; 20/20 positive; two-sided Wilcoxon $P = 1.91 \times 10^{-6}$). Restricting the analysis to the six route families and $K < 8$ removed the exact full-rank boundary and gave 0.639 (0.610–0.665; 20/20 positive; $P = 1.91 \times 10^{-6}$; Source Data, Table S12). This association is descriptive and post hoc. In particular, ancestry, depth-interleaved and random sparse partitions had similar address capture at matched K , so the residual does not explain the small trained ancestry advantage at $K = 4$.

3.6 Learning coefficients from an explicit local context cue

To test whether coefficients could be learned from a supplied cue, we used a four-route dictionary on fresh versions of the balanced eight-context task. We retained its 64 forward weights, paired initialization, 2,048-example train and test sets, and 80 full-batch task-training updates. The dictionary was fixed before task training: $\Phi_{bk} = \mathbf{1}[k = \lfloor b/2 \rfloor] / \sqrt{2}$ for zero-indexed blocks $b = 0, \dots, 7$ and routes $k = 0, \dots, 3$. For trial i with context c_i , the supplied local cue was $\mathbf{h}_i = \mathbf{e}_{c_i} + \sigma_h \boldsymbol{\varepsilon}_i$, with $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(0, I_8)$. Here $\mathbf{e}_{c_i}$ is the one-hot context vector and σ_h is cue-noise SD. The cue is an additional information resource; the experiment does not infer a biological signal.

Separate balanced calibration trials supplied the target route-activation vector $\mathbf{e}_{\lfloor c_i/2 \rfloor}$. A four-output softmax encoder with an intercept learned this mapping by minibatch delta-rule updates. Weights started at zero; calibration used 30 epochs, batch size 16, step 0.2 and weight decay 0.001 on non-bias weights. Calibration sets of 16, 64 and 256 trials were nested within each cue-noise condition. Because epoch count was fixed, increasing sample count also increased the number of encoder updates. This axis therefore changes both calibration data and computation. No task labels, exact task gradients or held-out task outcomes entered encoder training. Its parameters were frozen before the classification weights were trained.

At cue delays $d = 0, 1, 4$ trials, the encoder received the cue array circularly shifted by d . Predicted coefficients $\mathbf{c}_i$ produced the normalized field $\Phi \mathbf{c}_i / \|\Phi \mathbf{c}_i\|$. Delay is a mismatch between the current eligibility and an earlier

independent context; it is not a calibrated physiological time constant. Controls used exact one-hot route selection, a uniform frozen profile, or a cyclic permutation of the learned coefficient coordinates. Exact and frozen controls repeated across nuisance conditions are algebraic repetitions rather than independent evidence. Task updates used the original factor-eight context rescaling and step 0.3.

Twenty analysis seeds (52000–52019) followed the excluded implementation-check seed 51999. Crossing three calibration sizes, cue-noise SDs 0, 0.5, 1, delays and four rules produced 2,160 task fits and 10,800 checkpoint rows. At task epochs 0, 1, 10, 40 and 80, we recorded held-out accuracy/loss and delivered-versus- exact gradient cosine at each method’s own state. Held-out cue classification, coefficient mean squared error (MSE) and route-norm errors supplied separate encoder diagnostics. The protocol, fixed before new outcomes but not publicly preregistered, is `configs/review_completion/coefficient_encoder.json`.

The primary condition used 256 calibration trials, cue-noise SD 0.5 and no delay. Learned coefficients improved accuracy over the frozen profile by 7.16 percentage points (95% paired-seed bootstrap interval 6.26–8.07; 20/20 positive) and over mismatched coefficients by 7.10 points (6.19–7.99; 20/20). They remained 54.47 points below exact route selection (53.54–55.45; 20/20 lower). Two-sided Wilcoxon tests restored the exact 1/2048 accuracy-count grid before ranking and used Holm correction across these three primary contrasts. Adjusted P values were 5.72×10^{-6} versus the frozen rule and 1.77×10^{-4} versus each other control. Intervals used 20,000 paired-seed draws; other grid contrasts are descriptive sensitivities. All fits were finite. Learning from the supplied cue improved performance relative to fixed or misassigned coefficients, but left a large gap to the oracle. The experiment does not establish autonomous discovery of credit coordinates in a biological neuron. The complete outputs are in `source_data/review_coefficient_encoder/`.

An exploratory follow-up was specified after observing the soft-encoder outcomes. The same trained encoder, original paired seeds and task data were retained, but its maximum-probability route was delivered as a one-hot vector. No encoder parameter was refitted and no hyperparameter was selected from the task outcomes. This hard readout is a paired sensitivity, not a new independent replication or a replacement for the primary soft rule (Supplementary Fig. S10). At the primary noise SD 0.5, mean accuracy was 25.825% for soft coefficients, 64.031% for hard selection, 80.291% for exact selection and 18.669% for the frozen profile. Hard-minus-oracle accuracy remained -16.26 percentage points (95% paired-seed interval -17.83 to -14.83), while hard-minus-frozen was 45.36 points (43.69–46.92). At zero noise and 256 calibration trials, hard selection equaled the oracle numerically, whereas the soft rule reached 79.978%. With only 16 noise-free calibration trials, the soft encoder already identified every context’s route correctly, yet its graded output gave 19.392% task accuracy; selecting its maximum-probability route gave the oracle’s 80.291%. This sensitivity isolates leakage into incorrect routes from failure to identify the correct route. All hard-readout intervals and unadjusted tests are descriptive; these reused conditions do not define a new primary family. Both learned readouts failed when cues were delayed across independent contexts. Thus accurate route identification does not guarantee that graded coefficients are suitable for the learning rule: readout and timing are separate constraints. All exploratory fits and unchanged encoder settings are in `source_data/review_coefficient_hard_readout/`.

3.7 Route overlap and task interference

We finally examined how overlap between addressed regions and leakage outside an intended route determine interference with an already learned task. Branch-specific plasticity during motor learning suggests that different tasks can recruit different dendritic regions [4]. Let $\mathbf{w}_A$ be an optimum of a quadratic Task-A objective with Hessian H_A , and let an approximate Task-B teaching field produce update $-\eta M_B \mathbf{g}_B$, where η is the step size, $\mathbf{g}_B$ is the Task-B gradient and M_B is its routing operator. Expansion about the Task-A optimum removes the first-order term, leaving the exact loss increase

$$\mathcal{L}_A(\mathbf{w}_A - \eta M_B \mathbf{g}_B) - \mathcal{L}_A(\mathbf{w}_A) = \frac{\eta^2}{2} \mathbf{g}_B^\top M_B^\top H_A M_B \mathbf{g}_B. \quad (75)$$

In the equal-route example, opposing updates on the shared fraction ω have magnitude two, whereas leakage onto the Task-A-only fraction has magnitude λ . Dividing by route size gives $\eta^2[4\omega + \lambda^2(1 - \omega)]/2$. Numerical vector

894 updates matched this identity to 4.2×10^{-17} absolute error (Eq. 75). Separated routes therefore protect a learned
895 task only when both anatomical overlap and teaching-field leakage are small. The published somatostatin (SST)
896 perturbation is consistent with the resulting prediction that loss of branch-selective inhibition increases effective
897 leakage and cross-task interference [4], but it does not uniquely identify the credit-routing mechanism. This static
898 identity does not model dendritic calcium, spikes, eligibility timing or the SST intervention.

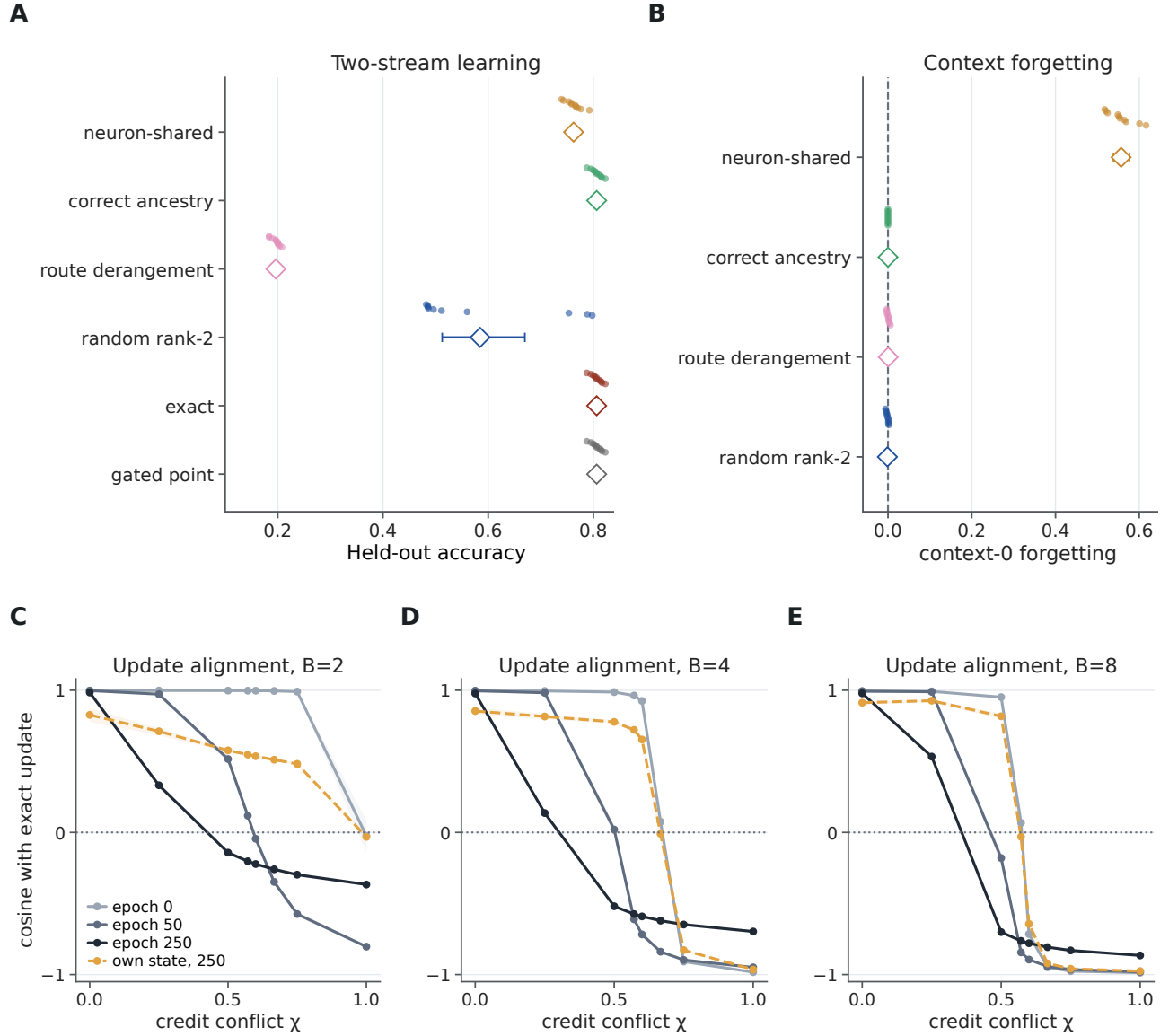


Figure S9: **Branch-selective credit limits interference and preserves update direction.** **A,B**, Two-branch learning and context-0 forgetting in ten paired seeds. Small forgetting after deranged feedback reflects its poor initial solution. Points show seeds; large symbols and bars are means and 95% paired-seed bootstrap intervals. Exact transport and the equivalent gated-point rule supply branch selection. **C–E**, Shared-versus-exact gradient cosine in the separate branch-conflict experiment with two, four or eight branches. Solid curves evaluate common exact-learning states at epochs 0, 50 and 250; dashed curves evaluate the shared learner’s own final state. Curves and shading use twenty paired seeds and 95% bootstrap intervals. An exact quadratic interference calculation is given in the accompanying derivation. Learned-state geometry can depart from the initialization mean-field approximation.

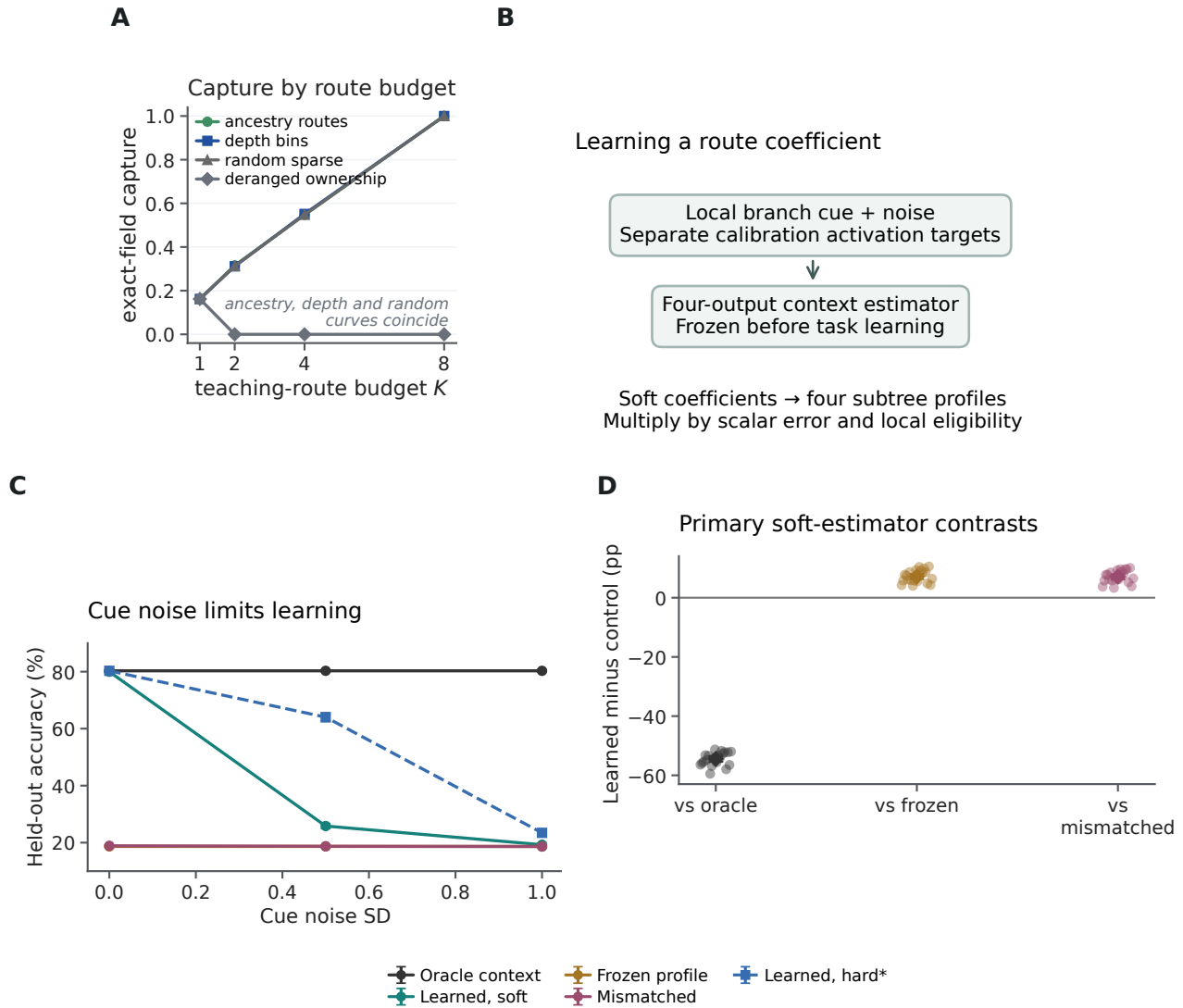


Figure S10: **Available route span and learned route coefficients are different constraints.** **A**, Trained field capture by route budget in the eight-context factorial. Ancestry, depth and random partitions have coincident capture at matched bandwidth; derangement removes the target coordinate. Thus capture does not explain the small ancestry-specific advantage. **B**, An explicit context cue trains a four-output encoder, frozen before task learning. **C**, Cue-noise sensitivity; dashed starred curves use a post hoc hard readout of the same encoder. **D**, Primary soft-encoder-minus-control accuracy contrasts. The primary encoder is about 54.47 points below the oracle. With noisy calibration cues, hard selection narrows but does not eliminate this gap; it matches the oracle in the noiseless condition. All curves/differences retain twenty paired seeds and 95% bootstrap intervals. The calibration cue and activation targets are supplied resources. No serial forward dendritic computation is present.

S4 Interaction structure, compatible trees and spatial credit

4.1 Interaction structure at fixed input-sensitivity rank

A tree constrains representation only if its units have restricted input access. A dense trainable leaf-input map can compensate for generic route rotations: if $W = \Phi V$, with orthonormal K -column routes Φ , and $Q_r^T H_T \Phi$ has row rank r , a suitable V reproduces every noiseless context in the linear task of Section 10.1. The algebraic study below instead fixes each input to one named leaf. Its exhaustive capacity analysis was formulated after the prospective-selector outcomes and is a population diagnostic.

To impose local input access, we placed eight independent Rademacher inputs (each equally likely to be -1 or $+1$) at named leaves of a binary tree. For internal node v , let u_L and u_R be its child outputs and let a_v, b_v, c_v, d_v be its four trainable coefficients. Each of the seven internal nodes computes

$$u_v = a_v + b_v u_L + c_v u_R + d_v u_L u_R. \quad (76)$$

Only the root is read out. Every candidate has 28 free coefficients and 14 edges, with no dense input mixing, intermediate readout or access to inputs outside a node's subtree. The fixed pool combines three shapes—a balanced tree, a comb and a tree with a $3 + 5$ root split—with the same four predetermined leaf permutations, giving twelve candidates. Equal edge and coefficient counts do not imply equal cable length, latency or biological energy cost. These signed algebraic nodes are not conductance compartments.

The primary target census contains all 105 perfect matchings M and all 35 unordered four-plus-four partitions A of the eight inputs:

$$f_M(\mathbf{x}) = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j, \quad f_A(\mathbf{x}) = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right). \quad (77)$$

The targets are enumerated independently of the candidate trees. Evaluating all $2^8 = 256$ equally weighted inputs gives target variances 1 and $1/2$, respectively. For the continuous derivative of each target's unique multilinear extension,

$$\mathbb{E}[\nabla_{\mathbf{x}} f \nabla_{\mathbf{x}} f^T] = I_8/4. \quad (78)$$

Thus both algebraic and participation-ratio effective rank are eight, and the complete input-sensitivity spectrum agrees. This is a property of the target, not equality of changing model parameter-gradient or synaptic-credit moments. Normalized MSE (NMSE) divides population squared error by the stated target variance.

Interaction-tensor descriptions have previously been used to analyze structured neural function classes [5]. The centered constant-feature criterion below is derived for the specific local model in Eq. 76.

4.2 A necessary and sufficient centered-cut condition

For each subset U of input indices, define the Walsh function $\chi_U(\mathbf{x}) = \prod_{i \in U} x_i$, with $\chi_\emptyset = 1$. These functions form an orthonormal basis under the uniform input distribution. Writing c_U for the coefficient of χ_U gives $f = \sum_U c_U \chi_U$. If $\mathbf{c}_f$ and $\mathbf{c}_g$ collect the coefficients of two functions f and g , Parseval's identity gives $\mathbb{E}[(f - g)^2] = \|\mathbf{c}_f - \mathbf{c}_g\|^2$. For the leaf set S of an actual subtree, reshape the target coefficients into C_S , with row index a subset of S and column index a subset of its complement $\bar{S}$. Let C_S° denote this matrix after removing the row for the empty subset of S .

Proposition 8 (Exact representation by a scalar multi-affine tree). *For a fixed binary tree with one named Rademacher input at each leaf, a target is represented exactly by Eq. 76 and a root-only output if and only if $\text{rank}(C_S^\circ) \leq 1$ at every proper subtree cut.*

Proof. Every ancestor is affine in a particular child's output, and sibling inputs are disjoint. Consequently a represented function can be written $g = A(\mathbf{x}_{\bar{S}}) + h_S(\mathbf{x}_S)B(\mathbf{x}_{\bar{S}})$. Its full cut matrix has rank at most two; removing the empty- S row leaves rank at most one. This proves necessity.

For sufficiency, call a cut active when $C_S^\circ \neq 0$. At each active cut, choose a centered function h_S whose Fourier coefficients span the column space of C_S° , and define $V_S = \text{span}\{1, h_S\}$. If $C_S^\circ = 0$, take $h_S = 0$ and $V_S = \text{span}\{1\}$. At a leaf use $V_i = \text{span}\{1, x_i\}$. The condition implies $f \in V_S \otimes \mathcal{H}_S$, where $\mathcal{H}$ is the complete finite-dimensional function space on the indicated inputs.

For sibling leaf sets L and R , the orthogonal projections onto V_L and V_R commute because they act on disjoint input factors. Hence $f \in V_L \otimes V_R \otimes \mathcal{H}_{L \cup R}$. For a non-root parent $P = L \cup R$ with nonzero centered component, write $f = 1_P \otimes A + h_P \otimes B$, with $B \neq 0$. Contracting the outside factor against a linear functional that maps B to one gives $h_P + c1_P \in V_L \otimes V_R$, where c is a scalar contributed by A . Since the constant function also belongs to this product space, so does h_P . It is therefore a linear combination of $1, h_L, h_R, h_L h_R$, exactly the four allowed features. Inactive parent features are zero and cause no exception. At the root, the complete target itself belongs to the product of its children’s spaces. Induction supplies all local coefficients and proves sufficiency. $\square$

Retaining the constant feature in every subtree makes this criterion stronger than an uncentered rank-two constraint. Let $\mathcal{T}_{\text{proper}}$ be the leaf sets of the tree’s proper subtrees, and let $\sigma_j(C_S^\circ)$ be the singular values in decreasing order. The optimal low-rank approximation identity and Parseval then give

$$\text{MSE}(f, g) \geq \max_{S \in \mathcal{T}_{\text{proper}}} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (79)$$

Taking the maximum preserves a lower bound. Summing tails from overlapping cuts instead gives a heuristic score, without the same guarantee. A positive maximum-cut bound need not equal the true optimum. The condition applies to actual subtree cuts; the complement of a subtree need not itself be a subtree. For the matching (12)(34)(56)(78), an even–odd root split has centered tail 0.75, whereas the full rank-two tail is 0.5. For the two-quartet target with supports 1234 and 5678, a five-leaf subtree 12345 has centered tail 0.25 despite a zero full rank-two tail. Supplementary Fig. S11 compares both bounds with fitted population error.

The proof requires ancestors to be affine in each child’s output. A general nonlinear scalar ancestor can generate higher cut rank from one scalar feature. Neither the necessity condition nor Eq. 79 therefore applies automatically to a tanh tree or a conductance model.

4.3 Population fitting, complete-tree construction and depth

All twelve candidates were fitted to every primary target by 32 alternating least-squares (ALS) sweeps with four fixed random restarts. Conditional on the other nodes, the design for one node is $q(\mathbf{x})[1, u_L, u_R, u_L u_R]$, where q is the derivative of the root with respect to that node’s output. The offset is the current root output minus qu_v . Sweeps alternate postorder and preorder. Hidden mean/scale gauges are normalized with exact parent compensation. Every restart and checkpoints at 0, 2, 8 and 32 sweeps are retained: 6,720 fits and 26,880 checkpoint rows. The endpoint is the smallest achieved NMSE among the four restarts. The difference between this endpoint and Eq. 79 can reflect both incomplete optimization and a loose bound, so it is not a certified optimization gap. Full-domain fitting has no held-out generalization interpretation.

For score-based selection among the twelve fixed candidates, values within 10^{-10} are treated as ties and resolved by lexicographic candidate identifier. A separately retained summary applies this tolerance to the raw floating-point minima without changing any fit. The two-sweep pilot fits every candidate; the best-fixed baseline chooses a single candidate per family in hindsight. The uniform-random baseline is its exact candidate expectation. Family averages and paired shape/assignment counts are descriptive censuses, with error comparisons at tolerance 10^{-7} and no sampling confidence intervals over the enumerated tasks. The lower bound is attained within this tolerance in 1,407/1,680 endpoints. Although some restarts develop large cancelling coefficients, every case selected by a deterministic policy has an existing equivalent restart within 10^{-7} NMSE whose coefficient norm is at most 2.645752. All original fits, including the remaining poorly conditioned case, remain included.

To search beyond the fixed pool, we used an exploratory dynamic program over all input subsets. Let $c(S)$ be the normalized cut tail from Eq. 79, with zero cost at the root and leaves. Let $b(S)$ be the smallest achievable

980 maximum cut cost for a subtree on leaf set S , with base case $b(\{i\}) = 0$. Then

$$b(S) = \max \left\{ c(S), \min_{L \cup R = S} \max[b(L), b(R)] \right\}. \quad (80)$$

981 Every binary tree decomposes into one root split and two independent children, so this recurrence covers all trees. A
 982 second pass restricts every cut to the globally minimal threshold, then minimizes total tail, depth and deterministic
 983 split ties. Costs are rounded to 12 decimals for numerical ties. This corrected two-pass version is retained separately
 984 from the first exploratory construction and agrees with exhaustive enumeration of all 945 six-input trees on four
 985 random cost tables. At a zero threshold, the theorem makes the minimum-depth solution the shallowest exact tree
 986 in this class; for a positive threshold only the cut score, not fitted error, is optimized.

987 Dominant left singular vectors of the centered cut matrices define local scalar features, and four-feature least
 988 squares recovers each node’s coefficients. All 105 matching and 35 quartic targets reconstruct at minimum depth
 989 three. A separate positive control uses 24 unique seeded permutations π of the nested target

$$f_{\pi}(\mathbf{x}) = \frac{1}{2} \sum_{k \in \{2,4,6,8\}} \prod_{j=1}^k x_{\pi(j)}. \quad (81)$$

990 Its rank is eight but its spectrum $(1/4, 1/4, 1/2, 1/2, 3/4, 3/4, 1, 1)$ differs from the primary families; it is constant
 991 only within this control family. Every nested target requires depth four. An independent depth-constrained
 992 recurrence gives minimum centered-cut NMSE bound 0.146446609407 at depth at most three and zero at depth
 993 four. All 164 constructions have NMSE below 7×10^{-30} and maximum absolute coefficient one up to rounding
 994 (Supplementary Fig. S11). The algorithm uses the complete target Fourier tensor: these are exact-target capacity
 995 certificates, not predictions from limited calibration data.

996 4.4 Boolean interactions, scalar tree structure and conditional credit

We used four-input Boolean functions to make the structural and credit questions explicit. The complete domain
 comprises sixteen equally weighted patterns. Writing $b_i = (x_i + 1)/2$ for Rademacher inputs $x_i \in \{-1, 1\}$, the
 raw labels $y \in \{0, 1\}$ are defined below. The symbols $\wedge$, $\vee$ and $\oplus$ denote AND, OR and exclusive OR (XOR),
 respectively:

$$y_{\text{AND}} = b_1 \wedge b_2 \wedge b_3 \wedge b_4, \quad y_{\text{OR}} = b_1 \vee b_2 \vee b_3 \vee b_4, \quad (82)$$

$$y_{\text{parity}} = b_1 \oplus b_2 \oplus b_3 \oplus b_4, \quad (83)$$

$$y_{\text{OR(AND)}} = (b_1 \wedge b_2) \vee (b_3 \wedge b_4), \quad (84)$$

$$y_{\text{XOR(AND)}} = (b_1 \wedge b_2) \oplus (b_3 \wedge b_4), \quad (85)$$

$$y_{\text{AND(XOR)}} = (b_1 \oplus b_2) \wedge (b_3 \oplus b_4), \quad (86)$$

$$y_{\text{nested}} = b_1 \wedge [b_2 \vee (b_3 \wedge b_4)]. \quad (87)$$

997 These are specified mechanistic examples rather than a representative sample of natural tasks. The associative AND,
 998 OR and parity families control for claims that greater depth or a particular input grouping is inherently necessary.
 999 Task permutations preserve a family’s truth-table distribution; comparisons between families need not preserve its
 1000 input-gradient spectrum.

1001 Every candidate has three scalar internal units, twelve trainable coefficients and six edges. For child outputs ℓ, r
 1002 and trainable coefficients a, b, c, d , each unit computes $u = a + b\ell + cr + d\ell r$. Only the root supplies the output.
 1003 All local coefficients can change during learning. There is no additional dense decoder or access to inputs outside
 1004 a unit’s descendant subtree. This is the multi-affine algebraic class of Proposition 8, not a positive-conductance
 1005 neuron. An independent enumeration of all fifteen unordered, rooted, full binary trees on four labeled inputs
 1006 supplies the structural analysis: three balanced trees have depth two, and twelve comb trees have depth three.
 1007 An actual subtree cut is evaluated by removing the Fourier row for the constant function inside that subtree and

retaining the singular-value tail beyond rank one. The maximum cut tail is a lower bound on regression MSE for the entire tree; zero tails on every actual cut are necessary and sufficient for exact representation in this class. These capacity statements do not by themselves establish convergence of a particular optimizer or attainability under an additional parameter box.

We distinguish exact truth-value regression from threshold classification. Let $\mu_y = \mathbb{E}[y]$ and $\sigma_y = \sqrt{\text{Var}(y)}$ be the mean and standard deviation over all sixteen patterns. Training targets are $t = (y - \mu_y)/\sigma_y$. Mean squared error on t therefore equals raw-output MSE divided by $\text{Var}(y)$. A predicted normalized output $\hat{t}$ corresponds to the raw Boolean decision $\hat{t} \geq (0.5 - \mu_y)/\sigma_y$. High classification accuracy does not imply small regression error, and a positive regression lower bound does not establish a classification impossibility. AND and OR are imbalanced: a constant majority prediction has accuracy 15/16. Balanced accuracy averages the true-positive rate (TPR) and true-negative rate (TNR), $(\text{TPR} + \text{TNR})/2$, and distinguishes this trivial result from accurate decisions on both classes. All sixteen truth patterns contribute to each reported population endpoint; they are not independent biological replicates.

Proposition 9 (An explicit Boolean grouping obstruction). *Let $f = (b_1 \wedge b_2) \text{ XOR } (b_3 \wedge b_4)$ on uniform independent bits, and let $z = (f - 3/8)/\sqrt{15/64}$. A scalar multi-affine balanced tree grouped as $(b_1, b_3) \mid (b_2, b_4)$ or $(b_1, b_4) \mid (b_2, b_3)$ has population NMSE at least 8/15 for every choice of its coefficients. The aligned grouping $(b_1, b_2) \mid (b_3, b_4)$ represents z exactly with coefficients in $[-2, 2]$.*

Proof. For the crossed cut $b_1 b_3 \mid b_2 b_4$, the raw-target centered Walsh matrix, with nonconstant inside rows ordered as $x_1, x_3, x_1 x_3$, is

$$C = \frac{1}{8} \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{pmatrix}.$$

The columns correspond to $1, x_2, x_4, x_2 x_4$. Its three orthogonal rows each have squared norm 1/16, so the rank-one tail is exactly 1/8. Dividing by $\text{Var}(f) = 15/64$ gives 8/15; the other crossed pairing is symmetric. For the aligned tree, set

$$u = 2b_1 b_2 - 1 = \frac{-1 + x_1 + x_2 + x_1 x_2}{2}, \quad v = 2b_3 b_4 - 1 = \frac{-1 + x_3 + x_4 + x_3 x_4}{2}, \quad z = \frac{1 - 4uv}{\sqrt{15}}.$$

The largest absolute coefficient is $4/\sqrt{15} < 2$. □

Exact rational Walsh coefficients were checked for all 105 family–tree pairs. All 49 zero-bound cases were reconstructed by signed Boolean slices and four-corner interpolation: AND, OR and parity admit all fifteen trees; each mixed-gate target admits one. Every normalized construction lies within $[-2, 2]$, with maximum absolute coefficient $4/\sqrt{15}$. This verifies existence within the experimental box, independently of optimization.

The OR-of-AND target and the nested target each use two AND gates and one OR gate, yet their minimum exact depths are two and three. For the nested function, the smallest lower bound among all three balanced trees is $(22 - 2\sqrt{57})/55 \simeq 0.12546$ NMSE; the stated comb tree is exact. All seven functions have input-gradient rank four, but their full spectra generally differ. This is a rank-versus-interaction example, not an isospectral comparison between the mixed-gate families. The maximum over cuts remains a lower bound; we do not assume every incompatible whole tree attains it.

Gate coordinates illustrate conditional derivatives. The elementary gates can also illustrate how another branch affects a local derivative. In raw Boolean branch-output coordinates $u, v \in \{0, 1\}$, their multilinear extensions are

$$F_{\text{AND}}(u, v) = uv, \quad \partial_u F_{\text{AND}} = v, \quad (88)$$

$$F_{\text{OR}}(u, v) = u + v - uv, \quad \partial_u F_{\text{OR}} = 1 - v, \quad (89)$$

$$F_{\text{XOR}}(u, v) = u + v - 2uv, \quad \partial_u F_{\text{XOR}} = 1 - 2v. \quad (90)$$

The other branch thus gates a derivative’s magnitude or sign in these canonical coordinates. The XOR-of-AND diagram illustrates this dependence and a specific input-grouping constraint. Its gate labels are explanatory; the learned units are not constrained to remain Boolean gates or to preserve that representation. A broadcast learner can change internal coordinates, so the derivative diagram is not a proof that broadcast cannot learn the task. Positive affine normalization of the raw target changes derivative scale but not these signs. Actual learned update fields and finite-training outcomes require the separate experimental comparisons below.

Frozen learning comparisons. Four candidate trees were tested: the three balanced pairings and the comb $b_1 \mid [b_2 \mid (b_3, b_4)]$. These are four of the fifteen trees in the independent capacity analysis. Each seed applied one common physical input permutation to both the semantic task variables and candidate leaves. Per-family streams supplied initialization, a 256-observation training cache, 1,024 independent test observations and minibatch indices. All paired tree, optimizer, credit and rate conditions shared these streams within family. Data were drawn uniformly with replacement from the sixteen patterns; training and test labels received independent Gaussian noise of SD 0.15 in normalized-target units. Independent draws could repeat the same input pattern.

Every coefficient began from an independent mean-zero Gaussian with SD 0.5, except zero biases; no target gates, coefficients or exact-construction weights were used. Initial weights and every subsequent update were clipped to $[-2, 2]$. Training used 2,048 updates, minibatches of 32, and gradient norm clipping at ten. Exact credit multiplied each local eligibility by the current complete path derivative. Broadcast set the two nonroot internal sensitivities to one; root sensitivity remained one under both rules. All twelve coefficients were trainable. Stochastic gradient descent (SGD) and Adam were tested separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$.

Five development seeds (2026110–2026114) selected a rate separately for each optimizer and rule from 0.003, 0.01 and 0.03, minimizing mean final clean population NMSE over all seven families and four trees. Ties within 10^{-12} selected the smaller rate. Adam selected 0.003 for exact credit and 0.01 for broadcast; SGD selected 0.03 and 0.01, respectively. These choices were recorded with cryptographic hashes before twenty new seeds (2026210–2026229), which retained all three rates. The resulting 6,720 fresh fits supplied 53,760 records at updates 0, 1, 16, 64, 256, 512, 1,024 and 2,048. An implementation check using seed 2026000 exercised all 336 conditions and was excluded from inference. These seeds vary initialization, permutation and sampled noisy data for seven fixed logical templates; they do not sample new Boolean function families.

The primary endpoint was clean population NMSE evaluated on all sixteen patterns. No sampled test variance or observed class-frequency denominator entered this endpoint. Noisy-test NMSE, raw threshold accuracy and balanced accuracy were secondary. Gradient cosines compare the delivered and exact aggregate clean-population gradients at each learner’s own trained state, before clipping or optimizer transformation; they are not mean per-example cosines or common-state comparisons between separately trained rules.

Two primary Adam contrasts concerned XOR-of-AND: the mean of the two crossed balanced-tree errors minus aligned-tree exact-credit error, and broadcast minus exact error on the aligned tree. The latter compares the prespecified development-selected recipes, not identical learning rates. Ten thousand resamples of twenty paired seed blocks supplied two-sided Bonferroni-adjusted 97.5% intervals for these two contrasts. A positive adjusted lower limit and mean improvement of at least 0.01 NMSE were both required. Other task, optimizer and common-rate contrasts have descriptive pointwise 95% intervals.

The grouping improvement was 0.56317 NMSE $[0.54914, 0.58275]$, satisfying both primary criteria. The credit difference was 0.002507 $[0.001780, 0.003318]$: its adjusted interval excluded zero, but it did not meet the prespecified 0.01-NMSE mean-effect margin. Both recipes learned the aligned XOR-of-AND task well and classified all sixteen patterns correctly in every fresh seed. All three same-rate Adam credit differences were also below 0.003 NMSE. The low-rate SGD contrast had a wide interval and is retained. The larger parity credit differences are descriptive outcomes of a prespecified control task, not primary discoveries. Neither the gate illustration nor these observations establish an impossibility theorem for broadcast learning, or show that a particular change of internal coordinates caused its success. Full-rate outcomes, constant-mean and random-candidate references, clipping records, weights and source/selection hashes are retained under `source_data/boolean_morphology`.

1088 **Post hoc diagnostic of local target information.** After examining the fresh learning outcomes, we asked how
 1089 much target information was available within each proper input subset S . In this diagnostic, f denotes a raw
 1090 Boolean target, $z = (f - \mathbb{E}[f]) / \sqrt{\text{Var}(f)}$ its normalized version, and $\hat{f}_U$ its Walsh coefficient on input subset U .
 1091 The squared L^2 norm below averages over the uniform input distribution. We defined

$$P_S = \|\mathbb{E}[z \mid x_S]\|_{L^2}^2 = \frac{1}{\text{Var}(f)} \sum_{\emptyset \neq U \subseteq S} \hat{f}_U^2.$$

1092 We cross-checked all ninety-eight task–subset values by exact rational evaluation of conditional means on the full
 1093 truth table. Parity4 has $P_S = 0$ for every proper subset. XOR-of-ANDs instead has $P_S = 1/5$ for every two-input
 1094 subset, including compatible and crossed pairs. Thus this diagnostic quantifies accessible lower-order target
 1095 information rather than grouping compatibility.

1096 Let F denote the model’s normalized prediction. At fixed parameters, a local eligibility $e_S(x_S)$ at a nonroot
 1097 subtree is a function only of that subtree’s inputs. Conditional expectation gives $\mathbb{E}[ze_S] = \mathbb{E}[\mathbb{E}[z \mid x_S]e_S]$; when
 1098 $P_S = 0$, the direct target contribution to the population broadcast update $\mathbb{E}[(F - z)e_S]$ vanishes. This identity does
 1099 not imply an inability to learn: label-dependent changes in root coefficients can subsequently affect F , and finite
 1100 training samples and stochastic updates need not exhibit exact population cancellation. The diagnostic was added
 1101 after observing the learning results and did not change the training protocol or its predeclared criteria.

1102 4.5 Learning compatible trees with exact and restricted credit

1103 To separate credit assignment from uncertainty in structure estimation, this experiment supplied a tree compatible
 1104 with the full target. Three development seeds and twenty fresh seeds generated independent signed and permuted
 1105 matching, quartet and nested-interaction targets. The complete target Fourier tensor supplied a compatible
 1106 tree through the preceding dynamic program; no target-derived coefficients initialize learning. An independent
 1107 permutation of leaf labels preserved the exact shape, 28 coefficients and 14 edges. Chance compatible permutations
 1108 were retained. The primary matching and quartet families retained identical input-sensitivity spectra, while the
 1109 nested family had the previously stated anisotropic spectrum.

1110 All seven internal units initialized their non-bias coefficients independently from $\mathcal{N}(0, 0.5^2)$, with zero biases.
 1111 The same initial coefficients, training cache and minibatch draws were used across paired assignments, credit rules
 1112 and optimizers. Local features were $(1, u_L, u_R, u_L u_R)$ and the exact path derivative $q_v = \partial u_{\text{root}} / \partial u_v$ supplied credit.
 1113 Exact gradients agreed with central finite differences for every coefficient in all three families. Root sensitivity
 1114 was one under every rule. The six nonroot internal units received either the exact field, a unit root broadcast, the
 1115 orthogonal projection onto a single constant profile, or the projection onto two proximal-subtree indicator profiles.
 1116 A size-matched permutation of the latter profiles served as a control. All projected coefficients were computed from
 1117 the current exact field and are therefore oracle coefficients; the experiment did not learn a circuit estimator for them.

1118 Training used 2,048 independently sampled Rademacher examples with output noise SD 0.15 and 1,024
 1119 minibatch updates of size 64. The gradient corresponded to half-MSE divided by the known clean target variance.
 1120 Stochastic gradient descent (SGD) and Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$) each tested learning rates 0.003,
 1121 0.01 and 0.03. Update gradients were clipped at norm 10 and parameters projected to $[-2, 2]$, which contains all
 1122 exact constructions. Every clipping/projection count was recorded. These are constrained finite-horizon training
 1123 results, not estimates of unrestricted optimization optima.

1124 A separate learning rate for each optimizer and rule minimized mean development population NMSE over all
 1125 families and both assignments. A common rate per optimizer was also selected on development. All choices were
 1126 recorded with cryptographic hashes before any fresh fit; all three rates remain in the released outcomes. Every
 1127 SGD rule selected the upper tested rate, so the grid does not establish its best attainable optimizer setting. The
 1128 independent test set contained 4,096 examples with fresh SD 0.15 output noise. Test NMSE, normalized by the
 1129 clean target variance, was primary; noiseless population NMSE on all 256 inputs was secondary. Evaluation at
 1130 steps 0, 1, 16, 64, 256 and 1,024 retained 21,600 checkpoint rows from 3,600 fresh fits. The endpoints at the
 1131 development-selected rates contained 1,200 fits. No run was excluded. Ten thousand bootstrap samples of twenty

whole paired seed blocks were drawn for pointwise 95% intervals, averaging families within seed for pooled contrasts. Multiple subgroup intervals are descriptive, not separately multiplicity-adjusted discoveries.

Exact credit strongly improves learning of compatible quartet and nested targets under both frozen optimizer recipes. Broadcast is already effective for matching targets and outperforms exact credit there under SGD. Restricted two-subtree projections help in some Adam conditions but do not consistently beat shuffled profiles, and have essentially no pooled advantage over broadcast under SGD. For compatible trees trained with exact credit and Adam, mean test NMSE was 0.0329; permuting the input labels increased it to 0.6080. At the common Adam rate 0.01, the compatible exact/broadcast averages remain 0.0329/0.5973. Supplementary Fig. S12 reports the full family, optimizer, profile, assignment and rate controls. Numerical outcomes and source/selection hashes are in `source_data/morphology_credit/`.

4.6 Matched fixed-profile learning and credit-field geometry

This experiment separates forward compatibility, spatial profile calibration and learning. This matched cohort retains the seven-unit model, 28 coefficients and compatible constructions above, specializing hierarchical tensor representations to scalar messages [6, 7]. Pairwise and quartic targets share a balanced tree and seed-specific leaf permutation. Independent coefficient signs have magnitude one-half; both input moments equal $I_8/4$, while target variances are one and one-half. The nested target uses its separate compatible tree and anisotropic spectrum.

Let $q_n(x) = \partial u_0 / \partial u_n$ be the exact path derivative for unit n , and let $\delta(x)$ be the scalar derivative of normalized half-squared error with respect to root output. Local coefficient eligibilities are $1, u_L, u_R, u_L u_R$. Exact credit multiplies these by δq_n . Unit broadcast replaces q_n by one. An initially calibrated profile uses $a_n = 256^{-1} \sum_x q_n(x)$ over 256 unlabeled initialization examples, then keeps a_n fixed. The sign control retains only $\text{sign}(a_n)$. Profiles require no target labels. Learning-rate choices are separate from the profile calculation. Fixed positive coordinate scaling largely cancels under Adam, so calibrated and sign-only Adam trajectories are expected to be nearly identical; they are not independent mechanistic confirmations.

The protocol preceded three disjoint development seeds, with the same rate grid for each rule and optimizer. Separate rule-specific and common rates minimized final validation NMSE across families and were sealed before twenty paired fresh seeds. Matching/quartic conditions share initialization, calibration, examples, noise and minibatches. The Adam primary contrast is the quartic-minus-pairwise difference in initial-profile-minus-exact held-out error. All selected/common-rate unions are retained: 432 development, 1,200 fresh fits, and 48 excluded short checks.

Algebraic train/validation/test sizes are 2,048/1,024/4,096; minibatches have 64 examples. Nonbias coefficients initialize independently with Gaussian SD 0.5 and biases zero. Both SGD and Adam test 0.003/0.01/0.03; Adam uses (0.9, 0.999) and $\epsilon = 10^{-8}$. Fits retain norm clipping ten, box $[-2, 2]$, all contacts and steps 0, 1, 16, 64, 256, 512 and 1,024. Label-noise SD 0.15 gives NMSE floors 0.0225 for matching/nested and 0.045 for quartic; normalized half-MSE uses each family’s clean variance. All poorly fitted exact endpoints remain. Pointwise intervals use 10,000 whole-seed draws.

The separate positive-conductance comparison uses the compatible sixteen-parameter teacher of Section 5.3, not the algebraic quartic target. Data sizes are 1,024/1,024/4,096 without label noise, box $[-5, 5]$, SGD rates 0.001/0.003/0.01 and Adam rates 0.003/0.01/0.03.

For each saved state, let Q have one row per evaluation example and six columns containing q_n . For a dictionary A , energy capture is $\|QP_A\|_F^2 / \|Q\|_F^2$, where P_A is its orthogonal projector. The best fixed rank-one profile is the leading right singular vector of Q , with capture $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$. Its profile and per-example amplitudes are oracle fits. The uncentered second moment is $C = Q^T Q / N$, and participation-ratio effective rank is $(\text{tr} C)^2 / \text{tr}(C^2)$. We additionally evaluated unit broadcast, the fixed initial profile, ancestry spans, full-rank capture, and loss-error-weighted fields $\text{diag}(\delta)Q$. The latter has its own optimal profile; applying the path-optimal profile after error weighting answers a different question.

Eligibility-weighted reconstruction multiplies exact and projected path fields by their local coefficient features before comparing parameter updates. Site-normalized spectra divide each path coordinate by its root-mean-square magnitude, exposing sensitivity to fixed coordinate scaling. Raw capture energies are not invariant to arbitrary

hidden-unit rescaling. At identical weights and inputs, Q is independent of the target; a scalar residual also cancels from normalized per-example projection capture whenever it is nonzero. Eighty common-state relabeling controls verify that identity. These diagnostics characterize learned-state geometry, rather than infer a task-only credit rank at initialization.

An independent saved-state audit checked 20,160 metrics and 120 matching–quartic pairing identities. For the earlier cohort whose states had not been stored, replay reconstructed 3,600 checkpoints and matched 28,800 archived metrics within 9×10^{-16} . These are reconstructed states, distinguished from the directly archived matched-cohort checkpoints. Source Data retain the complete numerical audit.

The positive-conductance comparison uses the prior compatible teacher across three paired input permutations and the same credit definitions. It does not implement the algebraic quartic target. All four rules learn it accurately with Adam; exact credit has mean NMSE 9.39×10^{-6} , compared with 1.35×10^{-6} for the fixed-profile controls. Positive path derivatives make sign and unit broadcasting identical. Both models’ local updates were independently checked by finite differences. Protocols, selections, checkpoints, full trajectories, numerical audits and scripts are supplied in `source_data/credit_rule_bridge/`, `source_data/credit_resolution_bridge/` and their corresponding script directories.

4.7 Balanced extension of the matched credit-rule comparison

After examining the released 1,024-update outcomes, we froze and committed a longer-budget protocol before running any extended trajectories. This is a protocol-led extension of previously observed seed blocks, not a fresh confirmatory cohort. All 720 algebraic trajectories were retained: twenty seed blocks, three task families, four delivery rules, Adam and SGD, and the union of the original development-selected and common rates. Matching and quartic targets retain the same balanced tree, initialization and example streams; nested targets retain their separately compatible trees. No rates were retuned.

Every trajectory was replayed from initialization and run for exactly 16,384 updates, preserving examples, minibatches, local rules, gradient clipping and coefficient bounds $[-2, 2]$. All original checkpoint arrays, numerical curves and diagnostics through 1,024 updates reproduced. Additional observations occurred every 256 updates after 1,024. Terminal outcomes at 1,024, 4,096, 8,192 and 16,384 updates preserve the original endpoint definition. Secondary states minimized validation NMSE over the declared observations within each budget, with ties assigned to the earlier checkpoint. Test and full-domain population errors were used only for reporting. Independent recomputation of all 5,760 budget-outcome rows from saved weights reproduced test, validation and population NMSE exactly; all 2,880 validation-only selections were checked.

All twenty selected-rate Adam exact quartic fits reached the test noise floor by 4,096 updates, resolving three slow 1,024-update outcomes. By that budget exact pairwise credit also had slightly lower mean error than the initial profile, with eighteen of twenty exact wins. The original pairwise mean ordering reflected slow exact seeds, despite exact credit winning fifteen of twenty initial pairs. Compact endpoints are in Supplementary Table S8.

At 16,384 updates, the selected-rate calibrated-minus-exact quartic-minus-pairwise interaction was 0.9516 (95% paired interval, 0.8829–1.0123) for terminal states and 0.7903 (0.6915–0.8797) for validation-selected states. Corresponding common-rate effects were 0.9028 and 0.7952. All four contrasts were positive in every seed. Intervals use 10,000 whole-seed bootstrap draws and are descriptive because the original outcomes were already known.

The comparisons remain constrained by time, parameter bounds and historical rates. Every selected-rate Adam initial-profile trajectory had contacted a coefficient bound by 16,384, whereas none of the corresponding exact trajectories had. Validation-selected initial-profile states still had substantial quartic and nested error, and the full record retains clipping and bound events. Extended exact-trained path fields remained almost one-dimensional for pairwise targets but multidirectional for quartic and nested targets; the original main-figure capture analysis is retained unchanged. These results strengthen the fixed-budget contrast and explain the slow exact runs, without proving that a bounded fixed profile fails under every optimizer or that full compartment resolution is uniquely necessary.

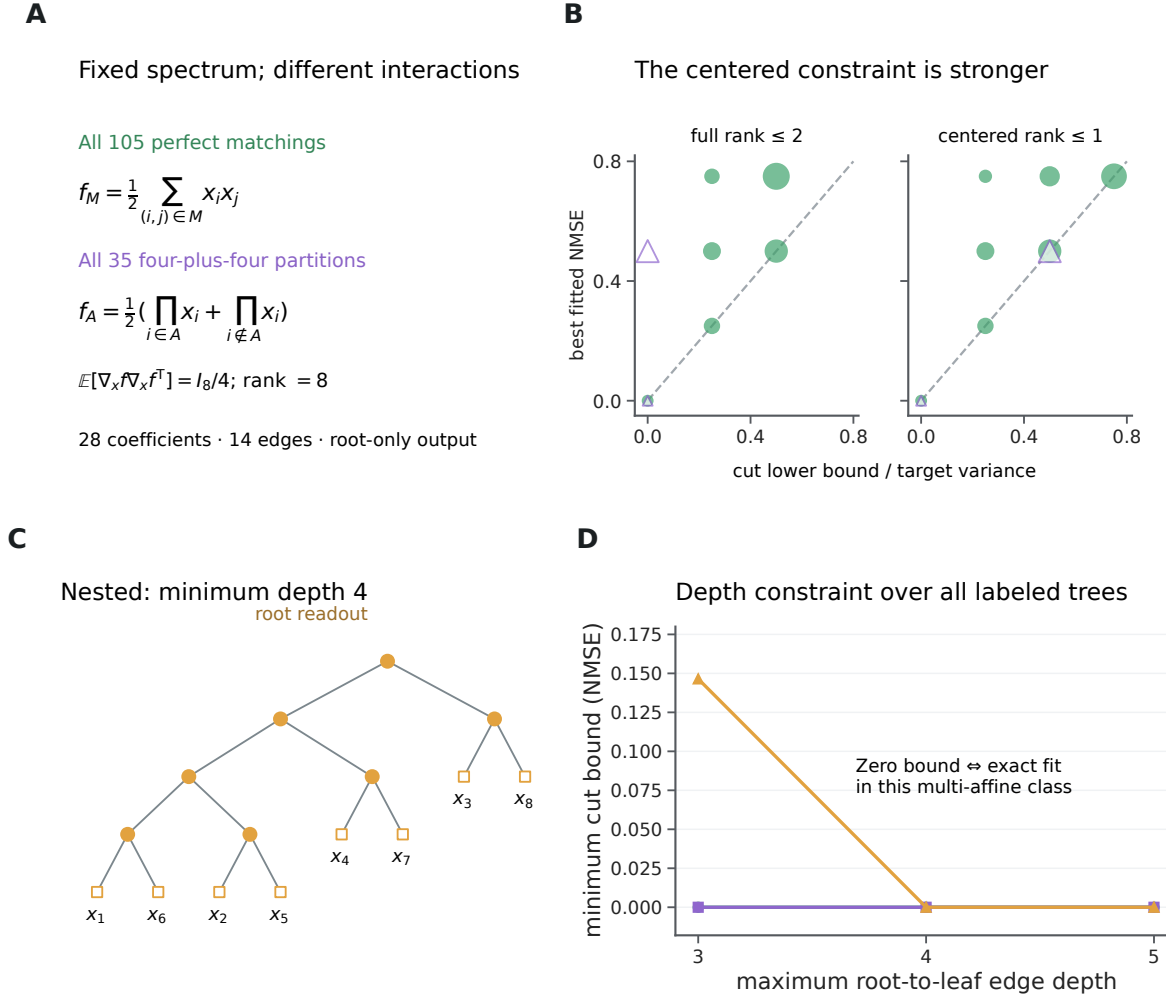


Figure S11: **Interaction structure constrains scalar trees at matched input spectra and resources.** **A**, Exhaustive pairwise matching and two-quartic target families on eight independent binary inputs; both have input-gradient second moment $I_8/4$. Each candidate has seven scalar multi-affine nodes, 28 coefficients and fourteen edges. **B**, Full rank-two and centered rank-one cut-tail lower bounds versus achieved population NMSE in four-restart alternating least squares. **C**, Exact compatible construction for the first enumerated nested target. **D**, Best centered-cut bound over all labeled trees at each depth limit. Matching/quartic targets admit exact depth-three trees; nested controls require depth four and have a different input spectrum. All 164 targets construct with error below 7×10^{-30} and coefficients of magnitude at most one. These exhaustive capacity results have no sampling error bars. The theorem concerns scalar multi-affine composition, not arbitrary conductance dendrites.

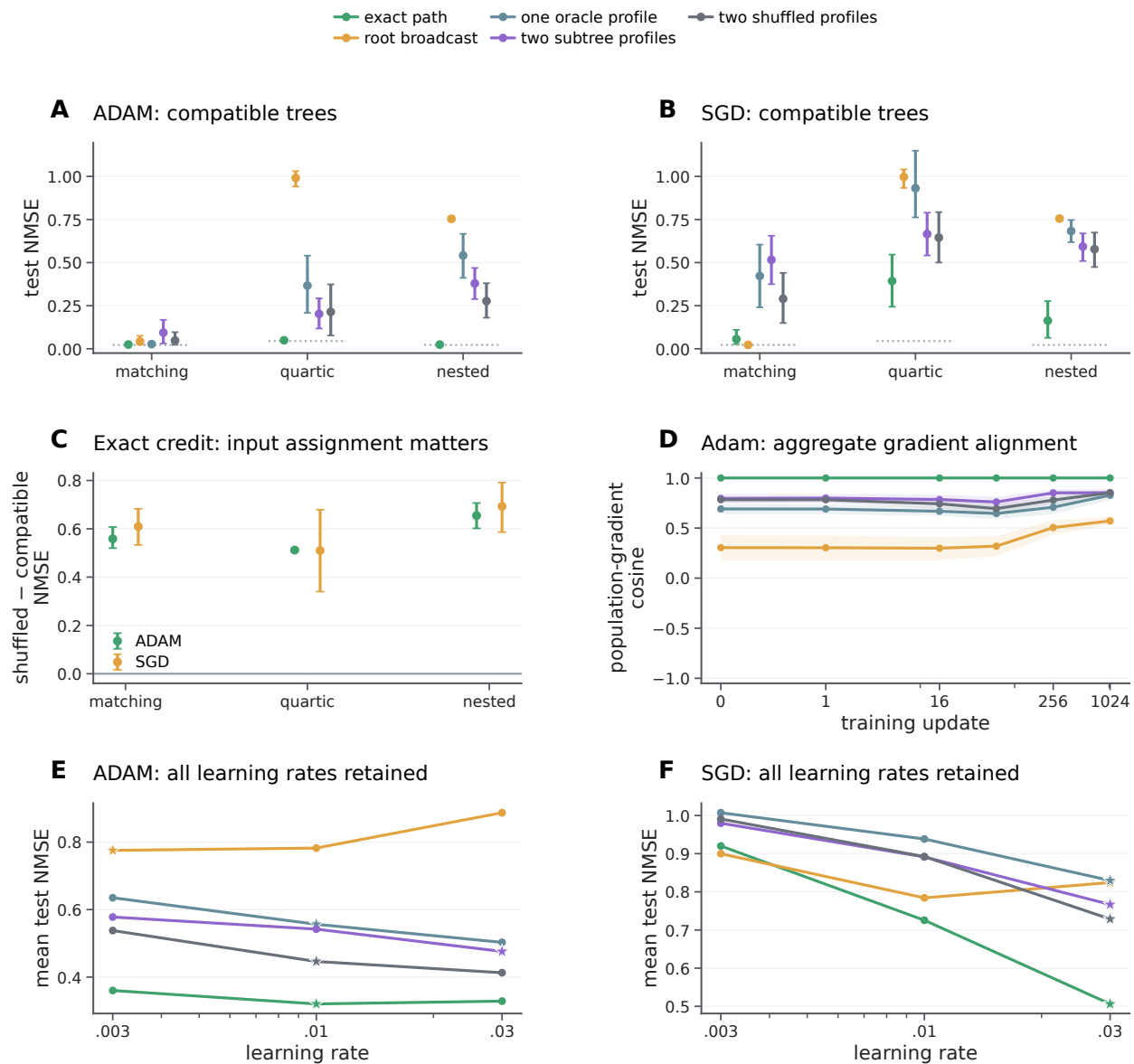


Figure S12: **Task-dependent credit requirements in bounded multi-affine trees.** **A,B**, Final test normalized mean squared error (NMSE) on oracle-compatible trees under Adam and stochastic gradient descent (SGD), respectively. Exact path, root broadcast, one oracle profile, two subtree profiles and two shuffled profiles share the forward model, initialization, examples and minibatches. Dotted segments mark known label-noise floors: 0.0225 for matching/nested tasks and 0.045 for quartic tasks. Projection coefficients access the exact current path field and are oracle diagnostics; root broadcast supplies the same scalar error to each internal node. **C**, Paired exact-credit NMSE increases after shuffling leaf input assignments while preserving tree shape and parameter count. Chance-compatible shuffles are retained. **D**, Adam gradient cosine between the exact and delivered aggregate clean-population updates, evaluated at each rule's own trained state. Each seed averages the three task families before aggregation; these are not comparisons at a common model state or mean per-example cosines. Points and whiskers in **A–C** and bands in **D** are means and pointwise 95% intervals from 10,000 whole-seed bootstrap draws, with twenty independent fresh seeds per family. **E,F**, Final NMSE at all three learning rates, averaged over both compatible and shuffled assignments and all three families. Stars mark rates selected separately for each rule and optimizer using only development seeds. These means therefore differ from compatible-only panels **A,B**. Every SGD choice was the upper tested rate; the result is a comparison of the stated training recipes, not a fully optimized SGD limit. Matching and quartic tasks share the input-gradient second moment $I_8/4$; nested tasks have a different spectrum. The oracle-compatible architectures here are independent of the noisy morphology-selection experiment. All 3,600 fresh fits and all tested rates are retained.

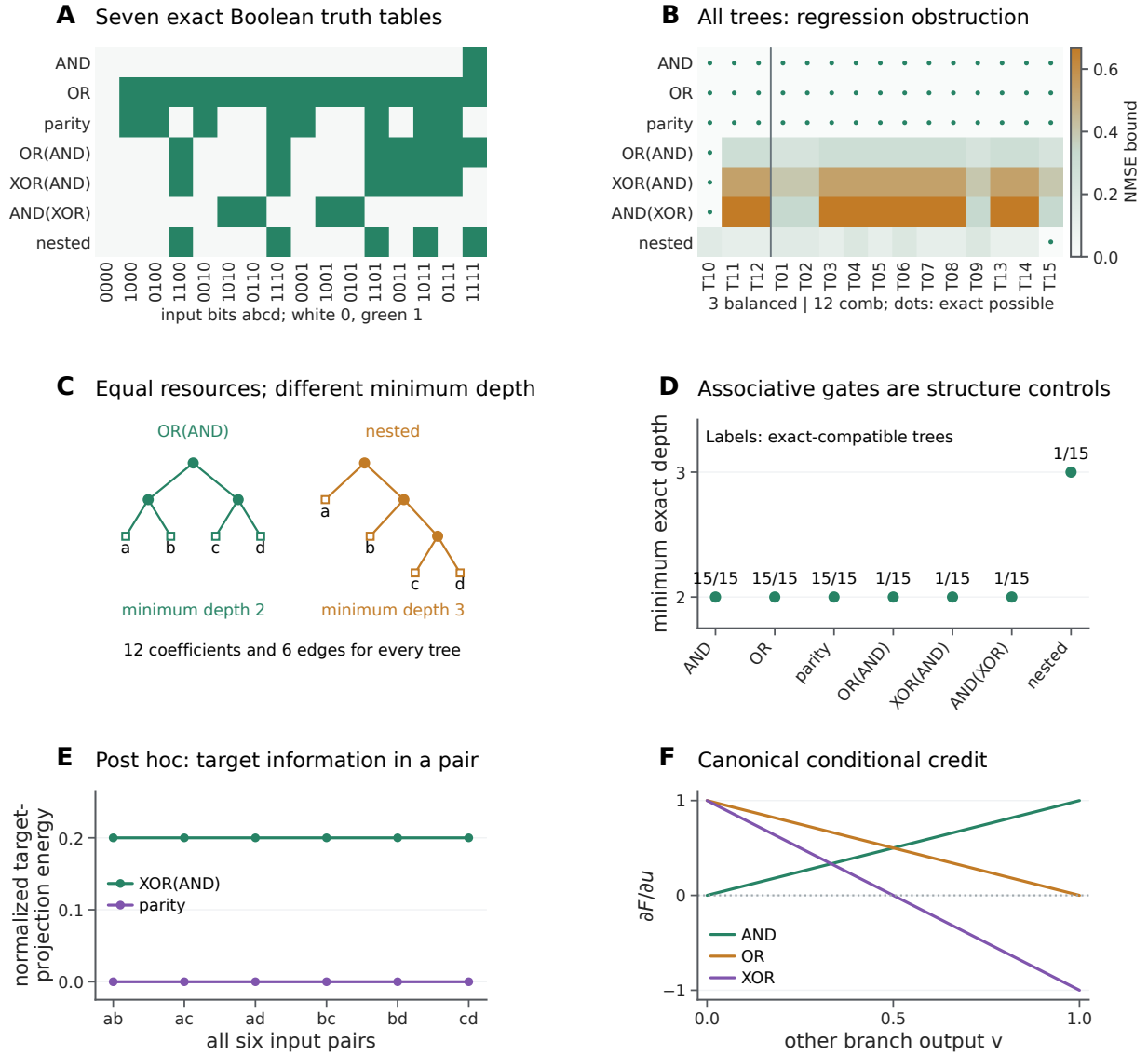


Figure S13: Boolean interactions distinguish structure, depth and canonical credit. **A**, Complete truth tables of seven four-input templates, with sixteen equally weighted patterns (white, 0; green, 1). Mixed targets are $(a \wedge b) \vee (c \wedge d)$, $(a \wedge b) \text{ XOR } (c \wedge d)$, $(a \text{ XOR } b) \wedge (c \text{ XOR } d)$, and nested $a \wedge [b \vee (c \wedge d)]$. **B**, Centered-cut normalized-mean-squared-error (NMSE) regression lower bounds for all fifteen labeled binary trees. Green dots mark 49 exactly representable family-tree pairs, independently certified with coefficients in $[-2, 2]$. The largest coefficient magnitude in these constructions is $4/\sqrt{15}$. All trees have twelve coefficients and six edges; T identifiers and cut sets are supplied in Source Data. A positive bound concerns truth-value regression, not threshold classification. **C**, OR-of-AND and the nested function each use two AND gates and one OR gate, but require minimum exact depths two and three. **D**, Minimum exact depth and compatible-tree counts. AND, OR and parity admit all fifteen trees; each mixed-gate target admits one. All seven input-sensitivity matrices have rank four, but their spectra generally differ. **E**, Explicitly post hoc target-projection energy $\|\mathbb{E}[z | x_S]\|^2$ for all six two-input subsets, with normalized target z . Parity has zero energy on every proper subset; XOR-of-AND has energy $1/5$ on every pair, including crossed pairs. This describes local target information and does not predict grouping compatibility or prove broadcast-learning failure. **F**, Canonical derivatives $\partial F / \partial u$ with respect to left-branch output u , versus right-branch output v : AND, v ; OR, $1 - v$; XOR, $1 - 2v$. These raw Boolean coordinates are illustrative; training can change every coefficient and need not preserve them. No learning outcome is inferred from a derivative sign alone. Panels are exhaustive analytic diagnostics, with no sampling error bars; the multi-affine and independent-input assumptions are explicit in Supplementary Note 4.4.

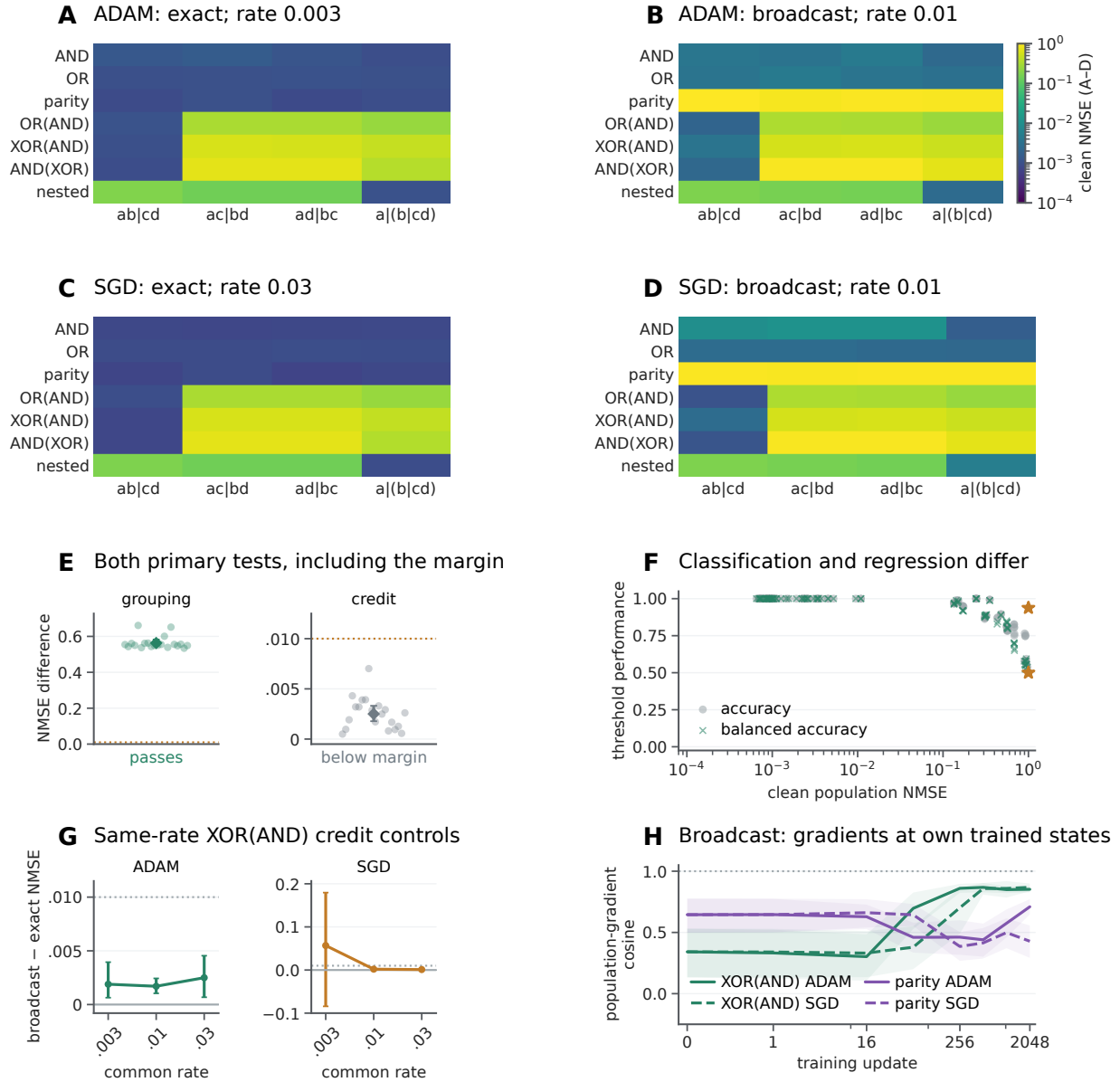


Figure S14: **Boolean grouping benefit and a small primary credit effect under frozen recipes.** **A–D**, Clean population normalized mean squared error (NMSE), averaged over twenty fresh seeds: Adam exact (**A**), Adam broadcast (**B**), stochastic gradient descent (SGD) exact (**C**) and SGD broadcast (**D**). Heatmaps share a logarithmic scale; columns show three balanced groupings and one nested tree. Rates were fixed on five development seeds; seeds vary permutations, initialization and noisy observations. **E**, Prespecified Adam XOR-of-AND contrasts: mean crossed-minus-aligned exact-credit error and aligned broadcast-minus-exact error. Dots, twenty paired seed differences; diamonds/whiskers, means and Bonferroni-adjusted 97.5% intervals (10,000 seed-bootstrap draws). Dotted lines: 0.01-NMSE mean-effect margin. Grouping passes both criteria; the positive credit difference is below the margin. Both compatible rules classify all sixteen patterns perfectly in every seed. **F**, Accuracy (gray) and balanced accuracy (green) versus NMSE for 112 condition means. Raw-output classification threshold: 0.5. Orange stars show the AND/OR constant-majority references: NMSE one, accuracy 15/16 and balanced accuracy 1/2. **G**, Same-rate credit contrasts on aligned XOR-of-AND trees, with descriptive 95% intervals; Adam and SGD axes have different ranges. The uncertain low-rate SGD result is retained. **H**, Broadcast gradient cosine on the aligned balanced tree for XOR-of-AND (green) and parity (purple), with solid Adam and dashed SGD curves, evaluated on all sixteen clean patterns at each learner's own state, before clipping or optimizer transformation. Bands are pointwise 95% seed-bootstrap intervals. The large parity credit effect is descriptive. All 6,720 fits and 53,760 checkpoints are retained.

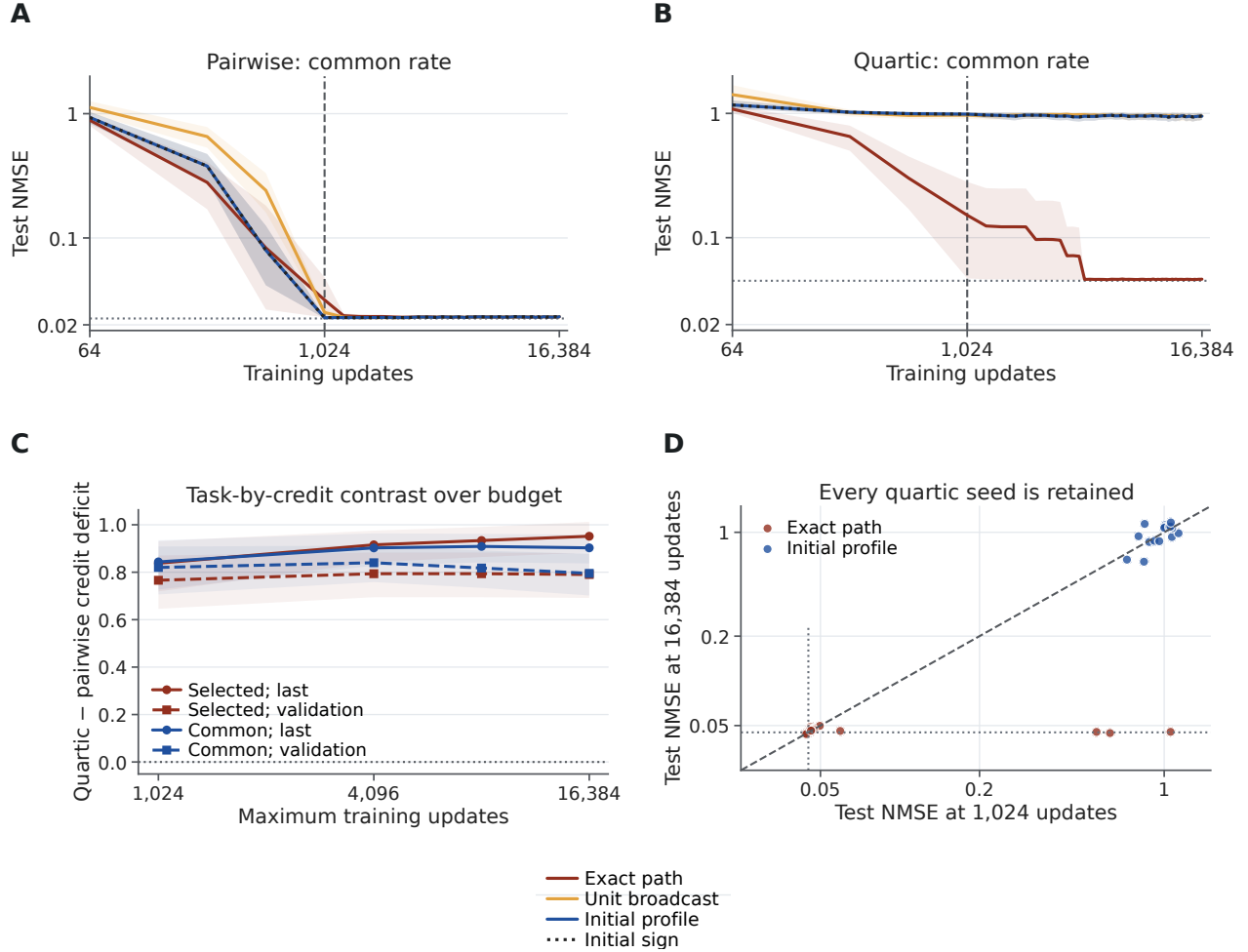


Figure S15: **Longer matched budgets retain the task-dependent fixed-profile deficit.** **A,B**, Pairwise and quartic targets under the common Adam rate, with exact, unit-broadcast, initial-profile and initial-sign credit. **C**, Quartic-minus-pairwise difference in calibrated-minus-exact NMSE across four budgets: circles use terminal states and squares validation-selected states, at selected and common rates. **D**, Every exact and calibrated quartic seed at the original 1,024-update and extended 16,384-update endpoints. Curves show means and descriptive 95% whole-seed bootstrap intervals. All 720 trajectories retain the previously observed twenty seed blocks, three tasks, four rules, two optimizers and original rate choices. The complete original endpoints replay exactly. Slow exact quartic fits resolve with more updates; restricted profiles retain a large deficit. Rates remain fixed and coefficients bounded; this is a retrospective-cohort extension under a prospective protocol, not fresh confirmation.

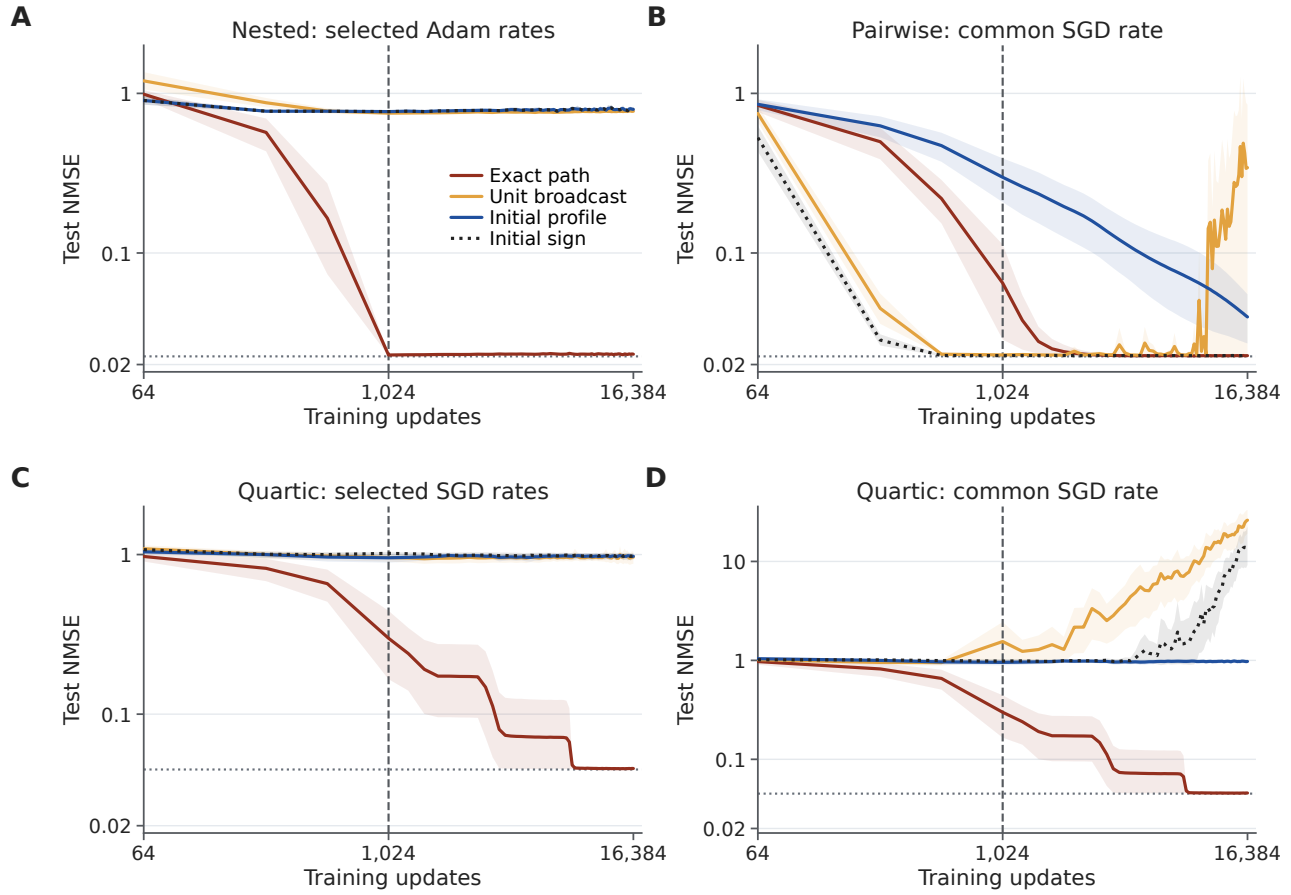


Figure S16: **Nested targets and SGD controls in the balanced budget extension.** **A**, Nested targets at selected Adam rates. **B**, Pairwise targets at the common SGD rate. **C,D**, Quartic targets at selected and common SGD rates. The same previously observed twenty seeds, fixed rules and rate choices are retained. Curves show mean test NMSE from update 64 onward; bands give descriptive 95% whole-seed bootstrap intervals. Vertical lines mark the original 1,024-update cap and horizontal lines the relevant noise floor. Panels state the task, optimizer and rate view; no outcome is excluded or used to retune rates. These controls retain optimizer dependence and bounded-parameter scope.

S5 Credit delivery in conductance-tree learning

5.1 Conductance-tree tasks with aligned or opposed feature tuning

We tested whether context-dependent branch selection can create a learning advantage for spatially resolved credit in a conductance tree. The model contains four terminal compartments, two proximal compartments and a soma. It uses DendriNet’s directed steady-state excitatory/inhibitory balance with identity reactivation; it is not a reciprocal cable model. All 24 positive conductances are trainable, including inhibitory gains and daughter and somatic couplings.

For each example, two independent latent variables $z_1, z_2 \sim \mathcal{U}[-2, 2]$ generate positive features $x_j^+ = \exp(z_j)$ and $x_j^- = \exp(-z_j)$. Both subtrees receive the same latent features: terminals 0 and 2 receive z_1 , and terminals 1 and 3 receive z_2 . Each terminal has two excitatory and two inhibitory conductances. Its effective excitatory and inhibitory conductances and voltage are

$$E_j = g_{E+,j}x_j^+ + g_{E-,j}x_j^-, \quad I_j = g_{I+,j}x_j^+ + g_{I-,j}x_j^-, \quad v_j = \frac{E_j}{1 + E_j + I_j}. \quad (91)$$

This normalization sets leak conductance to 1 and excitatory, inhibitory and leak reversal potentials to 1, 0 and 0. An independent binary context c is sampled with equal probability. Proximal compartment $p \in \{0, 1\}$ receives inhibitory activity $a_p = A\mathbf{1}[c \neq p]$, with $A = 10$ for the primary task pair. Its voltage and the somatic voltage are

$$u_p = \frac{\sum_{j \in \text{ch}(p)} g_{pj}v_j}{1 + \sum_{j \in \text{ch}(p)} g_{pj} + g_{I,p}a_p}, \quad v_s = \frac{g_{s0}u_0 + g_{s1}u_1}{1 + g_{s0} + g_{s1}}, \quad (92)$$

where $\text{ch}(p)$ denotes the two terminal daughters of proximal compartment p . Here $u_p = v_{4+p}$ denotes a proximal voltage; thus $v_j, j = 0, \dots, 5$, indexes all six nonsomatic compartment voltages. All conductances are parameterized by their logarithms.

Labels are the noiseless somatic responses of a teacher with this exact model, establishing a constructive zero forward-error floor. In the aligned task every teacher terminal has nominal gains $(8, 0.25, 0.25, 8)$, ordered as $(E+, E-, I+, I-)$. In the opposed task terminals 2 and 3 instead have $(0.25, 8, 8, 0.25)$. Teacher log gains receive independent Gaussian perturbations of standard deviation 0.15, shared across the paired tasks. Parent inhibitory gains are nominally 15, daughter couplings are $(20, 0.2)$ in each subtree, and somatic gains are 4. Students start with terminal gains 1 and these same other nominal gains, with independent Gaussian perturbations of standard deviation 0.4 applied to all 24 log gains. No exact symmetry is imposed. Paired tasks have identical student initializations, input samples, contexts and minibatches; their teacher tuning differs. Identical input samples do not imply identical task input-sensitivity spectra.

Training minimizes half the mean squared error divided by training-target variance. Writing $\delta = (v_s - y)/\text{Var}_{\text{train}}(y)$, the exact update uses the path factor $q_j = \partial v_s / \partial v_j$ and local eligibility $\partial v_j / \partial \theta_j$. Here θ_j is a local log conductance. Excitatory eligibilities retain the factor $(1 - v_j)/D_j$, inhibitory eligibilities retain $-v_j/D_j$, and daughter-coupling eligibilities retain $(v_{\text{child}} - v_{\text{parent}})/D_{\text{parent}}$, together with their corresponding input and conductance factors; D is the local conductance denominator. Thus changing the credit rule leaves all local voltage derivatives intact. Somatic couplings receive the exact somatic factor 1 under every rule.

The six non-somatic path factors are delivered exactly, replaced by 1, replaced by a fixed initial profile, or projected into three fixed spatial patterns. The initial profile is the mean initial exact path field on 512 independently generated, unlabeled calibration examples. The three-pattern dictionary retains this profile on terminals $(0, 1)$, terminals $(2, 3)$ and proximal compartments $(4, 5)$, respectively, with zero entries elsewhere. If A is the resulting six-by-three matrix, the oracle rule delivers $A(A^\top A)^{-1}A^\top \mathbf{q}$ for the current example. Its coefficients use the current exact field. It tests a restricted spatial dictionary with an oracle encoder; it is not a locally estimated routing rule or evidence that three independent external errors are necessary. At fixed parameters the binary context generates at most two path profiles, so this experiment concerns branch selection rather than a six-dimensional or quartic-interaction requirement.

Each task/seed has 2,048 training, 1,024 validation, 4,096 test and 1,024 diagnostic examples, independent of the calibration sample and of one another. No noise is added to labels. Across the 20 fresh teachers, mean test-target variance is 0.024108 for aligned tuning and 0.014691 for opposed tuning. Test NMSE divides mean squared error by each test set’s clean target variance.

Development comprised three seeds, four regimes, four credit rules, two optimizers and three rates, for 288 fits. Every outcome is retained. The regimes were aligned strong gating, opposed strong gating, opposed moderate gating ($A = 1$), and opposed ungated ($A = 0$). Before development outcomes, the primary strong-opposed regime had to satisfy mean teacher rank-one path capture at most 0.90 and mean exact-Adam validation NMSE at most 0.02, with a declared fallback order. Selection did not use the between-rule gap. Strong opposed tuning qualified, and a separate selection freeze preceded the 20 fresh paired blocks. Adam rates were 0.003, 0.01, 0.03, and SGD rates were 0.03, 0.1, 0.3. Each task/rule/optimizer selected the rate with lowest mean development best-validation NMSE; a common rate minimized the mean over rules. In the fresh task pair all selected rates equal their common rates: 0.03 for Adam and 0.3 for SGD. The common-rate results therefore reuse the same fits. The fresh cohort comprises $20 \times 2 \times 4 \times 2 = 320$ fits.

Minibatches contain 128 examples sampled with replacement. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$; SGD has no momentum. The original budget is 4,096 updates, with global gradient norm capped at 10 and log conductances projected into $[-7, 7]$. Validation checkpoints are updates 0, 64, 256, 1,024, 2,048 and 4,096; the lowest-validation-loss checkpoint supplies the primary test endpoint. Every fresh fit continues, with its original optimizer and minibatch state, to 16,384 updates, adding validation checkpoints 8,192, 12,288 and 16,384. This extension was declared before fresh outcomes and retains earlier checkpoints when selecting its endpoint. These are fixed optimization budgets, not claims of convergence. No fresh or extended fit undergoes gradient-norm clipping. Exact and oracle fits never reach a parameter bound, whereas broadcast fits can do so.

The primary interaction is the calibrated-minus-exact NMSE gap for opposed tuning minus the corresponding gap for aligned tuning. Confidence intervals are paired percentile bootstrap intervals from 20,000 resamples of the 20 teacher/initialization seed blocks. Full rule-specific, seed-specific and checkpoint-specific outcomes are supplied. Post-fit diagnostics report squared-energy capture for the path field and the loss-weighted field, their best rank-one projections, and projection into the frozen initial profile after weighting by squared local eligibility, with or without squared somatic error. All are oracle diagnostics rather than additional training rules.

To identify cancellation across contexts, each rule is evaluated at the same initial, 256-update, primary-best and extended-best states of each selected Adam fit. For a given state, the cancellation ratio is the norm of the probability-weighted sum of context-conditional distal gradients divided by the probability-weighted sum of their norms. Lower values indicate stronger cancellation. All 2,560 audited mean gradients reproduce the training adapter exactly. Since path factors and fixed profiles are positive, per-example approximate gradient estimates have nonnegative inner products with exact per-example gradients; the failure being tested concerns averaging conflicting examples.

A separately timestamped posthoc sensitivity restarts every fresh condition from its original initialization for 16,384 updates with wider log bounds $[-20, 20]$ and unchanged rates. It also replays every original-bound trajectory to record first contact with each parameter bound. These 640 trajectories reuse the same 20 seed blocks. All 320 original-bound replays reproduce every final parameter exactly. The wider-bound interaction remains positive in all 20 seeds for both optimizers. At the last saved checkpoint before each calibrated Adam fit’s first original bound contact, its opposed-task gap is already positive in all 20 seeds. Some broadcast parameters also reach the wider bounds, so this control establishes robustness across a much wider conductance range rather than unbounded convergence. The retained moderate-gating development regime also shows a substantial broadcast deficit despite teacher rank-one path capture near 0.98. Raw path-energy capture is therefore not a sufficient criterion for learning: a small spatial component can distinguish the gradient contributions of competing contexts.

5.2 Local inhibitory gating in the conductance task

This study was specified after exposure to a referee panel’s exploratory gate results. Those runs are excluded here. The protocol and numerical adapter were committed before execution on twenty new seed blocks, 2026090801–

2026090820. Within each seed, aligned and opposed targets and all nine rules shared teacher perturbations, initial students, features, contexts and minibatch streams. The 24-conductance model, nominal gains, input streams and independent perturbations are defined in Section 5.1; every conductance remains plastic.

Let $\delta = (V_s - y) / \text{Var}_{\text{train}}(y)$ denote the somatic error and $a_p \in \{0, 10\}$ the inhibitory activity at parent p . The hard local rule multiplies terminal eligibilities by $\delta \mathbf{1}[a_p = 0]$, while proximal and somatic eligibilities receive δ . It uses neither exact paths nor exact-field projections during learning. A secondary continuous gate uses $\delta / (1 + g_p^I a_p)$ at terminals, again retaining unit proximal and somatic multipliers. Both receive an imposed inhibitory context and an output error; neither generates those signals autonomously.

Exact, uniform, calibrated and three-profile oracle rules retain the previous comparators. Calibration averages initial exact path derivatives on 512 unlabeled examples. Two weighted leaf-subtree profiles with oracle projection amplitudes and unit proximal credit test the context distinction separately from proximal provision. The earlier three-profile oracle also includes a weighted proximal-pair column. The new control therefore has two distal profiles plus a fixed proximal rule, rather than an entirely two-dimensional six-site dictionary. Further controls reverse the distal gate or apply it at the proximal compartments as well. In the latter, inhibitory eligibility is nonzero only when inhibition is present, while its credit multiplier is then zero; both proximal inhibitory gradients vanish identically.

Data sizes, half-NMSE objective, Adam parameters, batch size 128, clipping threshold ten and log bounds $[-7, 7]$ were unchanged. The primary rate was 0.03; 0.01 and 0.1 were symmetric robustness conditions for all rules. All $20 \times 2 \times 9 \times 3 = 1,080$ trajectories continued to 16,384 updates. Minimum validation NMSE selected a state within the primary 4,096-update and extended windows, while fixed endpoints and every saved observation remain available separately.

Two prespecified paired contrasts used exact two-sided sign-flip tests: opposed-task broadcast-minus-local-gate NMSE, and the difference in this gap between opposed and aligned tasks. Both were positive in all twenty seeds, with Holm-adjusted $P = 3.81 \times 10^{-6}$ in the two-test family. Intervals use 20,000 whole-seed bootstrap draws. Other rates, budgets and contrasts are descriptive. The local-minus-exact opposed-task difference was 1.49×10^{-6} (95% interval, -1.60×10^{-6} to 4.86×10^{-6}), and the maximum local-gate seed error was 5.47×10^{-5} . These satisfy the frozen practical precision reference without establishing asymptotic equivalence.

At the primary rate, exact, proper local gates and both oracle dictionaries contacted no bounds; opposed broadcast contacted bounds in 18/20 fits by 4,096 and 20/20 by 16,384. No fit clipped gradients. Ten mechanism tests verify the derivatives, absence of exact-path access in the hard local update, and the proximal eligibility failure. Excluded historical-seed replays reproduced prior exact, unit and calibrated trajectories bit for bit and the independently projected dictionary to floating-point association error. Protocols, fresh-seed outcomes, states, curves and validation records are in `source_data/conductance_local_gate/`.

Retained first conductance study. A separately frozen first study used the same seven-compartment topology but one excitatory and one inhibitory feature per terminal, giving 16 trainable conductances. An independent context selected one subtree. Its two terminal latent inputs were $z_1, z_2 \sim \mathcal{U}[-2, 2]$; the inactive subtree received $-\rho z_j + \sqrt{1 - \rho^2} u_j$, where u_j were independent draws from the same distribution and $\rho \in \{0, 1\}$ set independent or opposite distractors. Each terminal received $\exp(z)$ through its excitatory contact and $\exp(-z)$ through its inhibitory contact, with the corresponding selected or inactive latent value. The four development regimes were ungated/independent, strong-gated/independent, strong-gated/opposite and moderate-gated/opposite, using inhibitory activities $A = 0, 10, 10, 1$, respectively. Nominal teacher excitatory gains were (0.25, 8, 0.25, 8) and inhibitory gains (8, 0.25, 8, 0.25); nominal student gains were 2 and 0.7, respectively. Parent inhibition, daughter couplings and somatic gains were 15, (20, 0.2) per subtree and 4, as above. Teacher and student log gains had independent Gaussian perturbations of standard deviations 0.15 and 0.4. Labels were noiseless outputs of the same teacher circuit. Dataset sizes, calibration, optimizers, learning-rate grids, batch size, validation cadence and the two update budgets matched those specified above. Three development seeds, four regimes, three rules, two optimizers and three rates gave 216 development fits. The declared teacher-geometry and exact-learnability criteria selected strong-gated/opposite inputs, paired with the ungated/independent control; this comparison changes

both gating and distractor dependence. Twenty fresh seeds gave 240 fits, all continued to the larger budget. All outcomes are retained. After 16,384 updates, its gated-task calibrated-minus-exact gap was 0.000502 [0.000267, 0.000772] NMSE, positive in all 20 seeds; under Adam the mean broadcast error remained below 0.0006 NMSE. This small precision effect motivated the separately declared opponent-tuning model above. The first result and all unsuccessful development regimes remain part of the evidence.

Expanded learning-rate screen. Because the original development selection lay at the upper edge of its rate grid, we performed a separately specified post hoc screen using the original three development seeds (201–203). Each of the four credit rules received the same six-rate search on both the aligned and opposed tasks: $\{0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ for Adam and $\{0.03, 0.1, 0.3, 1, 3, 10\}$ for SGD. We reran the original rates alongside the added rates, giving every fit 16,384 updates and selecting its minimum validation NMSE among checkpoints at 0, 64, 256, 1,024, 2,048, 4,096, 8,192, 12,288 and 16,384 updates. All other task, initialization, optimizer and conductance-bound settings remained those of the original opponent study. The screen comprises 288 retained fits; no test outcome entered rate selection.

Before inspecting these outcomes, we specified that an improvement of at least 0.01 NMSE and at least 10% in mean opposed-task validation error for either broadcast rule would trigger a follow-up on the 20 previously used fresh seed blocks. The first criterion compared the best added rate with the best original-grid rate at the same update budget; a pre-outcome addendum also compared the best rate across the whole grid with the originally selected rate (0.03 for Adam and 0.3 for SGD). Any triggered follow-up would retune all four rules on both tasks using development validation alone and would be reported as reuse of existing blocks. Neither criterion was met. No added rate improved the opposed-task broadcast minimum; within the original grid, reducing Adam’s rate to 0.003 lowered mean validation NMSE by approximately 0.00647 (0.89%), and reducing the unit-broadcast SGD rate to 0.1 lowered it by 0.03348 (4.46%). Calibrated-broadcast SGD retained rate 0.3. We therefore performed no additional fresh-block fits. These results address a finite grid at the stated update budget and do not establish optimizer-independent convergence.

5.3 Input grouping and restricted credit in directed conductance trees

We used a minimal directed shunting model to test input grouping while holding the branching shape and parameter counts fixed. Four terminal compartments fed two proximal compartments and a soma. Every model had seven compartments, six child couplings, four excitatory contacts, six inhibitory contacts and sixteen trainable conductances. Leak conductance was one; excitatory and inhibitory reversal potentials were one and zero. All compartment activations were the identity, and the soma had no direct synaptic inputs or additional decoder. Let $i \in \{0, 1, 2, 3\}$ index terminal compartments and $j \in \{1, 2\}$ proximal compartments. Write T_i , P_j and S for terminal, proximal and somatic voltages. The pair $\mathcal{S}_j$ contains the terminals assigned to proximal compartment j . Terminal i has synaptic conductances g_i^E and g_i^I , and b_{ji} couples it to proximal compartment j . The maximal proximal shunt conductance is h_j , and a_j couples that compartment to the soma. For terminal activities x_i^E, x_i^I and proximal shunt input z_j ,

$$T_i = \frac{g_i^E x_i^E}{1 + g_i^E x_i^E + g_i^I x_i^I}, \quad (93)$$

$$P_j = \frac{\sum_{i \in \mathcal{S}_j} b_{ji} T_i}{1 + \sum_{i \in \mathcal{S}_j} b_{ji} + h_j z_j}, \quad (94)$$

$$S = \frac{a_1 P_1 + a_2 P_2}{1 + a_1 + a_2}. \quad (95)$$

The positive local input z_j controls shunt conductance; it carries no teaching signal. Conductances were parameterized as $g = \exp(\theta)$, with $\theta \in [-5, 5]$. The matched resources were compartment, contact, coupling and parameter counts and the admissible conductance range; total learned conductance was not constrained to be equal. No fresh run reached a parameter bound. The NumPy implementation was checked compartment by compartment against

the repository’s production `forward_branch_dynamics` function in its identity-activation, zero-numerical-offset configuration. All sixteen log-conductance derivatives and ten input derivatives agreed with autograd to the numerical tolerances reported in the accompanying tests. This is a directed steady-state model, not a reciprocal cable or spiking model.

The three groupings were (01 | 23), (02 | 13) and (03 | 12). A planted teacher used the same conductance equations, with independent Gaussian perturbations of standard deviation 0.25 in log conductance around $g^E = 2$, $g^I = 0.7$, $h = 3$, $b = 3$ and $a = 4$. Independent student perturbations had standard deviation 0.4 around these centers and were shared across paired candidate groupings, credit rules and optimizer settings. Student initialization did not use teacher coefficients. Teacher tasks permuted terminal excitatory/inhibitory input pairs together; the proximal inputs retained their identities. Paired task inputs were inversely permuted so that the realized target values were identical across the three tasks. Thus each task had the same input distribution and the same spectrum of the input-gradient second moment. The latter was verified from 2,048 paired probes per seed; its matrix entries could differ by the corresponding coordinate permutation. Compatibility meant that the student’s input grouping matched the teacher’s grouping. These tasks are planted mechanistic positive controls within one model class, rather than tests of transfer to an unseen class of biological computations.

Each input was $\exp(\sigma Z - \sigma^2/2)$ for independent standard-normal Z . We used $\sigma = 0.7$ for terminal excitatory and inhibitory inputs and $\sigma = 1.2$ for proximal shunt inputs. All activities and conductances were strictly positive, all voltages lay between the reversal potentials, and there was no label noise. Each task used 1,024 training, 1,024 validation, 4,096 test and 256 calibration examples from independent random streams. Within a task, every candidate and credit rule received the same examples and minibatch indices. Training minimized half-MSE divided by the training target variance. Reported NMSE is MSE divided by the target variance of the evaluation split; it is dimensionless and is not a percentage.

Four feedback conditions shared the forward model. Let V_n be the voltage of compartment n and L the training loss. Write $\rho_n = \partial S / \partial V_n$ for its current exact path sensitivity and $\delta_0 = \partial L / \partial S$ for the somatic loss derivative. Exact transport supplied $\delta_0 \rho_n$. A fixed calibrated broadcast supplied $\delta_0 \gamma_n$, where γ_n was the mean initial path sensitivity of that student’s own forward states on the label-free calibration inputs. The profile was frozen during training. A one-profile projection used the same γ and supplied its best Euclidean fit to the current nonsomatic field. The two-subtree condition restricted γ to each proximal compartment and its two terminal descendants, giving disjoint columns $\mathbf{A}_j$. Its coefficients were

$$c_j = \frac{\mathbf{A}_j^\top \boldsymbol{\rho}}{\mathbf{A}_j^\top \mathbf{A}_j}. \quad (96)$$

The one-profile projection used the analogous coefficient with $\mathbf{A} = \boldsymbol{\gamma}$. These per-trial coefficients had access to the current exact student path field; both projection rules are oracle diagnostics of restricted feedback, not learned coefficient estimators. Soma error remained exact in every condition. Each conductance update multiplied the delivered compartment error by its local conductance eligibility, including the chain derivative of the exponential parameterization. For a child coupling, the local conductance derivative was $(V_{\text{child}} - V_{\text{parent}}) / G_{\text{parent}}^{\text{tot}}$, where $G_{\text{parent}}^{\text{tot}}$ is the parent’s total open conductance. Credit geometry was evaluated at the corresponding exact-learning model’s common checkpoint state; the reported path-capture normalization excluded the soma.

We used 1,000 minibatch updates of size 128, with checkpoints at 0, 10, 100, 300 and 1,000. Stochastic gradient descent (SGD) and Adam were evaluated separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. One learning rate per optimizer was chosen by averaging final validation NMSE over five development seeds (15300–15304), all task/student groupings and all four credit rules. The original SGD grid was 0.001, 0.003, 0.01; because its optimum was at the upper boundary, an additional development-only bracket tested 0.03, 0.1, 0.3, 1. Every result, including unstable high-rate trajectories, was retained. Adam tested 0.003, 0.01, 0.03. The selected rates were 0.3 for SGD and 0.01 for Adam. These settings were frozen before the twenty fresh seeds (15400–15419), producing $20 \times 3 \times 3 \times 4 \times 2 = 1,440$ fits and 7,200 checkpoint outcomes. No learning rate, horizon or credit rule was selected using fresh training outcomes. The two primary contrasts were the Adam exact-gradient grouping effect and the Adam calibrated-broadcast-minus-exact effect within compatible groupings. Other contrasts are descriptive optimizer and feedback-resolution sensitivities.

1440 Inference used the seed as the independent unit. Each seed’s compatible mean averaged its three teacher-
 1441 task permutations; its incompatible mean additionally averaged the two mismatched student groupings per task.
 1442 Confidence intervals used 10,000 bootstrap resamples of these paired seed summaries. Intervals are pointwise
 1443 rather than simultaneous across all reported subgroup comparisons. All sources, rates, states, physical-domain
 1444 checks and numerical tests are retained under `source_data/morphology_conductance`; execution code is under
 1445 `scripts/morphology_conductance`.

1446 **An interaction lower bound for this conductance model.** Let R_i denote a teacher terminal response, $A_j =$
 1447 $(1 + \sum_i b_{ji} + h_j z_j)^{-1}$ its proximal attenuation, and $\beta_j = a_j / (1 + a_1 + a_2)$. The sum in A_j runs over the teacher
 1448 terminals assigned to proximal compartment j . The teacher output is a sum of terms $\beta_j b_{ji} R_i A_j$. Let $p_{\text{teacher}}(i)$
 1449 and $p_{\text{student}}(i)$ denote the proximal parent assigned to terminal i in the teacher and student, respectively, and write
 1450 $p(i) = p_{\text{teacher}}(i)$. For terminal i , set $j = p(i)$. Because terminal inputs and proximal shunt inputs are independent, a
 1451 term whose terminal and teacher shunt belong to different *student* proximal input blocks has the centered interaction
 1452

$$H_i = \beta_j b_{ji} (R_i - \mathbb{E}R_i)(A_j - \mathbb{E}A_j). \quad (97)$$

1453 Every admissible student output is additive across its two proximal input blocks. Each such H_i is orthogonal to all
 1454 functions additive across those blocks, and the omitted H_i are mutually orthogonal. Therefore, for every choice of
 1455 the student’s conductances,

$$\mathbb{E}[(S_{\text{teacher}} - S_{\text{student}})^2] \geq \sum_{i: p_{\text{student}}(i) \neq p_{\text{teacher}}(i)} (\beta_{p(i)} b_{p(i)i})^2 \text{Var}(R_i) \text{Var}(A_{p(i)}). \quad (98)$$

1456 The bound is zero for compatible groupings. It is a lower bound for the larger class of arbitrary functions additive
 1457 across the student’s proximal blocks, not a claim that the constrained rational student attains it. Its validity relies on
 1458 the specified independent inputs, identity somatic activation and absence of direct somatic synaptic inputs; it does
 1459 not extend automatically to arbitrary conductance architectures or nonlinear readouts. An equivalent obstruction is
 1460 the absence of a mixed derivative between a terminal input and a shunt input assigned to the other student subtree.

1461 We specified this diagnostic before the fresh training outcomes and retained the learner and rate-selection
 1462 procedure unchanged. Population lognormal moments were evaluated with 32-, 64- and 128-node Gauss–Hermite
 1463 quadrature. The primary bound used 128 nodes; the maximum change from 64 to 128 nodes was below 8×10^{-17}
 1464 in normalized-MSE units. This is a converged numerical evaluation of an analytic bound, not a formally certified
 1465 quadrature error. Across the fresh teacher draws, the mean incompatible population-NMSE lower bound was
 1466 0.013222, with a minimum of 0.005367 across task/candidate pairs. These are population lower bounds, distinct
 1467 from held-out sample NMSE.

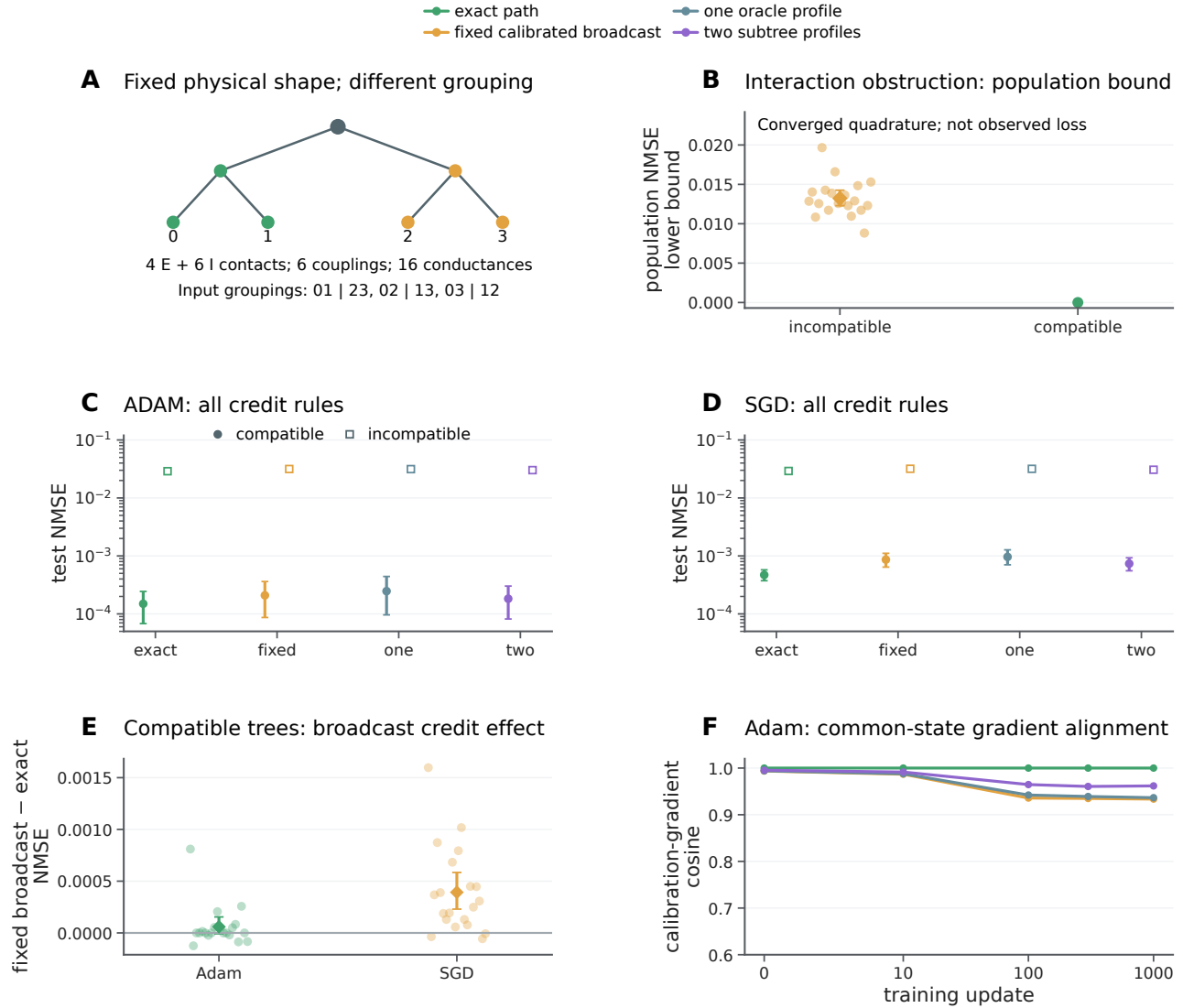


Figure S17: Input grouping and restricted feedback in positive-conductance trees. **A**, Seven-compartment directed shunting model with the same physical shape, sixteen positive conductances (four excitatory and six inhibitory contacts plus six couplings) and soma-only readout under all three input groupings. Teacher-task permutations preserve the input-gradient spectrum. These are planted mechanistic controls within the stated model class. **B**, Analytic population lower bound from interactions crossing the student's two proximal input blocks, evaluated by converged quadrature. Dots average the six incompatible task/grouping pairs within each of twenty seeds; the mean bound is 0.013222 population normalized mean squared error (NMSE). This is neither an observed loss nor a certified numerical integration bound. **C,D**, Final held-out sample NMSE for exact path, fixed calibrated broadcast, one oracle profile and two subtree-profile credit rules under Adam and stochastic gradient descent (SGD), respectively. Circles average three compatible task permutations within seed; open squares also average the two incompatible groupings per task. Rates were selected on development seeds and frozen before fresh outcomes. The fixed broadcast profile is calibrated from initial student path sensitivities; projection coefficients access the current exact field and are oracle controls. **E**, Paired compatible-tree fixed-broadcast-minus-exact NMSE differences. The Adam interval includes zero; the small SGD effect does not establish a universal need for precise credit. Dots are the twenty independent seed means. Means and whiskers in **B–E** use 10,000 whole-seed bootstrap draws and pointwise 95% intervals. Population-normalized bounds in **B** and split-variance-normalized test errors in **C–E** have distinct denominators. **F**, Mean aggregate gradient cosine on 256 calibration examples for compatible Adam models, evaluated at common exact-learning checkpoints; no uncertainty band is shown. All 1,440 fresh fits and 7,200 checkpoint records are retained. The interaction bound relies on independent input blocks, identity somatic activation and no direct somatic synapses; it is separate from the multi-affine cut-rank theorem.

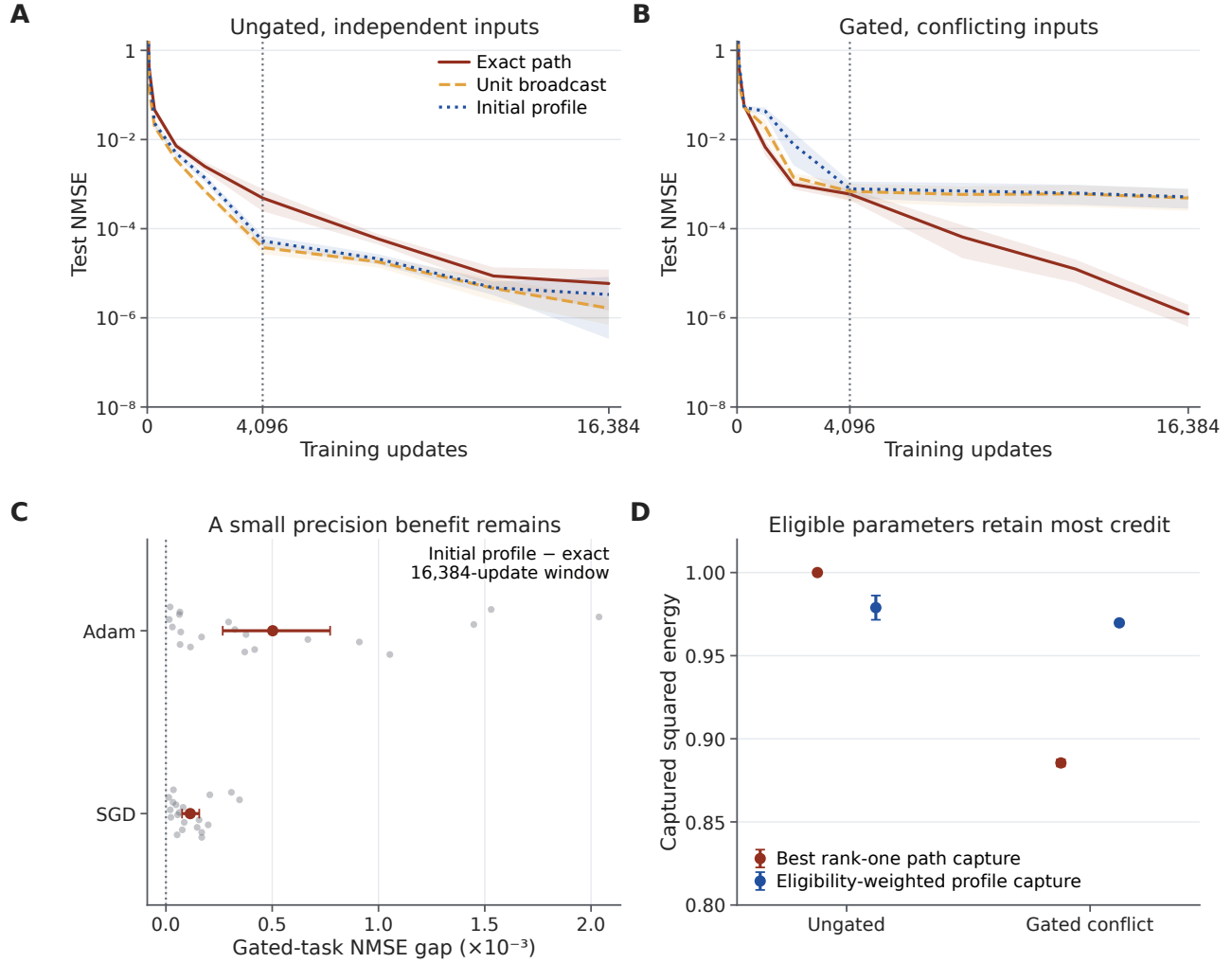


Figure S18: A first conductance task produces a small precision benefit from exact credit. **A,B**, Test normalized mean squared error (NMSE) at fixed checkpoints under Adam, for ungated independent inputs and gated conflicting inputs in the 16-conductance model. Curves show means and 95% paired-seed bootstrap intervals across 20 fresh seeds; the vertical dotted line marks the original 4,096-update budget. All fits continue to 16,384 updates with unchanged optimizer and minibatch state. Unit and calibrated profiles differ here because global gradient clipping can break Adam’s invariance to fixed positive coordinate scaling. **C**, Gated-task NMSE differences between the initial calibrated profile and exact credit, evaluated at each fit’s best validation checkpoint in the extended window. Points are paired seed differences; symbols and bars show their means and 95% bootstrap intervals. The Adam gap is 0.000502 [0.000267, 0.000772], positive in all 20 seeds; under Adam, mean broadcast error nevertheless remains below 0.0006 NMSE. **D**, Diagnostics at the exact-rule best-validation states distinguish the best rank-one capture of the six-compartment path field from capture by each student’s fixed initial profile after weighting by local parameter eligibility. Both are squared-energy projection ratios with one freely fitted amplitude per example. High eligibility-weighted capture helps explain the small learning difference. The gated and ungated tasks change both inhibition and distractor dependence, so this first study is a task-family comparison. All 216 development fits, 240 fresh fits and their continuations are retained in Source Data.

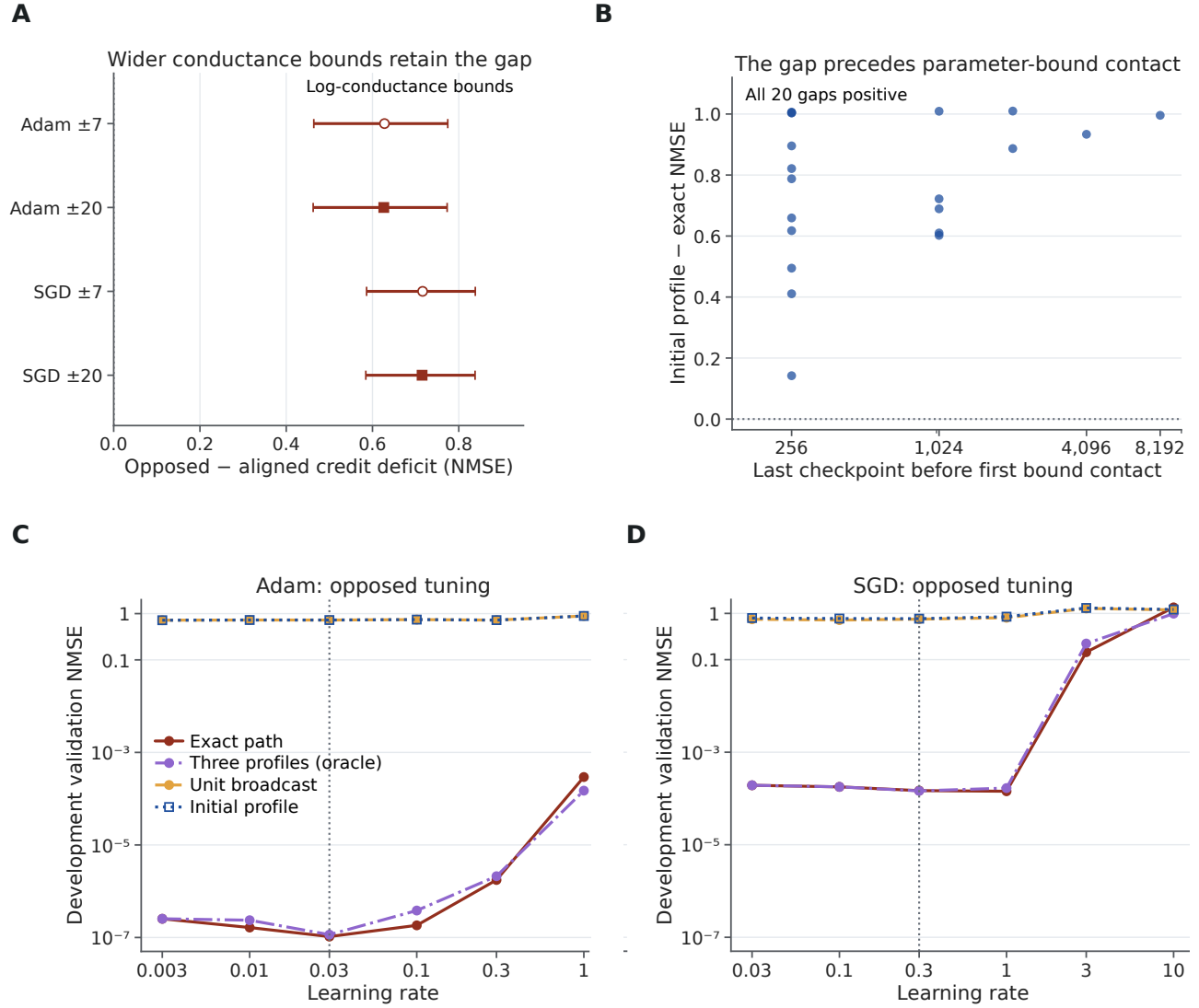


Figure S19: **Parameter-range and rate controls retain the opposed-task broadcast deficit.** **A**, Task-by-credit interaction under original $[-7, 7]$ and wider $[-20, 20]$ log-conductance bounds, with Adam and SGD. All fits retain their original initialization/minibatch streams and extend to 16,384 updates; endpoints use validation-selected checkpoints. Dots are twenty paired seeds and intervals are 95% paired bootstrap intervals. Wider-bound interactions are positive in all seeds. **B**, Initial-profile-minus-exact NMSE before each calibrated-profile fit's first original bound contact; all twenty differences are already positive. **C,D**, Equally budgeted six-rate development sweeps for opposed tuning under Adam/SGD; corresponding aligned-task outcomes are included in the full condition table. Each point averages three development seeds; all 288 outcomes are retained. The predefined improvement trigger was not reached, so rates were not retuned on fresh test outcomes. Some broadcast fits also reach wider bounds; these are finite-budget controls, not an asymptotic convergence proof.

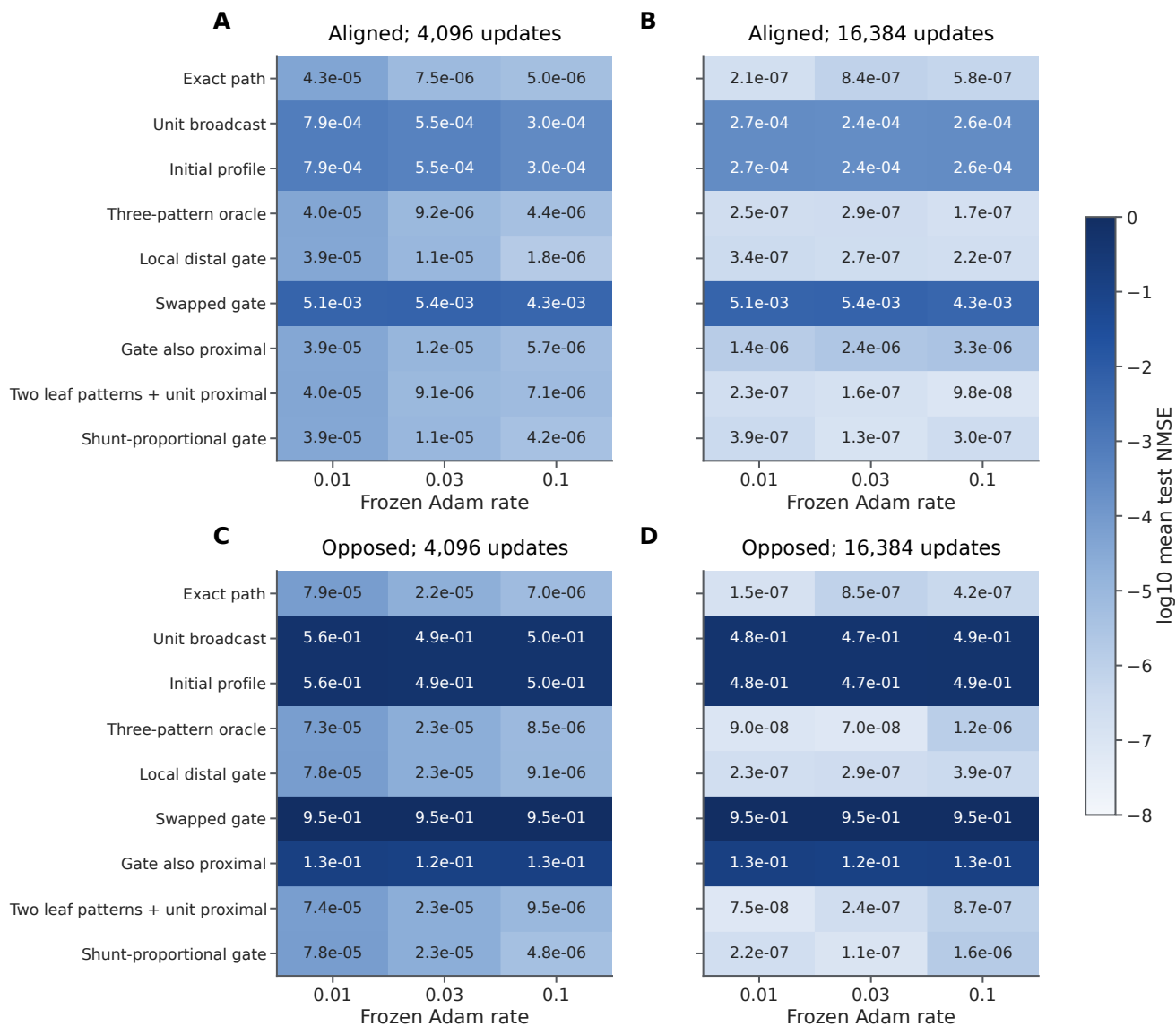


Figure S20: **Local conductance gating is robust to rate choice but depends on where it is applied.** **A,B**, Aligned tuning at 4,096 and 16,384 updates. **C,D**, Opposed tuning at the same budgets. Every rule receives every displayed Adam rate; 0.03 is primary. Entries are mean test NMSE at validation-selected checkpoints across twenty new paired seeds; the color scale shows base-ten logarithm. Exact credit, the three-pattern oracle, the local distal gate and the shunt-proportional gate learn accurately. Swapping the gate fails; extending the gate to proximal compartments leaves a substantial opposed-task deficit. The two leaf patterns with unit proximal feedback isolate this placement effect. These are supplied-context, bounded-conductance experiments; the local gate does not estimate arbitrary route coefficients. All rates, seeds and budgets remain included.

1468 S6 Serial physical computation and training budgets

1469 6.1 Generator, architecture and primary comparison

1470 The hierarchical_gain_load task supplies 64 positive excitatory and 64 positive inhibitory rates. Fine, coarse
1471 and global Gaussian node fields generate three mean-corrected lognormal gain factors (hierarchy H3). The product
1472 of these factors modulates distal class-bearing excitation; separate inhibitory sensor blocks report each factor. For a
1473 signal-bearing coordinate,

$$x_E = \max\{b_E + y\Delta + \varepsilon_E, 10^{-4}\} \exp\left(\sum_{\ell=1}^3 \sigma_\ell z_\ell - \frac{1}{2} \sum_{\ell=1}^3 \sigma_\ell^2\right), \quad (99)$$

1474 where $y \in \{0, 1\}$ is balanced class, $\Delta = 0.80$, $\sigma_\ell = 0.25$, z_ℓ are independent standard Gaussians and ε_E is independent
1475 Gaussian noise of SD 0.08. The same noise-floor-gain order applies to inhibitory rates. Class-independent
1476 middle/proximal excitation has baseline 0.02, so flooring makes its effective rate noise nonzero-mean. Each dataset
1477 contains 10,000 balanced examples.

1478 Sensor alignment α changes correspondence without changing marginal Gaussian variance:

$$s_\ell(\alpha) = \alpha z_\ell + \sqrt{1 - \alpha^2} \xi_\ell, \quad \xi_\ell \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (100)$$

1479 with sensor noise independent of latent gains. Alignment uses $\alpha = 1$; independent sensors use zero. A nonzero
1480 cyclic trial shift instead removes trial-specific access while preserving the entire inhibitory field. Reversed placement
1481 exchanges fine/global feature ranges across inventory tiers, preserving their 21/22/21 widths. An earlier three-cycle
1482 assignment changed candidate slots and was excluded before inference.

1483 Branch factors run soma to distal. For factors $[b_1, \dots, b_D]$, the nonsomatic count is $\sum_{j=1}^D \prod_{r=1}^j b_r$. D1 [8], D2
1484 [2, 3] and D3 [2, 1, 2] each contain eight units per soma; D3's unary middle stage preserves that inventory. Structured
1485 masks place all tiers in D1, fine/coarse distally and global proximally in D2, and one tier per stage in D3. Every
1486 architecture has 66,178 trainable parameters, 14,336 active synapses, 21,760 candidate slots and 2,944 persistent
1487 state scalars per sample. Compartments use the directed positive-conductance division, identity reactivation and
1488 softplus-positive synapses.

1489 Excluded calibration seeds 10100–10141 tested signal contrast, gain shift and child conductance. Contrast 0.24
1490 with train/test gain SD 0.25/1.2 remained at chance despite improved training fits at depth. Coupling and contrast
1491 sweeps selected child conductance 16, contrast 0.80 and matched SD 0.25; contrast 0.72 missed the predefined D1
1492 accessibility threshold. All failed calibration settings remain in Source Data.

1493 The primary 270 fits crossed D1–D3, ten paired seeds 10200–10209 and nine conditions: aligned shunting
1494 BP; independent-sensor, shuffled-sensor or reversed-placement BP; aligned raw-additive BP; and aligned/reversed
1495 LocalCA with shared-soma or exact-path credit. The protocol required at least 9/10 aligned-BP seeds above 0.55 at
1496 every depth and every mean below 0.98. All conditions passed completeness, finite-metric, resource and no-fallback
1497 checks. Training used a 180-epoch cap and validation-loss early stopping with patience thirty. The BP recipe used
1498 Adam at 0.003; LocalCA retained split parameter groups with nominal rate 0.0008. Group rates, boosting, decay
1499 and update options are retained in the resolved configurations.

1500 At that budget, aligned-BP D3 minus D1 was 30.86 percentage points; alignment-minus-control depth interac-
1501 tions exceeded 30 points. Aligned additive BP declined with depth. These contrasts separate forward composition
1502 from a credit-rule effect, whose budget dependence is evaluated below. A post hoc common-checkpoint audit found
1503 exact-path LocalCA gradients collinear with autograd (minimum cosine 0.9999999999999999), while shared-soma
1504 cosine declined from 0.946 at D1 to 0.767 at D3. This gradient diagnostic does not establish a converged learning
1505 requirement.

1506 6.2 Nonserial and point-model controls

1507 An all-active grouped star retains branch layers, masks and parameters but lets each nonsomatic layer read external
1508 inputs independently; child-aggregation parameters contribute only once at the soma. D1 logits match exactly. A

literal grouped-point control additionally replaces child aggregation by direct projections to the soma, copying the corresponding initial scalar conductances. These controls retain resource counts while removing serial inheritance of branch states.

The star/point/credit-coordinate study comprised 200 fits on the same ten H3 seeds: 140 initial controls and sixty optimizer-matched soma-broadcast fits. Serial exceeded star by 30.81 percentage points at aligned D3 (95% paired interval, 30.41–31.26), but fell below it by 0.43 points after reversal. Two unconstrained point MLPs matched active parameters (15,055) or nearest realizable total parameters (66,200 versus 66,178). Their mean accuracies, 0.9861 and 0.9902, exceeded the tree. Thus the serial advantage is relative to matched branch resources, not arbitrary point-network expressivity.

Soma-broadcast autograd preserves local synaptic derivatives but replaces nonsomatic loss derivatives by each unit’s somatic derivative. Full BP exceeded it by 0.64 percentage points under the BP optimizer. Under the LocalCA optimizer, broadcast autograd and shared-soma LocalCA differed by only –0.06 points (–0.23 to 0.09), whereas changing the broadcast optimizer cost 13.47 points at 180 epochs. The optimizer-matched comparison was specified after discovering the recipe difference. It isolates that implementation contrast; autograd is not a biological mechanism. Full configurations and outcomes are in `source_data/point_dendrite_credit_controls/`.

Sixty additional literal grouped-point H3 fits used aligned/reversed placement and D1–D3 on the same seeds. Grouped-point means remained near 0.61 across depth. Serial-minus-grouped at aligned D3 was 30.87 points, reversing to –0.44 under placement reversal. These observations agree with the star control without retaining its child-aggregation modules.

6.3 Alignment, hierarchy and task-family sensitivities

The alignment interpolation was specified after endpoint outcomes but before intermediate fits. Ninety runs crossed $\alpha = 0.25, 0.50, 0.75$, three depths and the original ten seeds. D3-minus-D1 effects at $\alpha = 0, 0.25, 0.50, 0.75, 1$ were 0.019, 0.207, 0.820, 3.126 and 30.855 percentage points. The prespecified within-seed linear summary had mean slope 25.84 points per unit alignment, but the final interval contributed 27.73 points: the response is strongly nonlinear. Software records identify an unused optional-rule edit; all unique conditions passed resource checks (`source_data/physical_alignment_dose/`).

The H2 follow-up fixed two 32-feature tiers, inventory 4/4 and D1 [8]/D2 [4, 1] before outcomes. Ten new seeds 10300–10309 crossed aligned/reversed placement, serial or literal grouped-point BP, and shared-soma or exact-path LocalCA (160 fits). No inherited signal, conductance or optimization setting was retuned. Aligned serial BP gained 30.84 points from D1 to D2; the placement interaction was 32.40 points. Grouped-point depth effects remained near zero. This is replication within the same synthetic mechanism family. Together with the sixty H3 grouped-point fits, all 220 outcomes are in `source_data/remaining_physical_experiments/`.

H4 used four 16-feature tiers, inventory [4, 2, 1, 1] and D1–D4, adding D4 [1, 1, 2, 2]. All depths retain the same eight-unit resources. The 360 fits crossed aligned/reversed placement, serial/grouped-point BP, both LocalCA rules and aligned raw-additive BP over seeds 10400–10409. An automatic D3 partition error stopped ninety configurations before outcomes; explicit grouping [4], [2], [1 + 1] restored the intended counts, and only failed configurations were repeated. The main contrasts, 50,000-draw intervals, exact sign-flip tests and BH family were specified before analysis, with positive claims additionally requiring 8/10 concordant signs. Aligned BP accuracy peaked at D3: D4 minus D3 was –1.46 points (–1.74 to –1.16), although D4 minus D1 remained 25.63 points. This falsified the predefined simple depth-equals-hierarchy crossover. Exact-path LocalCA’s 0.60-point D4–D3 gain failed the direction criterion (7/10); the shunting-minus-additive increment was also inconclusive. All reversed and null endpoints remain in `source_data/physical_depth_h4_factorial/`.

A same-seed computational repeat under source commit a99c3a7 covered all 220 H2/grouped-point fits and 210 principal H3 fits. Median absolute accuracy drift was 0.030 percentage points, mean drift 0.154 and maximum drift 15.01 in a raw-additive control; 357/430 pairs were within 0.1 points. Principal shunting and local-credit directions persisted. Supplementary Table S9 retains this unfavorable reproducibility range; these reused seeds are not fresh replication.

The fixed-D3 task-family study used seeds 10500–10509 and crossed three generators, $\alpha = 0, 0.5, 1$, serial and

grouped-point composition and BP or exact-path LocalCA (360 fits). Nested factors use Eq. 99. Flat factors retain the same product and marginal variances but assign each independent factor to the same deterministic eight-bin spatial partition. Local-ratio inputs give excitation and inhibition at each level correlated cumulative hierarchical fields, making divisive information locally available. All families retain 10,000 balanced examples, distal-only class signal and identical model resources. Primary architecture-by-alignment and between-family contrasts used 50,000 paired bootstrap draws and an 8/10 sign criterion. BP alignment interactions were 30.41, 22.26 and 1.30 percentage points for nested, flat and local-ratio tasks; LocalCA effects were 27.70, 12.37 and -11.68 . The local-ratio reversal is retained in `source_data/task_family_alignment/`.

6.4 Longer training and metric dependence

All fifty aligned D3 fits in five relevant credit/optimizer conditions reached the original 180-epoch cap with decreasing validation loss. A protocol-led extension retained those conditions and ten exact-BP D1 references, using the same ten seeds, configurations, patience and validation-based selection. Only the cap (600 epochs) and output names changed. Historical files lacked optimizer and random states, so runs restarted from their original seeds with `a99c3a7`; they were not exact checkpoint continuations. Four seed blocks ran on GPU and six on CPU, held fixed across all six conditions within each seed.

Current and best weights, optimizer and all random states were saved at 180, 360 and 600 epochs or earlier termination. Final weights exactly matched validation-selected states. Budget comparisons use the best state within the first 180 epochs and within the full new trajectory; test accuracy and cross-entropy never select states. Stopped curves are carried forward to preserve ten seeds per mean and are marked separately. Source Data retain 34,159 observed epochs, 120 original/extension configurations and sixty complete run records.

At 600 epochs, exact-BP D3 minus D1 was 38.17 percentage points (95% interval, 37.98–38.39). LocalCA exact minus shared changed from $+10.86$ points at 180 to -1.52 (-1.83 to -1.23), reversing in all ten seeds. Exact LocalCA nevertheless retained lower test cross-entropy, 0.2544 versus 0.2758. The BP-recipe exact-minus-broadcast accuracy difference was 0.48 points; the broadcast BP-minus-LocalCA-recipe difference was 1.54. Intervals use 10,000 paired bootstrap draws and are descriptive.

All fifty D3 fits and two D1 fits still reached the 600-epoch cap; D3 validation loss continued decreasing over the final thirty epochs. Relative to historical trajectories, maximum validation-loss drift was 0.005885; median and maximum selected-accuracy drift were 0.070 and 1.830 percentage points. Comparisons therefore use windows of the same new run. The extension establishes budget and metric dependence, not convergence or fresh-seed confirmation.

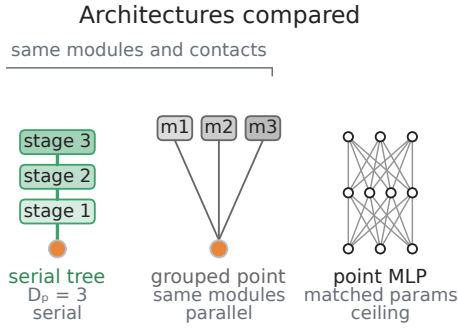
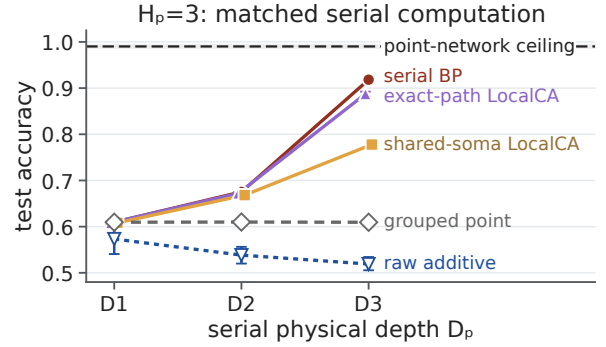
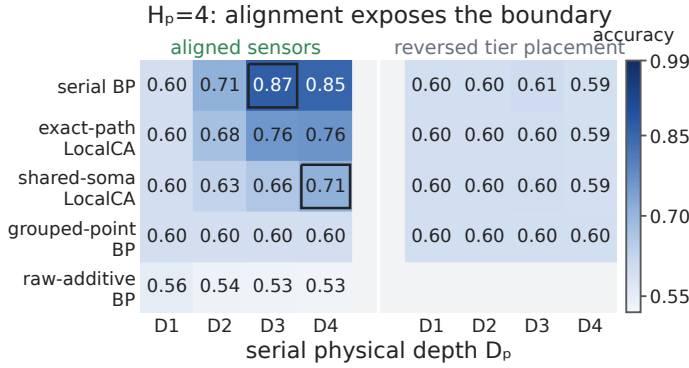
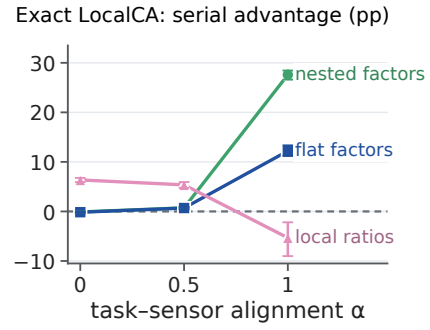
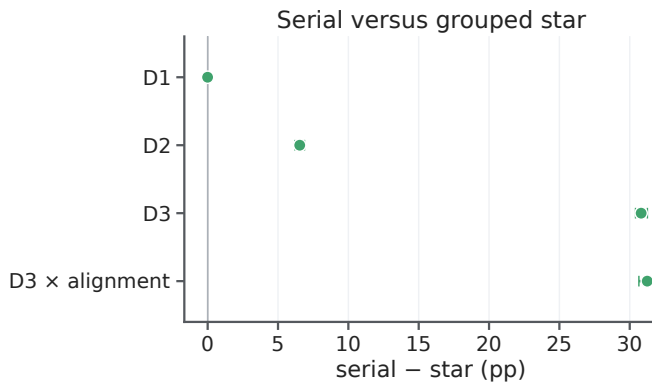
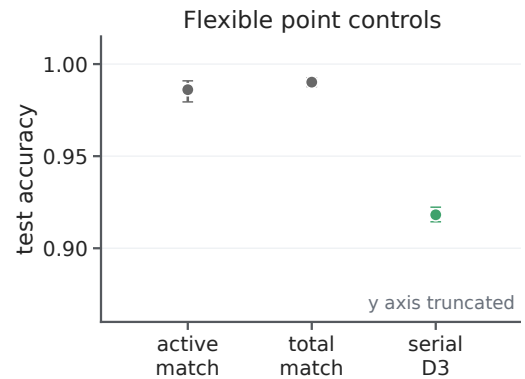
A**B****C****D****E****F**

Figure S21: **Task-matched serial computation differs from grouped and flexible point controls.** **A**, Serial D3, resource-identical grouped-point and approximately parameter-matched flexible point networks. Only serial/grouped models share modules and contacts. **B**, Original-budget three-level aligned task across D1–D3 and feedback rules. **C**, Four-level task across D1–D4 under aligned/reversed sensor placement. **D**, Serial-minus-grouped accuracy at fixed D3 across nested, flat and local-ratio task families, under exact-path LocalCA. **E**, Serial-minus-grouped-star effects and their alignment interaction. **F**, Flexible point controls matched to active or total parameter count. Curves and differences use ten paired seeds per cohort and 95% paired-seed bootstrap intervals. These task-calibrated, original-budget comparisons establish forward-computation dependence on task/placement, not an optimizer-independent requirement for resolved credit. Main Fig. 6 shows the longer-budget accuracy reversal and cross-entropy trajectory.

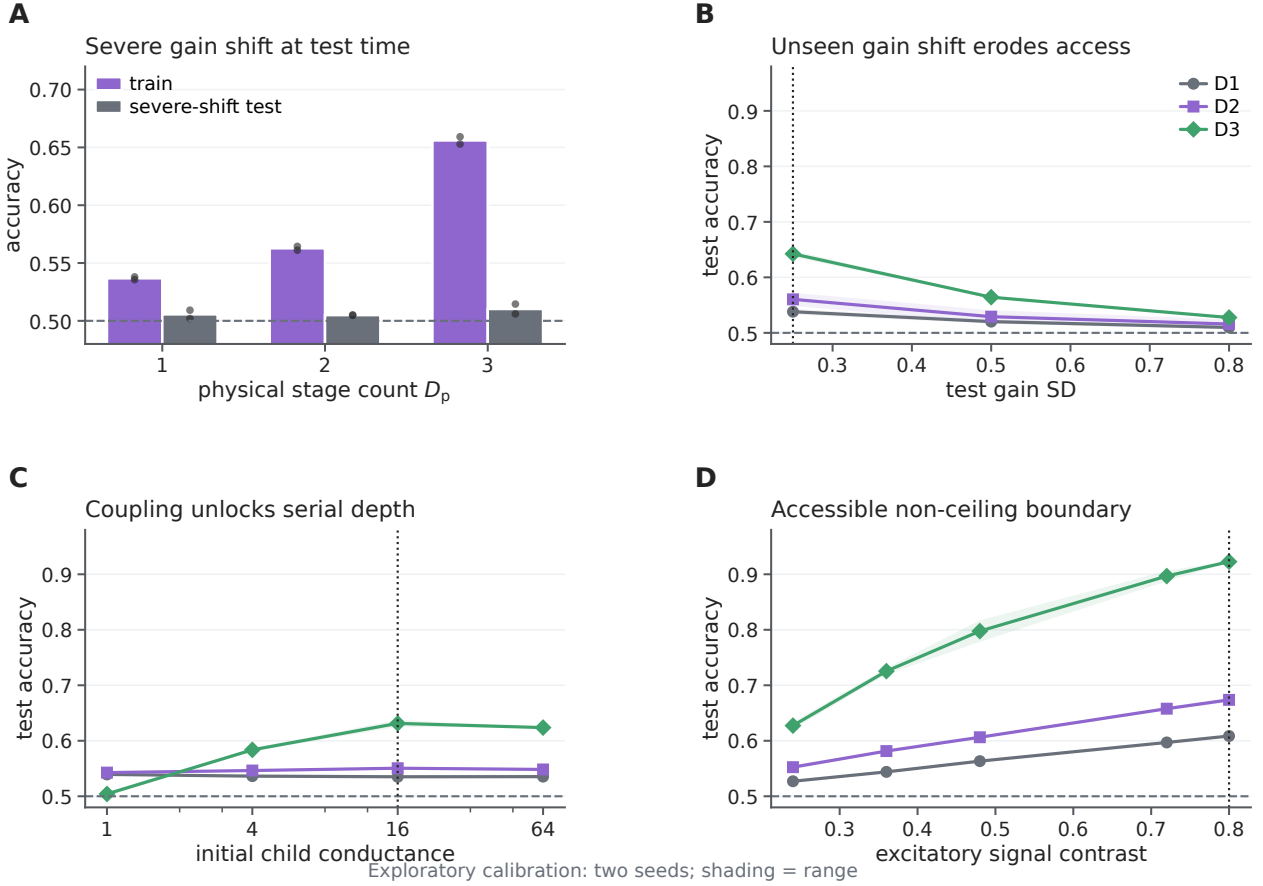


Figure S22: Transparent calibration of the nonlinear physical-depth operating point. All panels use the three-level hierarchical gain-load task and serial shunting morphologies D1 [8], D2 [2, 3] and D3 [2, 1, 2], each with eight nonsomatic compartments per soma. They test the prerequisite for the main hypothesis: whether the mechanism-matched divisive task is accessible at every depth before comparing depth benefits. **A**, The original aligned shunting-backpropagation (BP) pilot at signal contrast 0.24 fit the training distribution increasingly with depth but remained at chance under the prespecified severe gain shift (test gain standard deviation (SD) 1.2). Bars are two-seed means and dots are the individual exploratory seeds. **B**, With training gain SD fixed at 0.25, increasing unseen test gain variation eroded accessibility at every depth. The dotted line marks the matched train/test condition. **C**, At matched gain and signal contrast 0.24, stronger child coupling selectively made serial D3 computation accessible; the dotted line marks child conductance 16 used subsequently. **D**, At matched gain and child conductance 16, increasing excitatory signal contrast moved D1–D3 away from chance without saturating D1 or D2. The separately run 0.80 boundary (dotted line) met the predefined accessibility criterion and selected the operating point used for the analysis seeds. Lines show two-seed means; shading spans the two exploratory values and is not a confidence interval. All pilot, ladder and boundary seeds were excluded from the confirmatory contrasts.

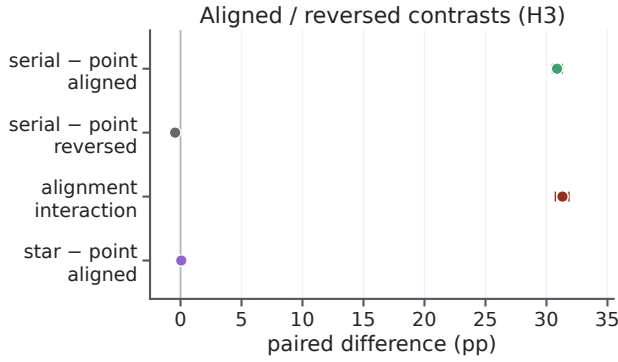
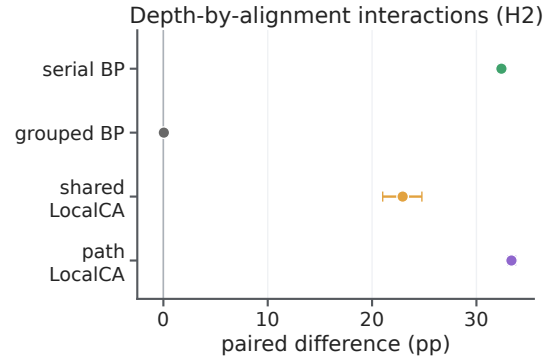
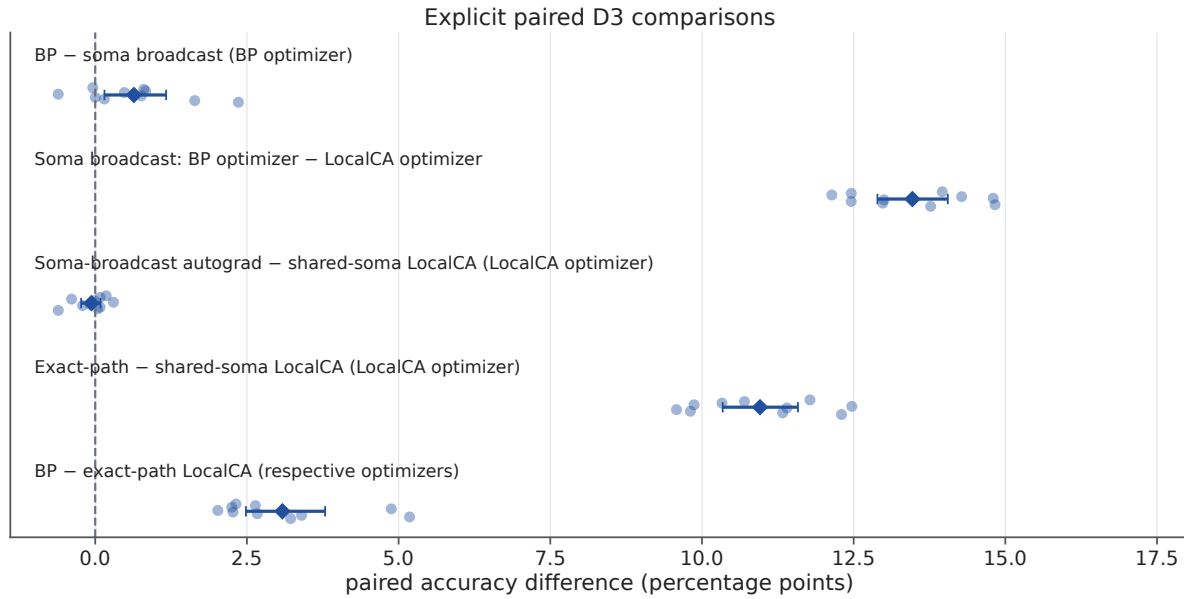
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Figure S23: Physical-depth conclusions depend on alignment, task hierarchy and optimization. **A**, Serial/point contrasts under aligned and reversed placement. **B**, Separate two-level task: depth-by-alignment interactions for serial/grouped-point backpropagation and shared/exact LocalCA. **C**, Original-budget D3 optimizer/credit-coordinate contrasts. Under the BP optimizer, full versus soma-broadcast differentiation differs by 0.64 points; under the split LocalCA optimizer, exact versus shared differs by 10.96 points; changing the broadcast optimizer costs 13.47 points. Means and intervals use ten paired seeds and 95% bootstrap intervals. These earlier recipe-specific effects do not imply a converged credit advantage: at 600 epochs the LocalCA accuracy difference reverses, while exact credit retains lower cross-entropy (main Fig. 6). All D3 continuations hit the cap; D1 usually stops early.

S7 Anatomical dictionaries and controlled field alignment

7.1 Reconstruction, cohorts and electrical normalization

The initial cohort comprises eight layer 2–5 intratelencephalic cells from MICrONS [8]. Stable nuclei were resolved in minnie65_public materialization 1822; skeletons came from the MICrONS skeleton service and incoming contacts from synapses_pni_2. Degree-two chains were collapsed to branch-to-branch cables. Contacts were assigned to the nearest dendritic node and retained within $5\mu\text{m}$. Across cells, 28,669 E/I-classified contacts passed, with mean mapping fraction 0.980. Root traversal checked acyclicity and a unique path to the soma. Cell identifiers, contact counts and source assets are indexed in Supplementary Table S12.

Presynaptic coarse type supplied E/I identity where available. Otherwise the in-development synapse_target_predictions_ssa_v2 table classified high-confidence spine targets as excitatory and shaft/soma targets as inhibitory. This proxy table lacks a stable public release identifier. Direct/proxy agreement was 73.01% among 2,012 jointly labeled contacts (E/E 489, E/I 84, I/E 459, I/I 980). Principal comparisons were repeated on the 2,077 directly typed contacts. Coverage and error asymmetry are observed properties, not missing-at-random assumptions.

A non-overlapping robustness cohort was defined from a 55-cell convenience export before outcomes, resolved in public v661 and excluding the original eight by stable nucleus. Its 47 retained cells are from the same mouse and an earlier reconstruction release. Direct presynaptic labels were available for 11,024 incoming contacts (4.71%) before mapping; 10,516 mapped contacts entered analysis. This layer-imbalanced cohort is not a population or independent-animal replication.

The second-animal cohort used published Pinky v185 mesh, soma-valence and dense-synapse data [9, 10]. Before mesh or selected-cell contact counts were inspected, 362 excitatory soma rows were ordered along the second volume coordinate and split into twelve equal-count strata. Each stratum contributed the root nearest the global first/third-coordinate medians, with root-ID tie-breaking. This deterministic rule spans the volume while reducing boundary truncation; it is not random sampling. Mesh processing retained the largest face-plus-link-edge component and used meshparty TEASAR with $7.5\mu\text{m}$ soma collapse and $12\mu\text{m}$ invalidation. Radius was half the opposite-surface ray distance at mesh anchors (embreex 2.17.7.post7). Only presynaptic roots labeled in the published soma-valence table supplied E/I identities.

Pinky inherited v661 compression, mapping, normalization, dictionary ranking and 500-shuffle settings. Inclusion required at least twenty mapped typed inputs and three inhibitory inputs. All twelve cells processed; ten qualified, with excluded cells having nineteen and thirteen inputs. Their route-generated $K = 4$ ancestry advantages over random routes, depth bins and shuffling agreed in direction with the other mouse. Cell-bootstrap intervals describe stability within each volume; two mice do not support a population-level animal test. In the equal-broadcast analysis below, only eight Pinky cells qualify at $K = 8$ and none at $K = 16$; lower-budget results retain the other eligible cells.

Raw axial coupling scales as radius squared over length, divided by its within-cell positive median and clipped to 0.05–20. Leak scales as radius times the larger of length and radius, similarly normalized and clipped to 0.1–10. Synaptic areas are divided by the median nonzero total area. These are normalized passive sensitivity weights, not fitted membrane conductances. The reconstructed model uses reciprocal coupling and a symmetric conductance matrix; the directed path product of Eq. 10 applies only to the artificial network.

For excitatory-bearing segment i and inhibitory-bearing route site k , define

$$\Gamma_{ik} = A_{ik}\beta_k, \quad (101)$$

where A_{ik} indicates that k lies on the path from i to soma and β_k is its normalized inhibitory fraction. This ancestry kernel approximates route weighting; it is not the reciprocal Green's-function derivative. Excitatory-area-weighted singular values s_j give participation rank $(\sum_j s_j^2)^2 / \sum_j s_j^4$.

7.2 Mapping, labels and cable compression

The eight-cell uncertainty analysis crosses mapping thresholds 2, 5 and 10 μm with direct-only or hybrid labels (proxy probability at least 0.8). Radius factors are $\exp(\sigma_{rz} - \sigma_r^2/2)$ for independent standard Gaussian z , with $\sigma_r = 0, 0.25, 0.5$ and five paired draws at each nonzero dose. Every condition reselects up to four routes by the same leverage. Jaccard overlap measures route changes; capture uses the fixed nominal field, sites and weights, isolating dictionary sensitivity rather than changing both sides of the comparison.

Compression preserves total length and length-radius product to rounding error, but mean-radius resistance $\ell/\bar{r}^2$ differs from the series sum $\sum_e \ell_e/r_e^2$. Median relative error was -0.00200 , with minimum -0.99717 . A second normalization uses inverse series resistance, retaining leak, synapses and clipping; radius perturbations rescale resistance inversely with squared radius. All 1,056 conditions remain. Nominal mean capture was 0.3293 under mean-radius and 0.3375 under series normalization; small mean changes do not negate substantial local resistance errors. Draws were averaged within cell and intervals used 5,000 cell resamples. Audits and cache hashes are in `source_data/review_morphology_uncertainty/`.

7.3 Route-generated capacity and spatial scales

Route-generated fields multiply independent amplitudes by Eq. 101. Dictionaries comprise a dense PCA oracle, selected real ancestry routes, random nonempty routes, cable-depth bins and independently shuffled ancestry columns preserving values and sparsity. Real routes rank by inhibitory fraction times descendant excitatory-area fraction. Fields and dictionaries are both transformed by square-root excitatory area before least squares (Eq. 33). Twenty Monte Carlo streams each contain 384 fields and 64 randomized dictionaries per cell and budget; streams are averaged within cell before inference.

At $K = 8$, original ancestry capture was 0.487 with 6.92% dense wiring. The same model-matched construction favored ancestry in the direct-label subset and all 47 non-overlapping cells. These positive controls measure internal compressibility of the route family. The older replication reports one minus residual norm, which differs from squared-energy capture; Source Data retain each endpoint's definition.

A deterministic unbalanced tree-Haar basis tests route scales on all 55 cells, with the 47-cell cohort primary. A region R is split at the descendant subtree closest to half its excitatory weight, giving $A \cup B = R$. For site weights w_i and region sums W_A, W_B, W_R ,

$$h_{R,i} = \begin{cases} \sqrt{w_i} \sqrt{W_B/(W_A W_R)}, & i \in A, \\ -\sqrt{w_i} \sqrt{W_A/(W_B W_R)}, & i \in B, \\ 0, & i \notin R. \end{cases} \quad (102)$$

With $h_{0,i} = \sqrt{w_i/\sum_j w_j}$ these contrasts form a complete orthonormal basis [11]. All bases were full rank, with maximum orthonormality error 4.44×10^{-16} . For $W = \text{diag}(w_i)$,

$$\tilde{\Gamma} = (I - h_0 h_0^T) W^{1/2} \Gamma, \quad p_r = \frac{\|h_r^T \tilde{\Gamma}\|^2}{\|\tilde{\Gamma}\|_F^2}. \quad (103)$$

Coarse supports exceed one quarter of total weight; intermediate supports exceed one sixteenth but not one quarter; others are fine. A scale with m_s of $n - 1$ nonconstant dimensions has isotropic reference $m_s/(n - 1)$. The anatomy null independently permutes raw route columns before weighting and centering, using 256 draws per cell.

Coarse modes contained 0.4065 of nonconstant energy but 0.1923 of dimensions in the 47-cell cohort. Enrichment over isotropic noise was 2.351 (95% interval, 1.995–2.880). Actual-minus-permuted coarse energy was 0.0345 (-0.0025 to 0.0713), and intermediate/fine permutation contrasts were also null. The original eight-cell positive permutation contrast did not reproduce in the larger cohort. Coarse enrichment therefore does not establish special fine-layout organization.

1667 7.4 Passive response fields and equal broadcast budgets

1668 Separately generated fields use focal shunts at inhibitory-bearing segments of the reciprocal cable. Each dose is
1669 0.001 times local leak-plus-synaptic conductance; a solved compensating current restores baseline somatic voltage.
1670 The response matrix H contains finite-difference changes in log absolute excitatory-conductance gradients. The
1671 inherited normalized model uses E/I scales 0.35 and reversals 1 and -0.2 . This generator is separate from Eq. 101,
1672 but remains ancestry-dependent through the passive identity in Section S8.

1673 The original dictionaries lacked a common field in selected subtrees while depth bins included it. The correction
1674 gives every family one all-site broadcast and $K - 1$ spatial columns at $K = 1, 2, 4, 8, 16$ when permitted. The initial
1675 eight cells served as development data; the protocol and inputs were frozen before new passive-field results on
1676 the existing 47-cell and Pinky cohorts. This is a frozen follow-up on previously used anatomy, not new animal
1677 sampling.

1678 Ancestry retains its outcome-independent leverage order. Random routes sample without replacement; site
1679 shuffles preserve each column's values/nonzeros and leave broadcast intact. Two hundred surrogate trees per cell
1680 preserve every node depth and parent out-degree by reassigning children among parent slots, keeping attributes
1681 with children, then recomputing ancestry and ranking. The depth dictionary uses broadcast plus the first $K - 1$
1682 inherited bins, preserving the original K -bin span. Equal column counts may have unequal numerical ranks or
1683 nonzero counts, which are reported.

1684 For $Y = W^{1/2}H$ and normalized weighted constant q , let $R = (I - qq^T)Y$. Total capture is $\|P_D Y\|_F^2 / \|Y\|_F^2$ and
1685 residual capture is $\|P_D R\|_F^2 / \|R\|_F^2$. Because every dictionary contains q , total capture equals broadcast capture plus
1686 one minus broadcast capture times residual capture. The constrained oracle adds the leading $K - 1$ left singular
1687 vectors of R .

1688 Randomized controls are averaged within cell before 20,000-draw bootstrap inference. Holm corrections
1689 cover four controls separately for each endpoint/cohort at primary $K = 8$; other budgets are descriptive. The
1690 65 operators and 149,331 dictionary evaluations remain in `source_data/anatomy_commonmode/`, including
1691 148,200 randomized realizations. Validation checks weighted identities, ceilings, depth-span equivalence, ranks,
1692 common-mode coverage and exact ancestry/shuffle nonzero matching. Small Pinky domains explain near-saturation
1693 at eight columns.

1694 7.5 Observed inhibitory contacts and descendant domains

1695 The inhibitory census uses the complete stable-nucleus intersection of v661 and v795: twenty targets, with no
1696 outcome selection. An affine raw-to-slanted transform reproduced paired published locations within $10^{-6} \mu\text{m}$; five
1697 cross-release root changes were joined by stable nucleus. Mapping within $5 \mu\text{m}$ retained 114,775 of 116,401 inputs
1698 and 2,637 of 2,789 known inhibitory contacts, from 114 of 116 inhibitory cells. No unreported proofreading status
1699 was inferred.

1700 One largest contact per published axonal clump avoids counting adjacent contacts as independent sites. Within-
1701 target/compartments candidate pools match log soma distance. For each site, a descendant-input indicator multiplied
1702 by square-root synapse size and normalized gives r_k ; $r_k^T r_l$ measures weighted domain overlap, not inhibitory activity
1703 correlation. The 402 connections with at least two clumps are nested within targets.

1704 Primary target-level spatial associations were retested by averaging within 92 presynaptic axons and by jointly
1705 matching 3D and path distance over 500 assignments. Coarse tree-distance association survived axon aggregation;
1706 shared-path and descendant overlap did not. No endpoint survived joint geometry matching. Thus the census
1707 establishes spatial co-targeting, not geometry-independent credit organization. Capacity columns use actual
1708 inhibitory-bearing segments weighted by contact size, with 200 random/shuffled controls. Their target is the full
1709 actual-route matrix, making this another compressibility diagnostic. All primary and sensitivity endpoints are
1710 indexed in Supplementary Table S12.

1711 7.6 Optimization with explicitly imposed alignment

1712 Eight-cell kernels supply eight leverage-selected routes. For their weighted orthonormal basis Q , paired unit
1713 directions parallel and orthogonal to its span define

$$\mathbf{g}(a) = \sqrt{a} \mathbf{u}_{\parallel} + \sqrt{1-a} \mathbf{u}_{\perp}, \quad a = 0, 0.2, 0.4, 0.6, 0.8, 1. \quad (104)$$

1714 Capture in the chosen span is therefore exactly a by construction. Eight random paths, depth bins and column-wise
1715 row shuffles are paired across alignment levels within each of forty streams per cell, each containing 256 fields.
1716 A positive diagonal quadratic uses curvature 0.5–1.5, permuted per stream. One-step updates match norm at 0.1
1717 times exact-gradient norm; iterative projected descent uses step 0.25 for twenty steps. Progress is normalized to
1718 full-gradient progress. The median within-cell capture/progress correlation was 0.990. This establishes usefulness
1719 conditional on imposed alignment and oracle coefficients, not observed biological alignment or autonomous
1720 coefficient learning.

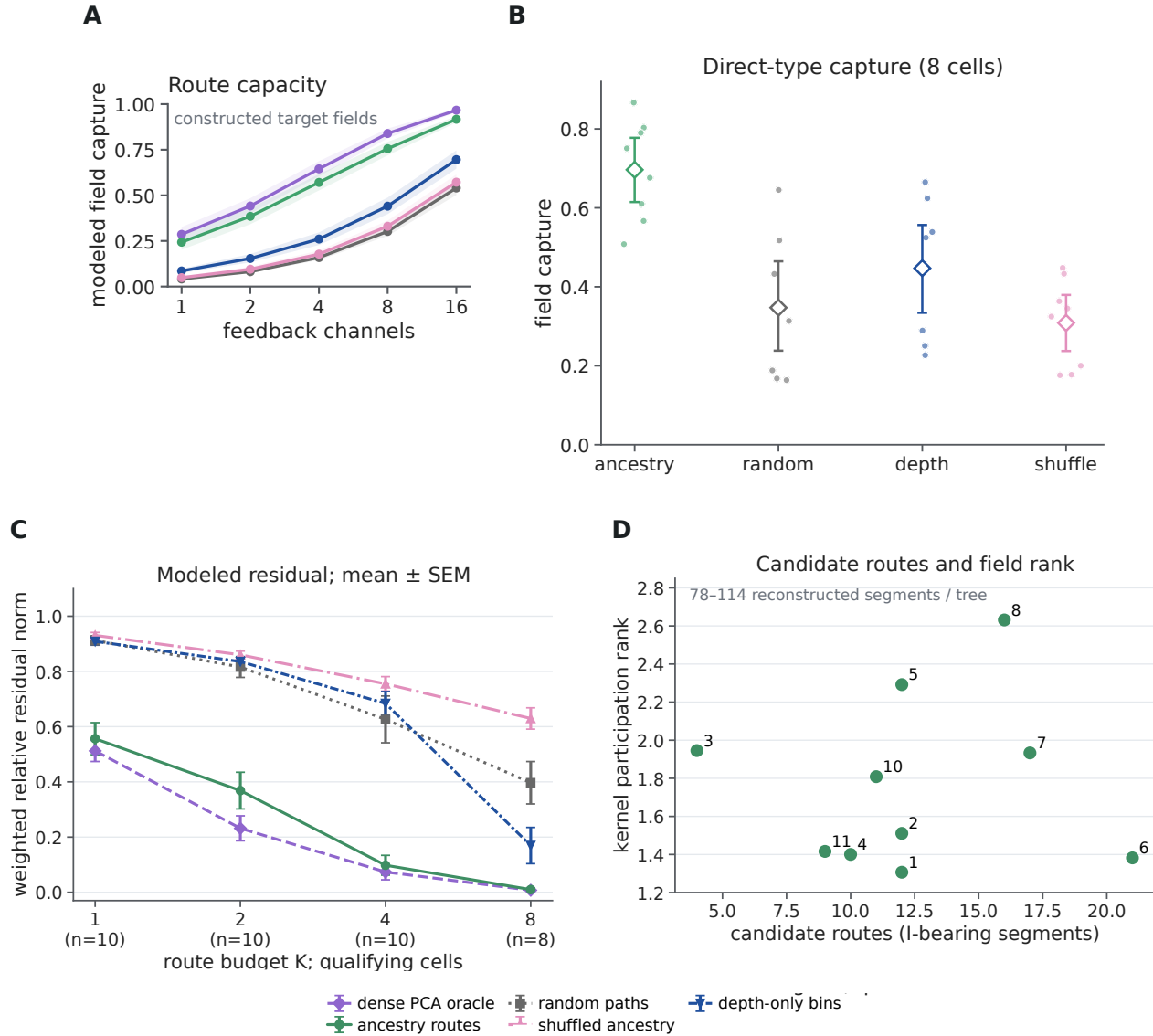


Figure S24: **Cohort, label and arbor-size controls bound modeled anatomical capacity.** **A**, Route-generated field capture versus budget in 47 disjoint same-mouse cells; morphology routes use 8.1% of dense-feedback wiring at eight channels. Twenty modeled streams are averaged within each cell; bands/bars are 95% cell-bootstrap intervals. **B**, Initial eight-cell cohort: capture restricted to directly typed presynaptic contacts. **C,D**, Second-mouse Pinky cohort: relative residual norm versus route budget, and candidate-site count versus effective field rank. Eligibility changes from ten cells at $K \leq 4$ to eight at $K = 8$; small arbors approach the rank ceiling. Panel C uses mean $\pm$ s.e.m., and its residual norm is not squared-energy capture. All fields are model generated. Main Fig. 7 equalizes a common broadcast and leads with actual-tree surrogate controls; these route-generated comparisons are structural positive controls, not measured biological teaching signals.

A

Inhibitory contacts: x-y projections

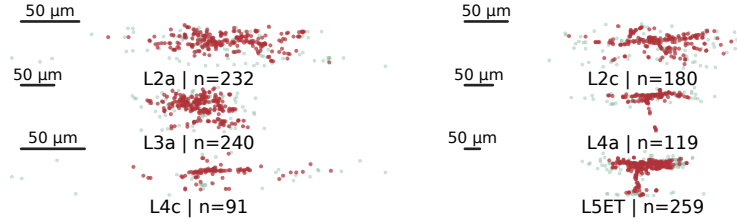
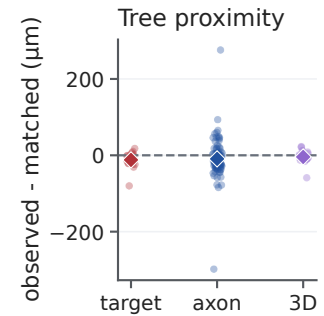
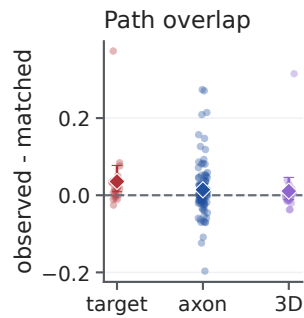
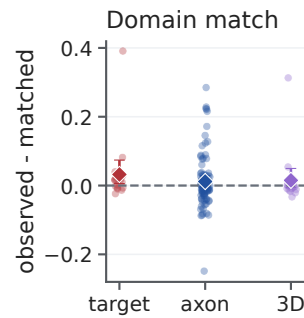
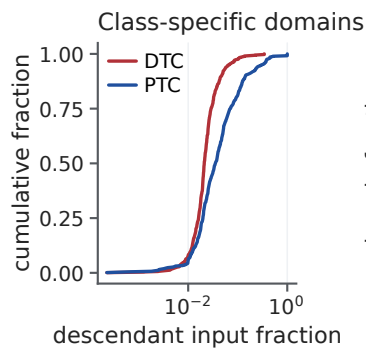
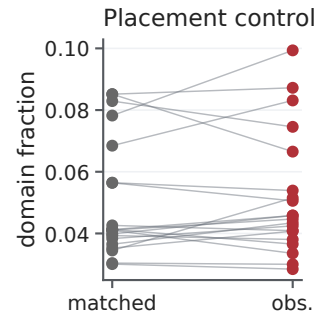
**B****C****D****E****F**

Figure S25: **Spatial matching limits apparent inhibitory targeting of ancestry domains.** **A**, Six layer-diverse reconstructions with mapped inhibitory contacts and physical scale bars. **B–D**, Observed-minus-matched tree distance, shared-path fraction and descendant-domain overlap in the twenty-target analysis, after aggregation within 92 presynaptic axons, and after joint path-distance/Euclidean matching. Symbols and intervals are means and 95% bootstrap intervals over the displayed inferential unit. Only tree distance persists after axon aggregation; no endpoint persists after joint 3D matching. **E**, Descendant-input fraction by distal-dendrite- versus perisomatic-targeting class, retaining one contact per axonal clump. **F**, Inhibitory-contact domain size versus matched locations; no reliable overall localization advantage is detected. Cells derive from one mouse. These structural comparisons do not show that contacts carried teaching signals.

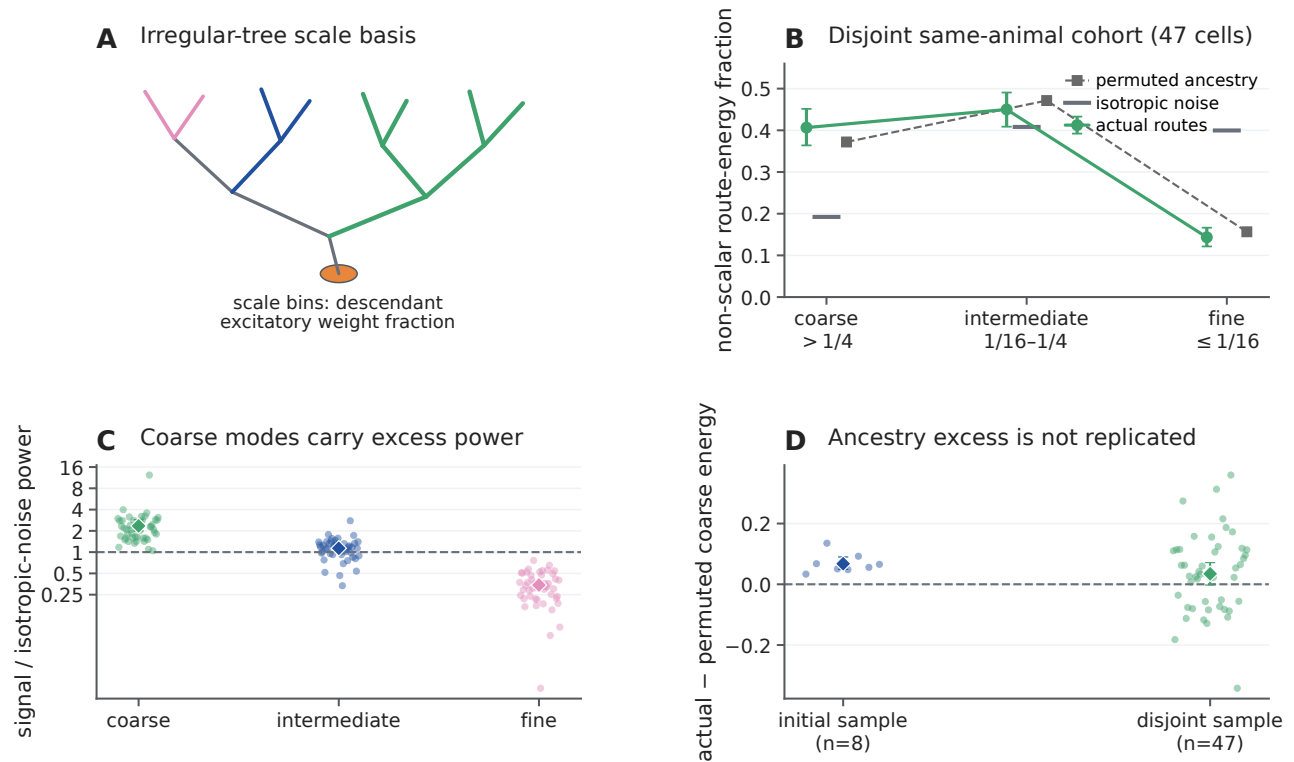


Figure S26: **Irregular reconstructed trees concentrate modeled route-field energy coarsely, but the anatomy-specific excess does not replicate.** **A**, Weighted unbalanced tree-Haar contrasts recursively partition each reconstructed arbor into coarse, intermediate and fine supports, defined by descendant excitatory-weight fractions $> 1/4$, $1/16-1/4$ and $\leq 1/16$, respectively. **B**, Non-scalar conductance-weighted ancestry-route energy fraction in the disjoint 47-cell cohort, compared with column-permuted ancestry and dimension-matched isotropic noise. Actual-route points and intervals are cell means and 95% cell-bootstrap intervals; null markers are cell means. **C**, Signal-to-isotropic-noise power by scale for all 47 cells; diamonds show means and 95% cell-bootstrap intervals. **D**, Actual-minus-permuted coarse energy in the original and disjoint cohorts. The positive original-cohort contrast is not reliable in the 47-cell cohort. Cells are the inferential units; all reconstructions derive from one mouse. Modeled route fields are not measured teaching signals.

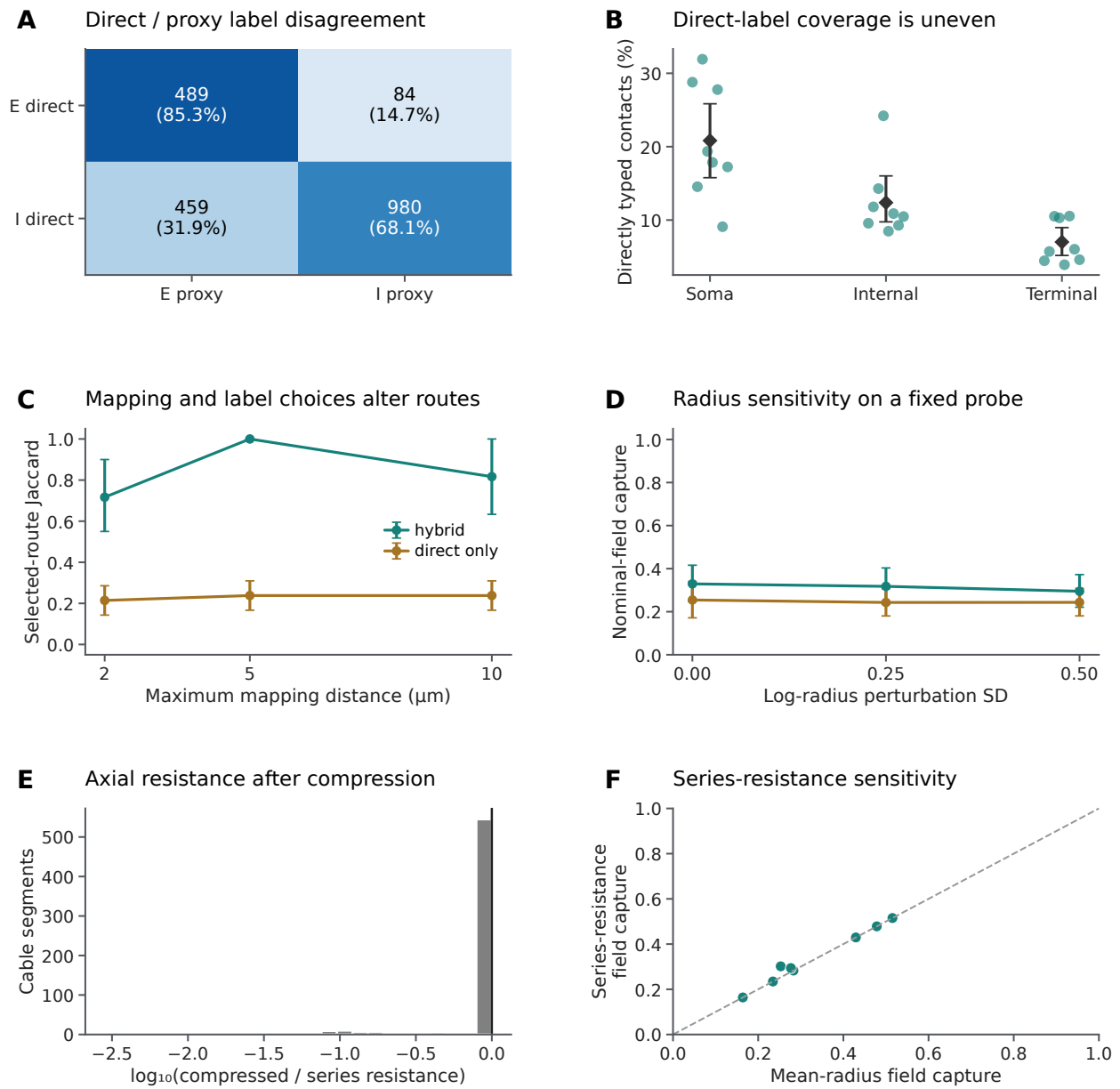


Figure S27: Label coverage and geometric preprocessing affect anatomical route dictionaries. All analyses use eight initial reconstructed cells from one mouse. **A**, Direct-presynaptic versus target-proxy excitatory/inhibitory (E/I) labels for 2,012 jointly labeled mapped contacts; counts and row percentages expose the asymmetric disagreement. Contacts are nested observations. **B**, Direct-label fraction by compartment: individual cells and cell means with 95% cell-bootstrap intervals. Label availability is not assumed random. **C**, Jaccard overlap of the four selected routes with the nominal hybrid-label, 5 μm dictionary across mapping thresholds and label choices. **D**, Capture of the fixed nominal model-generated field under heterogeneous log-radius perturbations, indexed by their standard deviation (SD); five draws per nonzero dose are averaged within cell. Curves and bars in **C,D** show means and 95% cell-bootstrap intervals (5,000 draws). **E**, Distribution of the mean-radius compressed axial resistance divided by the exact series sum over raw edges. Length preservation does not imply axial-resistance preservation. **F**, Paired cell capture under mean-radius and exact series-resistance normalizations, with leak terms held fixed. The diagonal marks equality. Across 1,056 retained sensitivity conditions, target fields, probe sites and weights remain those of the nominal model. These are preprocessing sensitivities, not calibrated geometric uncertainties or tests of endogenous task alignment.

S8 Focal shunting and spatial transport

8.1 Passive focal shunts attenuate a positive adjoint

Proposition 10 (No sign reversal in a passive grounded tree). *Let G be a symmetric positive-definite nonsingular M -matrix, with nonpositive off-diagonal entries. For a nonnegative adjoint source $\mathbf{s}$, define $\mathbf{q} = G^{-1}\mathbf{s}$. Adding a nonnegative shunt κ at compartment k gives $G' = G + \kappa \mathbf{e}_k \mathbf{e}_k^T$ and*

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa q_k}{1 + \kappa(G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (105)$$

Both G^{-1} and $(G')^{-1}$ are elementwise nonnegative, hence $0 \leq \mathbf{q}' \leq \mathbf{q}$ componentwise. A focal passive shunt attenuates but cannot reverse a positive adjoint field.

Proof. Equation 105 is the Sherman–Morrison identity. A nonsingular M -matrix has a nonnegative inverse; adding a nonnegative diagonal term preserves the M -matrix property [2]. The subtracted vector is nonnegative, and $\mathbf{q}' = (G')^{-1}\mathbf{s} \geq 0$. $\square$

This ordering does not by itself guarantee descendant selectivity relative to a matched current injection; that spatial comparison depends on the Green’s function, focal site and electrotonic regime and is evaluated numerically in the focal experiments.

Corollary 2 (A focal shunt applies gains to its ancestry partition). *Assume additionally that the nonzero off-diagonal entries of G form a connected tree rooted at the soma s , and that the adjoint source is $\ell \mathbf{e}_s$ for a fixed scalar somatic loss derivative ℓ . Write $R = G^{-1}$ and $\mathbf{h} = R \mathbf{e}_s > 0$. For a shunt at k , let $\mathcal{A}_k$ be the vertices on the path from s to k , including both endpoints. For each $a \in \mathcal{A}_k$, define the disjoint block $P_a = \{i : \text{LCA}(i, k) = a\}$, where LCA is the lowest common ancestor. Then*

$$q'_i = \eta_a q_i \quad (i \in P_a), \quad \eta_a = 1 - \frac{\kappa R_{ks} R_{ak}}{(1 + \kappa R_{kk}) R_{as}}, \quad \eta_k = \frac{1}{1 + \kappa R_{kk}}. \quad (106)$$

For finite $\kappa \geq 0$, every gain satisfies $0 < \eta_a \leq 1$. Let B_k have columns $\mathbf{1}_{P_a}$ and define the baseline-weighted partition dictionary $D_k = \text{diag}(\mathbf{h}) B_k$. With $\Gamma_k = \text{diag}(\eta_a)$,

$$\mathbf{q} = \ell D_k \mathbf{1}, \quad \mathbf{q}' = \ell D_k \Gamma_k \mathbf{1}. \quad (107)$$

Thus the adjoint effect is exactly a diagonal gain in this partition basis.

Proof. Removing a vertex a separates a tree into components. If a lies on the path from i to j , solving the source-free equations in the component containing i , with boundary voltage at a , gives $R_{ij} = R_{ia} R_{aj} / R_{aa}$. For $a = \text{LCA}(i, k)$, apply this identity to (i, k) and (i, s) ; it also holds when an endpoint equals a . Hence $R_{ik} / R_{is} = R_{ak} / R_{as}$. Substitution into Eq. 105 gives Eq. 106. Positivity of both tree inverse matrices gives positive ratios, and the preceding proposition bounds them by one. The formula also holds when $\ell = 0$, when both adjoints vanish. The disjoint blocks partition all vertices, yielding Eq. 107. $\square$

The block P_k is the whole subtree below the shunt. For an ancestor a above k , P_a contains a and its sister subtrees off the path to k . If c is the next vertex toward k , its indicator is $\mathbf{1}_{P_a} = \mathbf{1}_{\text{Desc}(a)} - \mathbf{1}_{\text{Desc}(c)}$, whereas $\mathbf{1}_{P_k} = \mathbf{1}_{\text{Desc}(k)}$. The gain field therefore lies in the span of nested ancestry indicators. The diagonal form requires the disjoint partition basis; an arbitrary selected set of overlapping routes need not have diagonal gains. Even for the ancestry chain, a change from partition to nested-subtree coordinates generally makes the gain matrix non-diagonal, while preserving that full span.

For the synaptic gradient, the local driving force remains a separate factor: $\gamma'_i = \eta_a q_i (E_E - V_i')$ for $i \in P_a$. Its within-block variation is therefore not constrained to a single gain relative to $\gamma_i = q_i (E_E - V_i)$. The corollary describes somatic-source adjoint transport on a passive tree, rather than the complete synaptic update, a distributed-error source or a nonlinear finite-dose response. It connects the morphology-defined dictionary to a conductance gain without implying a learning advantage.

1759 **Corollary 3** (Spectral size of a focal passive perturbation). *Under the preceding assumptions, let $\mathcal{R} = G^{-1}$ and*
 1760 *$\mathcal{R}' = (G + \kappa \mathbf{e}_k \mathbf{e}_k^\top)^{-1}$. The inverse-operator change has rank at most one, with rank one for $\kappa > 0$, and*

$$\|\mathcal{R}' - \mathcal{R}\|_2 = \frac{\kappa \|\mathcal{R} \mathbf{e}_k\|_2^2}{1 + \kappa \mathcal{R}_{kk}} \leq \frac{\kappa \|\mathcal{R}\|_2^2}{1 + \kappa / \lambda_{\max}(G)}. \quad (108)$$

1761 *Consequently, for any adjoint source $\mathbf{s}$, $\|(\mathcal{R}' - \mathcal{R})\mathbf{s}\|_2 \leq \|\mathcal{R}' - \mathcal{R}\|_2 \|\mathbf{s}\|_2$.*

1762 *Proof.* Sherman–Morrison gives $\mathcal{R}' - \mathcal{R} = -\kappa(\mathcal{R} \mathbf{e}_k)(\mathcal{R} \mathbf{e}_k)^\top / (1 + \kappa \mathcal{R}_{kk})$. The norm of this symmetric outer-
 1763 product matrix is its scalar coefficient times $\|\mathcal{R} \mathbf{e}_k\|_2^2$. Use $\|\mathcal{R} \mathbf{e}_k\|_2 \leq \|\mathcal{R}\|_2$ and $\mathcal{R}_{kk} \geq \lambda_{\min}(\mathcal{R}) = 1/\lambda_{\max}(G)$ for
 1764 the bound. $\square$

1765 Equation 108 provides only a global magnitude ceiling on the total adjoint change. It does not predict the
 1766 localization contrast or the observed crossover with membrane conductance. Those spatial effects depend on the
 1767 Green’s-function profile and electrotonic length constant of the cable.

1768 8.2 Focal shunting perturbations

1769 Route capacity does not show that conductance can selectively alter transport. We therefore compared a focal shunt
 1770 with a baseline-current-matched injection at the same site of each reciprocal reconstructed cable and quantified the
 1771 resulting change in the exact modeled gradient.

1772 8.3 Adjoint formulation and baseline-current-matched current injection

1773 For passive conductance matrix G and current-source vector $\mathbf{b}$, the voltage vector was $\mathbf{V} = G^{-1}\mathbf{b}$. Let $\mathbf{e}_s$ select the
 1774 somatic coordinate and let w_i denote the excitatory conductance at site i . For a loss depending on somatic voltage,
 1775 the adjoint error and conductance gradient were

$$\mathbf{q} = G^{-1} \mathbf{e}_s \frac{\partial \mathcal{L}}{\partial V_s}, \quad \frac{\partial \mathcal{L}}{\partial w_i} = q_i (E_E - V_i). \quad (109)$$

1776 Adding focal inhibitory conductance κ_k at node k , selected by the coordinate vector $\mathbf{e}_k$, gives

$$G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top, \quad \mathbf{b}' = \mathbf{b} + \kappa_k E_I \mathbf{e}_k.$$

1777 The shunt therefore changes both conductance loading and the forward source. Holding the somatic loss derivative
 1778 fixed, the Sherman–Morrison identity gives the transport change

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (110)$$

1779 The reversal E_I does not appear in Eq. 110 because the adjoint transport depends on G' , not on the forward source $\mathbf{b}'$.
 1780 It remains in the solved voltage and in every inhibitory driving force. Corollary 2 supplies the exact partition-gain
 1781 form. The full synaptic gradient additionally changes through its local driving force.

1782 Let $r_k = (G^{-1})_{kk} > 0$. Then

$$\Delta \mathbf{q} = -a_k G^{-1} \mathbf{e}_k, \quad a_k = \frac{\kappa_k q_k}{1 + \kappa_k r_k}, \quad \frac{\partial |a_k|}{\partial \kappa_k} = \frac{|q_k|}{(1 + \kappa_k r_k)^2} \geq 0. \quad (111)$$

1783 The derivative is strictly positive when $q_k \neq 0$; if $q_k = 0$, the perturbation vanishes. Its amplitude saturates at $|q_k|/r_k$.
 1784 For a nonzero perturbation, the normalized spatial profile is fixed by the Green’s-function column $G^{-1} \mathbf{e}_k$ in this
 1785 linear passive state. For a descendant set $D(k)$ and a prespecified matched comparison set $C(k)$, define

$$S_k(D, C) = \frac{\text{median}_{i \in D(k)} |(G^{-1})_{ik}|}{\text{median}_{j \in C(k)} |(G^{-1})_{jk}|}. \quad (112)$$

For $\kappa_k > 0$, $q_k \neq 0$ and a positive comparison median, the common amplitude $|a_k|$ cancels. Under these conditions, $S_k > 1$ is necessary and sufficient for a positive descendant-minus-comparison contrast in absolute adjoint change at fixed k . It is only a transport-level criterion. Relative or logarithmic changes additionally divide by the baseline q_i , and the full synaptic gradient also contains the post-perturbation driving force. This is why axial coupling, leak, background conductance and local input resistance must be varied explicitly rather than inferred from topology alone. The baseline-current-matched current injection applied $\kappa_k(E_I - V_k)$ at the same node without changing G . This equals the current carried by the added shunt at the unperturbed voltage; its local voltage effect matches the shunt only to first order in dose. An independently solved state-matching compensatory current injection restored baseline somatic voltage after both interventions. This is not a continuously adjusted voltage clamp. The somatic scalar error was therefore fixed. Shunting changed adjoint transport, whereas the current-injection control left it unchanged. Dendritic voltages were not clamped, so both interventions could also change local driving forces.

8.4 Prespecified endpoint and numerical validation

Before running the analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Focal sites were selected by a depth-stratified rule without access to gradient outcomes. A site required at least three excitatory descendants and three comparison sites. At most 16 eligible sites were sampled per cell across quartiles of topological depth. The primary model used excitatory and inhibitory synaptic scales of 0.35 and normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Synapses were classified as descendants, ancestors, sister-subtree sites or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g'_i , we defined $m_i = |\log[(|g'_i| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15} \max(1, \max_i |g_i|)$. For each site, the localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were first averaged within cell.

The somatic loss was one-half the squared difference between somatic voltage and a target one normalized unit below its baseline value. Conductance doses were 0.25, 0.5, 1, and 2 times the local leak-plus-synaptic conductance. The state-matching current was solved after each perturbation and then treated as fixed when evaluating the post-intervention conductance gradient. The topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it sampled another site in the same cell.

At unit dose, 101 sites in eight cells gave mean localization 0.104 for shunting and 0.035 for current injection. Their paired cell-level contrast was 0.069 log units (95% interval, 0.053–0.086), positive in all eight cells. True relations also exceeded the depth-stratified relation control. The contrast grew with relative dose; Source Data retain every site, dose and template.

We separated the two factors in the exact gradient algebraically. Writing $d_i = E_E - V_i$ and $d'_i = E_E - V'_i$, full shunting uses $q'_i d'_i$, adjoint-only substitution uses $q'_i d_i$, and driving-force-only substitution uses $q_i d'_i$. At unit dose, their mean localization indices were 0.104, 0.130 and 0.026, respectively, compared with 0.035 for the current-injection control. Adjoint-only minus driving-force-only localization was 0.104 (95% interval 0.080–0.125), positive in all eight cells.

Full shunting and the driving-force-only state provide a direct comparison of adjoint transport: they share the same post-shunt voltage field and differ only in whether the adjoint field is updated. Their localization difference was 0.079 and was positive in all eight cells. This is the exact adjoint-replacement contrast plotted in main Fig. 8C.

The gradient factors and localization statistic are nonlinear, so these four outcomes do not form an additive decomposition. To allocate the full-shunt-minus-current-injection contrast, we used two-factor Shapley values. For v_{00} equal to current-injection localization, v_{10} adjoint-only, v_{01} driving-force-only and v_{11} full-shunt localization,

$$\phi_q = \frac{1}{2}[(v_{10} - v_{00}) + (v_{11} - v_{01})], \quad (113)$$

$$\phi_d = \frac{1}{2}[(v_{01} - v_{00}) + (v_{11} - v_{10})]. \quad (114)$$

The allocations sum exactly to the full-shunt-minus-current contrast. Mean adjoint and driving-force allocations were 0.087 and -0.017 , respectively: driving-force change partly cancelled the adjoint contribution. This is an

algebraic allocation of a nonlinear statistic; the current-matched control is not a literal no-factor corner of a physical factorial. Repeating the allocation with the unperturbed field as its reference is retained in Source Data. Factor substitutions are not independent biological interventions.

All conductance matrices in the primary analysis were positive definite. The somatic state-matching residual was numerically zero. Analytic adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6} . The Source Data indexed in Table S12 retain prespecified conductance-scale and reversal-potential sensitivity runs. The direction of the shunting-minus-current-injection effect was positive in every condition, with seven or eight positive cells. These results establish the mechanism in a passive model on measured trees; they do not establish endogenous inhibitory teaching signals in vivo.

With proxy-classified contacts removed, the primary shunt-minus-current-injection contrast was 0.101 (0.075–0.129) across 55 sites and was positive in all eight cells. In the independent, non-overlapping v661 sample, 45 cells supplied 235 eligible sites. The same contrast was 0.138 (0.116–0.162), positive in 45/45 cells. Forty cells with at least two eligible sites supplied the reassigned-relation control; the true relation exceeded this control by 0.118 (0.094–0.144), with 39 positive cells and one numerical tie (Supplementary Fig. S29A,B). Cells, rather than sites or relation draws, were the units of inference.

To test sensitivity to electrical calibration, we also computed axial and leak conductances in physical units from the same compressed geometry. For segment radius r , length ℓ , axial resistivity R_a and specific membrane resistance R_m , a cylindrical non-root segment had axial conductance $\pi r^2 / (R_a \ell)$ and membrane leak $2\pi r \ell / R_m$; the root was assigned spherical area $4\pi r^2$. All conductances were converted to nanosiemens (nS). Because MICrONS synapse size is not an electrical conductance, synaptic sizes retained the prespecified relative scaling with the median physical leak as reference. We varied R_a and R_m while retaining the site-selection rule, relative dose, gradient definition and current-injection control. For each condition and intervention, we re-solved the compensating somatic current to restore that condition's baseline somatic voltage. The Source Data indexed in Table S12 retain this electrotonic dependence. The large contrast observed under the normalized proxy was recovered only in low- R_m , high-conductance settings; at $R_a = 150 \Omega \text{cm}$ and $R_m = 15,000 \Omega \text{cm}^2$ it was negligible or small. These are sensitivity settings, not fits to the recorded cells, and the model remains passive and steady-state.

A broader parameter matrix, specified before numerical validation, tested membrane resistance, background conductance and dose definition jointly. It used $R_a = 150 \Omega \text{cm}$ and crossed $R_m \in \{300, 1000, 15,000\} \Omega \text{cm}^2$, distributed leak-like background conductance of 0, 1 or 4 times baseline leak, and three dose definitions. Relative-local doses were 0.25, 1 and 2 times membrane-plus-synaptic conductance; fixed doses were 0.05, 0.5 and 5 nS; input-normalized doses were 0.25, 1 and 4 times $[(G^{-1})_{kk}]^{-1}$. The same 101 sites in eight cells supplied 16,362 site-condition-intervention rows. Every conductance matrix was positive definite (minimum eigenvalue 0.107 nS) and the maximum somatic-state residual was 3.33×10^{-16} .

At fixed absolute dose, the conductance-specific contrast did not follow the raw focal-shunt localization alone. At 0.5 nS, shunt-minus-current-injection localization was 0.0104 (0.0080–0.0126), 0.0073 (0.0044–0.0104), and -0.0039 (-0.0075 to -0.0004) at $R_m = 300, 1,000$ and $15,000$, respectively. At 5 nS the corresponding values were 0.0974, 0.0741 and 0.0061; the last interval included zero. Distributed background conductance restored the input-normalized effect in the high-resistance condition: at unit input-conductance dose, the contrast changed from -0.0004 with no background to 0.1210 and 0.2587 at background multipliers one and four. Across all 8,181 focal-shunt rows, every descendant gradient magnitude was attenuated, none was enhanced and no sign flip occurred. The passive mechanism therefore selectively attenuated gradients; sign reversal would require dynamics or active conductances absent from this phase (Supplementary Fig. S28).

8.5 Weak-channel steady-state linearization check

We checked the local-Jacobian calculation after adding weak voltage-dependent currents to the high-conductance cable. The physical reconstructed-tree system used $R_a = 150 \Omega \text{cm}$, $R_m = 1,000 \Omega \text{cm}^2$ and distributed background conductance equal to baseline leak. Steady-state sodium (Na), potassium (K), calcium (Ca), hyperpolarization-activated cyclic nucleotide-gated (HCN) and N-methyl-D-aspartate receptor (NMDA) current families had fixed reversal potentials, half-activation voltages, slopes and gate powers specified in the predefined parameter file. Global

and segment-level maximal conductances were drawn independently from prespecified log-normal distributions. Median maximal Na, K, Ca and HCN conductances were 0.06, 0.10, 0.025 and 0.05 times baseline leak, respectively; NMDA conductance was 0.15 times the excitatory conductance. Gate activation further scales these currents. These prespecified weak-channel draws are not cell-specific density estimates or a test of strongly regenerative operation.

For each draw, a residual-reducing Newton solve found the nonlinear steady state. The predefined acceptance criteria required all voltages to lie in the specified range, maximum absolute state residual below 10^{-10} , and minimum eigenvalue of the symmetrized local Jacobian above 10^{-8} . The implicit-adjoint calculation then used the exact inverse of this baseline Jacobian. For channel family a , let $\bar{g}_a$ be maximal conductance, $h_a(V)$ the steady-state gating factor and E_a the reversal potential. The current $I_a(V) = \bar{g}_a h_a(V)(V - E_a)$ contributed the diagonal term $\bar{g}_a[h_a(V) + h'_a(V)(V - E_a)]$; thus the baseline Jacobian includes voltage-dependent gate derivatives. Focal shunting and a baseline-current-matched current injection were evaluated at 0.25, 1 and 4 times local input conductance within this fixed local linearization. Sherman–Morrison updated the Jacobian inverse for the added focal conductance, and a compensating somatic current restored baseline somatic voltage in that linearized system. The active gates and their derivatives were not re-evaluated at a new nonlinear equilibrium after each finite dose. Up to 256 attempts were permitted to obtain 64 accepted draws per cell. All first 64 attempts were accepted in every cell, giving 512 equilibria, 101 focal sites and 38,784 site–draw–condition rows. All values were finite, the largest equilibrium residual was 9.51×10^{-11} and the smallest Jacobian eigenvalue was 1.084.

Channel draws and focal sites were averaged within each of the eight cells before inference. At doses 0.25, 1 and 4, mean shunt localization was 0.159, 0.529 and 1.339, respectively (Table S12). At unit dose, shunting exceeded the current-injection control by 0.382 (95% cell-bootstrap interval 0.356–0.409), positive in 8/8 cells (two-sided Wilcoxon $P = 0.0078$). Every modeled descendant gradient was attenuated, none was enhanced or changed sign, and descendant gradient energy fell to 0.330 of baseline. The dose response closely followed the passive response at the same electrical calibration. This agreement checks the weak-channel linearization; it does not establish robustness to strongly active dendrites, channel kinetics or spiking.

8.6 Driving-force cancellation and the ancestry dependence of passive fields

We reanalyzed the eight original cable operators used for the common-broadcast comparison. For each focal site, the perturbation was the original dose 0.001 with somatic voltage restored. The finite log-gradient difference divided by dose separates into an adjoint term and a driving-force term. Energy uses the original excitatory-area site weights and sums over focal perturbations. The full focal ancestry partition is defined separately for each shunt and is distinct from the seven selected routes in the budgeted anatomical dictionary.

Driving-force energy was 0.990–3.496 times full-field energy across cells; these are ratios of squared magnitudes, not amplitudes. Weighted centered correlations with the adjoint term were -0.979 to -0.913 . The negative cross term explains why large components yield a smaller full response. The full response retained 0.9711–0.9993 of its energy in the focal partition. Reconstruction agrees with the original operators to 10^{-8} ; numerical decomposition closes to rounding error. Cellwise energies, correlations and partition captures are supplied in Source Data.

The passive corollary depends on a tree Green’s function, a somatic adjoint source and a diagonal conductance change. Choosing other passive perturbations under the same assumptions does not make the anatomical field ancestry-neutral. The budgeted dictionary comparison measures compression within this response family. In the disjoint 47-cell cohort, actual ancestry exceeds the mean degree–depth surrogate in 38 cells, but in eleven cells more than half of 200 surrogates match or exceed the actual tree. This descriptive count uses tolerance 10^{-12} and retains the full within-cell distribution. A different error source, active dynamics or measured fields would be required to test beyond the present structural dependence.

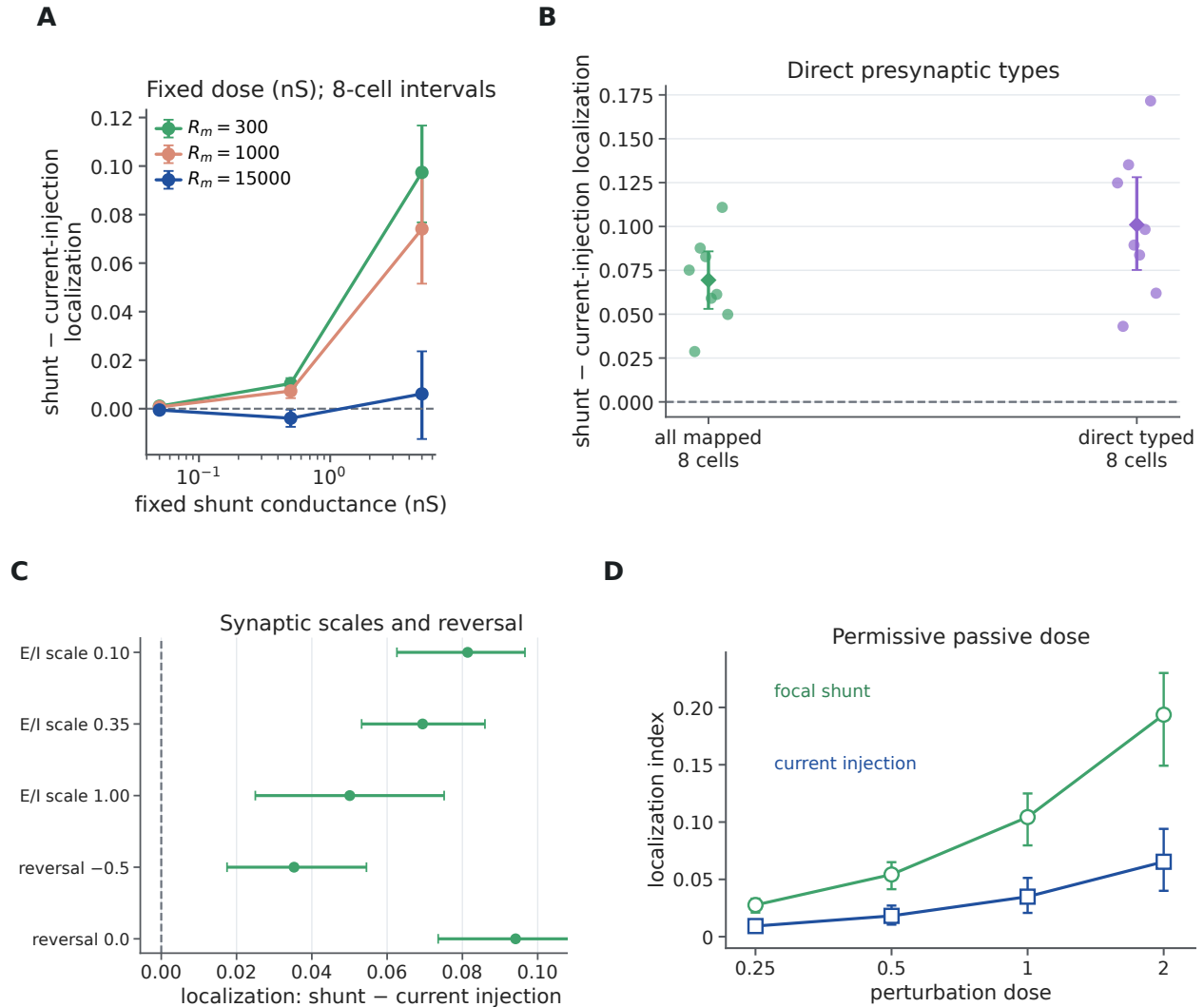


Figure S28: **Dose, cable parameters and input mapping determine focal-shunt selectivity.** **A**, Shunt-minus-current localization across absolute doses at three membrane resistances. **B**, All mapped versus directly typed presynaptic contacts. **C**, Excitatory/inhibitory scale and inhibitory-reversal sensitivity. **D**, Normalized-dose comparison of focal shunting with baseline-current-matched injection. Sites are averaged within the initial eight reconstructed cells; means and 95% cell-bootstrap intervals are shown in all panels. Positive localization is a larger descendant change, not a sign of enhancement; passive shunts attenuate descendant adjoints. Absolute dose, input-conductance dose and local nonaxial normalized dose are distinct. Electrical parameters are sensitivity settings, not cell-specific fits.

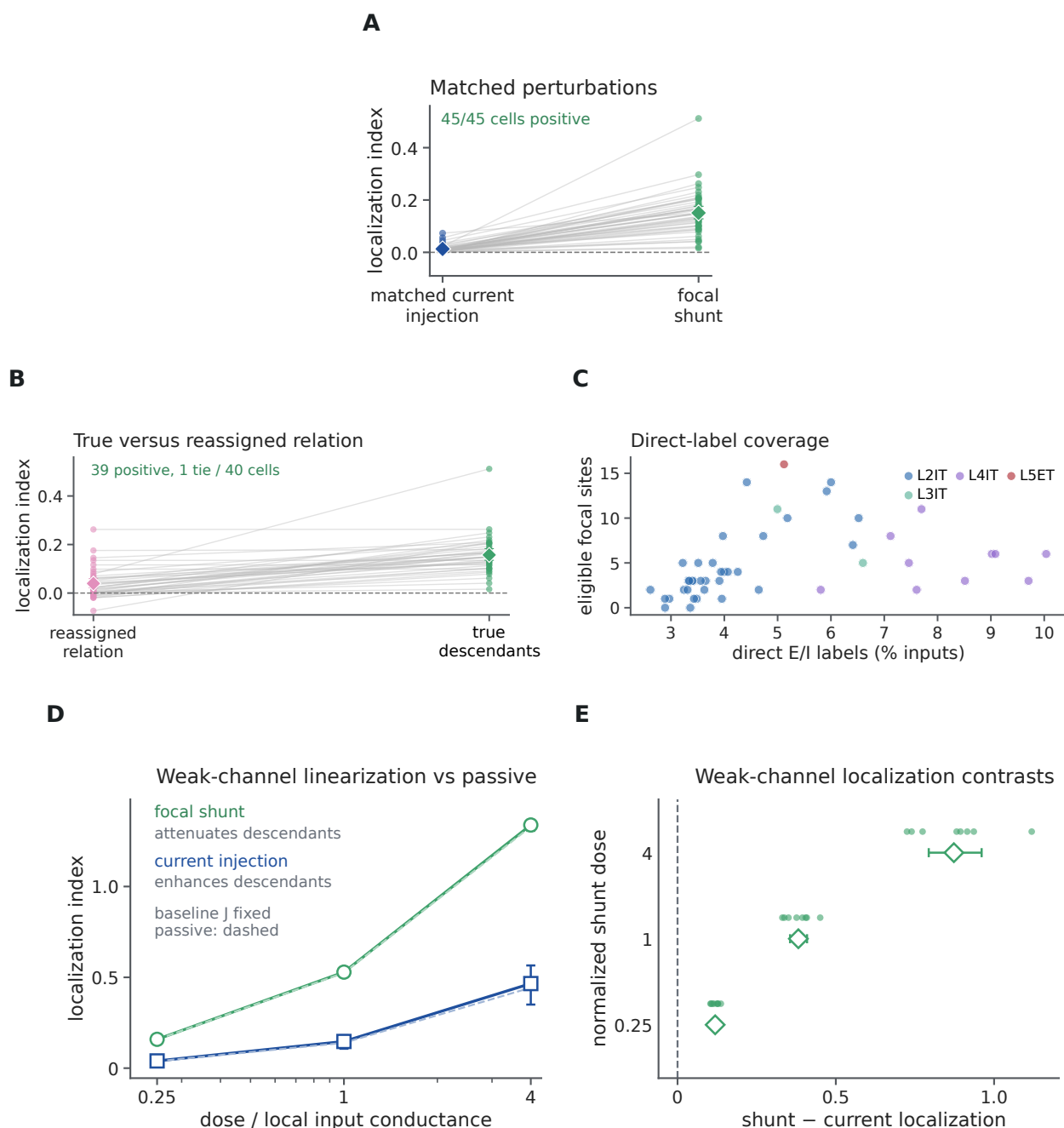


Figure S29: **Focal-shunt controls replicate structurally and under weak-channel linearization.** **A**, Focal shunt versus baseline-current-matched injection in 45 disjoint same-mouse cells (235 sites). **B**, True versus reassigned descendant relations in forty cells (230 sites). **C**, Direct E/I label coverage and focal-site eligibility. **D,E**, Separate initial eight-cell weak-channel ensemble: localization versus dose and paired shunt-minus-current contrasts. A fixed steady-state Jacobian is compared with the passive model; shunting attenuates descendants whereas matched current injection enhances them. Sixty-four channel draws and sites are averaged within each cell. All intervals bootstrap cells, not sites or channel draws. Weak conductances test local linearization near the passive response; they do not establish robustness to strongly regenerative dynamics, channel kinetics or a newly solved nonlinear equilibrium at each intervention dose.

1917 **S9 Measured responses, model diagnostics and detection sensitivity**

1918 **9.1 Functional topology and task-derived credit**

1919 Anatomical routes can be available without aligning with the credit required by a measured task. We therefore used
1920 recorded visual responses to define a prediction objective on reconstructed target cells. Within each model, we
1921 changed the backward field while holding the forward morphology fixed.

1922 **9.2 Direct anatomy–function join**

1923 Functionally imaged excitatory partners were linked to anatomical contacts through MICrONS/CAVE and DANDI
1924 mappings. The analysis contained 69 connected presynaptic partners across seven postsynaptic target cells. Au-
1925 tomatic matches came from `coregistration_auto_phase3_fwd_apl_vess_combined_v2` in materialization
1926 1822. We retained residual at most 10 and score at least zero, then deduplicated each presynaptic root by ascending
1927 residual and descending score. All mapped contacts from a retained presynaptic root contributed to its partner-level
1928 contact summary. For the one-site-per-partner model, contacts were first grouped by postsynaptic segment. The
1929 partner was assigned to the segment with greatest summed contact size, breaking ties by synapse count, and its
1930 major branch was the first segment below the soma on that segment’s ancestry chain. Thus a partner with contacts
1931 on several segments supplied one activity feature and one modeled learning site; its total contact size and synapse
1932 count still included all retained contacts.

1933 The target universe was the initial eight-cell morphology convenience sample. The analysis plan, specified
1934 before outcome inspection, included every target in that sample with a DANDI scan and directly connected,
1935 functionally imaged excitatory partners; seven targets qualified. Predefined inclusion criteria required exactly one
1936 postsynaptic target in an extract, agreement between partner and contact counts, at least five mapped partners for the
1937 topology screen, at least two sites after any requested task filter, and at least one nonempty inhibitory nested-subtree
1938 route. The primary task analysis applied neither a manual-match restriction nor a repeat-reliability threshold. The
1939 exact target/session/scan pairs are retained in the source records indexed in Table S12. The original analysis plan
1940 did not record a deterministic rule for choosing among multiple eligible scans of the same target. We therefore treat
1941 these scan choices as part of the initial selected-scan sample, not as the output of a coverage-ranking procedure.
1942 The omitted structural target, root 864691136043380566 (nucleus 292648), had one source-data row, for session
1943 8/scan 4, but `has_dandi_asset=False` and no DANDI asset identifier or path. No additional exclusion reason is
1944 inferred.

1945 A deterministic sensitivity retained every automatic-conservative target–session–scan cohort with at least five
1946 imaged connected presynaptic roots: thirteen scans across the same seven targets. Statistics were computed within
1947 scan, averaged within target and inferred over targets. The mean partial shared-path correlation was -0.0485 (95%
1948 target-bootstrap interval, -0.224 to 0.118), positive in three targets. The selected-scan and scan-complete analyses
1949 therefore both provide no evidence of preferential ancestry alignment. Full topology endpoints are indexed in
1950 Supplementary Table S12.

1951 Each target had 464 trials and 280 unique condition hashes, of which 136 were repeated. Trial families
1952 comprised 384 Clip, 40 Monet2 and 40 Trippy presentations. For each trial and region of interest, the response
1953 was the mean of the supplied `RoiResponseSeries` fluorescence trace over samples whose timestamps fell from
1954 trial start through trial stop, including both endpoints. Start and stop offsets were zero. These are the supplied
1955 fluorescence traces; our analysis did not deconvolve or apply additional raw-fluorescence preprocessing to them.
1956 No trials were dropped; every response used here was finite. The records indexed in Table S12 give target-level
1957 asset identifiers and sample counts.

1958 Functional similarity between two partners was the Pearson correlation of their vectors of condition-mean
1959 fluorescence over the 136 condition identities with at least two trials. Repeat reliability was computed separately
1960 for each region of interest: successive occurrences of every repeated condition were assigned alternately to two
1961 groups, responses were averaged within each group, and the two vectors of condition means were compared by
1962 Spearman correlation. No reliability correction was applied, and the primary cohort retained all partners regardless

of this diagnostic; thresholds of 0.1 and 0.2 were used only in sensitivity analyses.

The controlled shared-ancestry statistic was computed separately within each target. Functional similarity and shared-path fraction were rank transformed. The two controls were $\log(1 + d_{\text{Euclidean}})$ and $\log(1 + |d_{\text{soma},i} - d_{\text{soma},j}|)$, where the second term is the absolute difference in soma path length; each control was also rank transformed. We regressed each ranked variable on an intercept and the two ranked controls by ordinary least squares, then took the Pearson correlation between the two residual vectors. The mean target-level partial-rank effect was -0.069 with a target-bootstrap 95% interval of -0.255 to 0.108 ; three of seven targets had positive effects. Pairwise partner comparisons were not treated as independent replicates.

Several features limit the interpretation of this cohort. MICrONS calcium imaging measured excitatory neurons, the observed partners cover only a small fraction of each target’s complete inputs, and response reliability is modest. The volume comes from one animal, and the recordings were not made during learning.

9.3 Task-derived branch credit

A branch model predicted the standardized postsynaptic response from the standardized responses of its connected presynaptic partners. Let x_i be the response of partner/site i , w_i its input weight, and $b(i)$ the dominant major branch receiving its contacts. Branch b had summed input d_b , activity h_b and readout weight v_b ; c was an output bias. The fitted prediction $\hat{y}$ was

$$d_b = \sum_{i:b(i)=b} x_i w_i, \quad h_b = \tanh(d_b), \quad \hat{y} = \sum_b v_b h_b + c. \quad (115)$$

For N training trials with target responses y_t , the objective was

$$\frac{1}{2N} \sum_{t=1}^N (\hat{y}_t - y_t)^2 + \frac{\lambda}{2} (\|\mathbf{w}\|_2^2 + \|\mathbf{v}\|_2^2), \quad \lambda = 10^{-3}, \quad (116)$$

with no penalty on c . The exact voltage-space error at site i was $(\hat{y} - y) v_{b(i)} [1 - \tanh^2(d_{b(i)})]$.

Whole condition-hash stimulus identities, not trials of the same identity, were split: 20% of the 280 identities and all of their trials were held out. Input columns and the target were standardized separately using training-set means and sample standard deviations (ddof = 1); invalid or zero scales were replaced by one and remaining nonfinite standardized values by zero. A stable SeedSequence combined base seed 20260721, target root, stream and replicate. There were ten fixed split and initialization replicates per target.

With n_{site} input sites and n_{branch} occupied branches, input weights were initialized as $\mathcal{N}(0, 0.12^2/n_{\text{site}})$, branch readout weights as $\mathcal{N}(0, 0.25^2/n_{\text{branch}})$, and $c = 0$. Each method used 1,200 full-batch Adam steps with learning rate 0.015, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The exact model was trained first. Its exact site-error fields on training trials defined the dictionaries and its held-out exact field defined reported capture. All projected-learning methods then restarted from the identical initial state. Only the input-weight error was projected; $\mathbf{v}$ and c always received their exact gradients.

The primary budget requested four channels, reduced only when a target had fewer sites or nonempty candidate routes. Ancestry candidates were columns of ancestry multiplied by local inhibitory fraction g_i/g_{tot} ; routes were ranked by this fraction times their descendant-site fraction. Random paths sampled nonempty candidates uniformly without replacement. Each ancestry-shuffle column was permuted independently over sites. Depth controls were path-length quantile bins, scalar feedback was an all-ones column, and the dense oracle used the leading right singular vectors of the exact training error. Projectors used relative singular-value tolerance 10^{-10} . No held-out response was used to fit a dictionary.

Held-out normalized MSE divided a method’s held-out MSE by the MSE of predicting the training mean response. Replicates were averaged within target before inference; the seven target cells were the units. Intervals used 20,000 target bootstrap draws. Paired method comparisons used two-sided Wilcoxon tests, and the capture–utility analysis used method-label permutations stratified within target.

The four-channel primary outcomes are retained in the Source Data indexed in Table S12. The dense four-dimensional oracle nearly matched exact learning, establishing that feedback dimension alone was not limiting in this surrogate. Ancestry routes did not reliably outperform ancestry shuffling in capture or learning. The within-target association between capture and normalized error was $\rho = -0.150$ with a target-stratified method-label permutation $P = 0.419$. The target-bootstrap 95% interval for ρ was -0.449 to 0.144 . The coefficients are least-squares oracles and therefore upper bounds on the capacity of each feedback family, not biologically implemented encoders.

Sensitivity analyses at one, two and eight channels, with direct manual matches, with repeat-reliability thresholds, and with a linear branch model did not establish a reliable morphology-specific learning benefit. Some subsets improved morphology-specific capture without improving learning. The separation between capture and learning motivates the distinction between anatomical route capacity and task alignment.

9.4 Complete reconstructed-tree learning across all eligible scans

To retain the complete spatial state, we replaced the major-branch reduction with a segment-resolved model. Each complete compressed reconstruction was instantiated as a reciprocal passive conductance system. Each imaged partner supplied a trainable non-negative excitatory conductance at its dominant mapped segment. For trial-specific soma prediction error, the exact input gradient factored into the local term $x_i(E_E - V_i)$ and the exact soma-to-segment adjoint from the inverse conductance operator. Readout and bias parameters received exact gradients under every method.

Restricted rules received scalar output error, local eligibility and one fixed site profile lying in a four-route span. That profile was calibrated before training from a conductance-weighted topology dictionary, an equal-rank random selection of nonempty anatomical routes, or a column-value-preserving site shuffle. No target response entered this calibration. Each trial scaled the same profile by the scalar error; the rule did not infer four independent trial-dependent coefficients. The comparison therefore tests this frozen-profile rule within each dictionary, rather than all estimators its four-route span could support. Ten seed-fixed splits per scan held out complete stimulus identities. Runs were averaged within scan and then within target. The resulting design contained four methods, 13 scans and ten splits, for 520 fits. The exclusion criteria were missing routes, rank mismatch, nonfinite values, incomplete scan coverage or failed finite-difference validation. None of the runs met an exclusion criterion. One exact conductance coordinate per run agreed with finite differences to maximum relative error 3.05×10^{-7} .

Let δ_i^{out} be the scalar output error on trial i , $\mathbf{v}_0$ the baseline soma-to-site transfer profile and P_D the projector into dictionary D . The delivered nonsomatic field is

$$\hat{\delta}_{i,D}^V = \delta_i^{\text{out}} P_D \mathbf{v}_0. \quad (117)$$

This scalar-times-fixed-profile rule does not estimate four trial-dependent coefficients. Its prescribed updates U_D , with rows corresponding to trials and columns to trainable input coordinates, are compared with exact updates U_{exact} by

$$\mathcal{M}_D = 1 - \frac{\|U_{\text{exact}} - U_D\|_F^2}{\|U_{\text{exact}}\|_F^2}. \quad (118)$$

Because these coefficients are prescribed, $\mathcal{M}_D$ is normalized update match rather than an optimal projection-capacity bound. Table S6 retains all primary learning and geometry outcomes. Topology did not reliably improve learning over random or shuffled routes.

A retrospective geometry check separated this coefficient restriction from the capacity of the fixed route spans. Full checkpoint states were not archived, so we reexecuted the 130 exact-learning fits using their resolved settings, stimulus splits and recorded random-route identities. At each resulting checkpoint we compared the frozen profile, a trialwise exact-field projection, and an eligibility-weighted update oracle. For fixed dictionary Φ and trial eligibility vector ℓ_i , define $E_i = \text{diag}(\ell_i)$. The update oracle fitted

$$\mathbf{c}_i^* = \arg \min_{\mathbf{c}} \|\mathbf{u}_i - E_i \Phi \mathbf{c}\|_2^2, \quad \hat{\mathbf{u}}_i = E_i \Phi \mathbf{c}_i^*.$$

2040 This is the best attainable per-trial update inside that dictionary’s eligibility-weighted span. Its normalized update
2041 match equals projected energy capture. Split-level values were averaged within scan and then within target, with
2042 20,000 bootstrap draws over the seven targets.

2043 The update oracle increased ancestry match from 0.2322 to 0.2436, compared with oracle matches 0.3847
2044 for random routes and 0.4677 for shuffling. The unrestricted baseline transfer profile reached 0.9756 (95% target
2045 interval, 0.9641–0.9858). Thus most exact-update geometry was already carried by one nearly fixed profile;
2046 restricting it to the selected ancestry supports removed useful coverage. Selected routes covered about 48% of
2047 observed input coordinates, averaging 1.31 inputs per route, with six of thirteen scan dictionaries containing only
2048 single-input selected routes. Multiple observed inputs may map to the same physical segment. These geometric
2049 restrictions limit what the learning comparison can reveal about anatomical alignment.

2050 Three of 130 replayed exact fits differed from archived NMSE by more than 10^{-5} (maximum 5.74×10^{-4});
2051 maximum frozen-profile match drift was 2.57×10^{-4} . All fits are retained as reexecuted states, not asserted to
2052 reproduce every historical checkpoint exactly. `source_data/fulltree_within_span_oracle/` contains every
2053 geometry estimate and replay discrepancy. These common-state projections were not trained and do not establish
2054 eventual oracle-learning accuracy.

2055 9.5 Nested linear baselines and observed-input sensitivity

2056 To establish the predictive difficulty of the measured-response task, we fitted an intercept-only training-mean
2057 predictor, ordinary least squares and ridge regression on the same 130 outer stimulus-identity splits as the complete-
2058 tree analysis. All 13 scans and seven targets were retained. Three inner folds partitioned training stimulus identities;
2059 no outer test identity entered preprocessing, reliability filtering or penalty selection. The primary linear models
2060 used the training-quantile input transform from the complete-tree analysis (10th and 90th percentiles, clipped at
2061 three), followed by column centering and standardization using training data. These quantities were refitted inside
2062 each inner fold. Raw standardized inputs supplied a secondary ridge reference.

2063 Ridge regression minimized mean squared training error plus $\lambda \|\mathbf{w}\|^2$, with an unpenalized intercept. The
2064 penalty grid was $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$; the minimum mean inner-fold validation error selected
2065 the penalty, with ties resolved toward the earlier, smaller penalty. An eigendecomposition solved each fit. The
2066 selected primary penalties were 0.1 in 90 splits, 1 in 36 and zero in four. Held-out normalized MSE used the same
2067 training-mean denominator as the nonlinear models. Archived nonlinear outcomes were read unchanged and paired
2068 to the new linear fits; they were not refitted under the sensitivity masks.

2069 Observed-input sensitivity used three explicit restrictions. First, the manual-only mask retained partners with
2070 archived manual-match provenance. Second, repeat reliability was the Spearman correlation between condition
2071 means formed from alternating repeats; it was recomputed using only inner training identities during tuning and
2072 only outer training identities for the final fit. Partners passing thresholds 0, 0.1 or 0.2 were retained. Third, five
2073 independent partner orderings per split defined nested masks retaining 25%, 50% or 75% of observed inputs, with
2074 at least one input. Empty manual or reliability masks used the intercept-only model and were retained. A separate
2075 measurement-noise series added Gaussian predictor noise with SD 0.25, 0.5 or 1 in units of the training-standardized
2076 input, in both training and test examples. The same underlying noise draws were paired across doses. Noise was
2077 applied after the quantile transform and training standardization; the intercept was refitted after perturbation.

2078 The secondary baseline protocol was fixed before its outcomes but was not publicly preregistered. Its 3,380
2079 evaluations are in `source_data/review_response_baselines/`. Masks were averaged within split, ten splits
2080 within scan and scans within target; intervals use 20,000 resamples of seven targets. Input hashes and all
2081 reconstructed split counts matched retained records. A held-out-data substitution audit confirmed that tuning, masks
2082 and training reliability were unchanged. Historical split arrays were reconstructed with the unchanged helper and
2083 seed rule. These are descriptive sensitivities within one mouse.

2084 Mean normalized MSE was 1 for the training mean, 0.807 for least squares and 0.803 for nested ridge
2085 (Supplementary Fig. S32). Exact nonlinear learning reached 0.788, a paired nonlinear-minus-ridge difference of
2086 -0.0152 (95% target-bootstrap interval -0.0271 to -0.0042 ; six of seven targets lower). The nonlinear model
2087 therefore provided a small predictive improvement over this linear baseline on the measured task. The corresponding

subtree-minus-ridge difference was 0.0293 (−0.0130 to 0.0736), which does not establish a benefit for the frozen subtree rule. Manual-only inputs raised ridge MSE to 0.912, with 3.18 retained partners on average compared with 8.98 for all inputs. Retaining only 25% raised MSE to 0.917; adding unit-SD predictor noise raised it to 0.869. The strictest reliability mask retained 5.63 partners and reached MSE 0.837. These manipulations measure sensitivity to degradation of the observed inputs. They neither recover unobserved partners nor estimate the power of a complete-population experiment, and do not establish that missing inputs explain the absence of preferential ancestry alignment.

9.6 Sensitivity of the measured ancestry-response test

We estimated detection sensitivity conditional on the observed sampling, using a Gaussian response model frozen before simulation. The analysis retained 125 partner observations of 102 presynaptic roots across thirteen scans and seven targets, including twenty target–partner combinations repeated across scans and two roots shared between targets. Each scan contained 136 repeated stimuli: 130 presented twice and six ten times. Original condition hashes identified 316 distinct repeated stimuli across scans. Pair tables, contact geometry, recording identities and the required observed responses are included in Source Data.

For each target arbor, an ancestry covariance was defined by the shared soma-to-common-ancestor path length divided by the geometric mean of the two soma-to-contact lengths. This is a positive-semidefinite Brownian path kernel; zero-depth contacts contributed independent diagonal components. The seven kernels were embedded in common presynaptic-root coordinates, summed and diagonal-normalized, so each shared root had one consistent latent response. Reliable tuning mixed independent Gaussian unit-specific and ancestry-correlated components with variance fractions $1 - \lambda$ and λ , respectively. We tested $\lambda = 0, 0.05, \dots, 1$, retaining the actual partner sets and stimulus identities.

Measurement noise was calibrated to each recording’s odd/even split-half Spearman reliability. Values were clipped to $[0, 1]$ and converted to the Gaussian Pearson parameter $r_p = 2 \sin(\pi r_s / 6)$; the two negative estimates remained archived and contributed zero reliable variance. With h the mean inverse half-repeat count, trial responses had reliable variance r_p and noise variance $(1 - r_p)/h$. Condition-mean noise was divided by the actual repeat count. This matches the pooled Pearson second moment; its Spearman calibration is approximate because repeat counts vary. An independent 1,000-dataset audit found mean and maximum absolute Spearman discrepancies of 0.0020 and 0.0069, without retuning. A prespecified perfect-reliability comparison removed measurement noise.

In 4,000 simulated complete datasets, all pairwise similarities were computed from joint response matrices. The original partial-rank statistic controlled ranked Euclidean separation and contact-depth difference; scan effects were averaged within target. Positive detection required a positive mean effect and an exact two-sided signed-rank $P \leq 0.05$ over the seven targets. All 128 sign assignments were enumerated. For seven nonzero untied effects, the minimum two-sided P is 0.015625, and six of 128 null sign assignments reject at 0.05. This discreteness does not imply a universal power ceiling below one. Gaussian draws were paired across signal fractions and reliability scenarios; Wilson intervals describe Monte Carlo uncertainty.

With measured reliability, the first sustained tested 80% crossing was $\lambda = 0.60$, bracketed by 0.55–0.60. Descriptive interpolation gave $\lambda = 0.589$ and a mean simulated target-level partial-rank effect of 0.249. Detection at $\lambda = 0.60$ was 80.85% (95% Monte Carlo interval, 79.60–82.04%); the lower interval first exceeded 80% at $\lambda = 0.65$. At maximal specified signal, detection was 96.23% (95.59–96.77%). The zero-signal two-sided rejection rate was 4.95% (4.32–5.67%). Perfect reliability reduced the interpolated variance fraction to 0.206, with a similar mean partial-rank effect of 0.248 at detection. These sensitivity estimates assume the specified Gaussian ancestry covariance, reliability calibration and known shared-input dependence; they do not exclude weaker effects, other covariance patterns, unknown biological dependence or alignment involving unobserved partners.

9.7 External signed-credit analysis and task-interference identity

At the neuronal level, the framework gives a simpler prediction: when a causal task assigns opposite learning signs to two neurons, their measured plasticity-related contrasts should separate along the corresponding signed coordinate.

2134 The following reanalysis tests that prediction but does not address dendritic morphology. The Francioni et al.
 2135 workbook was downloaded from the source-data link of the published article; file identity was verified by SHA-256
 2136 be1b87481d0f1d1bffc85d8a4cabfd37b7055d640306afab5b3622de0ced8264. The animal-level contrasts in
 2137 sheet “Ext 13E” were used directly. The endpoint was the difference in soma–dendrite residual, in z-score units,
 2138 between error-reduction and error-increase epochs. P+ and P- denote populations assigned opposite causal signs in
 2139 the source brain–computer-interface mapping. For the source-defined P+ and P- groups, we required the table’s
 2140 signed-separation column to equal P+ minus P- to numerical precision. The orthonormal common and signed
 2141 projections sum exactly to total two-coordinate energy. Sheet “5E” supplied the four neuron-level distributions in
 2142 the display but was not used to increase the sample size of the animal-level test.

2143 For $n = 6$ animals, the one-sided exact random-sign test enumerated all 2^6 sign assignments. Bootstrap intervals
 2144 sampled animals with replacement. The directions were fixed by the experimental mapping: P+ positive, P- negative,
 2145 and P+ minus P- positive. This is a retrospective test because the prediction was formulated after publication of the
 2146 experiment, although it was evaluated from the numerical source table rather than the displayed figure.

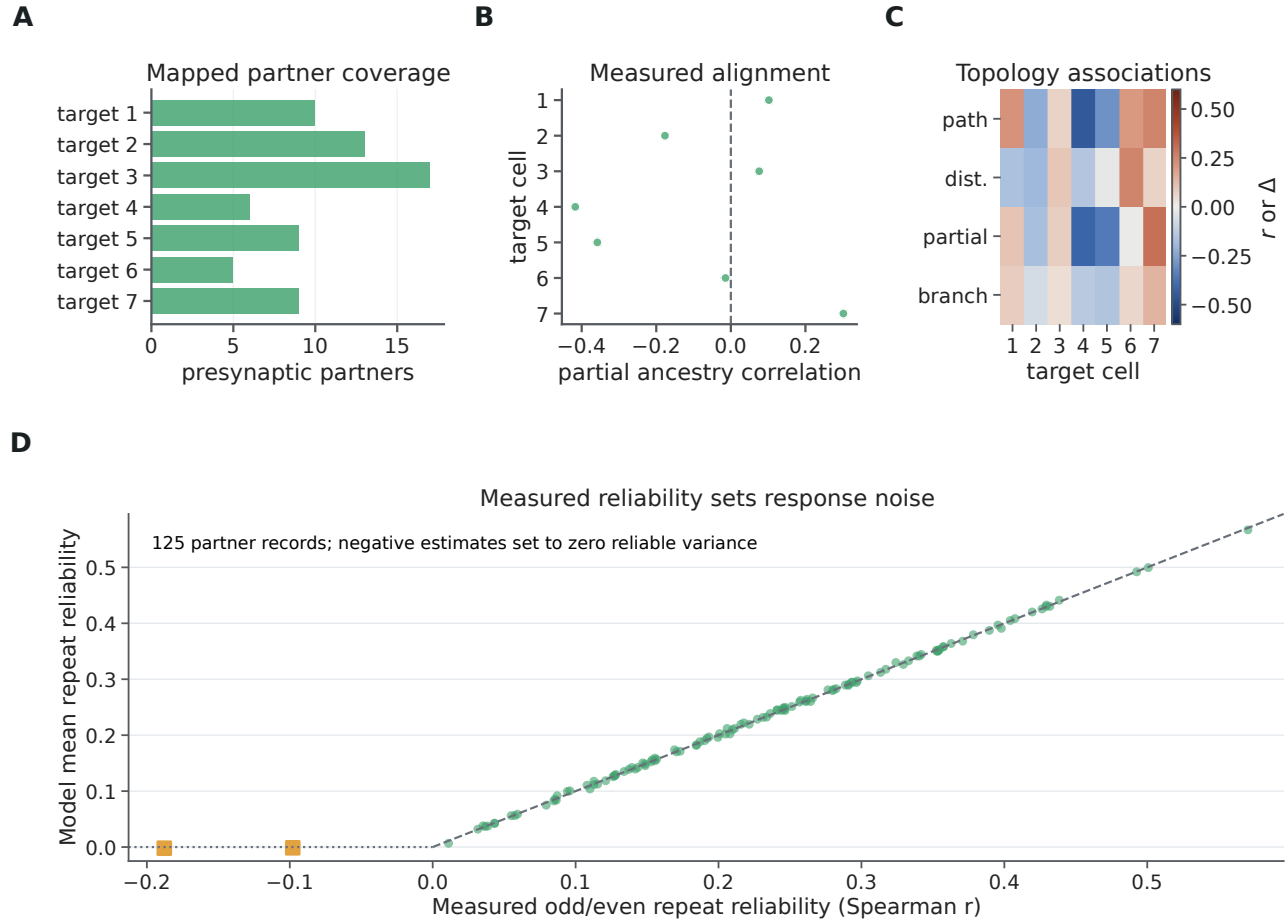


Figure S30: Measured topology associations have limited target and response reliability. **A**, Mapped presynaptic partner counts for seven targets. **B,C**, Target-level shared-ancestry/response association and alternative path, tree-distance, partial-correlation and same-major-branch measures. Partial correlations control Euclidean separation and soma-to-contact depth differences. The target, not the partner pair, is the inferential unit. **D**, Calibration of the sensitivity model against measured odd/even-repeat Spearman reliability for 125 partner records. Two negative estimates are assigned zero reliable variance; the identity line is not fitted. One thousand independent calibration datasets estimate model mean reliability. Main Fig. 9 reports alignment and detection sensitivity using all thirteen scans from these seven targets. The detected-effect scale is conditional on this covariance/noise model and sampling, not a universal detectable correlation.

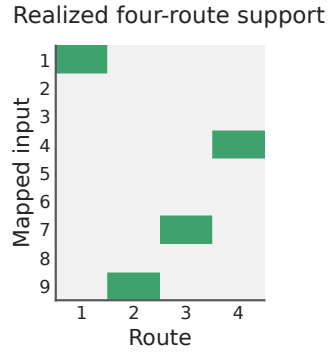
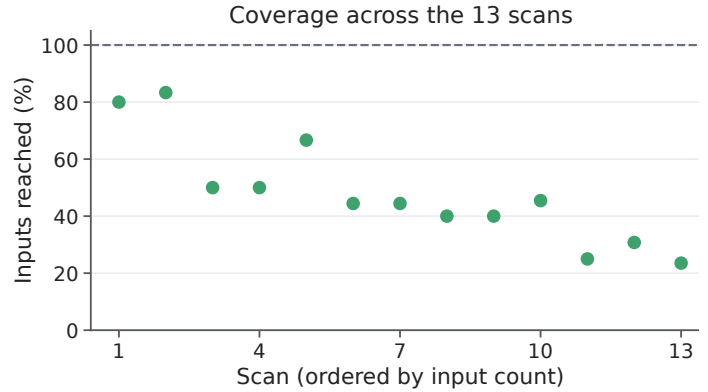
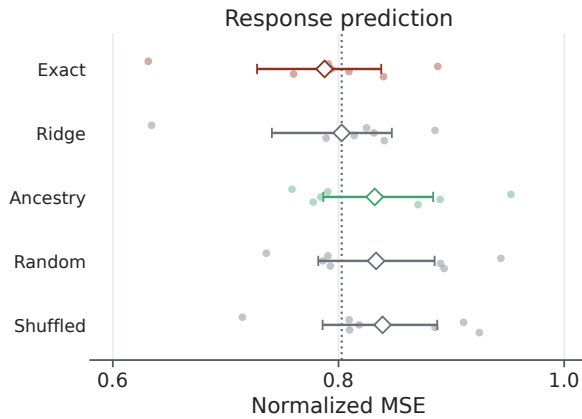
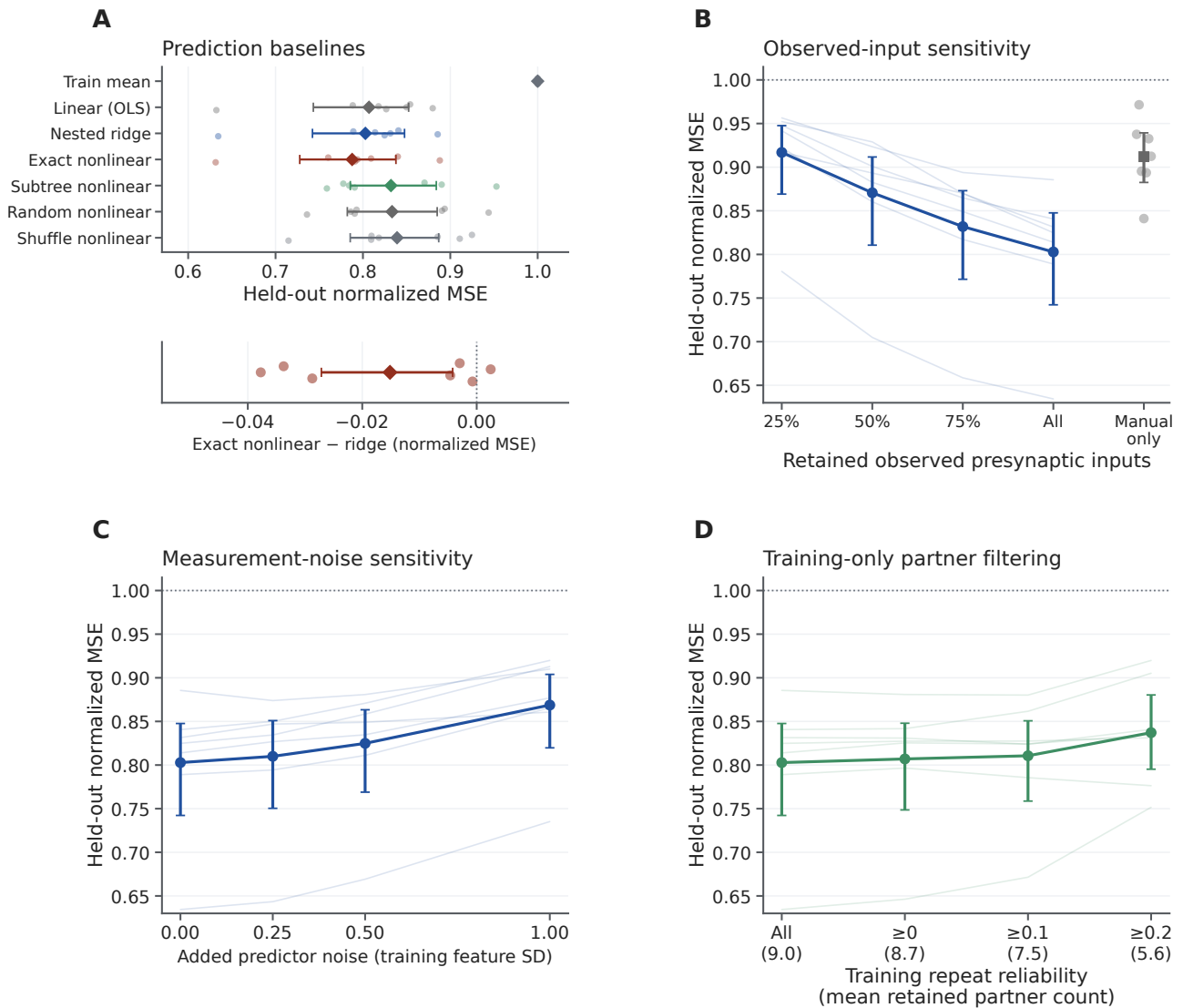
A**B****C****D**

Figure S31: The measured-response learning comparison is limited by transfer geometry. **A**, Actual four-route support for the scan selected by median mapped-input count, with identifier tie breaking; rows are input coordinates and columns selected routes. **B**, Input coverage in all thirteen scans, ordered by input count. The selected supports are mostly single distal sites, rather than illustrative nested bands. **C**, Response-prediction NMSE for exact learning, treeless ridge and restricted dictionaries; dots are seven target means, diamonds and bars their means and 95% target-bootstrap intervals. The dotted line marks ridge. **D**, Common-checkpoint update reconstruction by the unrestricted fixed transfer profile, the restricted ancestry profile and oracle trialwise ancestry amplitudes. The unrestricted fixed profile is nearly sufficient because somatic error mainly scales a stable transfer vector; restricted rules underperform ridge. Scans and splits are averaged within targets before inference. This is an offline passive-tree geometry diagnostic, distinct from the empirical ancestry-response association.



Seven targets; thin traces/dots: target means; diamonds/lines: means and 95% target-bootstrap intervals.

Figure S32: Linear baselines explain most measured-response prediction, and degrading observed inputs worsens performance. Thirteen scans retain the complete-tree analysis's 130 outer stimulus-identity splits. All preprocessing, reliability masks and ridge tuning use training identities only. **A**, Training-mean, ordinary least-squares (OLS) and nested ridge-regression baselines against the matched nonlinear fits. Lower normalized MSE is better. Below, paired exact-nonlinear-minus-ridge differences show the small additional nonlinear benefit: mean -0.0152 (95% interval -0.0271 to -0.0042). **B**, Ridge prediction when retaining nested random fractions of observed presynaptic inputs; five masks per split are averaged before inference. Manual-only partners are a separate provenance restriction, not a random retention dose. Empty masks use the training mean and remain included. **C**, Added Gaussian predictor noise in both training and test data, in units of training input standard deviation (SD) after the quantile transform. **D**, Reliability filtering based only on training repeats; parentheses give mean retained partner counts. Thin traces or dots show seven target means after averaging splits within scan and scans within target. Larger symbols and bars are means and 95% target-bootstrap intervals (20,000 draws). All targets come from one mouse. Only ridge is refitted under the sensitivity conditions. Removing observed inputs does not recover unknown partners, establish population power or explain away the absence of preferential ancestry alignment.

2147 **S10 Structure selection, statistics and reproducibility**

2148 **10.1 Prospective costed selection in tree-constrained linear learning**

2149 These secondary studies test structure selection separately from the main credit comparisons. The first asks
 2150 whether an initialization descent bound can select a model for later learning. Five explicit 15-node trees supplied
 2151 eight labeled leaves: four balanced hierarchies with distinct leaf orders and one contiguous comb. Each supplied
 2152 orthonormal subtree-cut dictionaries at $K = 1, 2, 4, 8$, giving 20 candidates per arm. Every candidate retained 64
 2153 trainable leaf-input weights. In the feedback-only arm, all candidates used identity forward transfer $H_T = I_8$. In
 2154 the joint arm, H_T was the leaf block of $(I_{15} + \kappa L_T)^{-1}$, where L_T was the full-tree weighted Laplacian, $\kappa = 1$, unit
 2155 leak was present at every node and edge conductance was inverse embedded length. Internal nodes were retained.
 2156 For input $\mathbf{x} \in \mathbb{R}^8$, trainable weights $W \in \mathbb{R}^{8 \times 8}$ and an imposed context decoder $\mathbf{a} \in \mathbb{R}^8$, the scalar prediction was
 2157 $\mathbf{a}^\top H_T W \mathbf{x}$. Here W denotes model weights. This passive-tree linear abstraction uses an imposed context readout,
 2158 not a somatic voltage or the nonlinear physical-depth model.

2159 A random orthogonal teacher mapped isotropic Gaussian inputs to labels. Context decoders were sampled
 2160 uniformly from the first r columns of an orthogonal basis Q , with $r \in \{1, 2, 4, 8\}$ specifying the reference credit
 2161 rank. Observation-noise standard deviation (SD) was $\sigma_y = 0$ or 0.75 . At zero initialization, the exact reference
 2162 decoder-space credit second moment was

$$S_0 = (1 + \sigma_y^2) Q \text{diag}(\underbrace{1/r, \dots, 1/r}_r, 0, \dots, 0) Q^\top.$$

2163 Rotation preserves this reference spectrum, but the joint arm’s transfer can change transported-credit and parameter-
 2164 gradient spectra. Angle and noise comparisons shared underlying random draws within seed and rank. Five
 2165 development seeds (6100–6104) used angles $0, \pi/4, \pi/2$; 20 separate confirmation seeds (7200–7219) used held-
 2166 out angles $\pi/8, 3\pi/8$. The 16 rank–noise–angle conditions within each confirmation seed were paired, giving 320
 2167 tasks per arm.

2168 For dictionary projector P , the actual synaptic descent direction was $\mathbf{d}_i = P H_T^\top \mathbf{a}_i \delta_i \mathbf{x}_i^\top$, where δ_i was the
 2169 scalar prediction error on example i . Inner products and covariances use the vectorized 64-coordinate update.
 2170 Coefficients receive the known decoder and transfer, rather than a learned local encoder. Each candidate had
 2171 64 fixed task-decoder coefficients, K channels, $8K$ possible context-to-route encoder coefficients, eight nonzero
 2172 delivery entries, 15 electrical nodes and 14 edges. Let ℓ_T be the total cable length in the fixed two-dimensional
 2173 embedding and $\ell_{\max}$ the largest candidate length. The normalized geometric cable cost was $\ell_T / \ell_{\max}$; the resource
 2174 cost was $C_T = K/8 + 0.1 \ell_T / \ell_{\max}$. This declared cost proxy does not measure biological expenditure. Candidates
 2175 with identical projectors but different cable costs remain identified as equivalence classes, not replicates.

2176 Independent calibration, training and test sets contained 1,024, 2,048 and 4,096 examples. Calibration
 2177 evaluated the actual routed update mean and empirical single-example covariance, giving $\mathbb{E} \|\mathbf{d}_{\text{batch}}\|^2 = \|\mathbb{E} \mathbf{d}_i\|^2 +$
 2178 $\text{tr Cov}(\mathbf{d}_i)/32$ for minibatches of 32 independent and identically distributed (iid) draws. The initialization score
 2179 was the bound at the actual step $\eta = 0.35$,

$$U_T = \eta \mathbf{g}^\top \mathbb{E} \mathbf{d}_i - \frac{\eta^2 L_T^{\text{cal}}}{2} \left(\|\mathbb{E} \mathbf{d}_i\|^2 + \frac{\text{tr Cov}(\mathbf{d}_i)}{32} \right).$$

2180 Here L_T^{cal} is the largest Hessian eigenvalue of calibration half mean squared error (MSE), global in W . The bound
 2181 concerns calibration loss, not independent-test or population loss. All weights started at zero. Paired candidates
 2182 received the same iid minibatch stream for 256 updates, with checkpoints at 16, 64 and 256. Only the final
 2183 checkpoint defined the primary endpoint, test half-MSE plus $0.08 C_T$.

2184 Development’s 4,800 fits fixed one nonnegative utility-to-final-loss scale $s \in [0, 256]$ per arm, using within-task
 2185 centered outcomes. The moment policy minimizes $\mathcal{L}_{\text{cal}}(0) - s U_T + 0.08 C_T$. Development also fixes the constant,
 2186 maximum-budget and known-rank policies; sampled and exact-expected random choice are references. All 3,200
 2187 confirmation choices were hash-sealed before training without retuning. The prospective protocol retains all 12,800
 2188 confirmation fits and 38,400 checkpoint rows, with no outcome exclusions.

2189 Regret subtracts the retrospectively best candidate’s costed loss on each task; this oracle is unavailable to
2190 selection. Intervals use 10,000 bootstrap draws over 20 whole-seed averages. Moment-policy regret was 0.08055
2191 (95% interval 0.07262–0.08781) for feedback-only selection and 0.10404 (0.09975–0.10791) for joint selection.
2192 Fixed maximum-budget regret was 0.04495 and 0.04269; rank-only regret was 0.00831 and 0.00821. The moment
2193 selector lost to both in every seed average in both arms. Rank-only improved over fixed by 0.03664 (0.03630–
2194 0.03697) and 0.03448 (0.03427–0.03468). The retrospectively best budget equaled the imposed rank on every
2195 confirmation task, while the contiguous balanced tree won 308/320 feedback-only and 303/320 joint tasks. Thus
2196 the constructed family supports rank-dependent capacity but little fine tree selection.
2197 Post hoc checks separate this failure from local prediction: the bound held in all 12,800 calibration conditions,
2198 and utility correlated 0.984/0.974 with first-update test improvement. Algebraic, covariance, derivative and source
2199 checks passed; 120 independent endpoints replayed within 5.55×10^{-17} . Complete matrices, policies and audits
2200 are in `source_data/prospective_morphology_selection/`.

2201 10.2 Finite, noisy calibration of interaction-based tree selection

2202 The finite-calibration study uses the eight-input, 28-coefficient model and twelve-candidate pool of Section 4.1, with
2203 four equally weighted families: four disjoint pairs, two complementary quartets, nested products of degrees 2/4/6/8,
2204 and a random-interaction control selecting four distinct degree-2–8 monomials. Input permutations, coefficient
2205 signs and magnitudes $\mathcal{U}[0.5, 1.5]$ vary by task; coefficients are normalized to unit squared norm. Targets have zero
2206 mean and unit variance on uniform Rademacher inputs, without matched spectra across families.

2207 The estimator receives input/output examples and all 255 nonconstant Walsh monomials, without family, planted
2208 support, true coefficients or derivatives. Lasso fits an unpenalized intercept and uses penalty $c \text{sd}(y) \sqrt{2 \log(256)/n_{\text{fit}}}$,
2209 cyclic coordinate descent, tolerance 10^{-8} and at most 20,000 iterations. Five development blocks select c from
2210 0.15/0.35/0.65, fixing 0.15. Calibration budgets 64/256/1,024 cross Gaussian label-noise SD 0/0.5. The first 75%
2211 of observations fit the estimator and pilots; the final 25% is a calibration gate. Primary calibration therefore uses
2212 192 fit observations out of 256 queries at SD 0.5.

2213 The selector minimizes the maximum normalized centered-cut tail beyond rank one, breaking 10^{-10} ties
2214 by summed tails and candidate identifier. Only true-coefficient maximum tails are bounds; estimated tails and
2215 summed-tail tie-breaking are scores. Secondary subset dynamic programming searches beyond the pool at the
2216 primary condition.

2217 Development fixes a constant candidate and an estimated-rank rule using the rank that explains 99.5% of
2218 estimated input-gradient energy. Both reduce to `balanced_p0`; sparse bins use the constant fallback. Other
2219 references are fixed shape representatives, exact uniform-random candidate expectation and a two-sweep ALS
2220 pilot. The pilot fits every candidate from one fixed initialization on the calibration-fit subset and chooses by gate
2221 MSE, then discards all pilot parameters. True-coefficient selectors and the retrospective best-trained candidate are
2222 privileged references.

2223 Twenty new seed blocks supply eighty tasks; calibration inputs, scores, choices and adaptive trees were sealed
2224 before fitting. Pre-fit implementation repairs preserved those choices and are recorded in Source Data. Final fitting
2225 uses separate 1,024-example training data with noise SD 0.3, 32 ALS sweeps, two paired random restarts and
2226 output-preserving gauge normalization. Independent 512-example noisy validation chooses the restart. Twelve
2227 fixed candidates, one estimated adaptive tree and one true-coefficient adaptive reference give 2,240 fits; fixed
2228 candidate outcomes are reused across calibration conditions.

2229 Primary evaluation uses clean-target MSE on 4,096 independent inputs and unit population variance; noisy-test
2230 and all-256-pattern errors are secondary. Streams can repeat patterns with independent observation identities/noise.
2231 Regret subtracts the best validation-selected fixed-pool result, retaining negative adaptive regret. Failed/nonfinite
2232 fits receive loss 10^6 , with warnings and counts retained.

2233 The primary 256-query/noise-0.5 comparisons against development-best fixed and the two-sweep pilot use
2234 10,000 whole-seed bootstrap draws, averaging families within seed, Bonferroni-adjusted 97.5% intervals and a
2235 0.01-NMSE margin. Other comparisons are descriptive. The exponential Walsh dictionary and ALS fitter constrain
2236 scalability and biological interpretation. Full outcomes and costs are in Source Data and Supplementary Fig. S35.

10.3 Direct confirmation from noisy morphology estimates to gradient learning

Twenty independent seed blocks 930910–930929 generate eighty new functions from the same four-family mixture. The fixed Lasso ($c = 0.15$) receives 192 of 256 calibration observations at label-noise SD 0.5; the remaining 64 are reserved, unused. All estimated trees are hash-sealed before learning. The earlier best fixed balanced tree and a post-seal true-coefficient construction are references. Construction coefficients never initialize training.

Each of three trees receives exact path credit or unit root broadcast, giving 480 fits. The seven multi-affine units, 28 coefficients and root-only output are unchanged; broadcast replaces six nonroot sensitivities by one. All conditions share Gaussian nonbias initialization of SD 0.5, zero biases and independent calibration/training/test/minibatch streams. Coefficients remain in $[-2, 2]$ and gradients are clipped at norm ten. Adam uses $(0.9, 0.999)$ and $\varepsilon = 10^{-8}$, with inherited rates 0.01 for exact and 0.003 for broadcast. Thus comparisons are between frozen recipes, not common rates.

Learning uses 1,024 updates of 64 examples from a 2,048-example cache with label-noise SD 0.15, minimizing half-MSE divided by known target variance. Separate 4,096-example test data use fresh noise. No validation selection, retuning or continuation from calibration fits occurs. Primary noisy-test NMSE has expected floor 0.0225; clean test and all-pattern errors are secondary. Within-seed family-averaged comparisons use 10,000 bootstrap draws, adjusted 97.5% intervals for the two primary contrasts and the same 0.01 effect margin. Fixed-minus-estimated exact-credit error was 0.45913 $[0.38489, 0.53227]$; broadcast-minus-exact on estimated trees was 0.60750 $[0.56916, 0.64610]$. Matching, quartet and nested tasks approached the floor with estimated-tree exact credit, while random interactions retained error and broadcast performed slightly better on matching.

Six full-horizon checks (seed 721000) are excluded; a documented pre-fit repair preserved settings and choices. Independent reconstruction reproduced all eighty trees and endpoints exactly. All 480 trajectories and validation records are in `source_data/morphology_calibration/end_to_end`. The pipeline demonstrates finite-data construction and learning within this class, not endpoint-optimal morphology.

10.4 Finite-horizon forecasts for the contextual linear learner

A finite-horizon predictor tests the failed scalar extrapolation while retaining the candidate pool, $\eta = 0.35$, minibatch size $B = 32$, 256-update horizon and cost. Five original development seeds informed its design; the predictor was frozen before fresh calibration or training. Twenty new seeds (9300–9319) crossed four ranks, two noise levels, two held-out angles, both transfer arms and both training regimes. Every candidate was trained, yielding 25,600 new fits and 76,800 checkpoint outcomes at updates 16, 64 and 256. Non-pilot scores were sealed before each seed’s outcomes without endpoint tuning. The predictor separates mode-dependent dynamics from a scalar initial slope, motivated by deep linear learning and task alignment with kernel eigenfunctions [12, 13].

To forecast learning, we express the weights in route coordinates. For orthonormal routes Φ , the learner stays in $W = \Phi V$. For context j with decoder $\mathbf{a}_j$, define the effective route vector $\mathbf{c}_j = \Phi^\top H_T^\top \mathbf{a}_j$. The scalar prediction is $\mathbf{c}_j^\top V \mathbf{x}$ and the actual update is minibatch descent in V . Let $\mathbf{v} = \text{vec}(V)$ stack the route-coordinate weights. For calibration example i with input $\mathbf{x}_i$, label y_i and effective route vector $\mathbf{c}_i$, define $\mathbf{z}_i = \text{vec}(\mathbf{c}_i \mathbf{x}_i^\top)$. With $\mathbb{E}_i$ denoting an average over the calibration-fit examples, set $H_{\text{emp}} = \mathbb{E}_i[\mathbf{z}_i \mathbf{z}_i^\top]$ and $\mathbf{b}_{\text{emp}} = \mathbb{E}_i[y_i \mathbf{z}_i]$. Write $H_{\text{emp}} = U \text{diag}(\lambda) U^\top$, with orthonormal eigenvectors in U and eigenvalues in λ . After T updates, full-batch descent from zero gives

$$\mathbf{v}_T = U \text{diag}\left(\frac{1 - (1 - \eta\lambda)^T}{\lambda}\right) U^\top \mathbf{b}_{\text{emp}}, \quad (119)$$

The diagonal expression is evaluated separately at each eigenvalue, with continuous value ηT when that eigenvalue is zero. The empirical selector fits the first 512 calibration examples and evaluates on the other 512; Eq. 119 is exact for that full-batch surrogate, not for the original cached-data SGD process.

For fresh inputs $\mathbf{x} \sim \mathcal{N}(0, I_d)$, where d is the input dimension, a smaller recurrence describes stochastic gradient descent (SGD). Let p_j be the context probabilities and $\mathbf{t}_j \in \mathbb{R}^d$ the target coefficient vector in context j . Suppose $y = \mathbf{t}_j^\top \mathbf{x} + \varepsilon$, with independent zero-mean noise of variance σ_j^2 . The exact recurrence closes on the mean

weights $M = \mathbb{E}[V] \in \mathbb{R}^{K \times d}$ and the route-space variance matrix $\Gamma = \mathbb{E}[(V - M)(V - M)^\top]$. The latter sums weight covariance over the input coordinates; it is not the full vectorized parameter covariance. Define

$$C = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top, \quad D = \sum_j p_j \mathbf{c}_j \mathbf{t}_j^\top, \quad \bar{G} = CM - D, \quad (120)$$

$$R_j = \|M^\top \mathbf{c}_j - \mathbf{t}_j\|^2 + \mathbf{c}_j^\top \Gamma \mathbf{c}_j, \quad A_\eta = I - \eta C, \quad (121)$$

$$J = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top [(d+2)R_j + d\sigma_j^2], \quad K_G = C\Gamma C + \bar{G}\bar{G}^\top. \quad (122)$$

Here $\bar{G}$ is the mean gradient, R_j is the expected squared residual excluding observation noise, J is the second moment of the single-example gradient Gram matrix, and K_G is the corresponding second moment of the conditional-mean gradient. For a fixed residual coefficient vector $\mathbf{e}$, Gaussian fourth moments yield $\mathbb{E}[(\mathbf{e}^\top \mathbf{x} - \varepsilon)^2 \|\mathbf{x}\|^2] = (d+2)\|\mathbf{e}\|^2 + d\sigma_j^2$. Conditioning first on the current weights, the Gram of a minibatch-mean gradient is its conditional-mean Gram plus $1/B$ times its conditional variance. Total covariance therefore gives

$$M_{t+1} = M_t - \eta \bar{G}_t, \quad (123)$$

$$\Gamma_{t+1} = A_\eta \Gamma_t A_\eta^\top + \frac{\eta^2}{B} (J_t - K_{G,t}), \quad (124)$$

$$\mathbb{E}[\mathcal{L}_T] = \frac{1}{2} \sum_j p_j [R_j(T) + \sigma_j^2]. \quad (125)$$

The forecast updates noise with the residual and weight variance. It requires Gaussian input fourth moments and fresh independent minibatches; noise needs only the stated first two moments. This closure does not cover nonlinear targets, anisotropic inputs, adaptive optimizers or correlated samples.

The primary plug-in predictor estimates context frequencies, conditional coefficient vectors and residual noise variances using 1,024 independent calibration examples. Coefficients use ordinary least squares without an intercept, and noise variances use residual degrees of freedom. Its recurrence is exact for the fitted Gaussian population and is a plug-in approximation for the generating population. Population-oracle versions instead receive true teacher coefficients and noise moments and are labeled privileged. The 16-step pilot performs real candidate updates and evaluates on calibration observations; its computation is included. An observed-context-count baseline chooses the smallest permitted K covering the number of distinct observed context vectors, followed by the cheapest tree. This and an observed-count-restricted conditioning baseline were frozen after development inspection, before new outcomes. Context count equals generating rank only in this constructed family.

We ran both fresh-example SGD and minibatches resampled from a fixed 2,048-example cache, retaining a common 4,096-example test set. Reusing a cache makes later weights dependent on those examples and violates the fresh recurrence's independence assumption. Population-oracle prediction minus exact population loss of the realized weights quantifies that discrepancy. All conditions are averaged within seed before 10,000 whole-seed bootstrap draws. Pointwise intervals for secondary comparisons carry no multiplicity-adjusted confirmation claim. Numerically constant cases retain undefined correlations with explicit defined-task counts, without outcome exclusions.

Supplementary Fig. S34 shows endpoint regret and strong baselines; panel D shows the joint-arm cache sensitivity. The complete timing and prediction outcomes are in Source Data (Table S12). The primary forecast reaches mean fixed-cache regret 0.000064 and 0.000042 in the feedback and joint arms, while the cheap observed-count baseline reaches 0.000134 and 0.000197. The scalar-score improvement therefore does not establish a large fine-selection advantage. Tests verify spectral forecasts against direct descent, stochastic moments against Gaussian quadrature, sealed choices and exclusion of training/test observations from predictors. Complete data, protocols and source hashes are in `source_data/morphology_structure/` and `source_data/morphology_finite_horizon/`.

2304 10.5 Statistical analysis, reproducibility and interpretation

2305 Central comparisons report paired effect sizes, the number of positive cell or seed pairs, and 95% intervals unless
2306 a different level is specified for that analysis. Tests were two sided unless a direction had been specified before
2307 analysis. Exact Wilcoxon signed-rank tests were used for small paired cell cohorts. Reconstructed-cell intervals
2308 resampled cells first and, where relevant, one Monte Carlo stream within cell. Measured-response intervals
2309 resampled target cells. Artificial-network summaries used independently initialized training runs. Inhibitory-census
2310 outcomes were first averaged within postsynaptic target. Signed-credit intervals and exact sign tests used animals;
2311 neuron-level source rows were descriptive only.

2312 Bootstrap intervals used 20,000 draws unless an experiment's protocol states otherwise; the Fashion-MNIST
2313 ladder, the physical-depth, H4 and task-family cohorts and the external animal reanalysis used 50,000 paired draws.

2314 Benjamini–Hochberg correction was applied within each prespecified comparison family—the archived four-
2315 budget subtree scan across $K = 1, 2, 4, 8$, the six bandwidth-matched routing contrasts and the H4 primary family.
2316 The additional four-budget ancestry-minus-best-control reanalysis enumerated all 2^{20} paired sign assignments
2317 for a two-sided mean-difference test and applied Holm correction across the four budgets. Accuracy differences
2318 were reconstructed on the exact $1/2048$ test-count grid before testing. Initialization-score correlation intervals
2319 used 5,000 whole-seed bootstrap draws. The local-context soft-encoder analysis used Holm correction across its
2320 three primary contrasts; the hard-readout follow-up remains exploratory. The prospective selection experiment
2321 resampled 20 whole seed blocks 10,000 times. The finite-horizon, morphology-calibration, credit and conductance
2322 studies also used 10,000 whole-seed draws, retaining task, candidate and optimizer conditions together within each
2323 seed. The two primary comparisons in each of the finite-calibration and separate end-to-end experiments used
2324 Bonferroni-adjusted 97.5% intervals and a prespecified 0.01 improvement margin in normalized MSE (NMSE).
2325 Other bridge intervals are pointwise and descriptive. Exhaustively enumerated structural targets have no biological
2326 sampling interpretation. Response-baseline sensitivity retained seven targets and 20,000 draws. Other parameter
2327 sweeps and sensitivity analyses received no multiplicity correction, are labeled secondary, and are not interpreted
2328 as independent confirmation.

2329 Cross-mouse contrasts are summarized within volume; cell-level tests do not provide an animal-level P value
2330 from two mice. The 12-cell Pinky cohort was sized to span 12 equal-count spatial strata while keeping complete
2331 mesh skeletonization and 500-shuffle processing feasible. The task-family factorial used ten paired seeds, matching
2332 the sample size used for the physical-depth experiments; positive directional language requires a paired 95%
2333 interval excluding zero and at least 8/10 seed signs.

2334 Cross-architecture conclusions exclude pooled error-field metrics that mix dendritic stages with a somatic stage
2335 whose reconstruction is exact by construction. They also exclude unweighted path-gain dispersion as evidence of
2336 gradient direction and post hoc comparisons between unmatched training cohorts. Table S10 records the supported
2337 interpretation and boundary of each evidence class.

2338 Each empirical main panel has a machine-readable table with values, independent and nested units, seeds,
2339 conditions and analysis version. Automated checks verify these figure inputs. Authentication tokens, restricted
2340 caches, and raw data already hosted by MICrONS or DANDI are not redistributed.

2341 Source Data retain configurations, outcomes, exclusions, selections and hashes; software contains implementa-
2342 tions, joins and figure builders. Manifests authenticate source identities and path translations without changing
2343 numerical settings. The source index maps compact tables to complete records.

2344 An exact package lock was not archived for the June exact-transport or backpropagation jobs, nor for the
2345 measured-response analysis. These versions are therefore not guessed from the current environment. The complete
2346 July feedback analysis records Python 3.10.13, its Git identity and selected source hashes. Its archived source state
2347 nevertheless includes tracked modifications. Table S3 and the accompanying reproducibility README distinguish
2348 exact, partial and unverifiable provenance fields.

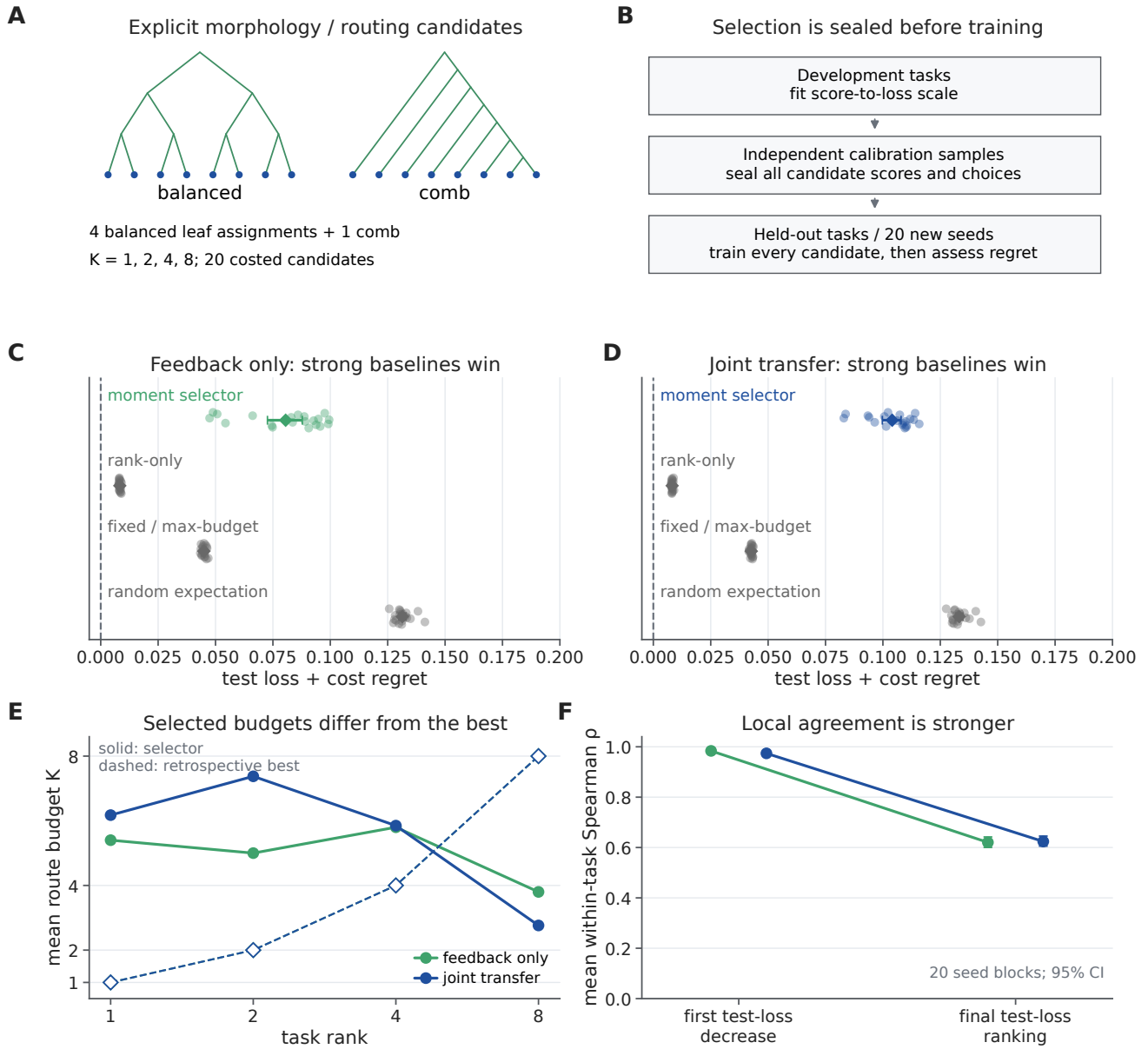


Figure S33: The original scalar initialization score predicts a first step but fails at prospective endpoint selection. **A**, Four balanced leaf assignments and one comb supply route budgets $K = 1, 2, 4, 8$: twenty candidates with prescribed cost; K counts feedback channels. **B**, Development, independent calibration, sealed choices and confirmation training. **C,D**, Final test half-mean-squared-error (MSE) plus cost, expressed as excess over the best trained candidate, for feedback-only and joint passive-transfer arms, respectively. Small points are twenty whole-seed means; diamonds and whiskers are means and 95% seed-bootstrap intervals (10,000 draws). The development-fixed/max-budget and privileged generating-rank baselines outperform the moment selector; random expectation averages all candidates uniformly. **E**, Selected mean budget versus imposed rank; dashed curves show the retrospectively best candidate's budget. **F**, Mean within-task Spearman correlation of initialization utility with actual first-step test-loss decrease and final test-loss ranking. Intervals resample whole seeds. All 12,800 confirmation candidate fits and 38,400 checkpoints remain included. Every model has a freely trained dense weight matrix; passive transfer changes conditioning and cost but does not necessarily restrict representation. This experiment is distinct from the locally connected multi-affine construction and the separately frozen finite-horizon follow-up.

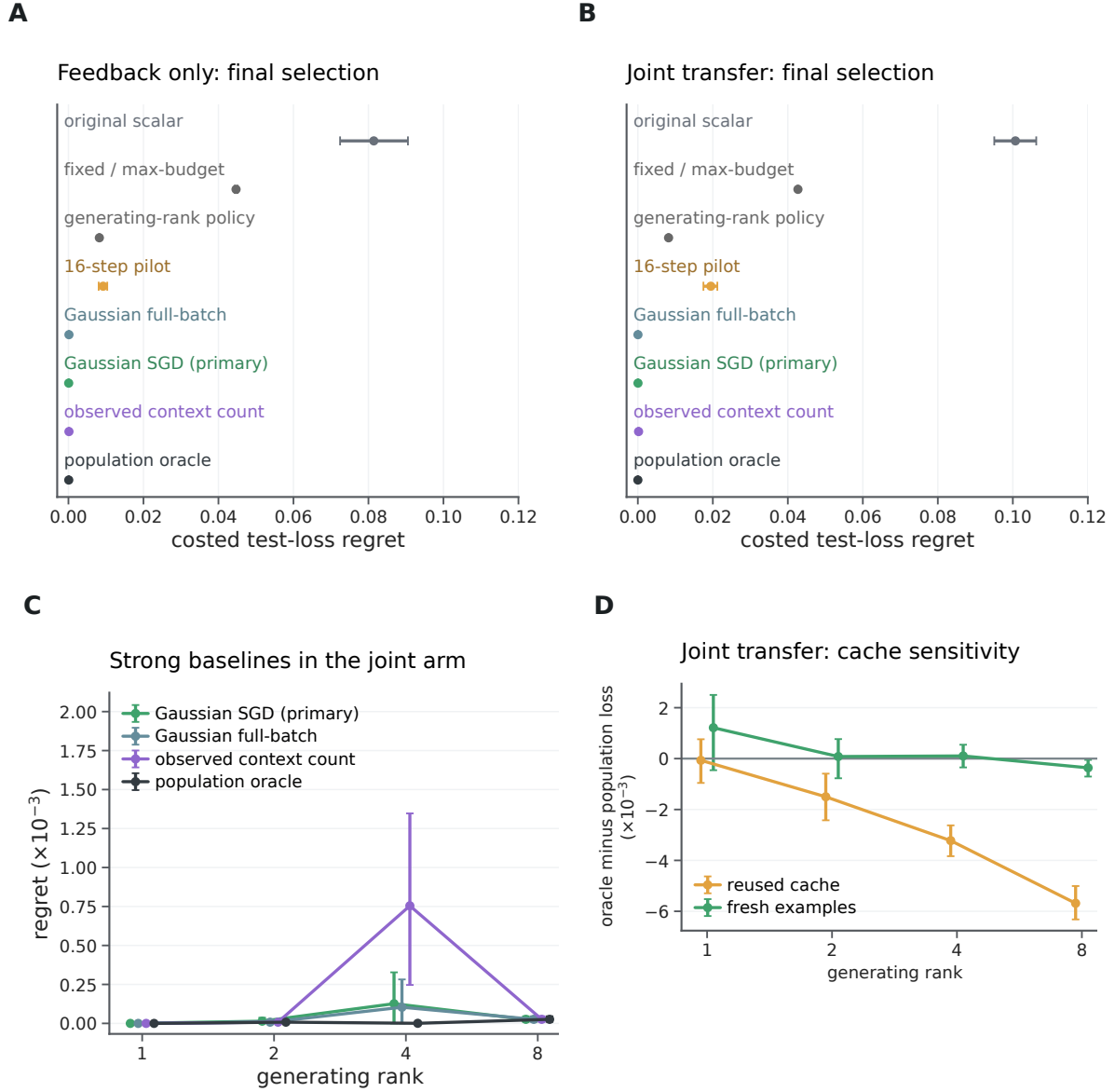


Figure S34: **Finite-horizon prediction improves selection but strong simple baselines remain.** **A,B**, Final costed-loss regret in feedback-only and joint-transfer fixed-cache learners. **C**, Strong selectors in the joint arm. The independently calibrated Gaussian SGD forecast improves over the original scalar utility; the full-batch and observed-context-count baselines remain competitive. All finite-horizon policies choose $K = r$ in this family, so improvement does not establish fine morphology selection. **D**, Population-oracle forecast minus realized population loss in the joint arm under fresh examples or finite-cache reuse. Curves and intervals resample twenty independent seed blocks with pointwise 95% bootstrap intervals. The fresh-example recurrence is exact under the stated Gaussian/conditional-linear assumptions; cache reuse violates independence. These are linear-learning forecast tests, not a general nonlinear dendritic morphology law.

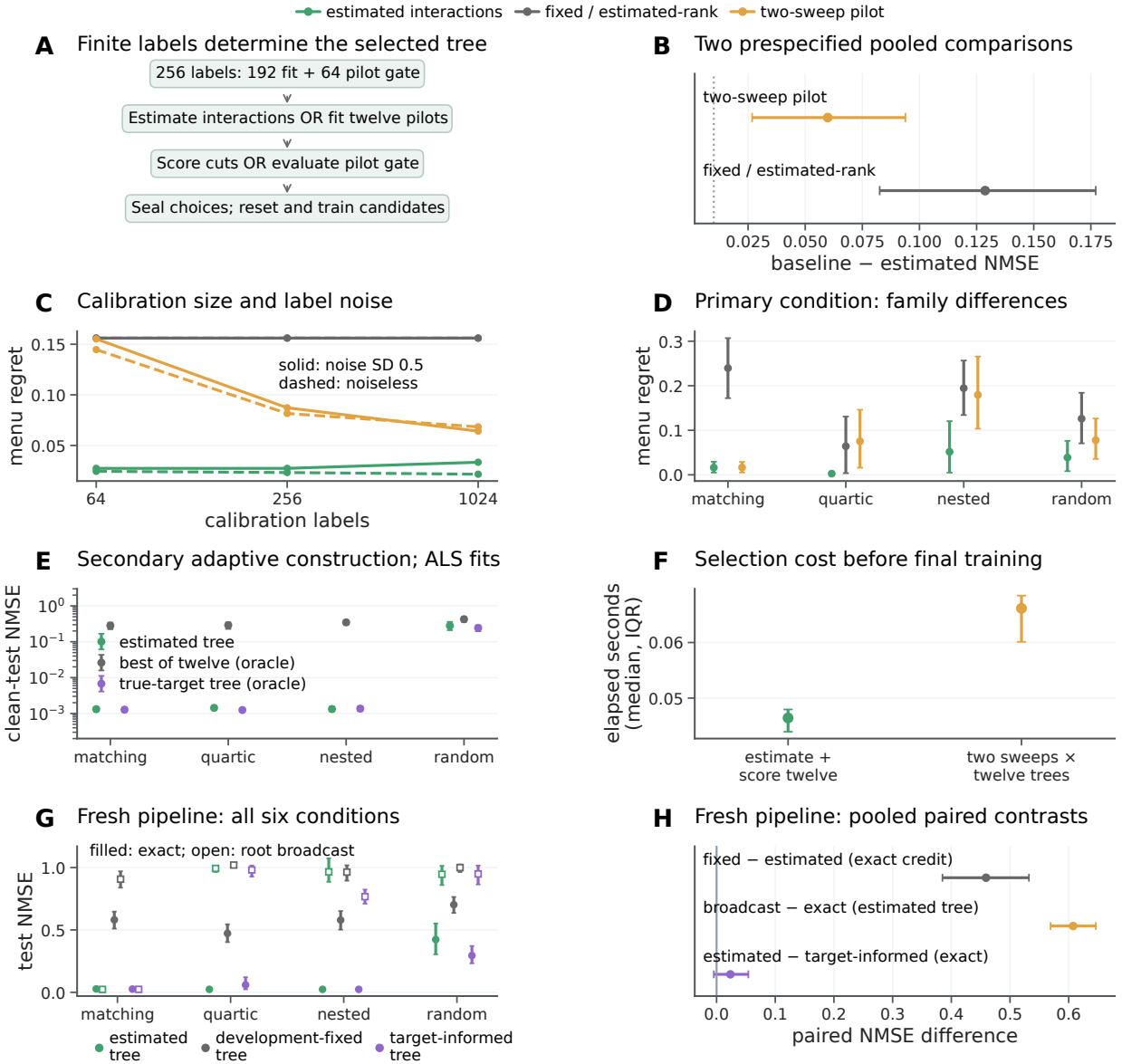


Figure S35: Finite calibration, adaptive construction and direct gradient learning. **A**, Primary calibration: 256 labels, noise standard deviation (SD) 0.5; 192 observations for estimation/pilot fitting and 64 for pilot gating. The 255-term Walsh estimator receives no family labels, true support or coefficients. The pilot fits all twelve candidates for two sweeps. Choices are sealed; training restarts independently. **B**, Prespecified pooled fixed-baseline-minus-estimated and pilot-minus-estimated normalized mean squared error (NMSE). Whiskers are Bonferroni-adjusted 97.5% paired bootstrap intervals; the dotted line marks the 0.01-NMSE meaningful-improvement margin. **C**, Excess clean-test NMSE above the best trained candidate across calibration sizes; solid curves use noise SD 0.5 and dashed curves no noise. **D**, Family-specific primary regret for estimated interactions, the development-fixed/estimated-rank baseline and the fitting pilot; pooled superiority is not uniform across families. **E**, Secondary adaptive trees trained with alternating least squares (ALS), versus the best trained twelve-candidate menu and a full-target construction oracle. Random-interaction errors remain. **F**, Elapsed time (median and interquartile range) on one CPU for eighty tasks: estimation/scoring of twelve trees versus pilot fitting/gating, excluding final training. **G,H**, Independent twenty-seed Adam cohort. **G**, Three structures and two credit rules in four non-isospectral families: filled symbols, exact credit (Adam 0.01); open, root broadcast (0.003). Label-noise floor: 0.0225; **E** uses clean targets. **H**, Fixed-minus-estimated error under exact credit and broadcast-minus-exact error on estimated trees use Bonferroni-adjusted 97.5% intervals; estimated-minus-target-informed error under exact credit uses a pointwise 95% interval. Target-informed construction is not a performance ceiling. Other whiskers: pointwise 95% intervals, 10,000 whole-seed bootstrap draws ($n = 20$); pooled statistics average families within seed. Both cohorts are independent of Supplementary Fig. S12.

Table S1: Core notation. Symbols are defined at first use; this index distinguishes quantities that recur in different coordinate systems.

Symbol	Meaning
$V_n, a_n = f_n(V_n)$	Compartment voltage and transmitted activity.
$g_i, x_i, g_i x_i; E_i^{\text{rev}}$	Maximal conductance, presynaptic activity, open conductance and reversal potential; E/I polarity does not require signed conductances.
$G_n^{\text{tot}}, R_n^{\text{tot}}$	Total open conductance and its reciprocal input resistance.
$\delta_n^V, \delta_n^a, q_n$	Errors in voltage/activity coordinates; root-to-unit output derivative. In reciprocal cables, q denotes the current-source adjoint.
Φ, K, c	Spatial-profile dictionary, column budget and delivered coefficients. Rank can be smaller than column count.
Γ	Route-gain matrix in the delivery equation; the rectangular anatomical kernel of Eq. 101 is separately defined.
$W, P_{\Phi, W}; \mathcal{C}_K$	Positive coordinate weights, weighted projector and explicitly specified per-field or aggregate energy capture.
g, d, m, V	Exact gradient, routed-update estimator, its conditional mean and covariance.
M, Σ, U	Fixed routing operator, estimator-noise covariance and one-step moment bound.
$H_c, D_r; H_p, D_p$	Task-credit hierarchy/routed resolution, and latent task hierarchy/physical stage count.
χ, α, a	Branch conflict, physical-task sensor alignment and imposed route-span energy fraction, respectively.
G, R_a, R_m, κ	Reciprocal conductance matrix, axial resistivity, specific membrane resistance and focal shunt dose.

Table S2: Delivery profiles and coefficient sources. A useful span and an implementable coefficient rule are separate requirements.

Comparison	Field, information source and model
Images	Twelve nonsomatic activation coordinates per $[3, 3]$ tree. Neuron-specific error is supplied by the exact linear readout; $K = 1/K = 3$ coefficients project the exact activation field. Local output derivatives remain in eligibility.
Quadratic screens	Sixteen abstract parameter coordinates; exact/noisy projections and supplied or paired-probe reliability estimates. No forward neuron is simulated in the Haar screens.
Context and ancestry	Eight terminal blocks receive $\delta_i \Phi_{c_i, \cdot}$. Context is supplied; the separate cue encoder learns coefficients before task training.
Multi-affine trees	Six nonroot unit errors receive exact paths, a unit or initial signed profile, or oracle-projected coefficients. Best-rank-one capture is a separate common-state diagnostic.
Conductance learning	Six nonsomatic voltages. Three-profile coefficients use current exact paths. Two leaf profiles plus unit proximal credit retain oracle amplitudes. The hard local rule instead gates distal somatic error by parent inhibition, retaining unit proximal credit; all 24 conductances learn.
Reconstructed arbors	Weighted input sites; broadcast plus ancestry supports. Free projection coefficients measure capacity for modeled passive fields. No encoder or biological learning is measured.
Measured-response models	Four-route spans contain one frozen baseline-transfer profile scaled by output error. Trial-wise field/update oracles are geometry diagnostics and are not trained.

Table S3: Regular-tree execution records. Missing historical software versions are not inferred from a current environment. Complete configurations, manifests and exclusion ledgers accompany Source Data.

Cohort	Retained provenance
120 MNIST feedback runs	Configurations, checkpoints, logs and hashes; Python 3.10.13; source identities 6c1aaa25/bd5f905e. Scientific signatures and executable sources match across fields within architecture. Heterogeneous NVIDIA devices.
60 gradient-diagnostic runs	Same records; Python 3.10.13, source 4ba38bde with archived tracked modifications and explicit source hashes; H200, 4,006 summed GPU-hours.
640 depth-feedback runs	480 conductance-valid retained; configurations, curves, resource records and the complete validity ledger. Later inactive-source edits were checked for numerical equivalence. Heterogeneous NVIDIA devices, 73.29 summed GPU-hours.
160 diagnostic checkpoints	120 valid; five fields/four step sizes per checkpoint, paired/clustered analysis and complete exclusions. No new training.
320 exact/BP repeats	Unmodified root 74792ca, nested generator 4f3612a5, configurations, seeds, checkpoints and scheduler records; H200. Signed additive and ReLU shunting inputs remain distinct.
20 exact-factorial / five BP references	Configurations, checkpoints, logs and grouped outcomes on A100-SXM4-40GB; 2.953/0.474 summed GPU-hours. Exact commit, Python and package versions were not recorded.

Table S4: Repeated exact-transport/BP outcomes, averaged over four depths within each of ten paired seeds. Accuracy is a percentage and the contrast is percentage points, with a 20,000-draw paired 95% interval. The full seed contrasts and descriptive tests remain in Source Data.

Task and core	Exact	BP	Exact minus BP (95% interval)
MNIST, raw additive	97.013	97.008	0.005 (−0.028 to 0.041)
MNIST, shunting	96.960	96.973	−0.013 (−0.047 to 0.022)
Noisy lines, raw additive	33.378	33.490	−0.112 (−0.465 to 0.247)
Noisy lines, shunting (ReLU)	36.690	36.788	−0.098 (−0.653 to 0.390)

The 320 runs form 160 method pairs. One depth-averaged MNIST additive seed was tied. Exact transport is an information-oracle implementation control; an interval containing zero is not an equivalence test.

Table S5: Outcome-independent conductance-validity accounting for the artificial-tree experiments. “Included” counts retained training runs, or checkpoints in the diagnostic row; these unit types are not added together. The entire inhibitory-dose family is excluded because its prespecified endpoint compares against invalid signed-input shunting cells.

Experiment	Planned units	Invalid shunting	Unmatched controls	Included units
Depth by feedback	640 runs	160	0	480
Matched routing	160 runs	40	0	120
Inhibitory dose	400 runs	200	200	0
Spatial topology	320 runs	80	0	240
Fixed-budget depth	320 runs	160	0	160
Checkpoint diagnostic	160 checkpoints	40	0	120

The exclusion rule is task plus core plus resolved transfer activation; no accuracy or gradient outcome enters it. The complete 1,840-run validity table and all included row tables are supplied as Source Data.

Table S6: Complete reconstructed-tree learning across all eligible scans. Means average ten held-out-stimulus splits within each of 13 scans, then scans within each of seven target cells. Mean squared error (MSE) is normalized to the training-mean predictor. Each restricted learning rule scales one fixed site profile in a four-route span by scalar prediction error. Common-checkpoint update match is one minus normalized squared update error for that prescribed profile, rather than the best attainable projection into the full span.

Teaching field	Normalized held-out MSE	Common-checkpoint update match	Fits
Exact compartment error	0.7877	1.0000	130
Topology routes	0.8321	0.2322	130
Random routes	0.8333	0.3584	130
Site-shuffled routes	0.8391	0.4392	130

Positive MSE improvement favors topology. Topology improved over site shuffling by 0.0070 (95% interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113). All restricted dictionaries had identical effective rank within run. None of 520 fits was excluded. A separate common-state diagnostic at 130 reexecuted exact-learning checkpoints gave per-trial eligibility-weighted within-span oracle captures of 0.2436, 0.3847 and 0.4677 for ancestry, random and shuffled routes, respectively. Those oracle values are geometric upper bounds at the evaluated states; no oracle-feedback learning accuracy was measured.

Table S7: Retrospective six-animal signed-credit reanalysis of Francioni et al. source sheet “Ext 13E”. The contrast is error-reduction minus error-increase soma–dendrite residual, in the source z-score units. P+ and P- denote populations with positive and negative causal decoder weights. Directions follow this published mapping; intervals bootstrap paired animal-level contrasts, retaining animals as the inferential unit.

Contrast	Mean	95% interval	Ordered animals
P+	0.0753	[0.0295, 0.1214]	5/6 positive
P-	-0.1498	[-0.2054, -0.0866]	6/6 negative
P+ minus P-	0.2251	[0.1416, 0.3108]	6/6 positive

Exact one-sided random-sign permutation P values were 0.0313, 0.0156, and 0.0156. The signed and common modes contained 83.7% and 16.3% of pooled squared projection energy across the six animal contrast vectors, respectively. The pooled fraction is a ratio of summed energies, not the mean of animal-level fractions. The signed fraction was 45.8–97.1% across individual animals, with an animal-bootstrap 95% interval of 60.6–95.7% for the pooled fraction and a leave-one-animal-out range of 76.5–90.2%. Animal-level values are provided in Source Data.

Table S8: Terminal outcomes in the balanced longer-budget follow-up. Rates are the original development-selected Adam rates. NMSE values are mean (median) across the same twenty previously examined seed blocks. The final column counts exact/profile fits with clean full-domain population NMSE below 0.02; this descriptive threshold was not used for stopping, selection or inference. Complete seedwise outcomes, validation-selected states, common-rate and SGD controls remain in Source Data.

Updates	Task	Exact NMSE mean (median)	Initial-profile NMSE mean (median)	Clean error < 0.02 exact/profile counts
1,024	Pairwise	0.0317 (0.0231)	0.0236 (0.0236)	19/20; 20/20
1,024	Quartic	0.1516 (0.0465)	0.9813 (1.0277)	17/20; 0/20
1,024	Nested	0.0230 (0.0229)	0.7690 (0.7829)	20/20; 0/20
4,096	Pairwise	0.0232 (0.0231)	0.0240 (0.0241)	20/20; 20/20
4,096	Quartic	0.0461 (0.0458)	0.9624 (1.0205)	20/20; 0/20
4,096	Nested	0.0233 (0.0232)	0.7913 (0.7853)	20/20; 0/20
8,192	Pairwise	0.0231 (0.0231)	0.0243 (0.0243)	20/20; 20/20
8,192	Quartic	0.0463 (0.0463)	0.9811 (1.0353)	20/20; 0/20
8,192	Nested	0.0233 (0.0233)	0.8039 (0.7918)	20/20; 0/20
16,384	Pairwise	0.0232 (0.0232)	0.0246 (0.0248)	20/20; 20/20
16,384	Quartic	0.0464 (0.0461)	0.9994 (1.0769)	20/20; 0/20
16,384	Nested	0.0232 (0.0232)	0.7944 (0.7826)	20/20; 0/20

Table S9: Same-seed physical-depth reproducibility. All 430 repeated fits used versioned source a99c3a7 and passed completeness, finite-metric and resource checks. Accuracy differences are percentage points. This is computational repetition of existing seeds.

Measure	Value
Repeated H2/grouped-point and principal H3 fits	220 + 210
Median absolute original/repeated accuracy difference	0.030
Mean absolute difference	0.154
Maximum absolute difference (raw-additive control)	15.01
Pairs within 0.1 points	357/430
Repeated H2/H3 aligned serial-BP depth effects	30.84 / 30.86
Repeated H2/H3 architecture-by-placement interactions	32.43 / 31.27

Table S10: Claim and evidence index. Each row identifies the question answered by the corresponding controlled comparison; scope and inferential details are stated once in its methods.

Question	Evidence and interpretation
Which spatial distinctions matter?	Image, conflict and ancestry comparisons hold the forward model fixed while changing delivery. Their scope is within-arbor credit with the stated somatic signal.
Can the tree represent the task?	Centered-cut theorem and Boolean constructions characterize the scalar multi-affine class; they do not prove optimizer convergence.
Does a fixed profile learn?	Matched algebraic tasks and longer budgets separate learned credit geometry and finite-training performance. Oracle capture is distinct from a trained rule.
Can a local rule supply selection?	Fresh conductance gates use parent inhibition and somatic error without exact paths. Both signals are supplied; autonomous error generation is unresolved.
Does physical depth help?	Serial/divisive controls support task-dependent forward benefit. Longer training reverses the LocalCA accuracy effect and retains metric dependence.
What does anatomy supply?	Budgeted broadcast-plus-ancestry dictionaries compress modeled passive responses; topology-preserving surrogates test finer structure.
Can conductance change route gain?	The passive partition identity and reconstructed-cable interventions establish modeled transport; the weak-channel ensemble checks a local linearization.
Are the routes used in measured responses?	Seven-target analyses provide no evidence of preferential alignment. Simulated sensitivity is conditional on recorded coverage/reliability and the specified Gaussian model.
Can an initial score choose morphology?	The original scalar selector lost to a fixed tree. Finite-horizon and interaction estimators answer narrower selection questions under their declared priors.

Table S11: Cohort and protocol index. Counts refer to runs within a study, not sums of independent replications. Detailed seed ranges, pairing, stopping and selection are given in the cited sections and source configurations.

Question	Units and retained design	Protocol section
Exact/rule screens	Fifty seeds per screen; fixed/adaptive conductance reliability 2,250/1,050 outcomes; same-span 4,800 checkpoint rows	S1; probes/checkpoints nested within seeds.
Images	Original MNIST/Fashion/CIFAR: 15/10/20 seeds, 90/60/80 fits; random feedback 120 new fits; fresh dictionary 108 development and 190 fresh fits	S2; twelve short dictionary checks excluded.
Depth/input controls	480/640 depth, 120/160 ownership, 240/320 spatial and 160/320 fixed-budget fits valid; separate 320-run exact/BP repeat	S2; Table S5.
Branch/ancestry	Twenty conflict seeds; ten two-branch seeds; twenty eight-context seeds and 2,700 fits; cue encoder 2,160 fits	S3; hard cue readout exploratory.
Interaction/Boolean capacity	Census of 140 primary +24 nested targets; 164 constructions. Boolean: seven templates, fifteen trees and 49 exact constructions	S4; complete-domain analyses.
Algebraic learning	Twenty fresh seed blocks each: 3,600 restricted-credit and 1,200 matched-profile fits; Boolean 6,720 fits. Extension reuses 720 original algebraic trajectories	S4; no extension retuning/new seeds.
Conductance learning	Grouping 1,440 fits; first context study 240 fresh; opposed tuning 320 fresh; wider bounds 640 same-seed trajectories; added-rate development 288 fits	S5; distinct models/cohorts.
Local conductance gate	Twenty new seeds, nine rules, two tasks, three rates: 1,080 trajectories to 16,384 updates	5.2; protocol frozen after prior exploratory exposure.
Physical composition	Ten seeds per cohort; H3 270, point/optimizer 200, dose 90, grouped H3/H2 220, H4 360, task families 360; same-seed repeat 430 and budget extension sixty	S6; finite budgets and metric dependence retained.
Anatomy	Eight initial +47 disjoint cells in one mouse; twelve selected/ten eligible Pinky cells in a second. Equal-broadcast: 65 operators, 149,331 dictionary evaluations	S7; separate cohort inference.
Inhibitory contacts/shunts	Census twenty targets, 92-axon sensitivity. Focal studies: eight initial cells, 45/47 disjoint cells eligible; 64 weak-channel draws per initial cell	7.5, S8; draws/sites nested in cells.

Question	Units and retained design	Protocol section
Measured responses	Seven targets/thirteen scans; ten splits, 520 full-tree fits, 130 geometry replays and 3,380 baseline evaluations	S9; target is inferential unit.
Detection sensitivity	4,000 complete simulated datasets across 21 signal levels/two reliability scenarios; 1,000 repeat-calibration datasets	9.6; simulations are not biological replicates.
Structure selection	Original scalar 12,800 confirmation fits; finite-horizon 25,600 fresh fits; noisy calibration 2,240 and end-to-end 480 fits	S10; sealed choices and independent observations.
Imposed/external controls	Eight arbors with forty streams each; six external animals	7.6, 9.7; imposed and retrospective outcomes.

Table S12: Source index for endpoint grids and metadata. All paths are relative to the released `source_data/` directory. These records retain the complete per-unit outcomes, contrasts and exclusions behind condensed or omitted print tables; figure-source mappings provide the individual panel joins.

Evidence	Machine-readable record
Two-branch controls	<code>trained_subtree_address/condition_summary.csv</code> ; <code>trained_subtree_address/paired_contrasts.csv</code>
Ancestry budgets	<code>trained_subtree_address_full_factorial/condition_summary.csv</code> ; <code>review_evidence_reanalysis/ancestry_control_contrasts.csv</code>
Moment bound/reanalysis	<code>credit_phase_theory/primary_contrasts.csv</code> ; <code>review_evidence_reanalysis/pooled_correlations.csv</code> ; <code>review_evidence_reanalysis/moment_scores.csv</code>
Fixed/adaptive reliability	<code>positive_conductance_reliability_step_consistent/paired_contrasts.csv</code> ; <code>adaptive_conductance_reliability/paired_contrasts.csv</code>
Same-span learning	<code>same_span_coefficient_learning/paired_contrasts.csv</code> ; <code>same_span_coefficient_learning/condition_summary.csv</code>
Original image outcomes	<code>mnist_feedback_ladder/condition_summary.csv</code> ; <code>figure2/feedback_gradient_runs.csv</code> ; <code>figure2/exact_transport_and_backprop_runs.csv</code>
Strict scalar audit	<code>mnist_feedback_ladder/strict_scalar_implementation_audit.csv</code> ; <code>mnist_feedback_ladder/strict_scalar_paired_contrasts.csv</code>
Valid depth/ownership	<code>prospective_input_validity/central_valid_condition_summary.csv</code> ; <code>prospective_input_validity/routing_valid_paired_contrasts.csv</code>
Spatial coverage	<code>spatial_topology_audit/owner_metrics.csv</code> ; <code>spatial_topology_audit/summary.csv</code>
Fixed-checkpoint step	<code>prospective_input_validity/mechanism_feedback_summary_valid.csv</code> ; <code>prospective_input_validity/mechanism_paired_contrasts_valid.csv</code>

Evidence	Machine-readable record
Operating-regime grids	regular_tree_regimes/competence_summary.csv; regular_tree_regimes/depth_scaling_summary.csv; regular_tree_regimes/broadcast_noise_summary.csv
Morphology inputs/positive controls	figure3/cell_metrics.csv; figure3/routing_capacity_summary.json; figure3/typed_only_compression_curves.csv
Passive-field controls	reciprocal_routing/cell_method_capture.csv; reciprocal_routing/summary.json; anatomy_commonmode/
Focal scale/reversal sensitivity	figure4/summary.json; figure4/scale0p1_summary.json; figure4/scale1p0_summary.json; figure4/irev0_summary.json; figure4/irevm0p5_summary.json
Physical/weak-channel shunts	physical_cable_sensitivity/cell_primary_contrasts.csv; focal_selectivity_active_ensemble/condition_summary.csv
Measured target/scan meta-data	figure5/functional_target_metrics.csv; figure5/functional_summary.json; functional_topology_all_scans/cell_metrics.csv
Branch-model restricted credit	figure5/task_target_method_means_ch4.csv; figure5/task_summary_ch4.json
Inhibitory census controls	microns_inhibitory_routes/target_metrics.csv; microns_inhibitory_routes/inhibitory_axon_metrics.csv; microns_inhibitory_routes/inhibitory_3d_matched_target_metrics.csv
Imposed alignment	alignment_controlled/cell_alignment_metrics.csv; alignment_controlled/cell_paired_contrasts.csv
Physical rerun outliers	physical_depth_clean_source_replication/historical_concordance.csv; physical_depth_clean_source_replication/seed_outcomes.csv
Algebraic budget endpoints	credit_rule_extension/summaries/all_budget_outcomes.csv; credit_rule_extension/summaries/compact_terminal_table.csv

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